Simulations and design of photonic circuits and setups with the SiN platform



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Abstract (English)

The silicon nitride (Si_3N_4) platform for photonic integrated circuits has been the subject of intensive research efforts. Over the years it has exhibited low propagation losses, large transparency window, high tolerance to temperature variations, and low fabrication and processing cost, coming from its compatibility with the CMOS technology for the realization of integrated optical circuits. For this reason, the platform has contributed to the demonstration of Photonic Integrated Circuits (PICs) for a very large variety of applications such as refractive index sensors, microwave filters, multiplexers, and others, based on its elevated passive optical characteristics. In parallel, SOI-based photonics has invaded intra-Data-Center transmission systems with transceiver devices that can support high-speeds featuring low power consumption and cost. Nevertheless, the fabrication of Si_3N_4 transmitters for intra-Data-Center communication systems, providing elevated characteristics in comparison with Si, would require the co-integration of Si_3N_4 passive waveguides with active optical elements such as modulators, lasers, and amplifiers based on III-V materials and the ferroelectric barium titanate oxide material $BaTiO_3$ (BTO).

This PhD Thesis deals with the simulation and design of photonic integrated devices and circuits on the Si_3N_4 platform, towards its co-integration with active devices, for intra-Data-Center optical interconnect applications. Initially, the Thesis presents the analysis and design of an optical coupling configuration for efficient light transfer from an InP-based Quantum Dot (QD) laser to a Si_3N_4 waveguide. The demonstrated structures are ideal for the monolithic co-integration of lasers and passive waveguides on the same chip. The optimum device is based on a tri-layer Si-rich nitride intermediate waveguide, exhibiting coupling losses near 1dB and back-reflections below -52.6dB. Then, efficient light coupling between Si_3N_4 waveguides and heterogeneously integrated plasmonic BTO phase-shifters utilizing CMOS-compatible materials such as Copper (Cu) is thoroughly analyzed and simulated. The two plasmonic waveguides under study that perform the phase modulation are based on a Hybrid Plasmonic Waveguide (HPW) and a Plasmonic Slot Waveguide (MIM). Extensive simulations revealed that the optimum structure utilizes the MIM waveguide, achieving propagation losses of 0.78dB/μm, a confinement factor of 56.18%, and coupling losses below 1.29dB per interface. Finally, with the employment of MIM phase shifters, the design analysis of plasmo-photonic Mach-Zehnder in Ring (MZIR) modulators based on noble metals, gold (Au) and silver (Ag), and CMOS-compatible metals, Copper (Cu) is demonstrated. The modulators exhibit a reduction to the required voltage in comparison with traditional Mach-Zehnder Interferometers, close to 46% for Ag, 39% for Au, and 30% for the Cu-based MZIR. The devices can deliver 112Gbaud PAM-4 signals with only $2x1.3V_{pp}$ driving voltage retaining signal Q-Factors above 5. The whole work enables the realization of a Si_3N_4 high speed transmitter as an alternative low-cost solution to legacy SOI-based solutions for co-packaged optics.

Abstract (Greek)

Η πλατφόρμα νιτριδίου του πυριτίου (Si_3N_4) για ολοκληρωμένα οπτικά κυκλώματα έχει αποτελέσει αντικείμενο αυξημένης έρευνας λόγω πλεονεκτημάτων όπως χαμηλές απώλειες διάδοσης, διαφάνεια σε πολύ μεγάλο συχνοτικό εύρος, μεγάλη ανοχή σε θερμοκρασιακές μεταβολές, καθώς επίσης και χαμηλό κόστος κατασκευής και επεξεργασίας, λόγω της συμβατότητας με τη τεχνολογία CMOS για την κατασκευή ολοκληρωμένων κυκλωμάτων. Η πλατφόρμα αυτή έχει οδηγήσει στη κατασκευή πολλών Φωτονικών Ολοκληρωμένων Κυκλωμάτων που έχουν συμβάλλει σε πληθώρα εφαρμογών όπως αισθητήρες δείκτη διάθλασης, μικροκυματικά φίλτρα, πολυπλέκτες και άλλες, βάσει των βελτιωμένων παθητικών οπτικών χαρακτηριστικών της. Παράλληλα, η φωτονική πυριτίου (Si) έχει εισβάλλει σε συστήματα μετάδοσης μέσα σε κέντρα δεδομένων με διατάξεις οπτικών πομποδεκτών που υποστηρίζουν υψηλές ταχύτητες με χαμηλή κατανάλωση ενέργειας και χαμηλό κόστος. Εντούτοις, η κατασκευή οπτικών πομπών για συστήματα επικοινωνιών μέσα σε ένα κέντρο δεδομένων στη πλατφόρμα Si_3N_4, που θα προσέφερε βελτιωμένα χαρακτηριστικά σε σχέση με το Si, απαιτεί την συν-ολοκλήρωση των παθητικών κυματοδηγών Si_3N_4 με ενεργά στοιχεία, όπως διαμορφωτές, lasers, και ενισχυτές, βασισμένα σε ενεργά οπτοηλεκτρονικά υλικά, όπως υλικά της ομάδας III-V και το σιδηροηλεκτρικό υλικό $BaTiO_3$ (BTO).

Η παρούσα διδακτορική διατριβή πραγματεύεται τη προσομοίωση και σχεδίαση φωτονικών ολοκληρωμένων κυκλωμάτων και διατάξεων στην πλατφόρμα νιτριδίου του πυριτίου (Si_3N_4), με στόχο τη συνολοκλήρωση της με ενεργές δομές, για εφαρμογές οπτικών διασυνδέσεων μέσα σε κέντρα δεδομένων. Αρχικά, η εργασία παρουσιάζει την ανάλυση και σχεδίαση ενός οπτικού συζεύκτη για τη βέλτιστη μεταφορά φωτός από ένα laser κβαντικών τελειών βασισμένο στο InP σε έναν κυματοδηγό Si_3N_4. Οι δομές που παρουσιάζονται είναι ιδανικές για την μονολιθική συν-ολοκλήρωση των lasers και των κυματοδηγών στο ίδιο chip. Η βέλτιστη δομή βασίζεται σε έναν τριών στρωμάτων κυματοδηγό νιτριδίου του πυριτίου με προσμίξεις πυριτίου (Si-rich nitride), παρουσιάζοντας απώλειες σύζευξης κοντά στο 1dB και ανακλάσεις στο laser κάτω από -52.6dB. Έπειτα, αναλύεται και προσομοιώνεται η σύζευξη φωτός μεταξύ κυματοδηγών Si_3N_4 με ετερογενώς ολοκληρωμένους πλασμονικούς διαμορφωτές φάσης βασισμένους στο υλικό BTO και αξιοποιώντας CMOS-συμβατά υλικά, όπως ο χαλκός (Cu). Οι δύο πλασμονικοί κυματοδηγοί υπό μελέτη που εκτελούν τη διαμόρφωση φάσης βασίζονται στον υβριδικό πλασμονικό κυματοδηγό (HPW) και τον πλασμονικό κυματοδηγό τύπου σχισμής (MIM). Η βέλτιστη δομή αξιοποιεί τον κυματοδηγό MIM επιτυγχάνοντας απώλειες διάδοσης 0.78dB/μm με παράγοντα δέσμευσης οπτικής ισχύος ίσο με 56.18% και απώλειες σύζευξης κάτω από 1.29dB ανά διεπαφή σύζευξης. Τέλος, με τη χρήση διαμορφωτών φάσης MIM, παρουσιάζεται η ανάλυση και σχεδίαση συντονισμένων πλασμο-φωτονικών διαμορφωτών Mach-Zehnder in Ring (MZIR) με βάση ευγενή μέταλλα, χρυσό (Au) και άργυρο (Ag), καθώς και CMOS-συμβατά μέταλλα, όπως ο χαλκός (Cu). Οι διαμορφωτές παρουσιάζουν μείωση στην απαιτούμενη τάση σε σχέση με παραδοσιακά συμβολόμετρα Mach-Zehnder κατά 46% για τον Ag, 39% για τον Au και 30% για τον Cu. Οι διατάξεις είναι ικανές να μεταδώσουν σήματα 112Gbaud PAM-4 με μόλις $2x1.3V_{pp}$ τάση, διατηρώντας τιμές του παράγοντα Q μεγαλύτερες από 5.

"In my work I now have the comfortable feeling that I am so to speak on my own ground and territory and almost certainly not competing in an anxious race and that I shall not suddenly read in the literature that someone else had done it all long ago. It is really at this point that the pleasure of research begins, when one is, so to speak, alone with nature and no longer worries about human opinions, views and demands. To put it in a way that is more learned than clear: the philological aspect drops out and only the philosophical remains."

Heinrich Hertz, In Davis Baird, R.I.G. Hughes and Alfred Nordmann, Heinrich Hertz: Classical Physicist, Modern Philosopher (1998), 157

"I am here to support the assertion that light of every kind is itself an electrical phenomenon—the light of the sun, the light of a candle, the light of a glowworm."

Heinrich Hertz, From Lecture (20 Sep 1889) delivered to the German Association for the Advancement of Natural Science and Medicine, Heidelberg, 'On the Relations Between Light and Electricity', Miscellaneous Papers (1896), 313, as translated by D.E. Jones and G.A. Schott.

Chapter 1: Introduction

1.1: Overview of Intra- and Inter- Data-Center communications

Optical communications have evolved to dramatic data rates the last few decades with the increasing demand for high capacity high-speed and low cost transmission and processing systems, for applications such as telecommunications, cloud computing, internet, Artificial Intelligence (AI) and others. The first form of optical communication came with the invention of the Optical Telegraph in the 1790s, developed in France by Claude Chappe [1]. The telegraph utilized networked towers with movable arms being able to transmit coded messages across long distances. In the 1950s research by Harold Hopkins and Narinder Singh Kapany, in combination with other works, eventually led to the invention of a thin strand composed of a transparent glass or plastic, the optical fiber. Its first experimental demonstration came with the contributions of Charles Kao and George Hockman in 1960, proposing it as a medium for optical communication [2]. The idea of the optical fiber, was realized with further works by industrial players such as Corning, gradually decreasing the propagation losses and realizing in the 1970s low-loss fibers capable of transmitting signal to long distances without significant degradation. The birth of the laser in 1962 with works by researchers such as Robert Hall, enabled coherent, focused, powerful optical beams that could be transmitted through optical fibers and detected by photodetectors.

Data center communications employing optics have attained increasing importance during the last few years. Data rate requirements have been significantly increased with the emergence of cloud apps and AI. When a user with a mobile connection uses a cloud app, location information is processed and stored with electronic high-speed computer boards in a Data Center and moved inside the Data Center itself from rack to rack, as well from the data center to the user mobile and vice versa, via optical fiber based connections. Intra-Data-Center (Intra-DC) Communications refer to the data flow in the network inside a data center with the distance usually limited below 10km. Because the distance is low and the cost is of the highest impact in the implemented technolgies, simple intensity modulation/direct detection (IM/DD) systems are deployed. Pure amplitude modulation with formats such as 2-level Pulsed Amplitude Modulation (PAM-2) is utilized and no amplification is imposed between the start and destination nodes. In the last few years to increase line rates beyond 100Gb/s per lane, PAM-4 modulation has been implemented, where each symbol accounts to 2 bits [3], offering higher data rates with lower required device bandwidth, but at the expense of higher required Optical Signal to Noise Ratio (OSNR) for error-free transmission. Higher N-level Pulsed Amplitude Modulation (PAM-N) (such as PAM-8) has also been proposed for intra-DC communications offering higher data rates, but the power budget has repressed its broad application. At all cases, the total data rate BR (in Gb/s per lane) is A times the symbol rate SR (in Gbaud), where $A = \log_2(N)$ with N being the number of levels. Apart from these well known techniques, co-integration of photonics with electronic Digital Signal Processing chips (DSP) the last few years has increased the achievable capacity and and distance with simple and more complex equalization schemes in intra-DC.

To increase the achievable bandwidth between the nodes, two solutions have been implemented: Parallel Single Mode Fiber (PSM) or Wavelength Division Multiplexing (WDM). While PSM incorporates four parallel transmitters and fibers, increasing the system cost, Coarse WDM (CWDM) for intra-DC can achieve modules without Thermo-Electric Coolers (TEC), being the most preferable option of the two. Wavelength division multiplexing (WDM) allowing the transmission of encoded information at different wavelengths in the same optical fiber was developed in the 1990s by David Payne and Rodney Alferness, further increasing the capacity of such systems [4]. Wavelength Division Multiplexing (WDM) is one of the most well known techniques, where N laser sources are co-integrated with passive waveguides and emit light at N different wavelengths. The light from the N waveguides is guided to the same final waveguide by passing through an integrated photonic multiplexer. Thus in the final waveguide, N wavelengths are transmitted simultaneously in the same waveguide section. The total data rate transmitted in the final waveguide, considering that each laser is directly modulated with a data rate BR (Gb/s) is BR_{tot} = N x BR. Nevertheless, WDM requires Multiplexer (MUX) and Demultiplexer (DEMUX) devices integrated with the transmitters. As a result, PICs co-integrated with the transmitter and the receiver emerge as the optimum solution to reduce cost.

On the other side, Inter-Data-Center (Inter-DC) Communications refer to the networks connecting the individual data centers with distances usually above 20km. For this type of interconnects, coherent optical systems have become the key technology for Tb/s scale connectivity [5]. In a coherent communication system, a spectrally pure, low-linewidth source (with a linewidth at the order of 100kHz or below) transfers a Quadrature Amplitude Modulated (QAM) signal where information is encoded both in the amplitude and phase of the light [6]. To receive the signal on the receiver side, the signal is mixed with a signal offered by a low-linewidth Local Oscillator (LO). The received signal from the balanced photodetector is amplified and processed with Digital Signal Processing (DSP) to improve its quality. In Inter-DC communications, the presence of the modulator element and the low-linewidth laser element is of paramount importance for transceivers in optical interconnects. Additionally, Dense WDM (DWDM) requiring a MUX or DEMUX element is a given to further extend the aggregate bandwidth. PICs emerge in both cases as the optimum low-cost, efficient and ultra-compact transmitter solution for inter and intra-DC communications. Even though coherent systems have been proposed in research to increase capacity in intra-DC communications, the power, complexity and cost requirements prohibit their practical use [7, 8].

With the exponential growth of AI and cloud services, future intra-DC transmitters should exhibit bandwidths large enough to be able to deliver over 100Gbaud symbol rates per lane. The design of the optical engine entails the design of the Photonic Integrated Circuit (PIC) interfacing the active and passive waveguide sections and providing the modulator device. To scale with the demands for increasing data rates, future optical interconnects based on Complementary Metal Oxide Semiconductor (CMOS) compatible integrated photonics, suitable for mass production, delivering over 100Gbaud symbol rates per channel should be realized. Electronic circuits co-integrated with them, in combination with the PICs for inter- and intra-DC should also be able to deliver high bandwidths to be able to transmit high data rates.

1.2: Integrated Photonics for Optical Interconnects

Photonic integrated circuits over the years have been implemented in a wide variety of applications such as sensing, neuromorphic photonic accelerators, signal processing, and Light Detection and Ranging Devices (LIDARs). In this Thesis the interest will focus on the application of PIC modules in intra-DC optical interconnects. Optical interconnects, in combination with their surrounding control, storage and computing electronics, are at the center of DC network nodes. Regarding tranceiver elements for datacom, PICs offer unmatched minaturization and integration capabilities. Deploying CMOS compatible processes, PICs can be fabricated in an electronics fab in mass numbers and with low cost. The most prevalent example is the semiconductor low-linewidth tunable laser that can exhibit properties similar to large fiber lasers but with a much smaller and more tolerant to external environmental conditions device. Utilizing the best technologies for passive and active components, PICs can surpass the performance and functionalities of discrete components, especially in terms of power consumption, size and bandwidth related properties. This is showcased in many publications such as [9], where a multi-layer III-V/SiN heterogeneous laser offered comparable performance with a bulky laser but with the form factor of the integrated device. Its relative intensity noise was lower than -155 dB Hz^{-1} outside the relaxation oscillation resonance near the 2 GHz offset frequency. Combining, components for different purposes, such as laser sources, modulators and photodetectors on the same chip, PICs containing large systems on a single chip can be realized, simplifying the overall system design.

The light carrier entering an optical interconnect needs to be modulated in amplitude and/or phase in very high speeds. This is achieved through two different approaches exchanging performance for cost. The first is Direct Modulation of a Laser Source, where the amplitude of the output of the laser is controlled with an electrical signal of a certain frequency f_{elec} that is directly applied to the laser. This option provides the lowest cost and is suitable only for short reach transmission up to 2km due to the low signal quality. The second route is the External Modulation where a laser needs to be co-integrated with an amplitude modulator element. The amplitude modulator can be an amplitude modulator that translates variations in the voltage of an applied electrical signal with a frequency f_{elec} to variations of the output power from the modulator, or an interferometric modulator that translates the phase-variations caused by variations of the voltage of the applied electrical signal to variations of the output power from the interferomer based PIC. This way in both cases an electrical bit-stream can be imprinted on an optical signal. Concerning the **Direct Modulation** of laser sources, a variety of works has been presented in the literature. The Directly Modulated Laser (DML) requires either the monolithic integration of the laser of a native platform such as InP or GaAs (which are direct bandgap semiconductors) [10] or hybrid, heterogeneous [11, 12] or even monolithic co-integration [13] of the laser/amplifier on a passive waveguide platform such as Si or Si_3N_4. With their integration in transceiver devices, they have provided line rates satisfying industrial standards.

Future transceivers for datacom will require **DMLs** with symbol rates over 100Gbaud per lane (or 200Gb/s per lane in PAM-4 modulation) featuring also low power consumption and high temperature tolerance to avoid extra power consumed in cooling. For this reason, in [14] a temperature tolerant InP-based Quantum-Dot (QD)

DML operating at a wavelength of 1550nm and achieving clear open eyes with speeds above 25Gb/s (in NRZ modulation) between 15°C and 50°C was reported. In [15], an InP-based QD stucture was grown on a 5°C off-cut silicon substrate, setting the route towards monolithic amplifiers and laser sources on Si. In [16] 25Gb/s NRZ operation was achieved for up to 4000h at 85°C with an O-band higher than 14GHz bandwidth (at 95°C with 60mA bias current) DML. In [17] a short cavity distributed reflector laser was able to achieve a 55GHz bandwidth leading to the demonstration of 112Gb/s (56Gbaud) transmission in the O-band. In [18], an 8-channel 8 x 56Gb/s transmission was presented with the co-integration of membrane laser sources with Si_3N_4 AWG multiplexers on Si. A two stage equalization scheme based on a reduced Volterra filter and an artificial neural network for 1.3μm-DML on Si signal transmission was shown in [19], promising the improvement of PAM-4 signal delivery. Having such equalization schemes in mind, >300Gb/s per lane per λ was demonstrated with >100GHz DMLs in a III-V on Silicon Carbide (SiC) platform [20]. For such data rates, the system employed an entropy loaded Discrete Multi-Tone Modulation (DMT) based on an adaptive algorithm. In [21] a DFB+R laser with >67GHz bandwidth a chirp parameter of only 0.6 and a reflection tolerance up to 12.5% was shown. A two-channel DML array, 200 Gbps consuming less than 0.3 pJ/bit over 2-km transmissions was brought forth in [22], while the authors examined the capabilities of such a device for 2x200Gb/s PAM-4 transmission.

Apart from bandwidth extension over 100GHz, chirp management, being a crucial issue in DMLs, has also been discussed in [23]. A more detailed review on beyond 100 Gbaud directly modulated lasers for short-reach applications can be found in [24]. DMLs with their very small size and easy control have been and will continue to be of interest in both academic and industrial research. However, as seen in the multitude of publications above, bandwidth extension over 70GHz is difficult and the system often requires advanced digital signal processing techniques to improve the quality of the received signals in >100Gbaud rates. Chirp is also a limiting factor in DMLs for long reach (>12km) links.

External modulation is the other route for high-speed signal delivery. Even though fiber attachement and edge coupling can be used to insert light from a laser source to an external modulator element, footprint considerations in transmitter elements for datacom applications require the eventual co-integration of the laser source with the passive waveguide and the modulator device. Also, if the design of a transceiver (TxRx) is targeted, the integration of a passive waveguide with a direct bandgap photodiode is also important. In external modulation, the most well known platforms are the following: a) Modulators based on Si-photonics, b) Modulators based on III-V elements such as InP or GaAs, c) Modulators based on bulk $LiNbO_3$ or thin film lithium niobate (TFLN), d) Modulators based on 2D-materials, e) Modulators with transparent conductive oxides, f) Modulators with Organic Materials, g) Modulators with other ferroelectric materials ($BaTiO_3$). The physical characteristics along with some demonstrations of each category are presented below:

a) **Si-Photonic Modulators:** Silicon being a low cost material, encountered on a daily basis in a CMOS fab, has been one of the most well utilized platforms for electro-optic modulators. Their physical operation is based on the carrier

injection or carrier depletion mechanism. The change in the carrier concentration of the semiconductor material results to the variation of the real and imaginary part of its refractive index leading seamlessly to phase and amplitude modulators [25]. In the first demonstrations the Si-based phase-shifter utilized the change of the real part of the modal effective index propagating in an interferometric or resonant circuit. In [26], a 12μm in diameter Si-photonic microring led to one of the first high-speed ring modulators (RMs) in the literature. In [27], the self-aligned formation of a pn-junction in Si brought forth Mach-Zehnder Modulators (MZMs) with operation up to and beyond 40 Gb/s and dynamic depth of 18dB at 10Gb/s. In [28], the limiting factors and optical and electrical trade-offs of Si MZMs were explored, while in [29] Si depletion type MZMs with an extinction ratio (ER) higher than 11dB at 28 and 32Gb/s OOK modulation for a differential voltage swing of 800mV_{pp} were demonstrated by the same authors.

Regarding RMs, in 2010 [30] 1Gb/s OOK signals were transmitted with microrings with 2.5μm radius for 150mV voltage swing and 0.96V bias. In [31], a ring with 7.5μm radius delivered 25Gb/s OOK signals with >5dB ER and 1V_{pp} driving. The bandwidth limit of microrings was surpassed with the introduction of a Mach-Zehnder in a Ring coupling modulation based device achieving twice the rates achieved with phase intracavity modulation [32]. In [33], Si MZMs and RMs were integrated with passive dielectric Si_3N_4 waveguides, achieving 22Gb/s and 13Gb/s NRZ signal delivery overall. An increase in the achievable line rates has been achieved with the utilization of both more complex modulation formats such as PAM-4 modulation and through signal processing techniques after reception. In [34], a packaged Si-photonic MZM delivered 56Gbaud PAM-4 (112Gb/s) signals with a Bit Error Rate (BER) below the Forward Error Correction (FEC) limit for distances up to 20km. In [35] aggregate rates of 400Gb/s were reached with a packaged four-lane transmitter based on Si MZMs modulated by 53Gbaud PAM-4 signals. In [36] 400Gb/s rates were delivered with an 8-channel transmitter (Tx) based on low-power RMs and an 8x8 Arrayed Waveguide Grating Router (AWGR). In [37] over 100Gbaud PAM-4 rates with Travelling Wave MZMs and Resonant RMs was reported. Yet the measurements included equalization steps or artificial neural network implementations. More recently, over 300Gb/s modulation with low-power microring circuits has been demonstrated [38]. The RM of [38] employed frequency detuning of the input laser to the resonance of the modulator. The BER at 110Gbaud PAM-4 modulation after 2km transmission was at the threshold of the Soft-Decision (SD) FEC.

Apart from pure Si-platforms, SiGe amplitude modulators can be grown easily on a Si substrate. These structures deploy the bending of the energy bands with the application of voltage through the Franz-Keldysh or Quantum Confined Stark Effects in Quantum Wells. In [39] simple SiGe Electro-Absorption Modulators (EAMs) achieving >50Gbaud PAM-4 modulation and more complex optical time division multiplexing EAM-based circuits delivering 104Gbaud PAM-4 signals over a single wavelength have been reported. In [38] >200Gb/s EAMs were presented, while more recently in [40] SiGe EAMs with a 3dB-bandwidth (BW) beyond 110GHz delivering 280Gb/s PAM-4 eyes were

experimentally demonstrated. Regardless of the latest very astonishing advances in terms of speed, more complex modulation formats such as PAM-8 are more prone to transmission impairments. Also, advanced signal processing techniques are more power hungry and introduce unreasonable cost to short-reach data-center applications, while Si phase-shifters obtain an almost metallic high-loss behavior to retain high bandwidth values. Carrier based phase modulators on Si are eventually restricted in terms of bandwidth, while in resonant structures (such as RMs) the roundtrip time introduces an extra restriction to bandwidth. On the other side, SiGe EAMs require considerably higher voltages than Si RMs and have a much higher footprint.

Another problem in RMs lies with their thermal stability of the resonant wavelength, since the ER of the modulator depends on the stable position of its resonance. To address this issue dithering techniques combined with non-invasive light probes to set the working point of modulators have been introduced [41]. In [42] such stabilization mechanisms were applied to a 50Gb/s microring modulator. Compensation of wavelength and thermal variations was demonstrated with the Field Programmable Gate Array (FPGA)-based control platform ensuring 30ms response times. In [43], a Contactless Integrated Photonic Probe (CLIPP) feedback circuit securing the operating point of photonic elements [44] was applied to demonstrate a temperature and wavelength shift tolerant transmission and routing interconnect system for multi-socket interconnects. Thermal stabilization schemes were also recently adapted to be compatible with any passive dielectric integrated photonic platforms [45].

b) **III-V based Modulators:** The integrated photonic platforms based on III-V materials have the inherent advantage of being able to integrate laser sources, amplifiers, passive waveguides, modulators and photodetectors on the same chip since they mostly have a direct bandgap. Modulators on the III-V platforms mostly employ the Franz Keldysh effect or Quantum Confined Stark Effect in Quantum Well (QW) or Quantum Dot (QD) electroabsorption (EAM) or electrogain (EGM) modulators [46]. The bending of the energy bands leads to the shift of the absorption peak of the semiconductor and thus to the modulation of the output light from such an optoelectronic device. A nice overview of III-V EAM devices for >200Gb/s IM/DD Tx for short reach optical interconnects has been provided in [47]. In most cases, III-V EAMs are integrated with laser sources to create a complete EML Tx with light emission and modulation capabilities. In [48], a 56Gbaud PAM-4 InP Tx module co-integrating a distributed feedback laser (DFB) with an electro-absorption modulator (EAM) was presented. The modulator had a 50GHz BW, a >13dB ER and up to 1.5mW output power. Reception was performed after transmission over 2km and a low-complexity three-tap equalization. In [49] a 59GHz Lumped Electrode EAM-DFB delivered 214Gb/s line rates over 10km single mode fiber (SMF) in the O-band. The BER had a value <3.8e-3 and a Feed Forward Equalization (FFE) was deployed. In [50] with a 59GHz LE EAM-DFB structure and Discrete Multitone Modulation (DMT) line rates over 300Gb/s were achieved with a BER<2.63e-2. More recently, in [51] an uncooled operation of an EAM-DFB in 224Gb/s PAM-

4 signal transmission was demonstrated. Line rates up to 420 Gb/s in PAM-8 modulation were also explored. In [52], high-speed EMLs with a hybrid waveguide structure transmitted 225Gb/s PAM-4 signals over 10km distances. Clear eye diagrams were obtained with a 5-tap equalization.

Apart from amplitude modulators, high-speed phase-shifters can be realized in III-V photonic platforms. The high mobility of the carriers contribute to the increase of the speed metrics compared to other semiconductors such as Si. Despite the tremendous speeds achieved by EAMs and EML composed of III-V materials, their footprint is much larger than that of Si ring modulators and their required voltages are also usually higher.

c) **Lithium Niobate ($LiNbO_3$) Modulators:** The lithium niobate ($LiNbO_3$) platform has been the traditional platform for electro-optic (EO) modulation over the years. These devices employ the Pockels effect to modulate the real part of the effective index of the propagating mode by inducing variations to the refractive index of the $LiNbO_3$ ferroelectric material with the application of an external voltage. In the first demonstrations $LiNbO_3$ waveguides were fabricated with Ti-diffusion [53] or Zn-diffusion [54] achieving extinction ratios over 20dB and switching rates over 40GHz. Yet their large modal sizes and loose electric-field confinement led to half-wave voltage length products V_π x L over 9-10 Vcm. The ability to etch $LiNbO_3$ and form high confinement thin film lithium niobate on insulator (LNOI) waveguides has sparked a renewed interest in the platform [55]. Thin film lithium niobate (TFLN) modulators have met the needs for ultra large bandwidths in high-baud rate short-reach IM/DD and long distance coherent systems [56]. In [57], >300Gb/s line-rates in short reach links with $500mV_{pp}$ driving signal and linear equalization were demonstrated with PAM-6 and PAM-8 modulation utilizing TFLN C-band MZMs. Over 300Gb/s data rates have also been presented with IQ TFLN modulators and 80Gbaud 16-QAM modulation achieving BER values <8.41e-3 and 1.8dB on-chip losses [58]. TFLN modulators exhibit the advantage of linearity with ultra-high spurious free dynamic range (SFDR) of 120.04dB x $Hz^{4/5}$ at 1 GHz in ring-assisted MZM structures [59]. Taking the above into consideration, electro-optic modulation with V_π<1V [60] and BW>100GHz can easily be realized in the TFLN platform boasting an EO coefficient between 30-33pm/V [61].

Nevertheless, the platform is not CMOS compatible and can also not facilitate amplification, lasing and photodetection. For this reason, the LNOI platform has been co-integrated with the Si and Si_3N_4 platforms to obtain the low-cost advantage of CMOS compatible platforms through hybrid or heterogeneous co-integration technologies. This has led to line rates over 112Gb/s in such platforms [62-64]. Lasing, amplification and photodetection capabilities technologies are coming through III-V lasers and PDs via transfer printing or hybrid integration [65, 66]. Also ultra-fast tunable laser sources in hybrid platforms have been demonstrated [66] rendering possible a full photonic link. Yet, as it will be seen later, there are materials with a much larger EO coefficient than $LiNbO_3$, exhibiting a more intense Pockels effect.

d) **Modulators with 2D-Materials:** The inherent advantage of 2D materials such as Transition Metal Dichalcogenides (TMD) and graphene is the high mobility of the carriers due to the 2D Density of States. Such modulators exploit the gate-tunable optical conductivity of graphene to modulate the light intensity, phase, or polarization. Electroabsorption modulators shift the Fermi Level of the material with the applied voltage to graphene, shifting the absorption rate. Thus suitable phase and amplitude modulators on graphene can be realized. Graphene can also be integrated with Si and Si_3N_4 waveguides thus rendering possible modulators that can be embedded in totally passive architectures with the desired physical characteristics. In [67], the resonator loss modulation at critical coupling in a Si_3N_4-Graphene RM allowed EO bandwidths >30GHz with modulation efficiency of 15dB per 10V. In [68], Si- and Si_3N_4- graphene waveguides brought forth 25Gb/s OOK modulation with an ER of 5dB in 20μm long devices. In [69], dual layer graphene integrated on Si photonic modulators with a 29GHz bandwidth delivered 50Gb/s NRZ open eye diagrams with the ER being 1.3dB at 50Gb/s. More recently, in [70] a >10dB ER was demonstrated at $4V_{pp}$ voltage with a dual layer graphene integrated with Si boasting a modulation depth of >0.2dB/μm. In [71], a hybrid plasmonic waveguide embedding three graphene layers and two Hexagonal Boron Nitride (h-BN) Layers led to the numerical demonstration of a modulation depth >19.9dB/μm with 2.9dB/μm loss at 1300nm. Finally, more complex modulation formats in graphene modulators have also been deployed. In [72] Si_3N_4 waveguides integrated with layers of graphene delivered 96.24Gb/s simulated data rates in PAM-4 modulation with $3V_{pp}$ maximum voltage and 144.4Gb/s in PAM-8 modulation. The simulated insertion losses were 12dB for PAM-4 and 11.5dB for PAM-8 modulation. In [73] Si-graphene IQ modulators led to the experimental demonstration of up 40Gb/s line rates in QPSK modulation.

From the above, it is seen that 2D materials are very promising for the future of Si-photonics. Yet the challenging reproducibility and large variations in fabrication have led to a hesitant industrial side in terms of inserting 2D materials to their standard CMOS processes.

e) **Modulators with Transparent Conductive Oxides (TCOs):** TCO materials are materials that are mostly encountered in applications such as flat panel displays and photovoltaics [74]. Epsilon near-zero materials are a special class of TCOs [75]. TCOs have high transparency and conductivity, while if they exhibit an epsilon near-zero behavior large phase-shifts can be experienced with very short lengths. With the application of external voltage, the permittivity of ENZ materials can be tuned, affecting the phase and amplitude of the output light field. For this reason these materials can lead to ultra-high speed EO modulation. Tin-doped Indium Oxide (ITO), Gallium-doped Zinc Oxide (GZO), Aluminum-doped Zinc-Oxide (AZO) are some common materials in the literature. These can be deposited at 300°C and are thus compatible with the CMOS processes. These have been embedded in CMOS plasmonic or hybrid plasmonic structures and simulated, exhibiting achievable ER values of 46dB/μm in electro-absorption [76]. In [77] an electro-absorption TCO plasmonic module was

experimentally demonstrated and employed for photonic neural activation function, boasting insertion losses <1dB for a 5μm long footprint and delivering 25Gb/s NRZ line rates with a 5dB ER. In [78] an ITO-based MZM achieved a half-wave voltage length product (V_π x L) of only 0.52Vmm. In [79], the Si-MZM deployed a subwavelength plasmonic phase shifter based on heterogeneously integrated ITO films on Si-photonics with a V_π x L of only 95Vμm, insertion losses of 6.7dB and bandwidths over 1.1GHz. The Si-rib and Si-slot structures of [80] co-integrated with ITO achieved simulated V_π x L products of 0.1Vmm, delivering also >50GHz switching speeds and sub-pJ/symbol energy consumption. In [81] the comparative study of ITO and graphene modulators led to impressive calculated >100GHz bandwidths with <1dB insertion losses. In recent presentations of [82], the experimental modulation efficiency was V_π x L < 1 Vmm, the device footprint was < 1 mm^2 and the bandwidth > 100 GHz. Nanophotonic modulating elements based on TCOs are very promising for telecom, datacom and neural network applications.

f) **Modulators with Organic Materials:** Polymer organic hybrid materials can be poled with the application of an external electrical signal and can exhibit the Pockels effect with high EO coefficients. In [83] an EO system composed of two chromophores HLD_1 and HLD_2 exhibited an EO coefficient r_{33} equal to 290±30 pm/V with a stability of 500h when the system is heated at 85°C. The high EO coefficient of polymer organic materials and their potential for integration with Si-photonic waveguides brought forth silicon organic hybrid (SOH) and plasmonic organic hybrid (POH) modulators [84].

Regarding SOH modulators, the work of [85] presented a 3dB-BW>100GHz, 500μm long electrodes and a V_π x L of 11Vmm. In [86] a SOH MZM based on 280μm long phase-shifters delivered 187Gb/s signals in PAM-4 modulation with 0.7dB insertion loss, a V_π x L of 0.41Vmm. More recently, in [87] 140Gbaud PAM-4 transmission was achieved with sub-1V driving signals and a temperature stable SOH while in [88] a cryogenic MZM featured a V_π of only 0.92V leading to 70Gbaud PAM-4 transmission.

On the other side, through the ultra-high confinement of the applied electrical and optical modal fields in small areas filled with polymer organic materials, POH modulators have exhibited elevated metrics. In [89], a POH MZM with phase-shifters only 29μm long experimentally demonstrated error-free OOK and BPSK modulation at data rates up to 40Gb/s and 70fJ/bit energy consumption. In [90] a plasmon assisted ring modulator operated at over 100GHz frequencies with insertion losses being as low as 2.5dB and ERs being >10dB, delivering 72Gb/s OOK signals. Recent POH technologies for high-speed modulators have been included in [91]. In [92] compact IQ modulators with a length of 25μm and sub-1V driving electronics allowed line rates of 400Gb/s with a 2fJ/bit energy consumption. In terms of IM/DD system applicability, in [93], Si-plasmonic MZMs transmitted ultra-high bit rates of 363Gb/s with PAM-8 modulation over 10km considering a Soft Decision (SD) FEC limit. Recent Si-plasmonic racetrack modulator demonstrations have allowed CMOS-level driving voltages, EO bandwidths of 176GHz and >400Gb/s line rates with 12.3fJ/bit energy

consumption in PAM-8 modulation [94]. This RM was also able to transmit 160Gbaud PAM-4 (320Gb/s) signals with equalization steps performed after the reception of the signal. Finally, POH modulators have been integrated with electronic drivers and led to symbol rates >100Gbaud in monolithic bipolar CMOS electronic–plasmonic high-speed transmitters [95, 96].

Nevertheless, there is a great restraint and hesitancy from industry regarding the adoption of organic materials in CMOS fabrication processes. The main concerns have to do with the stability of the material over time and with their resilience to high temperatures, necessary for subsequent processes related to the packaging of the final device.

g) **Ferroelectric Modulators with Barium Titatanate Oxide ($BaTiO_3$):** Due to the comparatively low EO coefficient of $LiNbO_3$ and the thermal stability issues of polymer organics, alternative ferroelectric materials have been explored for integrated photonic modulators. Barium titanate oxide ($BaTiO_3$ or BTO) exhibits 3x-10x higher EO coefficients than the LNOI platform and can be integrated with Si or Si_3N_4 waveguides leading to waveguides that can reach propagation losses <6dB/cm [97]. The EO coefficient r_{42} of BTO films can be as high as 1300±100pm/V in the stress-free and 730±100pm/V in the stressed case [98].

With these high values regarding the EO coefficient, various modulator and switch circuits have been demonstrated in the literature. In [99], active electrically tunable hybrid Si-BTO resonators exhibited tunability values over 4μW/nm. In [100], Si_3N_4 waveguides were integrated on top of BTO thin films. Resonant ring elements based on hybrid Si_3N_4-BTO waveguides boasted power consumption for tuning of only 106nW/FSR. In [101], Si-BTO MZMs worked at 25Gb/s speeds with a static power consumption of 100nW and a V_π x L of 0.2Vcm. In [102] BTO thin films were integrated with Si-passive waveguides and photonic and plasmonic BTO phase shifters were formed. The measured EO coefficient r_{42} from photonic thin films was close to 923pm/V. With such phase shifting elements, open eye diagrams from transmitted 40Gb/s NRZ in photonic MZMs and 50Gb/s NRZ signals in plasmonic MZMs were demonstrated. In [103], 116Gb/s PAM-4 and 72Gb/s NRZ signals were transmitted from Si-plasmonic MZMs with the ERs being >15dB. Very recently, in [104], 216Gbaud OOK data transmission was achieved with MZMs based on 15μm long plasmonic phase-shifters bonded on thick Si_3N_4 waveguides.

It is clear that the footprint, the insertion losses, the extinction ratios, the CMOS compatibility and the modulation efficiency and bandwidth of BTO phase-shifters constitute $BaTiO_3$ as one of the most favorable materials for future high-speed Si-photonic interconnects.

1.3. Co-Packaged Optics Si-photonics datacom transceivers

Co-Packaged Optics (CPO) has gained significant momentum in the last few years for Intra-Data Center (Intra-DC) Communications, targeting to save around 25dB of losses coming from high-speed signal transmission in the 6'' – 8'' long electrical co-planar waveguides connecting the switching ASIC with the front panel. This high loss value necessitates the deployment of an RF amplifier increasing the total power

consumption. Nevertheless, the power envelope for each switch is at max 300W. With CPO that power can be directed towards switching instead of signal propagation. Up to 400Gb/s Ethernet (400GbE), pluggable solutions with WDM and/or PSM had been the prevalent solution for Intra-DC optical interconnects. However, with data traffic grown by an approximate rate of 30% every year [105], leading to increased bandwidth, an escalating percentage of power is consumed at the switch-optical interface [106], and the bandwidth of the Serializer-Deserializer (SerDes) lanes does not scale equally with the bandwidth of the designed optical engines. In 2023, the more advanced and complex 5nm gate electronics technology can deliver >200Gb/s SerDes Lanes. Scaling the speeds further to >380Gb/s, achievable with 130Gbaud PAM-4 Optical Engines would require even more costly electronics technologies.

While a detailed review on Co-Packaged Optics can be found in [105], the OFC 2020 presentation by Facebook presents a nice visual overview on a CPO switch and CPO architecture. An image of a CPO sub-assembly can be found in Fig. 1.3.1 (a). All optical modules are placed directly on a common electronic CPO substrate. A switch IC resides at the center of the sub-assembly controlling the optical modules. In this demonstration, laser sources are external delivering light to the optical engines via fibers and interfaces designed for CPO. Fig. 1.3.1 (b) and (c) visually depict two methods for the delivery of the electrical signal to the optical module. The first employs an XSR/DSP interface between the switch and the optical module, while the second utilizes a direct-drive analog interface. The optical device is placed in both cases on top of an optics substrate. TIA (Transimpedance Amplifier) and driver electronics reside right next to or on top of the optical modules to reduce the RF lane length. This way >200Gb/s per lane and >1.6Tb/s for a 8-device based WDM transmitter are achievable with much less consumed power. Considering the close proximity of TIAs and drivers, optical transmitter and receiver modules with <20 μm phase-shifters elements are necessary for direct interconnection with the ASIC pins, while operating voltages <1V_{pp} are of interest for EIC amplifier-less operation.

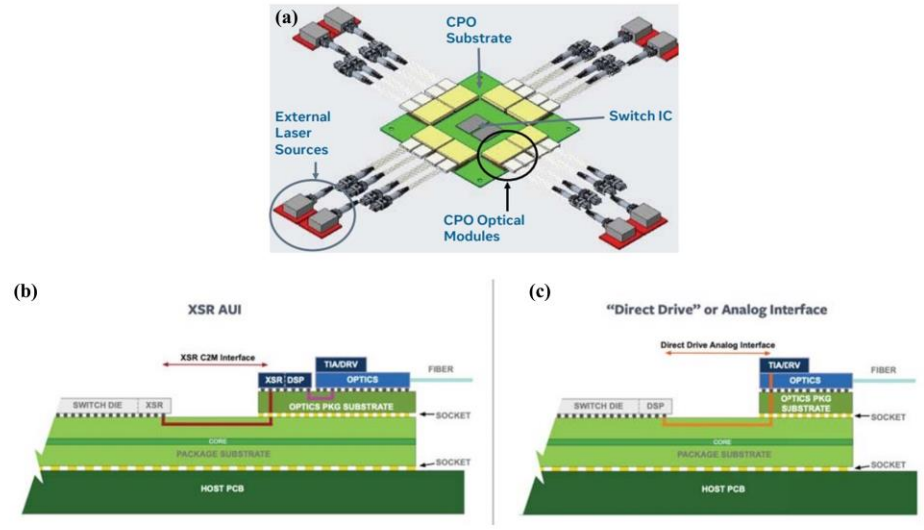


Fig. 1.3.1. (a) Schematic of CPO switch sub-assembly, (b) CPO Architecture Option with XSR C2M Interface and (c) CPO Architecture Option based on Direct Drive Analog Interface [106].

High-speed plasmonic modulators are able to satisfy these specifications offering small footprint values matching the ASIC's pin pitch, at the expense of increased optical propagation losses. To be able to provide the plasmonics benefits without the need to increase the insertion losses of the device, the evident solution is to interface the high-speed, low-footprint plasmonic phase-shifters with a low-loss integrated photonics platform. Si_3N_4 can provide the following desired characteristics and is the preferable PIC platform among other materials for passive waveguide based elements:

- Low cost
- Ultra-low propagation losses (below 0.2dB/cm)
- Low coupling losses coming from I/O interfaces with external optical fibers
- Low Thermo-Optic Coefficient (TOC) leading to less shifting of filtering elements, associated with the heat dissipation from the ASIC
- Large transparency window
- CMOS compatibility, suitable for mass-production

In summary, Co-Packaged Optics has prevalent position in Intra-Data Center Communications for >400Gb/s data rates, due to its ability to provide higher bandwidth, lower latency, improved power efficiency, space savings, scalability, and enhanced signal integrity. It addresses the challenges posed by the growing demands of modern data centers and supports the continued growth of data-intensive applications. Considering the voltages, data rates, and physical dimensions required by the optical engines in CPO, low-voltage plasmonic modulators on the Si_3N_4 platform are of interest in this thesis. In this Section, a variety of technologies for direct and external optical modulation in terms of materials and circuitries have been reviewed. Future dictations for high output power and high wall-plug efficiency in CPO also require the on-chip integration of the laser source [105]. The next Section will review some co-integration technologies to embed lasers and phase modulators inside passive chips based on Si_3N_4.

1.4: Co-integration of Active Components with Silicon Nitride Integrated Photonics

With its main advantages being the ultra-low propagation loss, its large transparency window, its low cost and its CMOS compatibility for potential mass-production [112, 113] Si_3N_4 is the ideal platform for applications such as sensing [114], quantum photonics [115] and microwave filtering [116]. In addition to its passive characteristics co-integrating such a low-loss waveguide with a Pockels material can lead to ultra-fast modulators in hybrid Si_3N_4-LNOI or Si_3N_4-BTO structures. This has been achieved either via hybrid or heterogeneous integration [117], or more recently via transfer printing [118]. Monolithic integration of BTO on an advanced silicon photonics platfom has also been performed [101]. Focusing on Si_3N_4 Pockels modulators promising the highest speeds among all phase-shifters with the lowest required voltages, in [119] 40Gbaud NRZ signals were delivered with a PZT on Si_3N_4 ring modulator requiring $4.2V_{pp}$ voltage. In [120], a Si_3N_4-LNOI MZI with phase shifter length of 7.8mm boasted a V_π x L of 22.4Vmm achieving 80Gbaud OOK modulation with ER>20dB with $2.8V_{pp}$ voltage. In [121], a slow-wave Si_3N_4-$LiNbO_3$ MZM boasted a 95GHz bandwidth and ER as high as 45dB. More recently, in [104], an Au-based Si_3N_4-BTO plasmonic MZM was able to deliver >216Gbaud 2-PAM signals with

a voltages of $3.6V_{pp}$ and phase shifter lengths of 15μm. However, a resonant plasmonic BTO on Si_3N_4 modulator could provide thermal stability unlike resonant POH platforms [110] with required voltages lower than simple plasmonic MZMs [104].

Considering the above, plasmonic enhancement of modulation in integrated plasmo-photonic transmitters is fundamental to obtain phase-shifters with length matching the ASICs' pin pitch and operating at CMOS sub-volt voltages at over 100Gbaud rates per lane. Plasmonic phase-shifters can strongly confine the field of the electrical applied signal and the field of the optical mode to an ultra-small area. This can lead to phase-shifters of ultra-small length at the expense of higher induced propagation and coupling losses. Plasmonic enhancement for low-footprint phase-shifters has already been used in material platforms such as $LiNbO_3$ [107-109], POH [89-96] and $BaTiO_3$ [102-104], all exploiting the Pockels effect, co-integrated with passive Si or Si_3N_4 photonic waveguides. Si- or Si_3N_4- plasmonic Mach-Zehnder (MZM), Ring or Racetrack (RM), and Mach-Zehnder in a Ring (MZIR) [110] modulator circuitries have been explored in the literature. The MZIR in $LiNbO_3$ delivered 240Gb/s line rates with a voltage of $0.75V_{pp}$ [111], much lower than that of MZMs.

Yet as seen in [104] most demonstrations in plasmonics employ noble metals that cannot oxidize such as Au. Introducing CMOS compatible plasmonics to Si_3N_4 passive circuits can enhance the potential of such devices for mass production and reduced fabrication cost. In [122] open cladded Cu slots have been co-integrated with Si_3N_4 photonic waveguides in simulations. Simulated propagation losses below 0.21dB/μm and coupling efficiencies above -1.15dB were demonstrated. In [123], coupling from Si to Si/TiN/Si_3N_4/GZO/Si modulators was theoretically studied. The propagation losses were close to 0.29dB/μm with the coupling losses being below 0.7dB. In [124], coupling from Si to open cladded Cu plasmonic slots led to experimental coupling losses of 1.55dB. In [125] the light transfer from Si_3N_4 to highly confining Si_3N_4-hybrid plasmonic waveguides (HPW) was experimentally demonstrated with the coupling losses being 2.84dB for waveguides with an 87% confinement factor. Finally, in [126] an Al-based coupled hybrid plasmonic amplitude modulator (CHPW) exhibited ultra-low propagation losses of 0.025dB/μm with coupling losses from Si-to-CHPW being 1.14dB. However, a coupling strategy from Si_3N_4 to CMOS plasmonic Pockels phase-shifters based on BTO could pave the way for CMOS plasmonic modulators on Si_3N_4 with their potential mass production.

Another front towards the realization of complete Tx modules on Si_3N_4 is the co-integration of lasers and amplifiers that are highly desired in this PIC platform. However, since Si_3N_4 is an indirect bandgap dielectric material, these components should be based on III-V materials and can only be integrated via hyrbid or heterogeneous integration [11, 127] or more recently, through micro-transfer printing for faster integration [128, 129]. Due to the ultra-low loss of the Si_3N_4 paltform, and its capability to tolerate high power levels without inducing non-linear effects, a large variety of spectrally pure tunable lasers on Si_3N_4, useful for applications such as sensing, microwave photonics, and coherent communications have been demonstrated during the last few years [130]. The monolithic integration of InP-based on Si laser sources can lead to Tx devices with mass fabrication capabilities [131]. Amplifiers based on the MBE-Grown QDs on 5° offcut silicon substrates boasted up to 20dB optical gain near the C-band [132]. In [133] III-V nanolasers deposited on amorphous

SiO_2 layers led to strong room temperature lasing under strong excitation. Very recently, demonstrated for the first time in [134], embedded InAs/GaAs quantum dot (QD) lasers were directly grown on trenched silicon-on-insulator (SOI) substrate, making possible the monolithic integration with a butt-coupled silicon waveguide. The estimated coupling efficiency was close to -6.7dB for continuous wave (CW) lasing up to 85°C, while the measured output power from the output of the end-fire coupled Si waveguides was as high as 6.8mW. To monolithically co-integrate grown III-V lasers on Si-substrates with passive Si_3N_4 waveguides proper low-loss coupling configurations that transfer light from the active to the passive elements should be designed. In [135], end-fire interfaces in hybridly integrated Si_3N_4 lasers led to 1.8dB coupling loss from the InP-based to the Si_3N_4 waveguide. In [129], heterogeneous micro-transfer printed lasers on Si_3N_4 boasted coupling losses from the amplifier to the passive waveguide below 0.98dB and back-reflections below -36dB. Finally, in [136], the heterogeneous integration of a membrane laser array with Si_3N_4 AWG exhibited coupling loss of only 0.2dB. The monolithic co-integration of III-V elements with Si_3N_4 waveguides has only been reported in [137], with the simulated coupling losses being <0.7dB and the back-reflections being <-30dB.

Therefore, towards a low-power transmitter (Tx) optical engine, compatible with the CPO technology, on silicon nitride (Si_3N_4), the PIC should incorporate the following: a) a low-loss coupling interface to transfer light from an III-V laser to a Si_3N_4 waveguide, b) a low-loss coupling strategy to transit from the passive Si_3N_4 platform to low-voltage short high-bandwidth CMOS compatible plasmonic phase shifters, and c) a CMOS plasmonic modulator circuitry supporting >100Gbaud rates in PAM-4 modulation, in combination with ultra-short phase-shifters requiring voltages achievable by CMOS driver Application Specific Integrated Circuits (ASICs), eliminating the need for RF amplifiers. All these problems were investigated in this PhD, trying to provide viable solutions compatible with existing fabrication and packaging processes through extensive simulations.

1.5: Contribution of the PhD Thesis and Structure

This thesis engages with the problem of the design of a complete integrated photonic transmitter device on Si_3N_4 featuring a monolithic co-integrated InP-based laser and a BTO-based plasmonic modulator targeting intra-DC connectivity at >100Gbaud/s rates. For this purpose, computational electromagnetic techniques along with other well-known computational methods for the analysis of Photonic Integrated Circuits (PICs) in the frequency and the time-domain have been employed targeting the design of monolithic and heterogeneous Si_3N_4 building blocks. These include the Si_3N_4 multimode interference couplers, the coupling interfaces from the InP-based lasers to the Si_3N_4 waveguides, the tri-layer coupling configurations from the Si_3N_4 waveguides to the plasmonic slot ferroelectric phase-shifters and the low-voltage plasmonic BTO ferroelectric high-speed MZIR modulator on the co-integrated Si_3N_4 platform. This PhD Thesis tackles various design issues in the monolithic co-integration of lasers with Si_3N_4 passives as well as the heterogeneous integration of BTO on Si_3N_4 for the formation of plasmonic BTO phase-shifters co-integrated with Si_3N_4. This PhD presents a novel ideal high-speed platform on Si_3N_4 for Tx modules featuring the best in class materials and devices, such as monolithic low-linewidth III-V QD-lasers for

the light carrier emitters and heterogeneous plasmonic ferroelectric BTO phase shifters for modulation.

Chapter 1, presented herein, provided a cohesive review on the photonic modulation material platforms and circuits in integrated Si and Si_3N_4 photonics. The chapter demonstrated several coupling configurations for efficient light coupling between the passive CMOS compatible Si_3N_4 waveguide platform and III-V based lasers towards monolithic light sources on Si_3N_4. Finally, several demonstrations of coupling strategies for the co-integration of Si or Si_3N_4 waveguides with plasmonic waveguides, being able to enhance light-matter interactions and confine the electric field in very small regions, have been showcased.

Chapter 2 lays the foundation of the numerical and semi-analytic methods deployed for the design of the devices in this Thesis. The chapters begin with Maxwell's Equations and some fundamental theorems in Electromagnetism. Then, the optical properties of materials such as metals and dielectrics are analyzed from a Solid State Physics point of view. Issues such as the Kramers-Kronig relations, and the dispersion relations of phonons and plasmons have been discussed. The Chapter transits to analytical methods for the physical interpretation of wave propagation in very simple perfectly metallic and dielectric waveguides. Finally, all the semi-analytic and numerical methods of this Thesis are introduced. These include the Finite Difference solution in Eigenvalue Problems (FDE), the Finite Difference Time Domain Method (FDTD), the Scattering Matrix method for the circuit analysis, and the Finite Element Method (FEM) for the numerical solution of electrostatic problems.

Chapter 3 deals with the design and analysis, via 2D-FDE and 3D-FDTD numerical techniques, of Si-rich nitride (SRN) interposer waveguides for the monolithic co-integration of InP-based QD laser sources with Si_3N_4 passive waveguides on a Si-substrate. The coupling strategy employs a two-step approach, an end-fire interface to transfer light from the laser to an ultra Si-rich waveguide, and a directional coupling step to couple light to the targeted mode in the final Si_3N_4 passive waveguide. From the computational analysis presented therein, three optimum devices have been defined, one with a strip, one with a rib and one with a tri-layer rib SRN waveguide exchanging performance for fabrication complexity. The strip-based configuration induced simulated total losses below 1.69dB and back-reflections of -54.4dB. The single SRN material rib waveguide structure boasted coupling losses below 1.41dB with -34.81 back-reflections. Finally, the tri-layer configuration exhibited 1.03 dB total losses (with 0.3dB being from the propagation in SRN waveguides). The back-reflections were lower than -52.6dB for this optimum layout. A tolerance analysis of the designed structures has been performed while more tolerant SRN-Si_3N_4 adiabatic couplers have also been simulated and tested yielding low experimental coupling losses close to 1dB. Studies of direct InP-QD-to-Si_3N_4 end-fire coupling interfaces have also been included in the Chapter. Finally, the design of thick Si_3N_4 Vernier ring reflector, acting as the main building block for monolithic Si_3N_4 low-linewidth tunable lasers is also demonstrated. The simple dual Vernier ring of this work exhibited a Free Spectral Range (FSR) over 38nm with Side-Mode Suppression Ratios (SMSRs) over 15dB. This led to experimental tunability values over 30nm. More, complex designs are also demonstrated with the triple ring based structure yielding simulated SMSR>20dB and the CROW-based design revealing a single wavelength peak in the (1500-1600)nm

wavelength range. Results from this Chapter have led to the first-authored journal publication [1] from the List of Publications.

Chapter 4 studies efficient multi-step coupling strategies between Si_3N_4 passive waveguides and CMOS plasmonic phase-shifters in the O-band. Initially, the Chapter lays out the fundamentals of plasmonic waveguides revealing also some important analytical expressions for some simple 2D thin film and plasmonic slot configurations. Then, co-integration strategies between Si- and Si_3N_4- photonics and plasmonics to construct low voltage and footprint phase-shifters are discussed. Finally, the technical work of this chapter is demonstrated, where 2D-FDE and 3D-FDTD numerical methods are used for the optimization analysis of this work. The two targeted plasmonic waveguides are the Cu-based Metal Insulator Metal or Plasmonic Slot (MIM) and the Cu-based Hybrid Plasmonic Waveguide (HPW). Both structures are based on a two-step tri-layer coupling approach. The MIM based configuration is composed of a Si_3N_4-to-aSi adiabatic coupling and then an aSi-to-MIM directional coupling step. The plasmonic slot exhibits propagation losses of 0.78dB/μm, a confinement factor over 56.18% inducing coupling losses below 1.29dB/interface. On the other side, the HPW coupling configuration is formed by a Si_3N_4-to-aSi/BTO adiabatic coupler and an aSi/BTO-to-HPW end-fire coupler. The HPW boasts lower propagation losses than the MIM, close to 0.5dB/μm sacrificing the confinement factor to near 31.49%. The induced coupling losses were below 1.75dB/interface. A tolerance analysis of all the designed structures has been included in this Chapter. Results from this work have led to the first-authored journal publication [2] from the List of Publications.

Chapter 5 employs the MIM phase shifters and coupling configurations from Chapter 4 to design an integrated Mach-Zehnder in a Ring (MZIR) plasmo-photonic circuit for >112Gbaud PAM-4 modulation in the O-band. For this purpose, 2D-FDE and 3D-FDTD numerical methods have been utilized for the design of the Cu-based tri-layer couplers but also to the design of equivalent couplers based on Au and Ag plasmonic slots. A 2D-FEM method for the electrostatic problem in combination with a 2D-FDE method has been implemented for the design of the plasmonic slot phase shifters. For this circuit analysis, the Scattering matrix method and the block-mode time-domain method have been applied in a circuit level commercial-grade solver. Apart from designing the circuits, the purpose of this Chapter is to compare the CMOS compatible Cu-based MZIR with noble metal Au and Ag based MZIR configurations, being the benchmark in terms of propagation losses and overall performance. The analysis in this work reveals that by opting for the MZIR layout there is a decrease in the V_π x L product of 46% for Ag, 39% for Au and 30% for Cu versus MZMs. Time-domain simulations signify the delivery of 112 Gbaud PAM-4 Signals from MZIRs with as low as 2×1.3 V_{pp} and 5μm long plasmonic phase shifters (PSs) with MZIRs providing a ΔQ signal improvement over MZMs of 2.9, 2.4, and 1.3 for Ag, Au, and Cu metals respectively. The operation of the MZIRs with $2x0.8V_{pp}$ voltages is also verified. The absolute Q factor is above 5.2 even for Cu metal indicating an error free link under Forward Error Correction (FEC). Additionally, the degradation of the electro-optic co-efficient due to fabrication induced variations is analyzed and the limits of the MZIR modulator for application to CPO technologies are revealed. For $r_{42}/2$ the voltage reduction values become >18% for Cu, > 28% for Au and >37% for Ag versus MZMs. Finally, Section 5.5 demonstrates the DC analysis of CMOS fabrication

compatible racetrack plasmo-photonic BTO modulators. It is clarified that while the insertion losses become lower than 3dB even for Cu the resonance bandwidth restricts the value below 66GHz, without sacrificing the ER to enlarge the bandwidth. This work is expected to lead to Si_3N_4-BTO based optical engines in co-packaged optics for next-gen 51.2Tb/s switching ASICs featuring 200Gb/s Ser/Des interfaces or beyond. Results from this Chapter have led to the first-authored journal publication [3] from the List of Publications.

Chapter 6, concludes the PhD Thesis providing a commentary on the results from each of the Chapters. The impact and future work following the results of this thesis are also discussed.

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Chapter 2: Theoretical Analysis and Numerical Modelling of Photonic Devices

2.1: Introduction

This thesis will focus on the design of Si_3N_4-based photonic integrated devices and circuits. The basis of modelling of such optical elements that usually have dimensions in the micron scale, and in some cases in the nanoscale (plasmonics), is the classical electromagnetic theory. The techniques that will be used throughout this thesis are semi-analytic and numerical methods. A well-known semi-analytic technique is the scattering matrix method used intensively for the frequency-domain modelling of photonic integrated circuits, making harmonic approximations for the propagating modal fields. On the other side, numerical methods employ techniques to discretize classical equations in electrostatics and electromagnetics, such as Poisson's, Laplace's and Maxwell's equations directly. Nevertheless, before going to the mathematical treatment of the physical basis equations, the fundamentals of electrodynamics will be analyzed in a very contained form, acting as a foundation of modern physics and engineering nowadays. The basis of this analysis in these chapters are references [1] and [2] and the target of this chapter is to transit from ideas of fundamental physics to practical applications seen in the next chapters in a smooth manner.

In this thesis, the classical electromagnetic field theory will be applied towards the design of photonic integrated devices and circuits in the Si_3N_4 platform. The scales studied in this thesis will all be classical meaning that all dimensions of the designed devices in space are much larger than the de Broglie wavelength [3] therefore a quantum electrodynamic treatment will not be needed. In the next chapters, the fundamental equations in electrostatics and electromagnetics will be set forth. There will be a constant effort to set up all equations and values in electromagnetics in the SI (Système International) metric system. All vectors will be displayed in bold (for example **a**). Nevertheless, for the study of certain systems, for example in quantum electrodynamics and for some calculations in theoretical condensed matter physics, the CGS (Centimetres Grams Seconds) metric system is very useful. In the Gaussian CGS metric system in electromagnetics one for the conversion of units from SI to CGS needs to apply:

$$\varepsilon_0 \to \frac{1}{4\pi} \quad (2.1)$$

Maxwell's equations are mathematically expressed differently in the SI and the CGS system. Therefore a (CGS) notification will be next to an equation set up in the CGS metric system.

2.2: Maxwell's Equations

Before Maxwell's equations were set forth in their known nowadays and applied differential form, various physicists had performed experiments and deployed models to interpret various phenomena in electromagnetism. Various parts of electromagnetism

were studied by Franklin (with his infamous and dangerous experiment of flying a kite in a storm that led to the demonstration of a connection between lighting and electricity and eventually to the invention of the lightning rod), Coulomb, Ampère, Faraday and others. In 1820, Hans Christian Ørsted observed that electric current could turn a magnetic needle. Not a long time after that, Ampère deployed the assumption that magnetic phenomena are caused by moving electrical charges. In 1831, Faraday discovered that a moving magnet could cause an electrical current. The attraction of ellipsoids was interpreted via Gauss's Law in 1835 (with its first formulation by Joseph Louis Lagrange in 1773). This law can be applied to derive Coulomb's law.

Maxwell and Lorentz made the final additions for the completion of the electromagnetic theory. In 1865, with his publication "A Dynamical Theory of the Electromagnetic Field" [4], James Clark Maxwell supported the idea that electricity and magnetism are different manifestations of the same phenomenon, a single electromagnetic force, and derived an electromagnetic wave equation with a light velocity close to experimental values, thus deducing that light is a wave and that the whole spectrum from gamma rays, x-rays, to visible light, to microwaves etc. are all electromagnetic waves in their basis. This is regarded as the second great unification in physics with the first being the classical mechanics theory by Newton. The main addition of Maxwell was the insertion of the Displacement "Current" in Ampere's Law, to satisfy the continuity equation of electric charge. More specifically, because currents are commonly used to generate magnetic fields, a changing electric field must be associated with a current. To satisfy this requirement, Maxwell extended Ampere's law by introducing the displacement current into the electric current term. Without the displacement current term, we have:

$$\nabla \times \boldsymbol{H} = \boldsymbol{J} \Rightarrow \nabla \cdot \boldsymbol{J} = 0 \ (2.2)$$

However, from the continuity equation this would mean that:

$$\nabla \cdot \boldsymbol{J} = -\frac{\partial \rho}{\partial t} = 0 \ (2.3)$$

which is wrong since the electric charge needs to be conserved.

Heinrich Rudolf Hertz, in 1888 generated an electromagnetic wave (radio wave) for the first time setting the decisive proof of Maxwell's theory for the first time. Of course, then, the concept of a quantum particle of charge did not exist and Maxwell considered matter as a continuum of charge that could be divided indefinitely. Additionally, Maxwell, at that time considered that electromagnetic waves propagated in a medium called aether. The concept of aether was rejected after the Michelson-Morley experiments. The target of these experiments was to detect the existence of the luminiferous aether. The experiment compared the speed of light in perpendicular directions to be able to sense the relative motion of matter through the stationary luminiferous aether ("aether wind"). However, the results were negative. Such experiments have been performed many times with increased sensitivity. Finally, the value of light velocity was experimentally measured by Wilhelm Eduard Weber and Rudolf Kohlrausch in 1856. They measured the ratio of the electromagnetic and

electrostatic units of charge, $1/\sqrt{\mu_0 \varepsilon_0}$, by discharging a Leyden jar, and found that its numerical value was very close to the speed of light as measured directly by Fizeau. Therefore, all the theoretical and experimental findings explained above have led the electromagnetic field theory to be prevalent in both physics and engineering with applications from telecommunications, optics, electrical motors and other electrostatic devices, microwaves, spectroscopy, condensed matter physics, fundamental physics and others.

Maxwell's equations initially were not set forth in their differential form found in electromagnetics textbooks found nowadays. Nevertheless, due to their usefulness in applications, this is their prevalent form nowadays. The fundamental equations coming from the Maxwellian theory of electromagnetism are presented and analyzed below. Combined with Ohm's Law, the Continuity Equation and Lorentz's force, they constitute the basis of the classical electromagnetic field theory. These are the main equations that are also encountered in the design of integrated photonic devices.

The first equation is Gauss' Law expresses that the divergence of the electric field is proportional to the local density of electric charge. The term **D** is called the electric displacement field.

$$\nabla \cdot \boldsymbol{D} = \rho \quad (2.4)$$

The second equation states that magnetic monopoles in nature cannot be found. Even though in some engineering textbooks the use of the term magnetic current and magnetic displacement current acting as a fictitious current for the $\partial \boldsymbol{B}/\partial t$ term is common, in this thesis a strict physical formalism will be followed and thus such terms will not be used. The term **B** is the magnetic induction.

$$\nabla \cdot \boldsymbol{B} = 0 \quad (2.5)$$

The third equation is Faraday's Law that states that a time-varying magnetic field coexists with a time-varying electric field in nature.

$$\nabla \times \boldsymbol{E} = -\frac{\partial \boldsymbol{B}}{\partial t} \quad (2.6)$$

The fourth equation is Ampere's Law with Maxwell's addition for the displacement "current" term. It states that a time varying magnetic field co-exists with an electric field. Additionally, an electrical current can generate a magnetic field. The direction of the magnetic field with respect to the current direction can be found with the right-hand rule.

$$\nabla \times \boldsymbol{H} = \boldsymbol{J} + \frac{\partial \boldsymbol{D}}{\partial t} \quad (2.7)$$

The electric displacement **D** is connected with the external electric field **E**. The magnetic induction **B** is also connected with the external magnetic field **H.** These relations are simple relations in linear materials and are called constitutive relations.

Even in non-linear materials, since the non-linear phenomena occur at high field values, we can initially treat them as linear. The constitutive relations for materials are shown below:

$$\boldsymbol{D} = \varepsilon \boldsymbol{E} = \varepsilon_0 \varepsilon_r \boldsymbol{E} = \varepsilon_0 \left(1 + \chi_e \right) \boldsymbol{E} \ (2.8)$$

$$\boldsymbol{B} = \mu \boldsymbol{H} = \mu_0 \mu_r \boldsymbol{H} = \mu_0 \left(1 + \chi_m \right) \boldsymbol{H} \ (2.9)$$

From these terms, ε is the dielectric permittivity and μ is the magnetic permeability. The terms χ_e and χ_m are the electric and magnetic susceptibilities. ε_0 is the dielectric permittivity in vacuum and has the value of 8.854 x 10^{-12} F/m approximately. The term μ_0 is the magnetic permeability in vacuum that is approximately 4π x 10^{-7} H/m. The terms ε_r and μ_r are the dielectric permittivity and permeability of the medium. In this thesis, only non-magnetic materials will be applied in photonic devices therefore at all cases $\mu_r = 1$. In anisotropic materials, such as ferroelectric materials ($BaTiO_3$, $LiNbO_3$, etc.) the dielectric permittivity ε_r is a tensor, with the propagation of light depending on the direction of propagation. Such materials have fast and slow optical axes, and the dielectric permittivity is a matrix. With the constitutive relations Maxwell's Equations take the following form:

$$\nabla \cdot \boldsymbol{E} = \frac{\rho}{\varepsilon} \ (2.10)$$

$$\nabla \cdot \boldsymbol{H} = 0 \ (2.11)$$

$$\nabla \times \boldsymbol{E} = -\mu \frac{\partial \boldsymbol{H}}{\partial t} \ (2.12)$$

$$\nabla \times \boldsymbol{H} = \boldsymbol{J} + \varepsilon \frac{\partial \boldsymbol{E}}{\partial t} \ (2.13)$$

As mentioned above three more equations are fundamental to the computation of electromagnetic fields with a classical field theory. The first is the continuity equation supporting that the electric charge must be preserved:

$$\nabla \cdot \boldsymbol{J} + \frac{\partial \rho}{\partial t} = 0 \ (2.14)$$

The second is the Lorentz Force expressing a force induced by an electromagnetic field:

$$d\boldsymbol{F} = \left(\rho \boldsymbol{E} + \left(\boldsymbol{J} \times \boldsymbol{B} \right) \right) d\tau \ (2.15)$$

The final is Ohm's law in classical electromagnetic field theory, indicating a current density induced by an applied electric field:

$$\boldsymbol{J} = \sigma \boldsymbol{E} \ (2.16)$$

The term σ is the conductance of the material (for example a metal). This becomes a complex quantity when there are bound and free charges in a material and a time-varying electric field is applied.

The above equations (2.10)-(2.13) exploited the assumption of non-dispersive materials to hold true. The dispersion relations and the physics of the optical properties of the materials used in photonic devices will be explained in the next section. A Helmholtz wave equation, a partial differential equation in the form $\nabla^2 f = -k^2 f$ can be obtained by applying the curl term to both sides of equations (2.12) and (2.13), considering the absence of current densities in (2.13), and using simple mathematical identities for the nabla operator:

$$\nabla \times (\nabla \times E) = -\mu \frac{\partial (\nabla \times H)}{\partial t} \Rightarrow \nabla^2 E = \mu\varepsilon \frac{\partial^2 E}{\partial t^2} \quad (2.17)$$

$$\nabla \times (\nabla \times H) = \varepsilon \frac{\partial (\nabla \times E)}{\partial t} \Rightarrow \nabla^2 B = \mu\varepsilon \frac{\partial^2 E}{\partial t^2} \quad (2.18)$$

Then,

$$\upsilon^2 = \frac{1}{\mu\varepsilon},\ c^2 = \frac{1}{\mu_0 \varepsilon_0} \quad (2.19)$$

The term υ is the light velocity in a material while c is the light velocity in vacuum and equal to 2.99792458 x 10^8 m/s. The impedance of vacuum can be calculated as:

$$Z_0 = \sqrt{\frac{\mu_0}{\varepsilon_0}} = 376.73\Omega \quad (2.20)$$

The impedance of a medium will have an equivalent value:

$$Z = \sqrt{\frac{\mu}{\varepsilon}} \quad (2.21)$$

The refractive index along with the basic condensed matter physics in a contained form interpreting the optical properties of the materials will be explained in the next section.

2.3: Gauge Theory in Electromagnetism and Energy Relations

In many problems the exploitation of the electric field-based equations with the appropriate boundary conditions induced by the material composition and the geometry of a device is not trivial and in many cases the expression of the electric and magnetic fields through the scalar potential V and the vector potential **A** is much more useful. These potential expressions should not distort the computed electric and magnetic fields. One can make use of the following identities of the nabla operator:

$$\nabla \cdot (\nabla \times \boldsymbol{A}) = 0 \quad (2.22) \quad \textit{and} \quad \nabla \times (\nabla F) = 0 \quad (2.23)$$

The analysis will begin from the electrostatics. Considering (2.22) and (2.12) for a non-time-varying problem one gets:

$$\nabla \times \boldsymbol{E} = 0 \Rightarrow \boldsymbol{E} = -\nabla V \ (2.24)$$

Combining (2.24) with (2.10) Poisson's equation can be derived:

$$\nabla^2 V = -\frac{\rho}{\varepsilon} \ (2.25)$$

If ρ = 0 then (2.25) is called the Laplace's equation. Therefore, it is easy to solve in electrostatic case for V instead of E and calculate E via V with (2.24). This is really useful in some specific problems.

For the magnetostatic case, one begins from (2.11) considering (2.23):

$$\nabla \cdot \boldsymbol{B} = 0 \Rightarrow \boldsymbol{B} = \nabla \times \boldsymbol{A} \ (2.26)$$

This expression is also true for electromagnetic fields as well since the basis of this expression is one of the generic Maxwell's four equations. Therefore, inserting (2.26) to (2.12) this yields:

$$\nabla \times \boldsymbol{E} = -\frac{\partial(\nabla \times \boldsymbol{A})}{\partial t} \Rightarrow \nabla \times \left(\boldsymbol{E} + \frac{\partial \boldsymbol{A}}{\partial t}\right) = 0 \Rightarrow \boldsymbol{E} = -\nabla V - \frac{\partial \boldsymbol{A}}{\partial t} \ (2.27)$$

The expressions (2.26) and (2.27) are the potential based expressions for the electric and magnetic field of an electromagnetic field. Therefore, in many problems due to an easier solution with potentials, first the scalar and vector potentials are computed and then the electric and magnetic field are computed through them. Through Gauge theory it is shown that any potentials V' and **A'** can be used without distorting the electric or magnetic field solutions computed through these transformed potentials. Such transforms are called Gauge transforms and for the above statement to be true, the following relation needs to be satisfied:

$$\left.\begin{aligned} \boldsymbol{A}' &= \boldsymbol{A} + \boldsymbol{a} \\ V' &= V + \beta \end{aligned}\right\} \Rightarrow \left.\begin{aligned} \boldsymbol{A}' &= \boldsymbol{A} + \nabla\lambda \\ V' &= V - \frac{\partial \lambda}{\partial t} \end{aligned}\right\} \ (2.28)$$

The most well-known Gauges are the Coulomb Gauge (2.29) and the Lorentz Gauge (2.30) shown below:

$$\nabla \cdot \boldsymbol{A} = 0 \ (2.29)$$

$$\nabla \cdot \boldsymbol{A} = -\mu_0 \varepsilon_0 \frac{\partial V}{\partial t} \ (2.30)$$

Having finished a contained analysis regarding the electric and magnetic fields, it is important to mention the fundamentals of energy transfer through electromagnetic

fields. This will also prove important in the practical simulations presented later in this thesis where the total power passing through a plane vertical to the wave propagation axis will need to be calculated. The energy per time unit and per area unit that is transferred by the electromagnetic fields is called the Poynting Vector given by the relation:

$$\boldsymbol{S} = \frac{1}{\mu}(\boldsymbol{E} \times \boldsymbol{B}) \ (2.31)$$

As it will be shown practically later on, since the Poynting Vector signifies the density of the energy flow, to calculate the power passing through a plane, transferred by an electromagnetic wave propagating vertically to that plane, one would need to calculate power by numerically (or analytically if possible) integrating relation (2.31) at both sides:

$$P = \iint \boldsymbol{S} \cdot d\boldsymbol{A} = \iint \frac{1}{\mu}(\boldsymbol{E} \times \boldsymbol{B}) \cdot d\boldsymbol{A} \ (2.32)$$

In some cases, the energy density that has been stored in the fields can be expressed as:

$$U_{em} = \frac{1}{2}\left(\varepsilon \boldsymbol{E}^2 + \frac{1}{\mu}\boldsymbol{B}^2\right) \ (2.33)$$

In some electrostatic problems the energy density stored in the electric field can be expressed as:

$$U_e = \frac{1}{2}\varepsilon \boldsymbol{E}^2 \ (2.34)$$

Therefore, electrical power stored in a device in three dimensional devices can be computed as:

$$P_{el} = \frac{\varepsilon}{2}\int\int\int \boldsymbol{E}^2 dV \ (2.35)$$

These expressions can be simplified with double integrals in 2D designed devices. These equations hold true in the static case and semi-static approximations, giving valuable information on the power consumption of devices used in optical interconnects on silicon nitride such as modulators.

2.4: Material Boundary Conditions

Before trying to solve Maxwell's equations and discussing about the analytical and numerical computation of electromagnetic fields in photonic devices, the fundamental relations that need to be satisfied at the interfaces of two materials with permittivities and permeabilities ε_1, μ_1 and ε_2, μ_2 should be tackled. The relations that need to be

satisfied at the interfaces between two materials in a device are called physical boundary conditions. These conditions are natural and need to be held true in any numerical method if predictability and close approximation of experimental results is the target. These conditions are different from the artificial boundary conditions in numerical methods that need to be applied at the edges of the computational domain to truncate the space where Maxwell's equations will need to be numerically solved. To derive the physical boundary conditions, one needs to integrate relations (2.10)-(2.13) at both sides and apply the Gauss and Stokes theorems. Therefore, this technique yields:

$$\left.\begin{aligned}\iiint \nabla \cdot \boldsymbol{E_1} dV &= \iiint \frac{\rho_1}{\varepsilon_1} dV \\ \iiint \nabla \cdot \boldsymbol{E_2} dV &= \iiint \frac{\rho_2}{\varepsilon_2} dV\end{aligned}\right\} \Rightarrow$$

$$\Rightarrow \oiint \boldsymbol{n} \cdot \left(\varepsilon_1 \boldsymbol{E}_1 - \varepsilon_2 \boldsymbol{E}_2 \right) dA = \oiint \rho_s \, dA$$

$$\Rightarrow \boldsymbol{n} \cdot \left(\varepsilon_1 \boldsymbol{E}_1 - \varepsilon_2 \boldsymbol{E}_2 \right) = \rho_s \quad (2.36)$$

In a similar way, the other boundary conditions ensue with the Gauss and Stokes theorems:

$$\boldsymbol{n} \cdot \left(\boldsymbol{H}_1 - \boldsymbol{H}_2 \right) = 0 \quad (2.37)$$

$$\boldsymbol{n} \times \left(\boldsymbol{E}_1 - \boldsymbol{E}_2 \right) = 0 \quad (2.38)$$

$$\boldsymbol{n} \times \left(\boldsymbol{H}_1 - \boldsymbol{H}_2 \right) = \boldsymbol{J}_s \quad (2.39)$$

The equations (2.36)-(2.39) signify that the component of the magnetic field vertical to the interface between two materials needs to be continuous while the component of the electric field parallel with the interface between two materials needs to remain continuous. These equations need to always be satisfied and this was a big issue when scientists developed a finite difference based eigensolver to study photonic waveguides since the discontinuities needed to be dealt with. Another important issue in computational methods for photonics is the inclusion of dispersion in simulations. For this reason, the next section will elaborate on how the electromagnetic waves interact with matter from a solid-state physics point of view.

2.5: Optical Properties of Materials

This section will make an attempt to present the fundamentals of the optical properties of materials and explain how light (electromagnetic waves) interacts with matter in a compact manner. Before that, it should be mentioned that the properties of most materials (mechanical, magnetic, electrical, thermal etc.) have their origin on the structure of the material. The crystal structure is fundamental for the determination of

many material properties (as it will be seen in a later chapter the structure of the material determines the strength of ferroelectricity in ferroelectric materials). The electronic structure of the material is the second parameter that results to almost every material property in solid state physics. In this section, a simple explanation of the electronic structure of a material will be provided, since certain parameters are fundamental in classical and quantum transport, and optical phenomena seen in this thesis. This Section is based on material from [5]-[7]. The results will be fundamental when one will need to add material dispersion in the Finite Difference Time Domain Method. An auxiliary differential equation will be able to insert dispersion there and its basis will be the relations from Solid State Physics analyzed herein. As this Thesis focuses on the macroscopic physical properties of photonic devices only the fundamentals of the Drude and Lorentz models will be provided. A reference with an in-depth study on the optical properties of materials can be found in [32]. This reference along with other books, for example on the FTIR spectroscopy [33], provide more advanced models beyond the simple Drude and Lorentz ones.

The fundamentals of quantum mechanical mathematical methods useful for solid-state physics can be found in references [8]-[9]. Herein, due to this Thesis not focusing on the physical phenomena occurring in microscopic non-classical regime, only the basis for the calculation of the bandstructure in solids will be given. We begin from the one electron approximation in band theory, and from the time-independent Schrödinger's equation considering a periodic potential V(**r**).

$$\frac{\hbar^2}{2m}\nabla^2\psi + \big(E - V(\boldsymbol{r})\big)\psi = 0 \text{ (2.40)}$$

Given the periodic boundary conditions of this system, the solutions will have the following form based on Bloch's theorem:

$$\psi_k(\boldsymbol{k},\boldsymbol{r}) = u_k e^{i\boldsymbol{k}\boldsymbol{r}} \; with \; u_k(\boldsymbol{r}) = u_k(\boldsymbol{r}+\boldsymbol{r_n}) \text{ (2.41)}$$

With the symmetry and the periodicity of the lattice one can calculate that $V(r) = \sum_G V_G e^{i\boldsymbol{G}\boldsymbol{r}}$ and $E_k(\boldsymbol{G}) = E_k(\boldsymbol{k}+\boldsymbol{G})$. Considering the periodicity of the lattice and an arbitrarily weak potential the eigenvalues of (2.40) get the form:

$$E = \frac{\hbar^2|\boldsymbol{k}+\boldsymbol{G}|^2}{2m} \text{ (2.42)}$$

The vector $\boldsymbol{G}$ is the reciprocal lattice vector. In a crystal structure with many atoms forming bonds, it is easy to see that the result will be energy bands instead of well-defined states. Bandgaps are formed due to the periodicity and the potential in the crystal structure. In reality, the electronic structure is an energy-wavenumber graph, and the wavenumber is expressed with respect to many directions in the graph. The first Brillouin Zone is defined between the wavenumbers –(2π/Λ) and (2π/Λ) where Λ is the period of a unit cell in the real space. The electronic structure is expressed in the k-space. A very good approximation of the electronic structure (bandstructure) of a material can be calculated via well-established computational methods such as Tight-

Binding, Non-local pseudo-potentials and the k.p method. There are also numerical methods such as Transfer Matrix Method (TMM) that can be used to more accurately calculate band structures. Also, Density functional theory (DFT) is a quantum-mechanical method used in chemistry and physics to calculate the electronic structure of atoms, molecules and solids, popular in computational solid-state physics since the 1970s. Using the single electron model and the Fermi-Dirac Distribution, the total electron population can be statistically described.

The lowest energy band is called Valence Band. The higher energy band is called the Conduction Band. Using the electronic structure of a material, materials can be classified to three categories in terms of their electrical properties: a) Metals, having a zero bandgap, b) Dielectrics, having a very large bandgap, and c) Semiconductors, where the bandgap is neither zero nor large. Semiconductors have great importance in electronics and photonics. They can be doped and actively control the flow of carriers with pn and pin junctions in practical electronic devices, such as mosFETs. Especially, the last few years industry has turned to nanoelectronics to achieve higher speeds and lower energy consumption, where the gate width has become comparable or lower than the de Broglie wavelength. In such devices, quantum transport phenomena need to be considered. In photonics, both direct and indirect bandgap materials are utilized. Materials of the III-V group (for example Ga, In, P etc.) have the maximum of the valence band at the same k-value as the minimum of their conduction band. For this reason, momentum can be conserved, and such devices can be used for light generation and detection. On the other side materials of the IV group (such as Si, Ge etc.) have the maximum of the valence band at different k-value from the minimum of their conduction band and therefore extra momentum is needed for the electron to fall from the conduction back to the valence band. Nevertheless, these materials are used a lot in devices such as modulators, waveguides etc. All materials have a Fermi energy level E_F which is the highest occupied energy state.

Finally, the Density of States is directly related to many optical and electrical properties in all materials. The Density of States (DOS) is a concept in solid-state physics that describes the number of states per unit volume that are available to be occupied by electrons at each energy level. It is a fundamental concept in condensed matter physics and is used to describe the electronic structure of materials. The DOS is typically represented as a function of energy and is used to calculate various properties of materials such as electrical conductivity, thermal conductivity, and specific heat capacity.

Having the solid-state physics concepts mentioned above, the dielectric properties of all materials will be interpreted. Initially, there are three ways to study the interaction of light with matter. The first is based on quantum mechanics in a microscopic view of matter, for example when studying the absorption of light via the generation of a phonon or an electron-hole pair. The second is macroscopic where all the analysis begins from the classical electromagnetic field theory equations. The third means of analyzing the optical propagation in a material is the optical ray theory where the paraxial approximation is used. This means that there is a hypothesis that light does not diverge off a certain axis. Even though the third means can offer a much-needed solution to many problems in optics, such as the design of lenses, and will be used for the study of dielectric waveguides, a much stricter formalism will be deployed to reveal

how light interacts with matter. More specifically, the basis of all the analysis will be the classical electromagnetic field theory while quantum calculations may be inserted only where needed. The materials that will be analyzed are considered linear.

The most fundamental viewpoint is that an external electric field results to the flow of "almost-free" carriers of charge, as indicated by Ohm's Law (equation (2.16)). The simplest case is an ohmic current induced by an applied voltage in a metal where classical transport theory can be exploited to derive a conductivity σ. In materials, such as semiconductors, whose dimensions in space are comparable or lower to the de Broglie wavelength, quantum transport phenomena and theory should be applied to interpret experimental measurements such as the quantized conductance, the quantum Hall effect and others. However, another interaction of an external electromagnetic field with matter consists of the spatial displacement of localized charges, such as the bound electrons feeling the attractive forces of the positively charged nuclei. The first phenomenon is interpreted through Ohm's law, and the current density is the result of carrier flow in partially filled bands induced by the field **E** in equation (2.16). The second can be described beginning from Maxwell's curl equations (2.6) and (2.7). Additionally, the Fourier transforms connecting the electric field expressions with respect to time t and frequency are considered, as shown below:

$$\boldsymbol{E}(t) = \int_{-\infty}^{+\infty} \boldsymbol{E}(\omega)\, e^{-i\omega t} d\omega \quad (2.43)$$

and

$$\boldsymbol{E}(\omega) = \frac{1}{2\pi} \int_{-\infty}^{+\infty} \boldsymbol{E}(t)\, e^{i\omega t} dt \quad (2.44)$$

Considering that the fields are harmonic, then through the constitutive relations:

$$\boldsymbol{D}(\omega) = \varepsilon_0 \varepsilon_r \boldsymbol{E}(\omega) \quad (2.45)$$

Maxwell's curl equations are then set up as:

$$\nabla \times \boldsymbol{H}(\omega) = \sigma \boldsymbol{E}(\omega) - i\omega\varepsilon_0 \varepsilon_r(\omega)\, \boldsymbol{E}(\omega) \quad (2.46)$$

and

$$\nabla \times \boldsymbol{E}(\omega) = -i\omega\mu_0 \mu_r(\omega)\, \boldsymbol{H}(\omega) \quad (2.47)$$

It is clear from (2.46) that light-matter interaction in a harmonic field of frequency f = ω/2π does not clearly lead to induction of current of free carriers or to the oscillation of bound charges. Therefore, two quantities, a generalized conductivity and a generalized permittivity can be defined as follows:

$$\tilde{\sigma} = \sigma - i\omega\varepsilon_0\varepsilon_r \quad (2.48)$$

and

$$\tilde{\varepsilon}_r = \varepsilon_r(\omega) + i\frac{\sigma}{\omega\varepsilon_0} \quad (2.49)$$

While in constant fields with ω = 0 the quantity σ becomes real and the free electron current is easily distinguishable from any bound charge displacements, the differentiation of free and bound carriers is very difficult in oscillating fields with $\omega \neq 0$. Also, as seen in eq. (2.49) the permittivity is a complex value that has a real part related with the phase of the oscillating field and an imaginary part related with the absorption of the field towards the transport of charge carriers. This macroscopic analysis remains true for electromagnetic fields whose wavevector **k** is much smaller than the vector **G** with |**G**|= 2π/Λ related with the period of the crystal's unit cell. This remains true for light interactions with wavelengths above the violet, therefore X-rays would need to take into account diffraction phenomena through the crystal structure. The function ε_r is a complex function defined in $-\infty < \varepsilon_r < +\infty$ *and* $\varepsilon^*(-\omega) = \varepsilon(\omega)$.

One. through the definition of polarization, can insert dielectric susceptibility as:

$$\boldsymbol{P} = \boldsymbol{D} - \varepsilon_0\boldsymbol{E} \quad (2.50) \Rightarrow$$

$$\Rightarrow \chi(\omega) = \varepsilon_r(\omega) - 1 \quad (2.51)$$

The function χ is an analytical complex function that has its poles at the negative complex semi-plane. From this the prime value of the integral $\oint \frac{\chi(\omega')}{\omega' - \omega} d\omega' = 0$ can be calculated. By separating the permittivity in a real and a complex part as $\varepsilon_r = \varepsilon_1(\omega) + i\varepsilon_2(\omega)$ (2.52) one can derive the Kramer's Kronig relations connecting the real and the imaginary part of the permittivity.

$$\left.\begin{aligned} \varepsilon_1(\omega) - 1 &= \frac{1}{\pi}\mathcal{P}\int_{-\infty}^{+\infty} \frac{\varepsilon_2(\omega')}{\omega' - \omega} d\omega' \\ \varepsilon_2(\omega) &= \frac{1}{\pi}\mathcal{P}\int_{-\infty}^{+\infty} \frac{\varepsilon_1(\omega') - 1}{\omega' - \omega} d\omega' \end{aligned}\right\} \quad (2.52)$$

The pair of equations (2.52) may seem non-trivial but has ample practical application. The physical meaning of (2.52) is that if the real part of the permittivity of a material is known at a certain frequency range, then its imaginary part can also be calculated. The pair (2.52) is fundamental in spectroscopy when one tries to measure the refractive index of a material.

Considering the transition of a simple plane electromagnetic wave through a thin film of thickness d, one can easily define the complex refractive index of a material as:

$$\tilde{n}(\omega) = n(\omega) + i\kappa(\omega) = \sqrt{\varepsilon_r(\omega)} \quad (2.53)$$

where

$$n^2 - \kappa^2 = \varepsilon_1 \ (2.54) \ \textit{and} \ 2n\kappa = \varepsilon_2 \ (2.55)$$

Solving for the electric field **E** through a material thin film, with simple mathematical actions one can get the transmission and reflection coefficients in a thin film:

$$T_1 = \frac{2}{\tilde{n}+1}, \ T_2 = \frac{2\tilde{n}}{\tilde{n}+1}, \ R_2 = \frac{\tilde{n}-1}{\tilde{n}+1} \quad (2.56)$$

These equations can also be used as an approximation in calculating the transmitted and reflected power in waveguide end-fire coupling interfaces. Nevertheless, in such cases the refractive index is a complex effective index $\tilde{n}_{eff}$ as it will be seen in the study of optical waveguides. Finally, considering that in optically dense materials one can make the approximation of $\left|\frac{\tilde{n}\omega}{c}d\right| \ll 1$ then following a linear approximation the light intensity exiting the material thin film:

$$I = I_0\left(1 - \frac{\varepsilon_2\omega}{c}d + \ldots\right) \Rightarrow K(\omega) = \frac{\omega\varepsilon_2(\omega)}{c} \quad (2.57)$$

Relation (2.57) is the absorption coefficient of a material. If one turns the term I/I_0 to a logarithmic quantity one can get the power efficiency and thus the power that has been absorbed from the material in dB. This is the method that is used for the extraction of the propagation losses in mode solvers. Nevertheless, again, in such cases ε_2 is the imaginary part of the effective complex permittivity of a mode in a waveguide, that is related to the imaginary part of the effective index. In the next sections, models for the refractive index of metals and dielectrics will be derived. In reality, for more complex amorphous materials there is a variety of models to fit the refractive index data and usually this is performed by the responsible person that fabricates the material and measures its complex refractive index. However, for a large variety of materials the metal and dielectric models for the refractive index hold true.

For the interaction of light with metals, the focus should be on how the free electron gas behaves under application of an external electromagnetic field. The dielectric behavior of metals and semiconductors with high levels of doping concentrations is dictated by the collective excitations of free carriers. The dielectric function can be created via motion equations for the electron gas as shown below:

$$nm\ddot{\mathbf{u}} + \gamma\boldsymbol{u} + neE = 0 \ (2.58)$$

The term n is the number of electrons, m is the electron mass, γ is the decay constant connected with the conductivity since for constant flow:

$$\ddot{\mathbf{u}}=0 \Rightarrow \boldsymbol{j} = -en\boldsymbol{u} = \frac{n^2e^2}{\gamma}\boldsymbol{E} \;\; thus\; \sigma = \frac{n^2e^2}{\gamma} \quad (2.59)$$

Considering the Fourier transform of (2.58):

$$\boldsymbol{u}(\omega) = \frac{ne}{nm\omega^2 + i\dfrac{n^2e^2}{\sigma(\omega)}}E(\omega)\; but\; \boldsymbol{P} = -en\boldsymbol{u}(\omega) \quad (2.60)$$

Therefore, considering (2.50) the dielectric function of metals can be expressed as:

$$\varepsilon_m(\omega) = 1 - \frac{{\omega_p}^2}{\omega^2 + i\dfrac{{\omega_p}^2\varepsilon_0}{\sigma(\omega)}\omega},\; where\; {\omega_p}^2 = \frac{ne^2}{m\varepsilon_0} \quad (2.61)$$

Relation (2.61) is the dispersion relation for metals in Drude's model. The term ω_p is defined as the plasma frequency. There is a solution for longitudinal oscillations to exist at $\varepsilon_m(\omega) = 0$. At this point transverse oscillations do not exist. These longitudinal oscillations that describe the coupling of the incident light with the incoming light are called plasmons. In some metals the inter and intraband transitions need to also be considered in the metal to have an accurate model of the dielectric permittivity of the material. However, it is clear that for frequencies below ω_p light will be reflected from the metal and decays inside the metal. For frequencies above ω_p the metal starts becoming transparent. For specific cases an electromagnetic wave can be coupled to a collective plasma oscillation that can propagate along the surface of the metal. Such an oscillation is a surface plasmon polariton, very useful for the creation of nanoscale metallic waveguides. The physics of surface plasmon polaritons along with the solution of Maxwell's equations in simple metallic waveguide geometries will be provided in Chapter 4.

For the dielectric function of common dielectrics, the collective oscillation of localized charge under the application of the external oscillating field is the main motion present. A displacement of the positive with respect to the negative charge will create a dipole moment which for an ion is defined as:

$$p = e^* u \quad (2.62)$$

The term e* is the effective ionic charge. Considering a reduced mass μ in an ionic based system one can get the motion equation of an oscillator under an external electric field **E**.

$$\ddot{\mathbf{u}} + \gamma\boldsymbol{u} = -{\omega_0}^2\boldsymbol{u} + \frac{e^*}{\mu}\boldsymbol{E} \quad (2.63)$$

The total polarization term in a material of volume V will be given by the relation:

$$\boldsymbol{P} = \frac{N}{V}e^*\boldsymbol{u} + \varepsilon_0\frac{N}{V}a\boldsymbol{E} \quad (2.64)$$

The term N/V is the numerical density of oscillators in the material, γ is the decay constant related with the finite lifetime of the normal modes, and a is the electronic polarizability term. In a precise approximation instead of **E** the localized electric field $\mathbf{E}_{loc}$ should be in equations (2.63) and (2.64). Since in this thesis ferroelectric materials such as barium titanate oxide (BTO) will also be encountered, the next section will deal with ferroelectricity. The localized electric field is the enhanced electric field due to the fact that dipoles created by the external electric field generate an electric field of their own, enhancing the external. In optically dense materials this term should not be omitted.

With the Fourier transforms of (2.63) and (2.64) the two equations turn to:

$$\boldsymbol{u}(\omega) = \frac{1}{\omega_0{}^2 - \omega^2 - i\gamma\omega}\frac{\varepsilon^*}{\mu}\boldsymbol{E}(\omega) \quad (2.65)$$

and

$$\boldsymbol{P}(\omega) = \frac{N}{V}e^*\boldsymbol{u}(\omega) + \varepsilon_0\frac{N}{V}a\boldsymbol{E}(\omega) \quad (2.66)$$

Therefore, from the definition of the polarization term **P** through the dielectric permittivity ε_r of a material, the dielectric function of a dielectric material can easily be derived:

$$\varepsilon_d(\omega) = 1 + \frac{N}{V}a + \frac{\frac{N}{V}\frac{e^{*2}}{\varepsilon_0\mu}}{\omega_0{}^2 - \omega^2 - i\gamma\omega} \quad (2.67)$$

Relation (2.67) is the Lorentz model for the dielectric permittivity of dielectric materials. By setting ε_{st} as the permittivity near zero frequency and $\varepsilon_\infty = \varepsilon(\omega \to \infty)$ at very high frequencies, one can get a simplified relation for (2.67):

$$\varepsilon_d(\omega) = \varepsilon_\infty + \frac{\omega_0{}^2\left(\varepsilon_{st} - \varepsilon_\infty\right)}{\omega_0{}^2 - \omega^2 - i\gamma\omega} \quad (2.68)$$

The relation (2.68) in its general form leads to a graph for ε_1 and ε_2 where ε_2 has a resonance near ω_0, where the absorption is maximum. Now going to the fundamental interaction of light with a dielectric, one can assume a plane wave induced term for the polarization **P** in the x- and y- axes:

$$\left.\begin{aligned} P_x &= P_{x0}e^{-i(\omega t - qx)} \\ P_y &= P_{y0}e^{-i(\omega t - qx)} \end{aligned}\right\} \quad (2.69)$$

The longitudinal waves should satisfy the following set of equations:

$$\left.\begin{aligned}\nabla \times \boldsymbol{P}_{\mathrm{L}} &= 0 \\ \nabla \cdot \boldsymbol{P}_{\mathrm{L}} &\neq 0\end{aligned}\right\} \quad (2.70)$$

The transverse waves should satisfy the equations:

$$\left.\begin{aligned}\nabla \times \boldsymbol{P}_{\mathrm{T}} &\neq 0 \\ \nabla \cdot \boldsymbol{P}_{\mathrm{T}} &= 0\end{aligned}\right\} \quad (2.71)$$

In a dielectric medium without sources or free carriers:

$$\nabla \cdot \boldsymbol{D} = 0 \Rightarrow \varepsilon_r(\omega)\frac{\nabla \cdot \boldsymbol{P}}{\varepsilon_r(\omega) - 1} = 0 \quad (2.72)$$

Therefore, since a longitudinal wave can only exist for $\varepsilon(\omega_L) = 0$ then:

$$\boldsymbol{E}_{\mathrm{L}} = -\frac{1}{\varepsilon_0}\boldsymbol{P}_{\mathrm{L}} \quad (2.73)$$

The longitudinal waves cannot interact with the transverse optical waves but interact with electrons. On the other side, considering the full set of Maxwell's equations, the solution for the transverse waves can be derived, since transverse waves couple with external electromagnetic fields. Considering a non-magnetic material:

$$\left.\begin{aligned}\nabla \times \boldsymbol{E} &= -\mu_0\frac{\partial \boldsymbol{H}}{\partial t} \\ \nabla \times \boldsymbol{H} &= \frac{\partial \boldsymbol{D}}{\partial t}\end{aligned}\right\} \Rightarrow (ins.\ 2.69) \Rightarrow$$

$$\Rightarrow \left.\begin{aligned} qP_{y0} - \omega\varepsilon_0\mu_0\left[\varepsilon_d(\omega) - 1\right]H_{z0} &= 0 \\ -\omega\frac{\varepsilon_d(\omega)}{\varepsilon_d(\omega)}P_{y0} + qH_{z0} &= 0\end{aligned}\right\} \quad (2.74)$$

The determinant of the matrix composed by the system of equations of (2.74) must be zero, which naturally leads to the dispersion relation of the phonon polaritons.

$$\omega^2 = \frac{1}{\varepsilon(\omega)}c^2q^2 \quad (2.75)$$

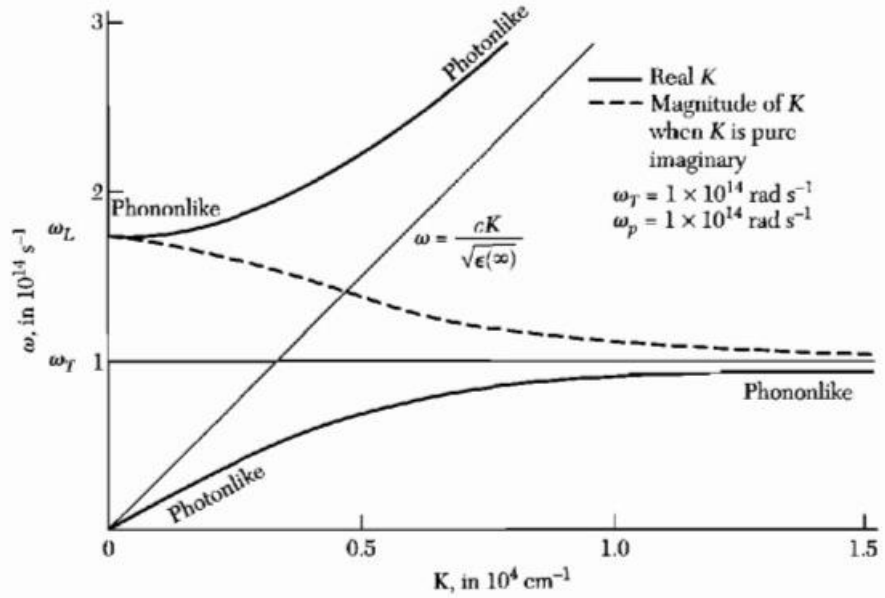


Fig. 2.5.1: Coupled modes of photons and transverse optical phonons in an ionic crystal [7]

Phonon polaritons are a type of quasiparticle that can form in a diatomic ionic crystal due to coupling of transverse optical phonons and photons. They are particular type of polariton, which behave like bosons. Phonon polaritons only result from coupling of transverse optical phonons, this is due to the particular form of the dispersion relation of the phonon and photon and their interaction. In many cases, phonon polaritons interpret the light-matter interaction. Since energy and momentum must be conserved, light is absorbed resulting in many cases to the oscillation of a crystal structure and the propagation of a phonon. Figure 2.5.1 presents the dispersion relation ω-K with the K value being the wavevector of the incident wave and ω being the cyclic frequency. The line ω represents the light line which is the dispersion relation of the optical waves in the medium without coupling to lattice oscillations. The heavy lines represent the dispersion relations considering coupling to collective oscillations of the crystal structure. Near the limits of the lines the propagating electromagnetic-mechanical wave obtains a photon- or phonon-like behavior while in the area between the values ω_L and ω_T the wave attenuates and there is not propagation of a phonon. An important relation is the Lydanne-Sachs-Teller (LST) relation that connects the natural frequency of longitudinal optic lattice vibrations with natural frequency of the transverse optical lattice vibration under the hypothesis of long wavelengths and is shown below:

$$\frac{\omega_L}{\omega_T} = \frac{\varepsilon_{st}}{\varepsilon_\infty} \quad (2.76)$$

A similar graph can be derived from the dispersion relation for the plasmons. These graphs prove very useful as they can be a measure of coupling of the external electric field with either mechanical (phonons) or electronic (plasmons) oscillations in a material. The graph for the dispersion relation of the plasmons will ensue again in Chapter 4, where plasmonic waveguides and coupling from photonic to plasmonic waveguides will be dealt with.

To sum up, in this section the fundamentals of light-matter interaction in solid-state physics were presented. The dielectric permittivity in a generalized medium is a complex function having a real and an imaginary part. The imaginary part of the dielectric function is related with the absorption of the material. The dielectric permittivity of the metals is related with the collective motion of the free-electron gas. The expression for the dielectric permittivity in metals is the Drude Model. An external electromagnetic field can couple with collective oscillations of plasma, the plasmons or plasmon polaritons. Plasmons can explain the reflectivity of metals. On the other side the permittivity of a dielectric is connected to the oscillations of ions in a crystal structure. The expression for the permittivity of a dielectric is the Lorentz model. An external electromagnetic wave can couple to collective oscillations that propagate in the crystal structure, the phonons or phonon polaritons. Phonons can interpret the light absorption of dielectrics. In the next Section we transit from materials to analyze practical devices that can transfer electromagnetic waves, the waveguides.

2.6: Analytical Expressions of Mode in Perfect Metal Based Waveguide

Going from materials back to the fundamentals of classical electromagnetic field theory, this Section will analyze the propagation of an electromagnetic wave through a waveguide. The waveguide is a device with symmetry along one axis, the axis of propagation, and a rectangular geometry in the two axes vertical to the direction of propagation. For many cases, the study of propagation through a waveguide is limited to the cross-sectional solutions of Maxwell's equations for this reason. The dimensions of the device in the vertical direction are comparable to the guided wavelength of the propagating electromagnetic wave, leading to confinement. The analysis will begin from the simplest form of a waveguide the perfectly metallic slab waveguide. In a slab

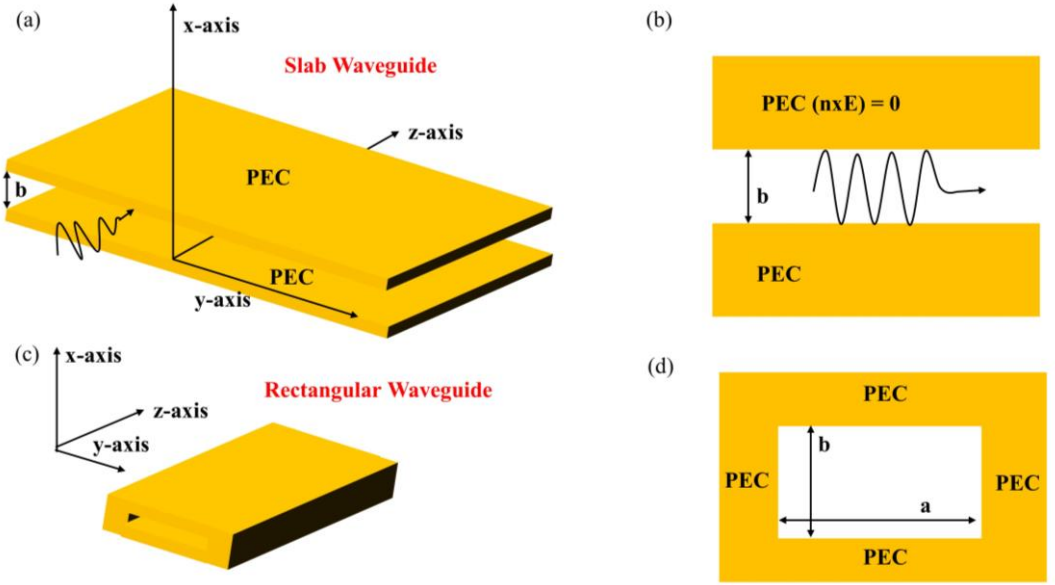


Fig. 2.6.1: (a) 3D-Schematic and (b) Cross-Sectional Design of Perfect Metallic Slab Waveguide, (c) 3D-Schematic and (d) Cross-Sectional Design of Perfect Metallic Rectangular Waveguide

waveguide the material is continuous in the two of the three directions and there is a material boundary condition only in one direction of the three. Such a slab is represented in Fig. 2.6.1(a) and (b) along with another perfect metallic rectangular waveguide design in Fig. 2.6.1 (c) and (d). As seen in Fig. 2.6.1(a) in a slab the two dimensions are considered infinite and continuous while in one dimension there is a confinement of the field to a value b comparable or close to the guided wavelength of the propagating wave. In a rectangular waveguide as in 2.6.1 (c) confinement in two dimensions with values a and b occurs. The results that will be derived for the slab waveguide will be generalized for the rectangular metallic waveguide. The material across all the material interfaces in the waveguides of Fig. 2.6.1 is a perfect metal which precisely imposes the boundary condition $\boldsymbol{n}\times\boldsymbol{E}=0$. Let the material at the core of the waveguide be air with dielectric permittivity ε_0 and permeability μ_0. Beginning from Maxwell's equations, after considering a field dependence on $e^{i\omega t}$:

$$\nabla \cdot \varepsilon \boldsymbol{E} = \rho_f$$

$$\nabla \cdot \boldsymbol{B} = 0$$

$$\nabla \times \boldsymbol{E} = -i\omega \boldsymbol{B}$$

$$\nabla \times \boldsymbol{H} = i\omega\varepsilon \boldsymbol{E}$$

In the space of the waveguide where there are no carriers or currents, Maxwell's equations take the form:

$$\left.\begin{aligned}\nabla^2\mathrm{E}+\beta^2\mathrm{E}=0\\ \nabla^2\mathrm{H}+\beta^2\mathrm{H}=0\end{aligned}\right\},\ where\ \beta^2=\omega^2\mu\varepsilon \quad (2.77)$$

From relation (2.77) analyzing the vector equations to all coordinates, six scalar second-order partial differential equations can ensue. Also, from the curl equations one can get:

$$\nabla \times \boldsymbol{E} = -i\omega\mu\boldsymbol{H} \Rightarrow$$

$$\frac{\partial E_z}{\partial y} - \frac{\partial E_y}{\partial z} = -i\omega\mu H_x \quad (2.78)$$

$$\frac{\partial E_x}{\partial z} - \frac{\partial E_z}{\partial x} = -i\omega\mu H_y \quad (2.79)$$

$$\frac{\partial E_y}{\partial x} - \frac{\partial E_x}{\partial y} = -i\omega\mu H_z \quad (2.80)$$

Also,

$$\nabla \times \boldsymbol{H} = i\omega\varepsilon \boldsymbol{E} \Rightarrow$$

$$\frac{\partial H_z}{\partial y} - \frac{\partial H_y}{\partial z} = i\omega\varepsilon E_x \quad (2.81)$$

$$\frac{\partial H_x}{\partial z} - \frac{\partial H_z}{\partial x} = i\omega\varepsilon E_y \quad (2.82)$$

$$\frac{\partial H_y}{\partial x} - \frac{\partial H_x}{\partial y} = i\omega\varepsilon E_z \quad (2.83)$$

Solving for E_x, E_y, H_x, H_y, the above equations can turn in a form where if E_z and H_z are known it is possible to calculate all the other field components through these two.

$$E_x = -\frac{\beta_z}{\beta^2}\frac{\partial E_z}{\partial x} - i\frac{\omega\mu}{\beta^2}\frac{\partial H_z}{\partial y} \quad (2.84)$$

$$E_y = -\frac{\beta_z}{\beta^2}\frac{\partial E_z}{\partial y} + i\frac{\omega\mu}{\beta^2}\frac{\partial H_z}{\partial x} \quad (2.85)$$

$$H_x = -\frac{\beta_z}{\beta^2}\frac{\partial E_z}{\partial y} + i\frac{\omega\varepsilon}{\beta^2}\frac{\partial E_z}{\partial x} \quad (2.86)$$

$$H_y = -\frac{\beta_z}{\beta^2}\frac{\partial E_z}{\partial x} + i\frac{\omega\varepsilon}{\beta^2}\frac{\partial E_z}{\partial y} \quad (2.87)$$

For the solution of this electromagnetic problem the method of the separation of variables will be employed. More specifically the component E_z is considered as the product of three functions, each depending on one spatial variable.

$$E_z(x,y,z) = X(x)\,Y(y)\,Z(z) \quad (2.88)$$

Inserting (2.88) in (2.77) one gets the equation:

$$\frac{1}{X}\frac{\partial^2 X}{\partial x^2} + \frac{1}{Y}\frac{\partial^2 Y}{\partial y^2} + \frac{1}{Z}\frac{\partial^2 Z}{\partial z^2} + \beta^2 = 0 \quad (2.89)$$

One can split (2.89) in three equations the need to be satisfied:

$$\frac{1}{X}\frac{\partial^2 X}{\partial x^2} = -\beta_x{}^2 \ (2.90)$$

$$\frac{1}{Y}\frac{\partial^2 Y}{\partial y^2} = -\beta_y{}^2 \ (2.91)$$

$$\frac{1}{Z}\frac{\partial^2 Z}{\partial z^2} = -\beta_z{}^2 \ (2.92)$$

The differential equations presented above can be solved considering the material boundary conditions of each case. In this particular problem, the field does not change with respect to the y axis, therefore Y(y) is constant. Its value can be considered equal to 1. The constant β_y is equal to zero. Therefore:

$$E_z(x,y,z) = X(x)\,Z(z) \ \ (2.93)$$

The solution of (2.92) is the sum of two terms that represent plane waves as shown in (2.94):

$$Z_E(z) = A_E e^{-\beta_z z} + B_E e^{\beta_z z} \ (2.94)$$

From the two terms, the second term represents a wave that propagates backwards along the z-axis and for this reason B_E is set to zero, since in this problem the electromagnetic wave propagates to the +z values. Therefore,

$$Z_E(z) = A_E e^{-\beta_z z} \Rightarrow \frac{\partial^2 X_E}{\partial x^2} + \left(\beta_z{}^2 + \beta^2\right) X_E = 0 \ (2.95)$$

The differential equation (95) has the solution:

$$X_E(x) = A'_E e^{-ihx} + B'_E e^{ihx} \ (2.96)$$

$$with\ h^2 = \beta_z^2 + \beta^2 \ (2.97)$$

Then, considering (2.96) and (2.97) the Dirichlet Boundary Condition is applied at the interfaces with the perfect metals:

$$\boldsymbol{n} \times \boldsymbol{E} = 0 \Rightarrow E_z = 0 \ for\ x = 0 \ and\ x = b$$

For this reason, at these two positions:

$$X_E = 0 \Rightarrow \left.\begin{array}{r} X_E(0) = A_E' + B_E' = 0 \\ X_E(b) = A_E' e^{-ihb} + B_{E'} e^{ihb} = 0 \end{array}\right\} \Rightarrow$$

$$D = \begin{vmatrix} 1 & 1 \\ e^{-ihb} & e^{ihb} \end{vmatrix} = 0 \Rightarrow 2i\sin(hb) = 0 \quad (2.98)$$

The condition (2.98) indicates that the parameter h must receive discrete values:

$$h = h_m = \frac{m\pi}{b} \; with \; m = 1, 2, 3 \ldots \quad (2.99)$$

Additionally,

$$A_E' = -B_E' \Rightarrow$$

$$X_E = 2iB_E' \sin\left(\frac{m\pi}{b}x\right) \quad (2.100)$$

Introducing (2.100) to the original expression of E_z, the final result is the following for the expression of the electric field E_z:

$$E_{zm}(x,z) = X_E(x) Z_E(z) = 2iB_E' \sin\left(\frac{m\pi}{b}x\right) A_E e^{-\beta_{zm} z} \Rightarrow$$

$$E_{zm}(x,z) = E_{ozm} \sin\left(\frac{m\pi}{b}x\right) e^{-\beta_{zm} z} \quad (2.101)$$

The physical repercussions of equation (2.101) are the following: every value of m corresponds to a different electric field distribution. Such a solution of Maxwell's equation in a waveguide-based system is called waveguide mode. The expression (2.97) becomes the criterion that determines whether a waveguide mode can propagate in a slab waveguide of thickness b. When the term β_{zm} is real, from (2.101) there is an exponential decay of the electric field, thus wave propagation is impossible. The term β_{zm} must be imaginary (or complex in a lossy waveguide for the wave propagation) to be possible. The part of the propagation constant that does not lead to an exponential decay of the electric field expresses the phase-term. The condition for β_{zm} to be imaginary is the following:

$$\beta_{zm} = \sqrt{h^2 - \beta^2} = \sqrt{\left(\frac{m\pi}{b}\right)^2 - \omega^2\mu\varepsilon} \quad (2.102) \Rightarrow$$

$$\Rightarrow \left(\frac{m\pi}{b}\right)^2 < \omega^2\mu\varepsilon \Rightarrow$$

$$\Rightarrow f > \frac{m}{2b\sqrt{\mu\varepsilon}} \quad (2.103)$$

The propagation constant under this condition becomes:

$$\beta_{zm} = \beta_m = i\sqrt{\omega^2\mu\varepsilon - \left(\frac{m\pi}{b}\right)^2} \quad (2.104)$$

The term m relates with the mode order. Considering that the lowest value for m is 1, then the fundamental mode has an order m-1. The fundamental mode is the mode having one maximum point as shown in (2.101). The first order mode is the mode with m = 2, having two maxima. The symmetry of the mode changes from even to odd as well for increasing mode orders. A waveguide supports a mode of order m if the condition (2.103) is true. The right term of (2.103) is the cutoff frequency of a waveguide mode, defined as follows:

$$f_c = \frac{m}{2b\sqrt{\mu\varepsilon}} \quad (2.105)$$

With the expression (2.104), the phase-velocity and the guided wavelength of a mode with a mode-order m are defined by the following expressions:

$$v_{ph} = \frac{\omega}{\beta} = \frac{\omega}{\sqrt{\omega^2\mu\varepsilon - \left(\frac{m\pi}{b}\right)^2}} \quad (2.106)$$

$$\lambda_g = \frac{2\pi}{\beta_m} = \frac{2\pi}{\sqrt{\omega^2\mu\varepsilon - \left(\frac{m\pi}{b}\right)^2}} \quad (2.107)$$

Similar to the material refractive index a waveguide effective index can be defined for each mode with order m:

$$\beta_m = \frac{2\pi}{\lambda_0} n_{eff} \Rightarrow$$

$$\Rightarrow n_{eff} = \frac{\lambda_0}{2\pi}\beta_m = \frac{\lambda_0}{2\pi}\frac{1}{\sqrt{\omega^2\mu\varepsilon - \left(\frac{m\pi}{b}\right)^2}} \quad (2.108)$$

The effective index expression in (2.108) shows that whether a mode with a mode order m can be supported by a slab waveguide of thickness b depends solely on the free-space wavelength comparison with the material properties μ and ε and the waveguide dimensions (thickness b). A waveguide group index can be defined as in the optical properties of materials. The group index value is very important in telecommunications since it is connected to the waveguide dispersion. The waveguide group index is defined as:

$$\upsilon_{gr} = \frac{d\omega}{d\beta_m} \quad (2.109)$$

Finally, the magnetic field expression of a waveguide mode with mode order m can be calculated with the same procedure as the one followed for the derivation of the electric field expression and is the following:

$$H_{zm} = H_{0m}\cos\left(\frac{m\pi}{b}x\right)e^{-\beta_{zm}z} \quad (2.110)$$

Going now to the geometries of Figure 2.6.1(c) and 2.6.1(d), a similar procedure can be followed for the solution of Maxwell's equations. In this case, the mode has two important values m and n and the order of a mode confined in two dimensions is defined as m-1 and n-1. The electric and magnetic field expressions are dependent on both dimensions in the xy-plane, a and b, and the cutoff frequency needs to have both values of m and n. Therefore, for a rectangular perfectly metallic (lossless) waveguide the following expressions can be defined:

$$E_{zmn} = E_{0zmn}\sin\left(\frac{m\pi}{b}x\right)\sin\left(\frac{n\pi}{a}y\right)e^{-\beta_{zm}z} \quad (2.111)$$

$$H_{zmn} = H_{0m}\cos\left(\frac{m\pi}{b}x\right)\cos\left(\frac{n\pi}{a}y\right)e^{-\beta_{zm}z} \quad (2.112)$$

$$\beta_{zm} = \beta_m = i\sqrt{\omega^2\mu\varepsilon - \left(\frac{m\pi}{b}\right)^2 - \left(\frac{m\pi}{a}\right)^2} \quad (2.113)$$

With respect to their polarization, the waveguide modes can be Transverse Magnetic (TM) modes where the magnetic field component parallel to the propagation direction is zero. For the slab waveguide of 2.6.1(a) and 2.6.1(b), using this condition and the expressions for the rest of the electric and magnetic field components (equations 2.84-2.87):

$$\boldsymbol{TM}: H_{zm} = 0 \quad (2.114)$$

$$E_{zm} = E_{0m}\sin\left(\frac{m\pi}{b}x\right)e^{-\beta_{zm}z} \quad (2.115)$$

$$E_{xm} = -E_{0m}\frac{\beta_{zm}}{\left(\frac{m\pi}{b}\right)}\cos\left(\frac{m\pi}{b}x\right)e^{-\beta_{zm}z} \quad (2.116)$$

$$H_{ym} = -E_{0m}\frac{i\omega\varepsilon}{\left(\frac{m\pi}{b}\right)}\cos\left(\frac{m\pi}{b}x\right)e^{-\beta_{zm}z} \quad (2.117)$$

Otherwise, waveguide modes can be Transverse Electric (TE) where the electric field component parallel to the propagation direction is zero. Similarly to the above conditions:

$$\boldsymbol{TE}: E_{zm} = 0 \quad (2.118)$$

$$H_{zm} = H_{0m}\cos\left(\frac{m\pi}{b}x\right)e^{-\beta_{zm}z} \quad (2.119)$$

$$E_{ym} = -H_{0m}\frac{i\omega\mu}{\left(\frac{m\pi}{b}\right)}\sin\left(\frac{m\pi}{b}x\right)e^{-\beta_{zm}z} \quad (2.120)$$

$$H_{xm} = -H_{0m}\frac{\beta_{zm}}{\left(\frac{m\pi}{b}\right)}\sin\left(\frac{m\pi}{b}x\right)e^{-\beta_{zm}z} \quad (2.121)$$

There are also special types of waveguide modes where both the electric and magnetic field components parallel to the direction of propagation are set to zero. These modes are called Transverse ElectroMagnetic (TEM) and the condition that is true is the following:

$$\boldsymbol{TEM}: E_{zm} = H_{zm} = 0 \quad (2.122)$$

A special type of a waveguide device that can support TEM modes is the transmission line, for example a coaxial cable that transfers the RF signal from a reception antenna to the TV. Microstrip lines in Printed Circuit Boards (PCBs) and Monolithic Microwave Integrated Circuits (MMICs) are also special cases of integrated transmission lines.

Finally, there are waveguide modes that do not have zero electric or magnetic field components parallel to the direction of propagation. Such is the case for all dielectric waveguides (for example optical fibers and integrated optical waveguides) that are of interest in this PhD Thesis. These modes are called hybrid HE or EH modes:

$$\boldsymbol{Hybrid\ EH\ or\ HE}:$$

$$E_{zm} \neq 0\,,\ H_{zm} \neq 0 \quad (2.123)$$

Even without completely zero components, integrated dielectric photonic waveguides usually have a rectangular cross-section supporting modes with a dominant E_x or E_y component (if z is the propagation axis, y is the horizontal and x is the vertical axis) or a dominant H_x or H_y component. Considering the axis definition of Figure 2.6.1, if E_y and H_x are dominant and the E_z approaches 0 then the supported mode is almost-TE. If E_x and H_y are dominant and H_z approaches 0 then the mode is almost-TM. From this point on, the almost-TE modes will be called TE and the almost-TM mode will be

called TM in integrated photonic waveguides. The direct mathematical expressions are true for the specific cases of the perfectly metallic slab and rectangular waveguide, but the fundamentals are also true for all types of waveguides.

2.7: Analytical Expressions of a Dielectric Slab Waveguide

In the above, for the calculation of the electric and magnetic field expressions propagating as waveguide modes in a waveguide, Maxwell's equations were solved imposing the boundary conditions at the material interfaces of the device and considering the designed geometry. In perfect metallic waveguides, the boundaries are closed since the boundary condition at all interfaces was $\boldsymbol{n} \times \boldsymbol{E} = 0$. Such is an ideal case of a waveguide for the delivery of RF signals. In dielectric waveguides, as those used in integrated photonics, the boundary conditions are open since there is always a middle material with higher refractive index n_2 and an upper and lower material with lower refractive indices n_1 and n_3 respectively. If $n_1 = n_3$ the dielectric waveguide is symmetric while if not the waveguide is asymmetric. The central area of the waveguide surrounded by the lower refractive index material (cladding) is the core of the waveguide (with a higher refractive index n_2) where wave propagation takes place. Unlike the case of metallic waveguides, dielectric waveguides support modes that at the area of the upper and lower cladding have evanescent decaying tails.

The solution of Maxwell's equations in dielectric waveguides because of the continuity that is imposed at the dielectric material boundary conditions is not as easy as the metallic waveguide case, where the Dirichlet Boundary Condition is imposed. Initially, the dielectric slab waveguide shown in a 3D manner and from its sideview in Fig. 2.7.1 (a) and (b) will be studied. Here, two routes can be taken for the analysis of the dielectric slab waveguide. The first is the solution of Maxwell's equations with the appropriate boundary conditions which as a task is more difficult, and the second is the utilization of the optical ray theory where a paraxial approximation is considered. The second option provides significant physical insight to the propagation phenomena in a dielectric slab waveguide but does not provide full information for the electric field of the optical modes at the core and the cladding of the dielectric waveguide. In this section, the second option will initially be followed, since in most cases with real

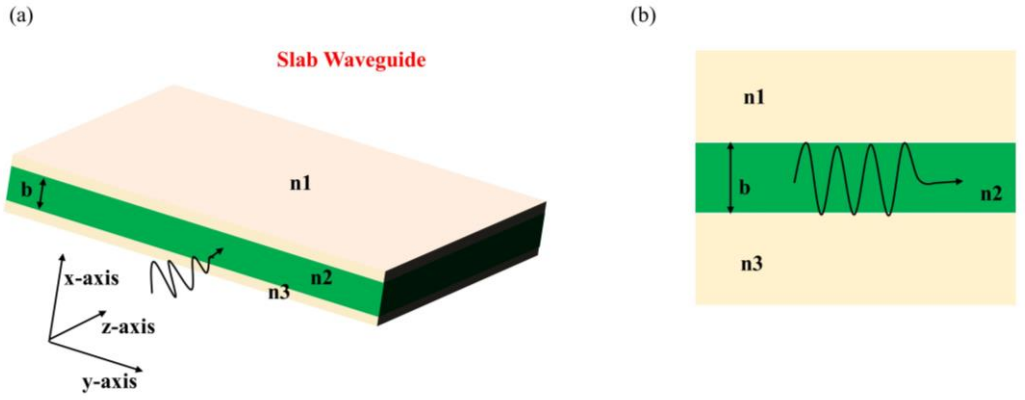


Fig. 2.7.1: (a) 3D-Schematic and (b) Cross-Sectional Design of Dielectric Slab Waveguide with core refractive index n_2 and cladding refractive index n_1 and n_3. If the slab waveguide is symmetric then $n_1 = n_3$.

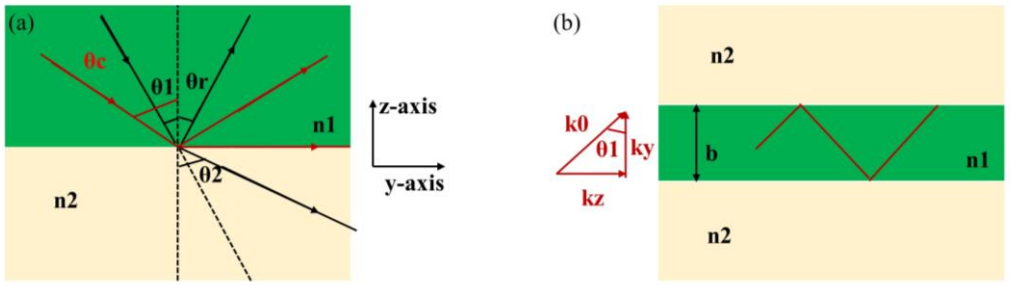


Fig. 2.7.2: (a) Fundamentals of Optical Waveguiding: Snell's Law and Brewster Angle, (b) Propagation in a waveguide through optical ray theory.

integrated photonic waveguides (with the exception of the optical fiber with a circular symmetry where analytical field expressions can be obtained) numerical solutions are necessary anyway. The basis of this optical ray approach is reference [10].

The optical ray approach for an optical waveguide begins from the formulation of Snell's law describing the propagation of an optical ray from a medium with a refractive index n_1 to a medium with a refractive index n_2. A schematic of such an optical ray transition is depicted in Fig. 2.7.2(a). An incident angle θ_1 is considered for the incident optical ray. Under the paraxial approximation, the electric field is dominant near the main propagation axis. As shown in the same figure, an amount of optical power is transmitted to the second medium with a refraction angle θ_2 and an amount of power is reflected with a reflection angle $\theta_r = \theta_1$. If the refractive indices of the materials of the two media have a relation $n_1<n_2$ (the second medium is optically denser than the first medium) then $\theta_1>\theta_2$, but as shown in Fig. 2.7.2(a), if the first medium is optically denser, thus $n_1>n_2$ then $\theta_1<\theta_2$. The relation that connects the values of n_1, n_2 with the incident and refraction angles θ_1 and θ_2 is Snell's Law as described below:

$$n_1 \sin\theta_1 = n_2 \sin\theta_2 \quad (2.124)$$

It can be proven that in the case of an optically denser first medium with $n_1>n_2$ there is a critical angle where the refraction angle is 90° and theoretically most amount of power is reflected. This happens at any incident angle θ_1 larger than the critical angle named the Brewster angle:

$$\theta_c = arc\sin\left(\frac{n_2}{n_1}\right) \quad (2.125)$$

Based on the definition of the Brewster angle, there is a cone with half-angle a_{max} where optical rays can propagate in a zig-zag manner in the core of a waveguide as in the case of Fig. 2.7.2(b):

$$n_0 \sin a_{max} = n_1 \cos\theta_1 = \sqrt{n_1^2 - n_2^2} \approx n_1 2\Delta \quad (2.126)$$

with

$$\Delta = \frac{n_1 - n_2}{n_1} \quad (2.127)$$

The amount of power that is reflected, when light transits from a medium with refractive index n_1 to a medium with a refractive index n_2 with an incidence angle θ_1, can be calculated through Fresnel's equations that can be proven for the TE and TM polarizations for the incident light. These equations can be modified considering relations (2.126) and (2.127):

$$r_{TE} = \frac{n_1 \cos\theta_1 - \sqrt{n_2^2 - n_1^2 \sin^2\theta_1}}{n_1 \cos\theta_1 + \sqrt{n_2^2 - n_1^2 \sin^2\theta_1}} \quad (2.128)$$

$$r_{TM} = \frac{n_2^2 \cos\theta_1 - n_1\sqrt{n_2^2 - n_1^2 \sin\theta_1}}{n_2^2 \cos\theta_1 + n_1\sqrt{n_2^2 - n_1^2 \sin\theta_1}} \quad (2.129)$$

$$R = \frac{S_r}{S_i} = \frac{E_r^2}{E_i^2} = r^2 \quad (2.130)$$

When the critical angle θ_c is surpassed, the values in the roots of (2.128) and (2.129) become negative and the reflection coefficients become complex. For this reason, phase shifts are imposed to the reflected waves with values:

$$\varphi_{TE} = 2arc\tan\left[\frac{\sqrt{\sin^2\theta_1 - \left(\frac{n_2}{n_1}\right)^2}}{\cos\theta_1}\right] \quad (2.131)$$

$$\varphi_{TM} = 2arc\tan\left[\frac{\sqrt{\left(\frac{n_1}{n_2}\right)^2 \sin^2\theta_1 - 1}}{\left(\frac{n_2}{n_1}\right)\cos\theta_1}\right] \quad (2.132)$$

Going back to Fig. 2.7.2(b), in the y-axis, light should form a standing wave pattern with phase-shifts indicating the back-and-forth propagation (distance 2b). So that the wave is sustained in the y-axis, the total phase-shift should have a value of multiple of 2π, therefore:

$$2\beta_0 n_1 b\cos\theta_1 - \varphi_u - \varphi_l = 2m\pi \quad (2.133)$$

The extra phase terms from equation (2.133) are imposed by the reflection of light to the upper and lower cladding of the slab waveguide. The value of m signifies the mode of propagation as in the case of the metallic waveguide studied with the classical electromagnetic field theory. A discrete set of values θ_1 can let optical rays propagate through the slab waveguide. If the waveguide is symmetric and the upper and lower cladding have the same refractive index n_2 then, considering a TE polarization, (2.133) can be rearranged to:

$$TE: \tan\left(\frac{\beta_0 n_1 b\cos\theta_1 - m\pi}{2}\right) = \frac{\sqrt{\sin^2\theta_1 - \left(\frac{n_2}{n_1}\right)^2}}{\cos\theta_1} \quad (2.134)$$

The relation (2.134) provides all the supported guided modes of the waveguide via all the possible discrete values θ_1. The maximum number of guided modes supported by a slab waveguide can be calculated for the critical angle θ_c from the relation:

$$m_{max} = \frac{\beta_0 n_1 b\cos\theta_c}{\pi} \quad (2.135)$$

The single-mode condition can also be obtained through (2.134) considering $\theta_1 = \theta_c$ and m = 1 and is provided by the following relation:

$$\theta_c \le arc\cos\left(\frac{\lambda_0}{2n_1 b}\right) \quad (2.136)$$

As in the case of the perfectly metallic waveguide, whether a waveguide mode is supported or not depends on the free-space wavelength λ_0 to the physical dimension b relation. From the above, a modal effective index n_{eff} (considering no losses thus $Im\{n_{eff}\} = 0$) based on the optical ray theory can be defined as:

$$n_{eff} = n_1\sin\theta_1 \quad (2.137)$$

and

$$\beta_z = \beta = n_{eff}\beta_0 \quad (2.138)$$

Based on the upper and lower limits of the discrete values of θ_1, an upper and lower value for the real part of the effective index can be calculated:

$$n_2\beta_0 \le \beta \le n_1\beta_0 \Rightarrow$$

$$\Rightarrow n_2 \le Re\{n_{eff}\} \le n_1 \quad (2.139)$$

As in the cases of the optical materials, in a case of a waveguide having propagation losses then the effective index n_{eff} is complex and is given by the relation:

$$n_{eff} = Re\{n_{eff}\} + iIm\{n_{eff}\} \quad (2.140)$$

The real part of the effective index is related with the phase of the propagating light while the imaginary part of the effective index in a waveguide is connected with the propagation losses of a propagating mode with number m.

The classical electromagnetic field calculation can provide complete expressions of **E** and **H**. Again, considering magnetization **M** = 0, current densities **J** = 0, and free charge densities ρ_{free} = 0, considering that the electric and magnetic fields have a dependency:

$$E = E(x,y)\, e^{i(\omega t - \beta z)} \quad (2.141)$$

$$H = H(x,y)\, e^{i(\omega t - \beta z)} \quad (2.142)$$

It is very easy to construct Helmholtz equation as performed in the easier case of a perfectly metallic waveguide:

$$\nabla^2 \boldsymbol{E} + n^2 \beta_0{}^2 \boldsymbol{E} = 0 \quad (2.143) \Rightarrow$$

$$\frac{\partial^2 \boldsymbol{E}}{\partial x^2} + \frac{\partial^2 \boldsymbol{E}}{\partial y^2} + \left[\beta_0{}^2 n_i{}^2 - \beta^2\right]\boldsymbol{E} = 0 \quad (2.144)$$

$$where\ \beta = \frac{2\pi}{\lambda_0} n_{eff},\ i = 1,2,3 \quad (2.145)$$

The values of E and dE/dy must be continuous for (2.144) to be solved. The boundary condition that needs to be applied to both interfaces to solve (2.144) considering the symmetric waveguide of Fig, 2.7.2 (b) is the following:

$$\boldsymbol{n} \times \boldsymbol{E}_1 = \boldsymbol{n} \times \boldsymbol{E}_2 \quad (2.146)$$

An eigenvalue equation as the one derived with the optical ray theory will ensue from the solution of the differential equation (2.144). The term β is the propagation constant related with the complex modal effective index given in (2.140) and is the eigenvalue calculated through (2.144). The vectors **E** and **H** calculated through the differential equations are the eigenvectors of such eigenvalue problems. Further details on the mathematical expressions are provided in [10] and [11].

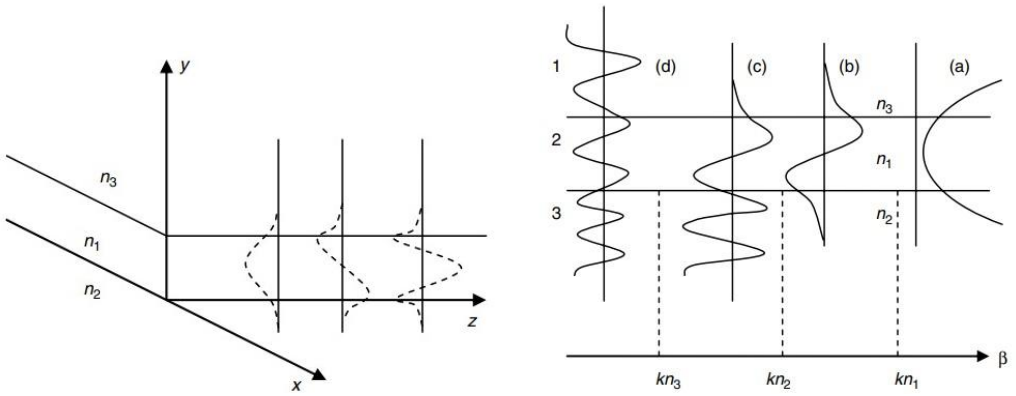


Fig. 2.7.3: (a) The first three modes of a dielectric slab waveguide, (b) Radiated and Guided Modes supported by a dielectric slab waveguide [10]

A physical insight on the modal electric field expressions in a waveguide can be seen in Fig. 2.7.3. Fig. 2.7.3(a) depicts the three first guided modes supported by a dielectric slab waveguide with sufficient thickness b, the fundamental, the first order mode and the second order mode, considering one of the two (almost-TE or almost-TM) polarizations. These modes correspond to the discrete values m=1,2,3 and as seen in Fig. 2.7.3(a) they have an alternating symmetry for an increasing number m. The fundamental mode can be named as TE_0 (if the polarization is TE), the first order mode is TE_1 and the second-order mode is TE_2. The fundamental mode TE_0 has one maximum and is even in terms of symmetry, the second mode TE_1 is odd and has two maxima, and the third mode TE_2 is even and has three maxima. Also, as shown in Fig. 2.7.3(b) apart from the guided modes with discrete numbers m a continuous set of radiated modes can be supported, extending to the cladding. All guided modes on the other side have exponentially decaying tails at the cladding sides and their mathematical expression is exponentially decaying. The guided and radiated modes in a waveguide form an orthonormal set from a mathematical point of view. For the optical ray theory to approach the accurate findings of the classical electromagnetic field theory the Goos-Haenchen shifts should also be considered. For this reason, in this thesis a classical computation of the electromagnetic fields will be applied towards the design of integrated photonic devices.

An important parameter coming as a result of the exponentially decaying tails of the guided modes in the cladding is the confinement factor in an integrated photonic waveguide given in its 1D expression as:

$$\Gamma(\%) = \frac{\int_0^b E_x^2(y)\,dy}{\int_{-\infty}^{+\infty} E_x^2(y)\,dy} \cdot 100 \quad (2.147)$$

In its 2D expression the terms of (2.147) are double integrals and considering x as the axis of propagation and y, z as the cross-sectional directions the expression for the confinement factor becomes:

$$\Gamma(\%) = \frac{\int_0^b \int_{-\frac{a}{2}}^{\frac{a}{2}} |E|^2 dydz}{\int_{-\infty}^{+\infty} \int_{-\infty}^{+\infty} |E|^2 dydz} \cdot 100 \quad (2.148)$$

The final consideration of the mode calculation that is also of interest in rectangular waveguides is the determination of losses in waveguides. In dielectric waveguides, the electric fields of the propagating modes have exponentially decaying expressions at the claddings. Electric field proportions of the mode exist in both the core and the cladding. For this reason, the main loss contributions come from the following phenomena: a) Scattering losses, b) Bending losses, c) Absorption losses, and d) Coupling losses [10]. The scattering losses are due to the electric field proportion that exist at the core-cladding material interfaces. Because the surfaces of these interfaces are rough, an amount of the optical power is scattered. Even though older and newer publications have made an attempt to model the scattering losses in waveguides [12, 13], it is very difficult to achieve excellent agreement with experimental measurements. For this reason, feedback from measurements is used as an addition to the other loss terms calculated in simulation. The bending losses are a direct result of power percentages that transit from the waveguide cladding and decay at the position of the bends. This loss term can easily be calculated in a mode solver [10]. Companies that provide a material platform for integrated photonics usually provide a minimum radius where this loss term is miniscule. The absorption losses come as a direct result of the imaginary part of the effective index $\mathrm{Im}\{n_{eff}\}$ of a waveguide mode, where the result of the absorption of all materials of a waveguide is inserted. Extra absorption losses will come directly as the result of the presence of extra carriers in semiconductors, for example in Si. These loss terms can be accurately calculated by a mode solver [10]. The last term relates with the amount of power that may be transferred through tunneling to another waveguide or device in close proximity to the initial waveguide. This is a direct result of the evanescent form of the electric field at the cladding (since the value is non-zero). A photonics device designer can easily calculate the minimum gap and length that is needed for considerable coupling to occur via a mode solver and a 3D-FDTD solver. This phenomenon needs to be avoided when only wave propagation is of interest but can be exploited in practical coupler designs.

The fundamentals for the dielectric slab waveguides presented above may have been calculated analytically thus the accurate expressions do not hold true for rectangular waveguides but the physical insight is the same going from the optical confinement in one axis to optical confinement in two axes.

2.8: The Finite Difference Method in Eigenvalue Problems

In Sections 2.6 and 2.7 the modes that can be supported by a perfectly metallic and a dielectric slab of thickness b were calculated. Even though the optical ray theory was utilized in Section 2.7 for an easier calculation of the propagating modal numbers m,

for both cases the employment of Maxwell's equations from the classical field theory and the solution of the Helmholtz equation were necessary for the calculation of the effective indices n_{eff} or propagation constants β and the electric and magnetic field expressions **E** and **H**. The values of β or n_{eff} were the eigenvalues of the Helmholtz equation while the expressions of **E** and **H** were the eigenvectors. The rigorous analytical solution of the eigenvalue problem was possible for the dielectric slab waveguide. Nevertheless, numerical or approximate semi-analytic solutions are necessary to deal with the eigenvalue problem in practical rectangular or complex geometry-based waveguides. The 3D-Schematic along with the cross-sectional view of a practical rectangular Si-based waveguide is shown below in Fig. 2.8.1. The waveguide of Fig. 2.8.1 is composed of a Si-based core, a SiO_2 based upper and lower cladding, and a Si-based substrate. Electromagnetic problems in practical photonic device design can be classified to two categories: a) Eigenvalue Problems: such is the case when one searches how many and which modes are supported by a waveguide composed of certain materials and having dimensions of width a and thickness b, and b) Excitation-based Problems: where one needs to simulate the evolution of the electric or magnetic field to time and space to determine the response of a certain device design for example to evaluate in simulation the coupling efficiency, imbalance, and bandwidth of a multimode interference coupler (MMI).

This section will deal with the solution of the eigenvalue problem in practical integrated photonic waveguides. In the past due to the limitations of computers approximate solutions of the eigenvalue problem were dominant in the literature. Two very known approximate methods for the eigenvalue problem solution are the Marcatili's method and the effective index method described in [10]. Marcatili's method was published by Enrique Marcatili in 1969 and uses an Ansatz on the shape of the electromagnetic fields in the waveguide. In the core of the waveguide, the mode is composed of a standing wave in the x- and y-directions. Outside the core, the field decays exponentially in horizontal and vertical directions. Marcatili's method can provide an approximate analytical solution for the modal calculation in optical waveguides. On the other side, the effective index method is a very efficient tool designed to transform a two-dimensional electromagnetic field problem to a one-

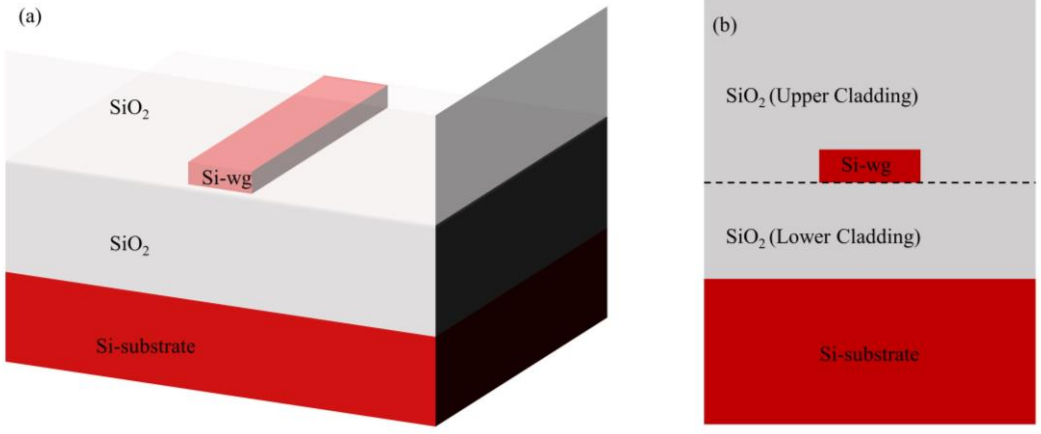


Fig. 2.8.1: (a) 3D-Schematic and (b) Cross-sectional view of a practical rectangular Si-photonic waveguide fabricated on a Si-wafer.

dimensional effective waveguide. This method was used as early as the late 70's applied to semiconductor lasers having a gradual lateral variation in the complex permittivity. The effective index method is an analytical method that is based on separating the spatial variables of the electromagnetic field, narrowing the rib waveguide problem to a set of slab waveguides analysis.

Numerical solutions of the eigenvalue problem for rectangular waveguides came much later when personal computers could handle calculations with large matrices. However, numerical methods provide a direct solution of Maxwell's equations and the eigenvalue problem that is formed through them. Thus, they are the workhorse of modern integrated photonic device design along with numerical methods used in problems having a source excitation. Herein, one of the simplest numerical methods to solve the eigenvalue problem will be presented: the Finite-Difference Eigenmode solution (FDE). Even though in the thesis a commercial solver from Ansys (Ansys-Lumerical) is used [15], it is beneficial to present the foundations of the algorithm behind the solver as well as the limitation of such a finite-difference based solver.

The eigenvalue problem is set up by an equation having the form:

$$Af = \lambda f \ (2.149) \ \Rightarrow (A - \lambda \mathbb{I}) f = 0 \ (2.150)$$

The Helmholtz equation can take the general shape of (2.150) in two forms. In the first form, the eigenvalue is frequency ω and the eigenvectors are **E** or **H**. For this form, the frequency ω is not known but an assumption for the propagation constant β is made. Such is the case when one needs to calculate the resonance frequencies and the resonant modes of an uncoupled disk or ring resonator [16]. In such cases, the eigenvalue problem takes the form:

$$L[f] = -\omega^2 f \ (2.151)$$

An even more important case for this thesis is when the fields are considered harmonic with respect to a certain frequency ω_0 or wavelength λ_0 and one needs to calculate the complex propagation constants β or effective indices n_{eff} as eigenvalues and the electric field **E** or magnetic field **H** as eigenvector [14]. In such a case, considering the electric field as eigenvector:

$$L[\boldsymbol{E}] = -\beta^2 \boldsymbol{E} \ (2.152)$$

From this point onwards, the finite-difference solution of such an eigenvalue problem will be presented based on the most important findings from reference [17]. For many years the condition that dictated that the electric field and its derivative must be continuous at the material interfaces of the rectangular waveguides restricted the discovery of an accurate numerical solution. This was also connected with the abrupt transition of the refractive index value across the material interfaces. The method described in [17] is the basis of the numerical method followed by the FDE solver by Lumerical [15] and overcomes such issues with the described techniques.

A variety of discretization schemes have been proposed that directly solve the Helmholtz equation for the transverse electric and magnetic fields, as in [18, 19]:

$$\left(\nabla_t^2+\beta_0^2\varepsilon_r\right)E_t+\nabla_t\left(\varepsilon_r^{-1}\nabla_t\varepsilon_r\cdot E_t\right)=\beta^2E_t \quad (2.153)$$

$$\left(\nabla_t^2+\beta_0^2\varepsilon_r\right)H_t+\varepsilon_r^{-1}\nabla_t\varepsilon_r\times\left(\nabla_t\times H_t\right)=\beta^2H_t \quad (2.154)$$

Nevertheless, possible discontinuities may exist on the mesh grid and a grid point may be connected with four refractive index values. In this eigenvalue problem numerical solution, the 2D orthogonal mesh grid discretization by Yee [20] is applied towards the calculation of modes supported by an optical waveguide. Yee's mesh is utilized in its 3D form in the 3D-FDTD method that will be presented in the next Section. An idea of a 2D-Yee mesh can be seen in Fig. 2.8.2. It can be seen that the electric and magnetic field components are calculated at different grid points in space. The position of the electric and magnetic field components calculated with the numerical 2D-FDE algorithm naturally satisfies Maxwell's curl equations since a curl of the electric field indeed provides a magnetic field component at the center of the square, considering that the other respective electric field components are at the edges of a square grid. For material interfaces a grid-dependent staircasing is performed. Nevertheless, near material interfaces an index or dielectric permittivity averaging may be performed. This is the only disadvantage of the 2D-FDE algorithm. On the other side, the eigenvalue problem solution with the Finite Element Method [21] may deal with more complex geometries without resorting to staircasing or index averaging due to its unstructured triangular (instead of rectangular) grid, but it is more computationally resource intensive.

The 2D-FDE numerical method for the mode calculations in optical waveguides will be mathematically described below. One should begin with the direct discretization of the Maxwell curl equations using Yee's method, having made the assumption that the electric and magnetic fields have a dependence on $e^{i(\beta z-\omega t)}$). Considering a free space impedance $Z_0=\sqrt{\frac{\mu_0}{\varepsilon_0}}$ one will get:

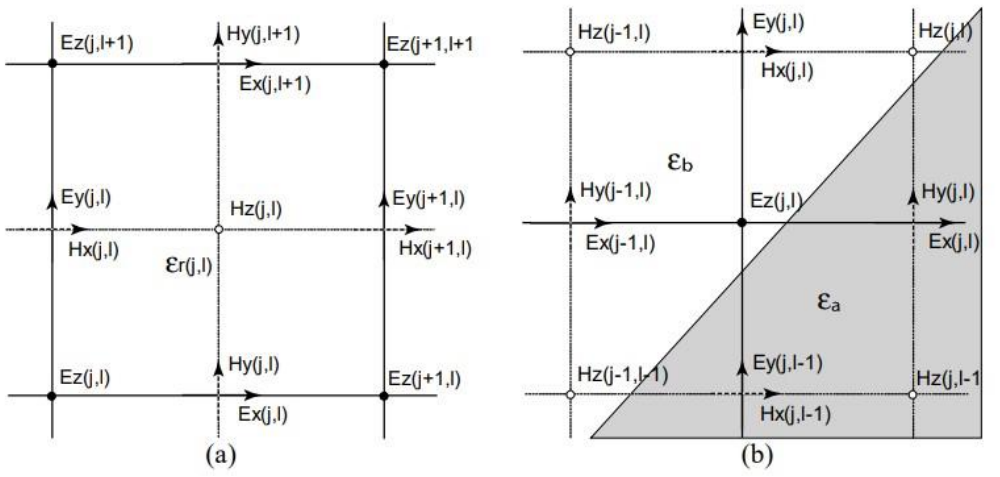


Fig. 2.8.2: (a) The 2D Mesh Grid by Yee for the space Discretization and (b) Yee's Mesh applied in curved material interfaces [17].

$$\nabla \times \boldsymbol{E} = -\frac{\partial \boldsymbol{B}}{\partial t} \Rightarrow \left\{ \begin{aligned} i\beta_0 H_x &= \frac{\partial E_z}{\partial y} - i\beta E_y \quad (2.155) \\ i\beta_o H_y &= i\beta E_x - \frac{\partial E_z}{\partial x} \quad (2.156) \\ i\beta_0 H_z &= \frac{\partial E_y}{\partial x} - \frac{\partial E_x}{\partial y} \quad (2.157) \end{aligned} \right\}$$

And

$$\nabla \times \boldsymbol{H} = \frac{\partial \boldsymbol{D}}{\partial t} \Rightarrow \left\{ \begin{aligned} -i\beta_0 \varepsilon_r E_x &= \frac{\partial H_z}{\partial y} - i\beta H_y \quad (2.158) \\ -i\beta_0 \varepsilon_r E_y &= i\beta H_x - \frac{\partial H_z}{\partial x} \quad (2.159) \\ -i\beta_0 \varepsilon_r E_z &= \frac{\partial H_y}{\partial x} - \frac{\partial H_x}{\partial y} \quad (2.160) \end{aligned} \right\}$$

The differential equations (2.155)-(2.160) are then numerically discretized using the algorithm imposed by Yee's rectangular grid.

$$i\beta_o H_x(j,l) = \frac{E_z(j,l+1) - E_z(j,l)}{\Delta y} - i\beta E_y(j,l) \quad (2.161)$$

$$i\beta_0 H_y(j,l) = i\beta E_x(j,l) - \frac{E_z(j+1,l) - E_z(j,l)}{\Delta x} \quad (2.162)$$

$$i\beta_0 H_z(j,l) = \frac{E_y(j+1,l) - E_y(j,l)}{\Delta x} - \frac{E_x(j,l+1) - E_x(j,l)}{\Delta y} \quad (2.163)$$

$$-i\beta_o \varepsilon_{rx}(j,l) E_x(j,l) = \frac{H_z(j,l) - H_z(j,l-1)}{\Delta y} - i\beta H_y(j,l) \quad (2.164)$$

$$-i\beta_0 \varepsilon_{ry}(j,l) E_y(j,l) = i\beta H_x(j,l) - \frac{H_z(j,l) - H_z(j-1,l)}{\Delta x} \quad (2.165)$$

$$-i\beta_0 \varepsilon_{rz}(j,l) E_z(j,l) = \frac{H_y(j,l) - H_y(j-1,l)}{\Delta x} - \frac{H_x(j,l) - H_x(j,l-1)}{\Delta y} \quad (2.166)$$

The three points in each direction x (j, j-1, j+1) and y (l, l-1, l+1) are the stencils of the finite difference algorithm. The discretized equations of (2.161)-(2.166) lead to an iterative algorithm where the electric field at the next grid position comes as a result of the magnetic field at a previous grid position. To avoid discontinuities in equations (2.164)-(2.166) three averaged dielectric permittivity values are utilized expressed through the following relations:

$$\varepsilon_{rx}(j,l) = \frac{\varepsilon_r(j,l) + \varepsilon_r(j,l-1)}{2} \quad (2.167)$$

$$\varepsilon_{ry}(j,l) = \frac{\varepsilon_r(j,l) + \varepsilon_r(j-1,l)}{2} \quad (2.168)$$

$$\varepsilon_{rz}(j,l) = \frac{\varepsilon_r(j,l) + \varepsilon_r(j-1,l-1) + \varepsilon_r(j,l-1) + \varepsilon_r(j-1,l)}{4} \quad (2.169)$$

The discretized equation can be combined in a matrix form as follows:

$$i\beta_0 \begin{bmatrix} \mathbf{H}_x \\ \mathbf{H}_y \\ \mathbf{H}_z \end{bmatrix} = \begin{bmatrix} 0 & -i\beta\mathbb{I} & U_y \\ i\beta\mathbb{I} & 0 & -U_x \\ -U_y & U_x & 0 \end{bmatrix} \begin{bmatrix} \mathbf{E}_x \\ \mathbf{E}_y \\ \mathbf{E}_z \end{bmatrix} \quad (2.170)$$

And

$$-i\beta_0 \begin{bmatrix} \varepsilon_{rx} & 0 & 0 \\ 0 & \varepsilon_{ry} & 0 \\ 0 & 0 & \varepsilon_{rz} \end{bmatrix} \begin{bmatrix} \mathbf{E}_x \\ \mathbf{E}_y \\ \mathbf{E}_z \end{bmatrix} = \begin{bmatrix} 0 & -i\beta\mathbb{I} & V_y \\ i\beta\mathbb{I} & 0 & -V_x \\ -V_y & V_x & 0 \end{bmatrix} \begin{bmatrix} \mathbf{H}_x \\ \mathbf{H}_y \\ \mathbf{H}_z \end{bmatrix} \quad (2.171)$$

Apart from the identity matrix and the diagonal permittivity matrices, the values of U_x, U_y, V_x and V_y depend on the boundary conditions employed. A Perfect Electric Conductor (PEC) boundary condition will ensure that at the interfaces of the computational window $\boldsymbol{n} \times \boldsymbol{E} = 0$. This will lead to a faster calculation of the modes in the waveguide. However, apart from the physical modes of the waveguide, artificial or artefact modes will be supported. These modes are solutions supported by the metallic box of the computational window and not by the cross-section of the waveguide. To negate this effect an artificial perfectly absorbing material is imposed at the edges of the computational window in a way that in an eigenmode calculation most artefact modes vanish and only the physical modes of the waveguide remain. This artificial perfectly absorbing material is called Perfectly Matched Layers or PML and more details will be given on its formulation in the FDTD method as it is used in both eigenvalue and excitation based electromagnetic problems [22].

After some algebra from equations (2.170) and (2.171) one can derive:

$$P \begin{bmatrix} \mathbf{E}_x \\ \mathbf{E}_y \end{bmatrix} = \begin{bmatrix} P_{xx} & P_{xy} \\ P_{yx} & P_{yy} \end{bmatrix} \begin{bmatrix} \mathbf{E}_x \\ \mathbf{E}_y \end{bmatrix} = \beta^2 \begin{bmatrix} \mathbf{E}_x \\ \mathbf{E}_y \end{bmatrix} \quad (2.172)$$

$$P_{xx} = -\beta_0^{-2} U_x \varepsilon_{rz}^{-1} V_y V_x U_y + \left(\beta_0^2 \mathbb{I} + U_x \varepsilon_{rz}^{-1} V_x\right)\left(\varepsilon_{rx} + \beta_0^{-2} V_y U_y\right) \quad (2.173)$$

$$P_{yy} = -\beta_0^{-2} U_y \varepsilon_{rz}^{-1} V_x V_y U_x + \left(\beta_0^2 \mathbb{I} + U_y \varepsilon_{rz}^{-1} V_y\right)\left(\varepsilon_{ry} + \beta_0^{-2} V_x U_x\right) \quad (2.174)$$

$$P_{xy} = U_x \varepsilon_{rz}^{-1} V_y \left(\varepsilon_{ry} + \beta_0^{-2} V_x U_x\right) - \beta_0^{-2}\left(\beta_0^2 \mathbb{I} + U_x \varepsilon_{rz}^{-1} V_x\right) V_y U_x \quad (2.175)$$

$$P_{yx} = U_y \varepsilon_{rz}^{-1} V_x \left(\varepsilon_{rx} + \beta_0^{-2} V_y U_y \right) - \beta_0^{-2} \left(\beta_0^{2} \mathbb{I} + U_y \varepsilon_{rz}^{-1} V_y \right) V_x U_y \quad (2.176)$$

When the absorption is not significant, then the matrices P and Q that depend on the matrices of U and V are sparse and can be easily stored in a computer and can be solved with a simple iterative eigenvalue solver [23]. An equivalent eigenvalue formulation can come forth for the transverse magnetic field elements considering z as the axis of propagation, as seen in [17]. Considering the conditions above the effective index of each mode will be calculated finally as:

$$n_{eff} = \frac{\beta}{\beta_0} \quad (2.177)$$

If a cell is occupied by two media, the permittivity is modified with f_a being the fraction of one of the two media to the cell sized:

$$\varepsilon_{rz}(j,l) = f_a \varepsilon_a + \left(1 - f_a \right) \varepsilon_b \quad (2.178)$$

Similar modifications are performed for the rest of the diagonal permittivities. In the case of materials with losses for example the presence of metals and in the case of a PML application to the boundary conditions, a complex effective index $\tilde{n}_{eff}$ and $\tilde{\beta}$ ensue from the solution of the eigenvalue problem. In such cases because two values for β one with a positive and one with a negative real part may come up as a solution, the FDE algorithm suggests which of the two $Re\{\tilde{\beta}\}$ should be chosen. The imaginary part of the propagation constant is related with the propagation losses of the waveguide as mentioned above and thus the propagation losses may also be physical or artificial. An artificial imaginary part of the propagation constant $Im\{\tilde{\beta}\}$ may come as a result of the application of a PML boundary condition near the waveguide. Artificial propagation losses should be avoided, and a silicon photonics designer should be careful to place the PML far enough from the optical waveguide under study. Finally, from the discretized equations it is clear that the solution will be more accurate for a smaller grid size Δx or Δy but this also makes the iterative algorithm more computationally intensive requiring more computer resources. Equivalent eigenvalue equations can come up considering a harmonic dependence for the axis x or y as performed here for the z axis. The next section will study in depth the FDTD method for the study of excitation-based problems in integrated photonics. The FDTD algorithm will be the basis of the work in this thesis.

2.9: The Finite Difference Time Domain Method

The 3D Finite Difference Time-Domain (3D-FDTD) method constitutes the workhorse of modern integrated photonics device design. Even though such a technique had been studied in the 1960s by Kane Yee [20], the corresponding stability criterion of the paper was incorrect. The method emerged and gained its fame in 1975 and afterwards, while a researcher, Allen Taflove, was studying the microwave absorption

by the human eye. Taflove came up with a correct stability criterion for the numerical FDTD method [24, 25] and with the boom in the development of computers handling calculations with even larger matrices, the method has become one of the foundations of modern microwave and optical engineering. Taflove has written various books that are considered as the reference for the method [26, 27]. One could say however that the finalization of the method came with the invention of the Perfectly Matched Layers in 1994 [28]. Before that, in Taflove's FDTD method one should terminate the simulation (by setting a proper total simulation time) before electromagnetic waves were reflected by the PEC boundary condition at the edges of the computational window back to the area under study. Afterwards, various absorbing boundary conditions were proposed but, in many cases, they were dependent on either the frequency or the incidence angle of the electromagnetic wave. Jean-Pierre Berenger invented an artificial material layer (or a multitude of artificial layers) that could be placed at the edges of the computational window and absorb any incident electromagnetic wave regardless of the frequency (or wavelength) or the incidence angle [28].

The 3D Finite-Difference Time Domain method is one of the most famous numerical methods used in photonics, employed to study excitation-based problems. In this thesis, the FDTD software by Ansys-Lumerical was utilized [15]. Nevertheless, the fundamentals of a simple FDTD algorithm in the 3D-space will be discussed in this Section. In terms of references, [16] is a more introductory book for computational electromagnetics that describes the basics of the FDTD algorithm. Reference [14] provides the details of the method applied for the design of photonic devices, while [29] also provides a very good overview of the method. All these references were employed in this part of the thesis.

The whole FDTD method is based on the fact that central differentiation provides a second order-based error $O(h^2)$ instead of a first order error as in forward and backward numerical differentiation $O(h)$. This can be calculated by using a Taylor expansion on the terms produced by each type of numerical differentiation [16]. Central differentiation of a function f is mathematically expressed as shown below:

$$\frac{f(x+h)-f(x-h)}{2h}=f'(x)+O(h^2) \quad (2.179)$$

An even better alternative is to use a staggered grid in a way that the derivative is computed at half the step of the grid:

$$x_{i+\frac{1}{2}}=x_i+\frac{h}{2} \quad (2.180)$$

$$\frac{f(x+h)-f(x)}{h}=f'\left(x+\frac{h}{2}\right)+O\left(h^2\right) \quad (2.181)$$

For an equidistant grid, the second derivative can be expressed as:

$$\frac{f(x+h)-2f(x)+f(x-h)}{h^2}=f''(x)+O\left(h^2\right) \quad (2.182)$$

In all cases, the error is second order and not first order, dependent of course on the grid step h. The fundamental steps of a 3D-FDTD algorithm are the following:

1. The definition of material properties on the grid and the grid itself in time and space. These include the complex permittivity $\tilde{\varepsilon}$ definition in space, for example.
2. The definition of the Boundary Conditions. In this thesis mostly PML boundary conditions have been used. However, PEC or periodic boundary conditions may be used in some cases. An analysis of possible boundary conditions in FDTD will be later provided.
3. The definition of sources and detectors/monitors in the simulation. In the FDTD software of Ansys-Lumerical [15] in most cases first the eigenvalue problem is solved and one of the eigenmodes of a waveguide section is chosen as the source. However, there are cases when a dipole, Gaussian, or plane-wave source definition are required and Lumerical [15] can define such sources using functions interpolated as matrices on a grid [23].
4. Running the simulation - Evolution of the source electromagnetic fields in time and space. The algorithm for the evolution of the electric and magnetic field components will be provided below.
5. Recording and postprocessing of information received from the monitors of the simulation. Through discrete or fast Fourier transforms [23] with the help of (2.44) and (2.45) a single simulation in FDTD can give back, using a defined pulse in time as a source, the frequency response of device in a certain frequency range, i.e. the transfer function of a designed device. This is the main advantage of the FDTD method compared to other numerical methods such as the FEM method where a separate simulation is needed for each frequency point. An auxiliary differential equation can be set up for the inclusion of the dispersive characteristics of all the materials in a device. The basis of these auxiliary differential equations are the Drude and Lorentz model presented in a previous Section.

Below, details on certain sections of the FDTD algorithm are provided.

2.9.1: Electric and magnetic field evolution

Again, the starting point in the evolution algorithm will be Maxwell's Vector Curl equations. In contrast, with the FDE algorithm, no assumption is made on the electric and the magnetic field dependence. From Faraday's Law:

$$\nabla \times \boldsymbol{E} = -\mu \frac{\partial \boldsymbol{H}}{\partial t} \Rightarrow \left.\begin{aligned} \mu \frac{\partial H_x}{\partial t} &= \frac{\partial E_y}{\partial z} - \frac{\partial E_z}{\partial y} \\ \mu \frac{\partial H_y}{\partial t} &= \frac{\partial E_z}{\partial x} - \frac{\partial E_x}{\partial z} \\ \mu \frac{\partial H_z}{\partial t} &= \frac{\partial E_x}{\partial y} - \frac{\partial E_y}{\partial x} \end{aligned}\right\} \quad (2.183)$$

From Ampere's Law with Maxwell's modification, considering zero induced current densities **J** = 0:

$$\nabla \times \boldsymbol{B} = \mu\varepsilon \frac{\partial \boldsymbol{E}}{\partial t} \Rightarrow \left.\begin{aligned} \mu \frac{\partial E_x}{\partial t} &= \frac{\partial H_z}{\partial y} - \frac{\partial H_y}{\partial z} \\ \mu \frac{\partial E_y}{\partial t} &= \frac{\partial H_x}{\partial z} - \frac{\partial H_z}{\partial x} \\ \mu \frac{\partial E_z}{\partial t} &= \frac{\partial E_x}{\partial y} - \frac{\partial E_y}{\partial x} \end{aligned}\right\} \quad (2.184)$$

Since the centered differences can produce a second-order dependent error and the derivative can be better calculated in half-grid using the staggered grid. Considering a function having a form:

$$f(n,p,q,r) = f(n\Delta t,\ p\Delta x,\ q\Delta y,\ r\Delta z) \quad (2.185)$$

One should then use Yee's cell for the discretization of the systems of equations of (2.183) and (2.184) in time and space as follows:

$$\begin{aligned} &\varepsilon \frac{E_x\left(n+1,\ p+\frac{1}{2},q,r\right) - E_x\left(n,p+\frac{1}{2},q,r\right)}{\Delta t} = \\ &\frac{H_z\left(n+\frac{1}{2},p+\frac{1}{2},q+\frac{1}{2},r\right) - H_z\left(n+\frac{1}{2},p+\frac{1}{2},q-\frac{1}{2},r\right)}{\Delta y} \\ &- \frac{H_y\left(n+\frac{1}{2},p+\frac{1}{2},q,r+\frac{1}{2}\right) - H_y\left(n+\frac{1}{2},p+\frac{1}{2},q,r-\frac{1}{2}\right)}{\Delta z} \end{aligned} \quad (2.186)$$

$$\begin{aligned} &\varepsilon \frac{E_y\left(n+1,\ p,q+\frac{1}{2},r\right) - E_y\left(n,p,q+\frac{1}{2},r\right)}{\Delta t} = \\ &\frac{H_x\left(n+\frac{1}{2},p,q+\frac{1}{2},r+\frac{1}{2}\right) - H_x\left(n+\frac{1}{2},p,q+\frac{1}{2},r-\frac{1}{2}\right)}{\Delta z} \\ &- \frac{H_z\left(n+\frac{1}{2},p+\frac{1}{2},q+\frac{1}{2},r\right) - H_z\left(n+\frac{1}{2},p-\frac{1}{2},q+\frac{1}{2},r\right)}{\Delta x} \end{aligned} \quad (2.187)$$

$$\varepsilon \frac{E_z\left(n+1,\ p,q,r+\frac{1}{2}\right)-E_z\left(n,p,q,r+\frac{1}{2}\right)}{\Delta t}=$$
$$\frac{H_y\left(n+\frac{1}{2},p+\frac{1}{2},q,r+\frac{1}{2}\right)-H_y\left(n+\frac{1}{2},p-\frac{1}{2},q,r+\frac{1}{2}\right)}{\Delta x}$$
$$-\frac{H_x\left(n+\frac{1}{2},p,q+\frac{1}{2},r+\frac{1}{2}\right)-H_x\left(n+\frac{1}{2},p,q-\frac{1}{2},r+\frac{1}{2}\right)}{\Delta y} \quad (2.188)$$

Respectively, for the calculation of the magnetic field components:

$$\mu \frac{H_x\left(n+\frac{1}{2},p,q+\frac{1}{2},r+\frac{1}{2}\right)-H_x\left(n-\frac{1}{2},p,q+\frac{1}{2},r+\frac{1}{2}\right)}{\Delta t}=$$
$$\frac{E_y\left(n,p,q+\frac{1}{2},r+1\right)-E_y\left(n,p,q+\frac{1}{2},r\right)}{\Delta z}-$$
$$-\frac{E_z\left(n,p,q+1,r+\frac{1}{2}\right)-E_z\left(n,p,q,r+\frac{1}{2}\right)}{\Delta y} \quad (2.189)$$
$$\mu \frac{H_y\left(n+\frac{1}{2},p+\frac{1}{2},q,r+\frac{1}{2}\right)-H_y\left(n-\frac{1}{2},p+\frac{1}{2},q,r+\frac{1}{2}\right)}{\Delta t}=$$
$$\frac{E_z\left(n,p+1,q,r+\frac{1}{2}\right)-E_z\left(n,p,q,r+\frac{1}{2}\right)}{\Delta x}-$$
$$-\frac{E_x\left(n,p+\frac{1}{2},q,r+1\right)-E_x\left(n,p+\frac{1}{2},q,r\right)}{\Delta z} \quad (2.190)$$

$$\mu \frac{H_z\left(n+\frac{1}{2},p+\frac{1}{2},q+\frac{1}{2},r\right)-H_z\left(n-\frac{1}{2},p+\frac{1}{2},q+\frac{1}{2},r\right)}{\Delta t}=$$
$$\frac{E_x\left(n,p+\frac{1}{2},q+1,r\right)-E_x\left(n,p+\frac{1}{2},q,r\right)}{\Delta y}-$$
$$-\frac{E_y\left(n,p+1,q+\frac{1}{2},r\right)-E_y\left(n,p,q+\frac{1}{2},r\right)}{\Delta x} \quad (2.191)$$

The evolution equations can take an even simpler form to be able to have steps multiple of 1 and not ½ and can then be directly inserted in a programming language environment for calculations like Matlab [14]. The method to evolve the fields in time and space is also called a leapfrog algorithm because indeed due to the electric and magnetic field component calculation at different points in time and space the updating of the field elements resembles the leap of a frog. The visualization of the field evolution algorithm using Yee's cell in 1D and 2D can be seen in Fig. 2.9.1(a) and (b). Fig. 2.9.1(a) visualizes the leapfrog algorithm in time and space. The stencils indicate which field components are used in the discretized partial differential equations. On the other side, Fig. 2.9.1(b) displays the unit cell of a 3D-FDTD grid in space. It is clear that the electric field components are calculated at the middle of the edges of the cubic unit cell while the magnetic field components are calculated at the middle of the faces of the unit cell. For this reason, the curl equations by Maxwell and field continuity are naturally satisfied with the choice of such a mesh grid. Index averaging techniques as used in the FDE method may also be used in the FDTD method to deal with field discontinuities.

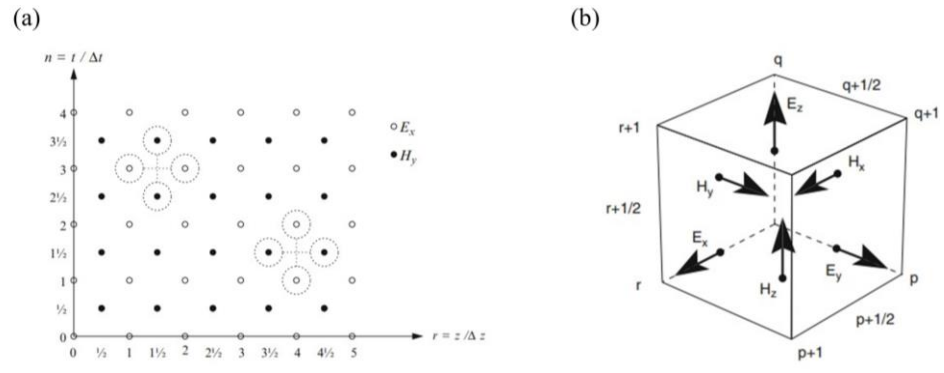


Fig. 2.9.1: (a) 1D-FDTD staggered grid in Time and Space and (b) The Unit cell of the 3D-FDTD Grid in Space for a certain value n for time [16].

2.9.2: Numerical dispersion and stability of the FDTD method

In this section the relations providing the artificial numerical dispersion and the stability criterion of the FDTD method will be analyzed. Indeed, regarding numerical dispersion, the wave propagation through an FDTD grid may cause dependence of the phase velocity to lattice discretization, propagation direction, wavelength and others. The numerical dispersion in 3D is provided by the following relation:

$$\frac{1}{c^2\Delta t^2}\sin^2\left(\frac{\omega\Delta t}{2}\right)=\left(\frac{1}{\Delta x}\sin\left(\frac{\beta_x\Delta x}{2}\right)\right)^2+\left(\frac{1}{\Delta y}\sin\left(\frac{\beta_y\Delta y}{2}\right)\right)^2+\left(\frac{1}{\Delta z}\sin\left(\frac{\beta_z\Delta z}{2}\right)\right)^2 \quad (2.192)$$

In order to have stable FDTD simulations, the time step should obey a criterion called the Courant stability criterion or the Courant-Friedrichs-Lewy Condition, connecting the time step with the grid step in all three dimensions in space. Instabilities would cause the electric field in the simulation region to increase indefinitely and the simulation to shatter. Instabilities in the FDTD simulation may also be caused sometimes by the interaction of the boundary condition (for example PML) with a certain material having an imaginary part in the effective index. This of course is a numerical artefact as well.

$$\Delta t \le \frac{1}{c}\left[\left(\frac{1}{\Delta x}\right)^2+\left(\frac{1}{\Delta y}\right)^2+\left(\frac{1}{\Delta z}\right)^2\right]^{-\frac{1}{2}} \quad (2.193)$$

In a perfectly cubic 3D grid, the stability criterion takes the simple form:

$$\Delta t \le \frac{\Delta x}{c\sqrt{3}} \quad (2.194)$$

2.9.3: Introduction of Material Dispersion to the FDTD method

The material dispersion for metals and dielectrics may be described using the Lorentz and Drude models as a starting point. An extra auxiliary differential equation will come up and can be introduced as an extra discretized equation to be able to take material dispersion into account in FDTD simulations.

The Drude and Lorentz permittivities were derived in Section 2.5 and from them the dielectric susceptibilities from the Drude and Lorentz models can easily be derived. The utilization of the dielectric susceptibilities to differential equations with polarization "currents" for metals and dielectrics will lead to the discretized auxiliary differential equations.

For the dispersive characteristics of metals, the polarization "current" using the susceptibility from Drude's model is:

$$\boldsymbol{J}_{\mathrm{p}}(\omega)=\frac{i\omega\varepsilon_0{\omega_p}^2}{i\omega(i\omega+\gamma)}\boldsymbol{E}(\omega) \quad (2.195)$$

Therefore, from (2.195) one can derive an equation in the frequency and time-domains:

$$i\omega \boldsymbol{J}_{\mathrm{p}}(\boldsymbol{\omega}) + \gamma \boldsymbol{J}_{\mathrm{p}}(\boldsymbol{\omega}) = \varepsilon_0 \omega_p{}^2 \boldsymbol{E}(\boldsymbol{\omega}) \Rightarrow$$

$$\Rightarrow \frac{\partial \boldsymbol{J}_{\mathrm{p}}}{\partial t} + \gamma \boldsymbol{J}_{\mathrm{p}} = \omega_p{}^2 \varepsilon_0 \boldsymbol{E} \quad (2.196)$$

Thus, when (2.196) is discretized, the updating auxiliary differential equation for the FDTD algorithm will ensue:

$$J_p(n+1) = \frac{2-\gamma\Delta t}{2+\gamma\Delta t} J_p(n) + \frac{\omega_p{}^2 \varepsilon_0 \Delta t}{2+\gamma\Delta t} (E(n+1) - E(n)) \quad (2.197)$$

Of course, the "polarization current" will exist in the right-hand side of Maxwell's fourth equation as:

$$\nabla \times \boldsymbol{H} = \varepsilon_0 \frac{\partial \boldsymbol{E}}{\partial t} + \boldsymbol{J}_{\mathrm{p}} + \sigma \boldsymbol{E} \quad (2.198)$$

In a similar way, a polarization current from the Lorentz model will easily come up as follows:

$$\boldsymbol{J}_{\mathrm{p}}(\boldsymbol{\omega}) = i\omega\varepsilon_0 \frac{\omega_l{}^2}{\omega_0{}^2 + i\gamma_l\omega - \omega^2} \boldsymbol{E}(\boldsymbol{\omega}) \Rightarrow$$

$$\Rightarrow \left(\omega_0{}^2 + i\omega\gamma_l - \omega^2\right) \boldsymbol{J}_{\mathrm{p}}(\boldsymbol{\omega}) = i\omega\varepsilon_0 \omega_l{}^2 \boldsymbol{E}(\boldsymbol{\omega}) \Rightarrow$$

$$\Rightarrow \omega_0{}^2 \boldsymbol{J}_{\mathrm{p}} + \gamma_l \frac{\partial \boldsymbol{J}_{\mathrm{p}}}{\partial t} + \frac{\partial^2 \boldsymbol{J}_{\mathrm{p}}}{\partial t^2} = \omega_l{}^2 \varepsilon_0 \frac{\partial \boldsymbol{E}}{\partial t} \quad (2.199)$$

In its discretized form (2.199) will become:

$$J_p(n+1) = 2\frac{2-\omega_0{}^2\Delta t^2}{2+\gamma_l\Delta t} J_p(n) + \frac{\gamma_l\Delta t - 2}{2+\gamma_l\Delta t} J_p(n-1) + \frac{\varepsilon_0 \omega_l{}^2 \Delta t}{2+\gamma_l\Delta t} (E(n+1) - E(n-1)) \quad (2.200)$$

The FDTD solver by Ansys-Lumerical [15] used in this thesis employs a multi-coefficient polynomial fitting technique for the inclusion of the material dispersive characteristics in the simulations, without employing the Drude and Lorentz model. Nevertheless, in most cases the dispersive characteristics of the materials are provided by the companies that have provided accurate fitting for their experimental ellipsometric data. For this reason, in most cases, the target is to approach the experimental refractive index values provided by the collaborating company after fitting using the multi-coefficient model by Lumerical. This way, accurate broadband simulations are possible.

2.9.4: Boundary Conditions and Power and Coupling Efficiency Calculations with Expansion Monitors

As pointed out also in the sections above, the boundary conditions of the FDTD simulations are crucial for the termination of the simulation region and proper treatment of the edges of the computational window. The FDTD boundary conditions should be differentiated from the source and expansion monitor boundary conditions, where an eigenvalue problem is initially solved for the determination of the source mode and the targeted mode in an integrated photonics device. A proper boundary condition should be able to terminate the simulation region by absorbing any incoming electromagnetic wave regardless of its frequency and incidence angle. This does not happen when the simulation region is terminated by a PEC boundary condition formulated as follows:

$$\boldsymbol{n}\times\boldsymbol{E}=0 \quad (2.201)$$

The Perfectly Matched Layers (PMLs) are a boundary condition first proposed by Berenger [22] to terminate excitation-based computational electromagnetic problems without the propagating wave being reflected from the boundaries back to the device under study thus distorting the simulation results. The main idea behind Berenger's PML lies with the introduction of an artificial material layer at the edges of the simulation region consisting of both an electric and a magnetic conductivity, σ and σ* respectively. Then Maxwell's equations become:

$$\varepsilon_0\frac{\partial \boldsymbol{E}}{\partial t}+\sigma\boldsymbol{E}=\nabla\times\boldsymbol{H} \quad (2.202)$$

$$\mu_0\frac{\partial \boldsymbol{H}}{\partial t}+\sigma^*\boldsymbol{H}=-\nabla\times\boldsymbol{E} \quad (2.203)$$

For a normal incidence, one can easily get the reflection coefficient back to the simulation region as:

$$\Gamma_0=\frac{Z_0-Z_{PML}}{Z_0+Z_{PML}} \quad (2.204)$$

$$where\ Z_{PML}=\sqrt{\frac{\mu_0+\left(\frac{\sigma^*}{i\omega}\right)}{\varepsilon_0+\left(\frac{\sigma}{i\omega}\right)}} \quad (2.205)$$

From (203) it is clear that for $Z_0=Z_{PML}\Rightarrow\frac{\sigma^*}{\mu_0}=\frac{\sigma}{\varepsilon_0}$ (2.206) no reflections should occur regardless of the operating frequency. The situation is a bit trickier when the propagating electromagnetic wave has an angle of incidence. However, Berenger found a trick to avoid reflections even in arbitrary angle of incidence by splitting the

electric and magnetic field to each direction. Considering z as the normal direction, Maxwell's equations give the following result:

$$\varepsilon \frac{\partial E_{xy}}{\partial t} = \frac{\partial \left(H_{zx} + H_{zy} \right)}{\partial y} \quad (2.207)$$

$$\varepsilon \frac{\partial E_{xz}}{\partial t} = - \frac{\partial \left(H_{yz} + H_{yx} \right)}{\partial z} - \sigma_z E_{xz} \quad (2.208)$$

$$\varepsilon \frac{\partial E_{yz}}{\partial t} = - \frac{\partial \left(H_{xy} + H_{xz} \right)}{\partial z} - \sigma_z E_{yz} \quad (2.209)$$

$$\varepsilon \frac{\partial E_{yx}}{\partial t} = - \frac{\partial \left(H_{zx} + H_{zy} \right)}{\partial x} \quad (2.210)$$

The layer therefore changes the propagation only in the z-direction which is why there are no reflections even in arbitrary angles of incidence. In reality, because the conductivity varies strongly with respect to the grid, a parabolic conductivity profile is usually chosen for a layer that extends from z_0 to z_0+L_z, having the form:

$$\sigma(z) = \sigma_0 \left(\frac{z - z_0}{L_z} \right)^2 \quad (2.211)$$

In most cases, more than a single PML layers are used for example N=8 layers. This helps decrease the artefact power reflected to the simulated device but of course increases resources needed for computation because of the PML needing extra space, for example some grid cells after the edges of the simulation region. An alternative tensorial PML is also usually employed with its form being as a matrix, causing changes to the time-stepping [30].

$$\overline{\varepsilon_r} = \overline{\mu_r} = \begin{pmatrix} 1 - \left(\frac{i\sigma}{\omega\varepsilon} \right) & 0 & 0 \\ 0 & 1 - \left(\frac{i\sigma}{\omega\varepsilon} \right) & 0 \\ 0 & 0 & \left(1 - \left(\frac{i\sigma}{\omega\varepsilon} \right) \right)^{-1} \end{pmatrix} \quad (2.212)$$

The final step of the simulation is the post processing of data. The calculation of total power at a plane vertical to the direction of propagation is based on the numerical double integration [23] of Equation (33) presented in the previous Sections. Nevertheless, in many cases such as in the design of couplers the amount of power coupled to a specific target mode from the incident normalized power is of interest. For this purpose, the mode expansion monitor is used. More specifically, one first calculates the total electric and magnetic field from a Discrete Fourier Transform (DFT) plane

monitor. Then an expansion monitor of the same dimensions as the DFT monitor should reside at the same position as the DFT monitor. Using the mode expansion monitor the eigenvalue problem can be solved at the specific plane and the targeted mode in an integrated photonic waveguide region can be selected. Therefore, the coupling efficiency will just be provided by the power overlap integral of the total field with the modal field of the targeted mode that had been selected.

The total power at a plane and the net power coupled to a specific mode m are provided below:

$$P_{tot} = \frac{1}{2} Re\left\{ \iint (\boldsymbol{E} \times \boldsymbol{H}^*) \cdot d\boldsymbol{S} \right\} \quad (2.213)$$

$$P_m(norm) = Re\left[\frac{(\iint \boldsymbol{E_m} \times \boldsymbol{H}^* \cdot d\boldsymbol{S}) \cdot (\iint \boldsymbol{E} \times \boldsymbol{H_m^*} \cdot d\boldsymbol{S})}{\iint \boldsymbol{E_m} \times \boldsymbol{H_m} \cdot d\boldsymbol{S}} \right] \frac{1}{Re(\iint \boldsymbol{E} \times \boldsymbol{H} \cdot d\boldsymbol{S})} \quad (2.214)$$

Numerical integration techniques as in [23] can be used. The vectors $\mathbf{E_m}$, $\mathbf{H_m}$ represent the electric and magnetic field of the targeted mode derived from the solution of the eigenvalue problem at the plane of the expansion monitor while **E, H** are the total electric and magnetic fields recorded at the DFT plane after running the FDTD algorithm.

To sum up, in Sections 2.8 and 2.9 the fundamentals of the 2D-FDE and 3D-FDTD methods were provided. The purpose of these theoretical findings was the application of the techniques to practical integrated photonics designs, therefore some details on the methods may have been omitted. More details can be found of course in references [14, 16, and 29].

2.10: The Scattering Matrix Method for Photonic Integrated Circuits

Transiting from photonic integrated devices to photonic integrated circuits, the abrupt numerical implementation of the FDTD algorithm when the wave propagation length is larger than 400μm-500μm is no longer sustainable. For this reason, the frequency-domain based scattering matrix method has widely been used for the design of photonic integrated circuits. Lumerical INTERCONNECT [15] is the software that utilizes such a technique in the frequency-domain and was used also in this thesis. The Scattering Matrix method has its roots in microwave theory and considers any photonic device as a black box, where the output fields come as a result of the input fields multiplied by a Scattering Matrix. In a tensorial form the relation for any photonic device in the frequency domain is:

$$b_i(\omega) = S_{ij}(\omega)\, a_j(\omega) \quad (2.215)$$

In Eq. (2.215) the term b_i is the wavevector including the fields exiting a bi-directional design and the term a_j expresses the fields entering a photonic device. The matrix S_{ij} is the scattering matrix of the designed device. The two simplest cases in the scattering matrix method can be found in Fig. 2.10(a) and (b), with the first representing

a unidirectional two-port device and the second depicting a bidirectional two-port device, such as a coupling interface. A harmonic dependence with respect to time is always assumed in the scattering matrix method. In the unidirectional case the output field b has a simple relation:

$$b_2(\omega) = S_{21}(\omega)\, a_1(\omega) \quad (2.216)$$

On the other side, considering also the reflections of a two-port device, (2.216) receives a matrix form:

$$\begin{pmatrix} b_1(\omega) \\ b_2(\omega) \end{pmatrix} = \begin{pmatrix} S_{11}(\omega) & S_{12}(\omega) \\ S_{21}(\omega) & S_{22}(\omega) \end{pmatrix} \begin{pmatrix} a_1(\omega) \\ a_2(\omega) \end{pmatrix} \quad (2.217)$$

The transfer function of a bidirectional device can be obtained, taking into account that the scattering matrix is expressed in term of field terms, thus the transmission will come as a result of the matrix element $|S_{21}|^2$, considering port 2 as the output. The back-reflections to port 1, on the other side, are calculated with the term $|S_{11}|^2$.Making these assumptions for every photonic device it is very easy to obtain the scattering matrix of a circuit by multiplying the scattering matrices of each photonic component it is comprised of. For a more complex circuit, for example one composed by four photonic components one can easily obtain the relation between the output and the input fields from the following equation:

$$b_i = S_{ijklm} a_m = S_{ij} S_{jk} S_{kl} S_{lm} a_m \quad (2.218)$$

Fast and Discrete Fourier Transform techniques are thoroughly used in Lumerical INTERCONNECT to transform any signal from the Time Domain to the Frequency Domain and vice versa [15]. For simulations in the Time-Domain finally, compact models are employed for the mathematical description of the laser, the photodiode, relying on models as the ones in [31]. In Time-Domain circuit simulations either block or sample mode simulations can be performed. In all cases, the convolution theorem is applied, and the output signal is the input signal convoluted with an impulse response:

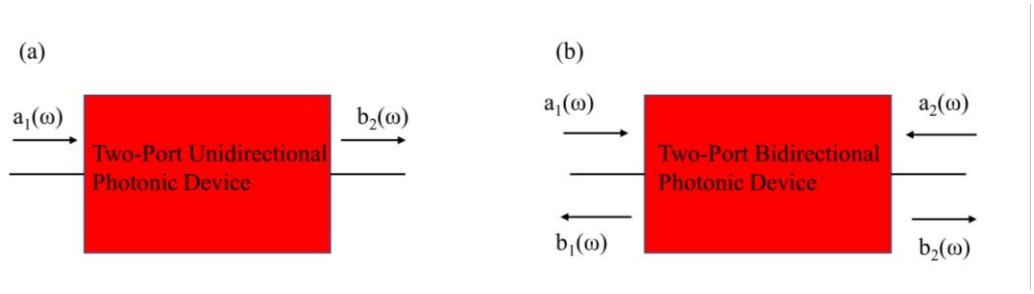


Fig. 2.10.1: (a) Unidirectional and (b) Bidirectional photonic device in the Scattering Matrix Method

$$y(t) = \int_{-\infty}^{+\infty} b(\tau) \cdot x(t-\tau)\, d\tau \quad (2.219)$$

Nevertheless, in a sample-mode simulation the convolution is discrete, and the signal passes many times back-and forth in the form of samples through the various components of the circuit:

$$y[n] = \sum_{i=0}^{N} b[i]x[n-i] \quad (2.220)$$

The sample-mode method is suitable for bidirectional propagation but in many cases the Finite Impulse Response Digital Filter employed may create numerical artefacts in the calculated transfer functions in resonator circuits. On the other side, the block-mode simulations are more suitable for unidirectional propagation. The signal passes only once through each device of the circuit so the numerical convolution is much simpler and direct than in the sample mode case.

Frequency and time-domain circuit simulations were utilized in Chapter 5 for the design of low-power plasmonic modulator circuits on Si_3N_4.

2.11: The Finite Element Method (FEM)

The Theoretical Section of this thesis is concluded with the presentation of the fundamentals of the 2D-FEM (Finite Element Method) for the solution of Poisson's and Laplace's Equation. This method is used in the plasmonic phase-shifter design of Chapter 5. There, a commercial 2D-FEM solver by Ansys-Lumerical (Lumerical DEVICE) was employed. The foundation of Section 2.11 were references [14] and [16].

In the FEM method, unlike the FDTD method, no explicit formulas can be found that update the electric and magnetic field quantities. In FEM, a system of linear equations is constructed and has to be solved every time to update the electric and the magnetic field values. The field solutions are expressed with respect to low-order polynomial functions that fulfill the differential equations and their boundary conditions in an averaged manner. Two techniques have been used over the years for the FEM method: (a) Employing test functions for the field solutions and setting their weighted residual to zero. If the test functions are equal to the weighting functions $w_i = \varphi_i$ then this method is named the Galerkin Method in FEM. (b) The utilization of a variational method to locate a stationary point of a quadratic form. The Variational Method is a generalization of the Galerkin method that uses a more general class of test functions. Nevertheless, in this Section a Galerkin-based approach will be presented.

The steps of the 2D-FEM Method are the following:

1. Divide the computational domain in triangular elements using an unstructured grid.
2. Approximating the solution f as the sum of a finite number of basis functions: $f(\boldsymbol{r}) \approx \sum_{i=1}^{n} \varphi_j(\boldsymbol{r}) f_j \quad (2.221)$. The functions φ_i are low order polynomials.

3. The formation of a residual with a weak form of the initial differential equation as $L(f) = s \Rightarrow r = L(f) - s$ (2.222).
4. Setting the weighting function $w_i = \varphi_i$ (2.223) in Galerkin's method and setting the weighted residual to zero as $\int_{\Omega} w_i r d\Omega = 0$ (2.224).

Moving now to the application of the 2D-FEM method for the solution of the Poisson and Laplace Equations, one should begin from Gauss's Law in the electrostatic case:

$$\nabla \cdot E = \frac{\rho}{\varepsilon} \Rightarrow -\nabla \cdot (\nabla V) = \frac{\rho}{\varepsilon} \quad (2.225)$$

The scalar function of the electrostatic potential can be approximated with its basis functions:

$$V \approx \sum_{j=1}^{n} f_j \varphi_j \quad (2.226)$$

$$\Rightarrow \iint w_i \left[\nabla \cdot (\nabla V) + \frac{\rho}{\varepsilon} \right] d\boldsymbol{S} = 0 \quad (2.227)$$

By applying the identity:

$$w_i \nabla \cdot (\nabla V) = \nabla \left[w_i \nabla V \right] - (\nabla w_i) \cdot (\nabla V) \quad (2.228)$$

The equation in its weak form can be obtained:

$$\iint (\nabla \varphi_i) \cdot (\nabla \varphi_j) f_j d\boldsymbol{S} = \iint \varphi_i \frac{\rho}{\varepsilon} d\boldsymbol{S} \quad (2.229)$$

The equation (2.229) is the weak form of the Poisson's Equation. It is now in a form:

$$A_{ij} z_j = b_i \quad (2.230)$$

In Poisson's equation, the parameters of equation (2.230) are provided below:

$$\left.\begin{aligned} A_{ij} &= \iint (\nabla \varphi_i)(\nabla \varphi_j) \\ z_j &= f_j \\ b_i &= \iint \varphi_i \frac{\rho}{\varepsilon} \end{aligned}\right\} \quad (2.231)$$

The Laplace equation will be the same as (2.229) but instead the right-hand side will be zero:

$$\iint (\nabla \varphi_i) \cdot (\nabla \varphi_j) f_j d\boldsymbol{S} = \boldsymbol{0} \quad (2.232)$$

$$\left.\begin{aligned} A_{ij} &= \iint \left(\nabla \varphi_i\right)\left(\nabla \varphi_j\right) \\ z_j &= f_j \\ b_i &= 0 \end{aligned}\right\} \quad (2.233)$$

Depending on the number of basis functions, a large linear system of equations may need to be solved. This is why unlike the FDTD method the FEM algorithm will consume more computer resources for the same problem. On the mathematical form of the basis functions and the triangular elements the reader is directed to references [21] and [29]. Instead, a qualitative discussion on the 2D-FEM unstructured mesh grid is provided herein. An image of such a triangular mesh grid is depicted below in Fig. 2.11.1. In both FDTD and FEM the mesh can be denser in a certain area of the simulation. In FDTD this is achieved with a second denser mesh grid in an area where an initial grid with a larger mesh grid step resides. However, FEM with its unstructured triangular grid can adapt to sharp angles and curved surfaces, while the FDTD algorithm will eventually lead to a staircasing or index averaging effect. In terms of Boundary conditions, again the Dirichlet ($V=0$) and Neumann ($\boldsymbol{n}\cdot\nabla V=0$) conditions can be used while for electromagnetic wave problems, PML boundaries can also be applied to a FEM numerical formulation [21].

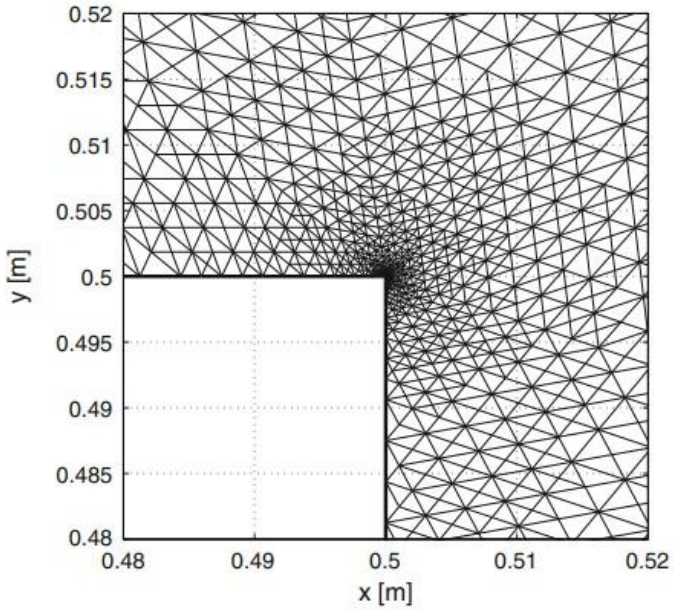


Fig. 2.11.1: The image of an adaptive 2D-FEM Mesh Grid for an object in a simulation [16].

2.12: Chapter Conclusions

This Chapter dealt with the theoretical background of most of the waveguide devices encountered in this thesis along with the analytical and numerical methods that can be employed for the solution of eigenvalue and excitation-based computational

electromagnetic problems in photonics. The next chapters will deal with the practical photonics device design on the Si_3N_4 platform using elements also from this Chapter.

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Chapter 3: Bridging monolithic InP-based laser sources with Si_3N_4 waveguides on a Si-substrate

3.1: Fundamentals of laser sources and low-dimensional semiconductor light emitters

Since their inception, lasers have contributed to the tremendous growth noticed in optical communication systems over the last few years. Laser sources are based on a gain medium, enclosed in an area limited by two mirrors. The lasing operation is based on the stimulated emission, where the gain medium achieves population inversion and incoming photons stimulate the emission of additional photons. These emitted photons have the same frequency, phase, and direction as the stimulating photons, resulting in coherent laser light. This is differentiated from spontaneous emission where atoms or electrons in the gain medium undergo a transition from an excited state to a lower energy state, emitting photons in a random direction and with random phases. Spontaneous emission can lead to noise in lasers, due to triggering quantum noise, intensity noise, and linewidth broadening. All these noise factors are related with the random timing, amplitude, and phase of the emitted photons. Efforts have been made to minimize the impact of spontaneous emission-induced noise in laser systems, even if spontaneous emission is an inherent property of a laser. Techniques such as optical feedback, active stabilization, and noise reduction schemes can be employed to mitigate noise and improve laser performance.

Crucial properties of laser sources are the wavelength of operation, the output power, the wavelength tunability of the laser if the device exhibits tunable characteristics, the electrical power consumption, the electro-optic bandwidth of the laser if the laser is

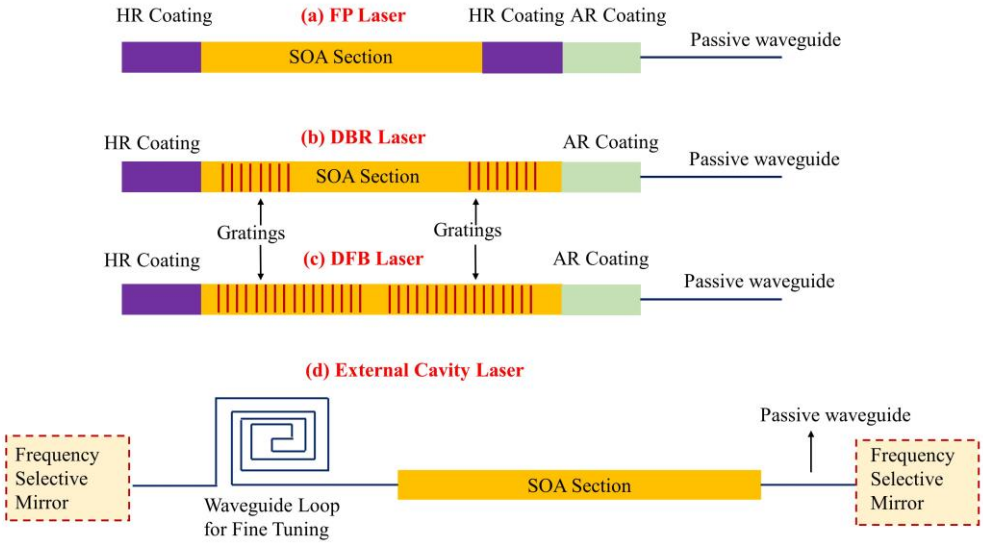


Fig. 3.1.1: (a) Topview Schematic of FP Laser, (b) DBR Laser, (c) DFB Laser, and (d) External Cavity Laser

meant for high-speed operation and others. Out of those, the linewidth is related with its coherence, and thus to the noise and phase-stability characteristics when it is applied in optical communication systems. It refers to the width or range of frequencies present in the output of a laser. A narrow linewidth means that the laser emits light at a very specific frequency or wavelength, while a wide linewidth indicates a broader range of frequencies. The InP-based photonic platform has been the platform most studied for the integration of laser sources in PICs for the C-band wavelength range (with a central free-space wavelength of λ = 1550nm), while the GaAs is most preferred for the O-band (with λ = 1310nm central wavelength). Because III-V based semiconductors have a direct bandgap [1, 2] they can easily facilitate light amplification and emission in PICs. On the other side, Si-photonic and Si_3N_4 photonic platforms employing semiconductors or dielectrics with an indirect bandgap are not suitable for active optical processes such as amplification or lasing. A variety of semiconductor lasers have been designed for PICs in the III-V based platforms. Most of them can be separated in four different categories, as seen in Fig. 3.1.1. Fabry-Perot lasers are the simplest form of a laser, consisting of the gain medium and two highly reflective (HR) interfaces forming the mirror of the laser. Light can be coupled to a passive waveguide in a PIC utilizing an anti-reflective (AR) coating. Even though this circuit is the simplest one, it leads to multiple wavelengths emitted simultaneously by the laser. Two categories that are based on grating structures are the Distributed Bragg Reflector (DBR) and the Distributed Feedback (DFB) lasers as seen in Fig. 3.1.1(b) and (c), respectively. The DBR device utilizes the Bragg structures as frequency selective mirrors and a phase-shifter at the active section can be used for electro-optic tuning of the laser. On the other side, DFB lasers have the distinctive feature of the presence of a grating structure within the waveguide. This grating consists of a periodic pattern of variations in the refractive index of the material and acts as a distributed feedback mechanism, that reflects a specific wavelength of light and provides the required optical feedback to sustain lasing at that wavelength. An electro-optic phase-shifter at the active section can provide phase-tuning as well in this case. Both the DFB and DBR lasers can provide single-wavelength operation with high Side-Mode Suppression Ratio (SMSR) values combined with larger tunability. The final category of lasers is the external cavity laser shown in Fig. 3.1.1 (d). In this design, an SOA device acting as the gain medium is placed at the center of the circuit interfaced with an AR-coating and a suitable designed interface with the passive waveguide section. The frequency selective mirrors are designed on the passive waveguide circuitry and phase-tuning can be achieved by placing a thermo-optic phase-shifter (or electro-optic phase shifter in the case of III-Vs) at a loop in the total cavity and at the frequency selective mirrors. This design may require a higher footprint but achieves a much lower linewidth value leading to spectral purity, combined with single-mode operation and high output powers.

Towards the ultra-low linewidth values, the layer stack configuration of the III-V gain medium is fundamental since the design target is to reduce the linewidth enhancement factor (Henry factor) and achieve high optical gain, low optical loss, and reduced resistance, followed by reduced Joule heating and thermal noise. In the beginning, mainly bulk laser devices based on III-V materials were used. The emission wavelength can be controlled by deploying alloys of III-V materials and layer stacks in a process called bandgap engineering [3]. On the other side, the linewidth of the laser

is related with the electronic Density of States (DoS) of the materials used in the gain medium. In bulk materials, electrons transit from the valence band to the conduction band during absorption, and from the conduction band to the valence band during the spontaneous and stimulated emission. In the first process, the emitted photons have random phase terms while in the second process all photons are in-phase and lead to coherence. The density of states is defined as the number of the available electronic states per energy and volume unit. In three dimensions it can be proven that the DoS is given by the relation [3]:

$$DoS(3D) = \frac{m\sqrt{2mE}}{\pi^2\hbar^3} \quad (3.1)$$

Bulk gain media will inevitably lead to large linewidth values due to the large number of available electronic states in the valence and the conduction band. For this reason, low-dimensional semiconductors have been proposed and applied as a solution for the development of low-linewidth lasers [4]. A material is defined as a quantum well if one of its three dimensions has a length value close to or lower than the de Broglie wavelength of the electron [3]. In this case, the transport of the electron in the low-dimensional layer is ballistic and resembles the propagation of an electromagnetic wave in a waveguide. The DoS in quantum wells takes the form:

$$DoS\ (2D) = \frac{m}{\pi\hbar^2} \quad (3.2)$$

The energy states in the quantum well get a distinct form and separate from one another, unlike bulk materials. Also, the high confinement of electrons in the quantum well layer can lead to higher velocities in the quantum transport. Quantum wells can enable enhanced carrier confinement, reduced threshold current, increased gain bandwidth, which is the reason why many light emitting devices are based on Multi-Quantum Wells (MQWs).

Finally, Quantum Dots are materials where three out of three dimensions have values comparable or lower than the de Broglie wavelength of the electron. This way, the quantum dot material exhibits atomic-like characteristics, leading to excellent carrier confinement in the quantum-dot. The optical properties of quantum dots can be explained by models that employ the time-dependent perturbation theory. Due to the very distinct electronic states, quantum dot lasers exhibit the best linewidth characteristics out of all laser devices. Layers of material filled with quantum dots can be fabricated layer by layer quantum dot by quantum dot on top of one another in a systematic manner, by utilizing the stresses developed in each layer [5].

Integrated photonic spectrally pure low-linewidth tunable lasers are greatly desired by the optical communications industry for transceivers for telecom applications. Light emitting devices with low-linewidth high output power and large tunability can deliver high-baud rate signals modulated by external modulators in a variety of modulation formats at extremely high distances with coherent systems. They have been employed in Optical Test and Measurement, Access Networks, and Transport. Especially, in the last category, tunable lasers have been detrimental for the rapid deployment of these devices in real networks. In those, integrated external cavity tunable lasers are utilized

as light sources in the transmitters and as local oscillators (LO) in coherent optical receivers. Coherent receivers use a tunable LO to match the received optical signal's frequency and phase, enabling accurate detection and demodulation of the transmitted data. Tunable lasers provide the flexibility to adapt to various transmission wavelengths and formats in coherent optical systems. In [6] an external cavity tunable laser with 40mW fiber coupled output power and 5kHz linewidth combined with >50dB SMSR was utilized in a 90Gbaud 64QAM coherent system achieving a Signal to Noise Ratio (SNR) close to commercial Tunable Laser Assemblies. In [7], a Si_3N_4-based Brillouin laser was used to demonstrate an aggregate 5.4Tbps coherent links using 15 wavelengths at 90Gbaud 16QAM modulation. Finally, in [8], a tunable hybrid InP-Si_3N_4 dual laser module (DLM) with low linewidth and relative intensity noise (RIN) was used to demonstrate a wavelength and frequency flexible optical fronthaul system for high-capacity wireless networks, achieving low Error Vector Magnitude (EVM) values in the received signal.

The above showcase the importance of external cavity low linewidth tunable lasers in a wide range of application areas. Going back to the monolithic III-V based platforms, the achievable linewidth value is limited by the high propagation losses in the III-V waveguides, free carrier light absorption phenomena and non-linear phenomena when very high light powers are confined in the laser cavity. The co-integration of the III-V with Si-photonic waveguides featuring low-propagation losses with values as low as 0.16dB/cm has been presented in the literature [9]. However, eventually, Si exhibits high non-linear losses due to two-photon absorption, free-carrier absorption and four wave mixing. For this reason, Si_3N_4 is the optimum passive waveguide platform for the development of external cavity tunable lasers, avoiding non-linear effects at high inserted optical powers, achieving low propagation losses, and much higher temperature intolerance than both the III-V and Si platforms [10]. Quantum dot lasers co-integrated with passive Si_3N_4 waveguides are expected to exhibit optimum performance.

3.2: Co-integration of lasers with Si-photonic and Si_3N_4 photonic waveguides

A very nice overview of the co-integration techniques between III-V based gain elements and Si-photonic and Si_3N_4 photonic waveguides has been demonstrated in [10]. Towards the co-integration of III-V gain media with passive Si_3N_4 waveguides, four main techniques have been proposed in fabrication: a) Hybrid integration, b) Heterogeneous integration, c) Transfer printing and d) Monolithic integration. This section will briefly analyze each category.

In hybrid integration the Si_3N_4-on-Si substrate and the III-V based wafer are pre-processed separately and are then combined through either in-plane butt-coupling or flip-chip technology. In the first case, the III-V chiplet is placed at the edge of the chip as the formation of a cavity on the Si_3N_4 wafer targeting a proper etching depth is a relatively difficult task. An AR coating is placed between the Si_3N_4 and the III-V waveguide, while the waveguides are also placed at an angle at the edge of the chip to minimize spurious reflections that would ruin the laser coherence from light back-propagation. Precise spatial alignment of the laser and the passive waveguide is a key

issue to achieve high coupling efficiency. In flip chip bonding, the III-V chiplet is reversed and is glued upside down to certain metallic contact pads with high precision. Hybrid integration is the most deployed technology for lasers, commercially available, providing very good results on lasers but requiring very high precision, fabrication time and having a higher cost.

In heterogeneous integration, the III-V layer stack is bonded on the Si_3N_4 preprocessed wafer. Then etching of the upper layers can take place in the next process steps. Wafer bonding can be performed with or without intermediate layers [10]. The first category can be conducted via adhesive bonding, glass frit bonding, and metal bonding. On the other side, the second category consists of anodic bonding and direct bonding. Adhesive bonding is the heterogeneous integration choice in many cases using a Benzocyclobutane (BCB) dielectric layer as the separating layer between the active and passive platforms. Heterogeneous integration can provide more massively produced laser wafers, but requires a more complex process than hybrid integration. Precise alignment is also needed to obtain devices with optimum performance.

Transfer printing, on its own, stands as a solution in between hybrid and heterogeneous integration. It entails a very short production, typically 30-45s. This technology can thus enable low-cost, large-scale integration, and high-quality devices. The steps of the transfer printing process steps are depicted in a 3D and cross-sectional manner in Fig. 3.2.1 (a) and (b). Flexible elastomer polydimethylsiloxane (PDMS) stamps are used to simultaneously transfer the patterned active III-V device on the passive Si or Si_3N_4 waveguide. In this case precise alignment is crucial, however, unlike hybrid integration, many co-integrated devices can be fabricated at the same time and there is a much larger allowable space for errors in the wafers. This is why it is currently under development even in pilot lines [67].

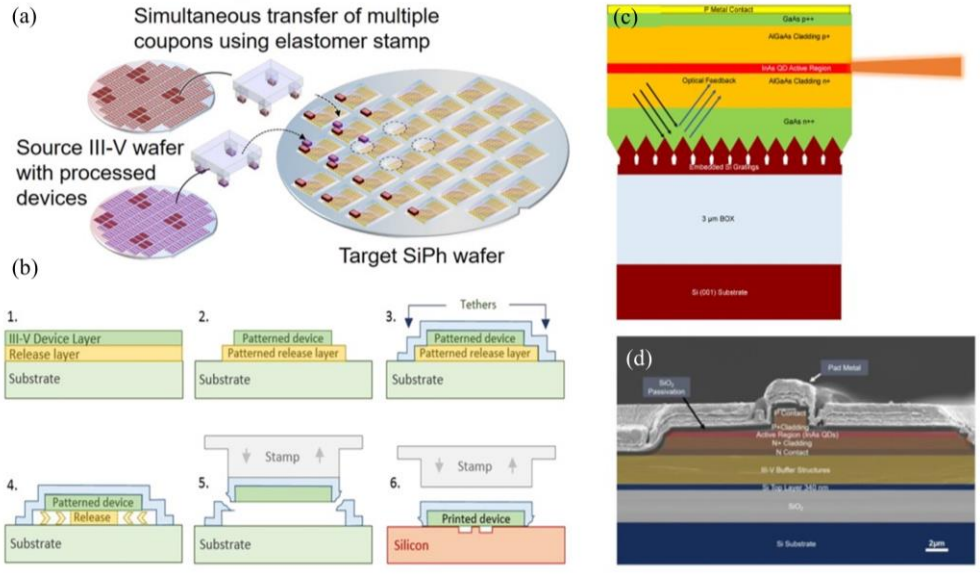


Fig. 3.2.1: (a) 3D Schematic of Transfer Printing Process of III-V on Si-photonic waveguides [11], (b) Process steps in a cross-sectional schematic for Transfer Printing [11], (c) Sideview Schematic of Monolithically integrated laser on Si [12] and (d) SEM schematic of fabricated monolithic laser on Si of [12].

Monolithic integration, finally, is the holy grail in Si-photonics targeting the direct growth of III-V-based active layers on a silicon substrate. Metal-Organic Chemical Vapor Deposition (MOCVD) is the main deposition technique in monolithic integration but also Molecular Beam Epitaxy (MBE) has been deployed in few publications [13]. The difference in the lattice constant between the III-V materials like InP and Si prohibits the direct growth of such active sources on Si. On the other side, monolithic GaAs lasers on Si are already in a commercialization stage. The lattice constant difference can lead to lattice dislocations propagating across the layer stack. Quantum dots (QDs) can elevate the laser performance since the charge is localized in the dots and thus QD lasers are more tolerant to lattice dislocations and optical feedback [68]. Monolithic integration of InP-based lasers is at an experimental stage, but tremendous growth has been observed the last few years. In [14], a monolithic GaP on Si based laser was presented employing regrowth processes, achieving a 480kHz linewidth, a SMSR of 50dB, and a 3.6mW maximum power. More recently in [12] a regrowth free laser boasting a 157kHz linewidth, SMSR over 44dB and 9mW maximum power was demonstrated. This laser is shown in a sideview schematic and a SEM image in Fig. 3.2.1(c) and (d). The higher thicknesses due to the buffer layers require multi-step coupling based techniques to transfer light to a passive Si or Si_3N_4 waveguide. Therefore, the next section will focus on the design of multi-step coupling interfaces for the monolithic integration of Si_3N_4 and InP-based laser sources.

3.3: Design of Si-rich nitride interposer waveguides for efficient light coupling from InP-based QD-emitters to Si_3N_4 waveguides on a silicon substrate

Advances in CMOS technology have led the field of Si photonics to flourish during the last few decades [15]. Nevertheless, one major technical challenge hindering the wider adoption of Si photonics to other applications comes from the incorporation of light sources in Photonic Integrated Circuits (PICs). Until recently, most silicon photonic circuits have relied on external light sources to couple light in and out of the circuit through novel fiber-PIC I/O interfaces [16]. This approach has the advantage of high yield during assembly and packaging at the expense of higher overall device cost and lower light output power stemming from the fiber-PIC interface losses. The advent of hybrid integration has enabled the next generation of PICs, where the laser source is embedded on the chip through bonding of III-V elements in die [17, 18] or wafer level [19]. The light transfer from the laser to the Si waveguide in these PICs is performed either with adiabatic [20], or end-fire coupling mechanisms [21], with both options providing pros and cons related to coupling loss performance versus alignment accuracy and overall assembly cost. However, the best PIC technology in terms of cost and performance combining active and passive elements is the monolithic co-integration of Quantum Dot (QD) III-V materials on the Si platform. QDs provide the inherent advantages of minimum damage at the active layers from the dislocation propagation formed during the growth of the III-V material at the III-V/Si interface [22], higher temperature insensitivity to output power [23] and higher lasing tolerance to back-reflections [24]. The first trials from this new integration technology have focused almost solely on the integration of QD GaAs lasers [25, 26] on Si, with recent

results indicating that commercialization is imminent [27]. Additionally, a design strategy for co-integrating monolithically grown GaAs lasers on Si with Si-passive waveguides has recently been presented in [28]. Going to monolithic integration of QD InP, the higher mismatch of the lattice constant relative to GaAs generates additional complexity for the InP/Si buffer, but first results are already reported in [29].

At the same time, the Si_3N_4 technology has gained ground versus traditional Si for fabrication of advanced PICs due to lower cost, lower propagation losses, lower PIC-to-fiber coupling loss [30] and higher temperature insensitivity [31]. The integration of the lasers in this type of PICs is performed through flip chip bonding at the Si_3N_4 waveguide facet and the light is coupled to the passive waveguide with the end-fire mechanism. This solution inspired by the thick (>1um thick) Si platform mandates sub-micron placement accuracy for the III-V die and provides coupling losses in the order of 1.8 dB [32]. In addition, it is not compatible with dual facet devices e.g. modulator and Semiconductor Optical Amplifiers (SOAs). An approach to overcome all these problems was presented in [33], where a PIC employing a crystalline intermediate Si layer for transferring the light from the InP to the Si_3N_4 low loss waveguide was demonstrated, relying on a rather complex fabrication process involving both die and wafer bonding, without providing any value about coupling loss. In [34] the integration is simpler, and the light is propagating from the laser to the Si_3N_4 adiabatically through a hydrogenated aSi waveguide acting as a medium and in this way a fully integrated laser on Si_3N_4 via transfer printing was shown. The coupling losses were 2 dB including propagation to the aSi interposer. Nonetheless, both techniques require multiple taper steps and taper tips widths of 150nm or lower, that render them incompatible with low cost 248nm DUV lithography, while the taper formations also mandate multiple lithography steps, degrading in this way the low fabrication cost advantage coming with the Si_3N_4.

In this work we present the results from the systematic investigation of a low-cost coupling scheme for seamless light transfer between QD InP active elements and the Si_3N_4 passive platform that is relying on a novel SRN material with tunable optical properties in terms of refractive index acting as an intermediate medium. The whole concept is compatible with Back-End-Of-Line monolithic integration process and 248nm DUV lithographic specifications for maximization of low-cost fabrication potential. Furthermore, although the analysis assumes an InP active medium, it can be applied also for monolithically grown GaAs based lasers and dual facet devices such as SOAs and electroabsorption modulators, demonstrating the wide applicability of the proposed concept.

The results of this Section complete the previous work presented in [35], [36] and [37], by providing the full analysis of the optimization process through simulations performed for three different interposer waveguide geometry designs exchanging performance for fabrication complexity. The best solution is a three-layer Si-rich nitride (SRN) waveguide stack composed of one layer with n=3.41 sandwiched between two layers of n=3.18 with a 1100 nm total thickness providing 0.9dB coupling losses and back reflection of -52.4dB, when the InP-SRN interface is angled at 15^0. In addition, a fabrication tolerance analysis is presented at the QD InP-SRN angled interface for misalignments of 300nm in the y-axis and 200nm in the z-axis, where the tri-layer SRN exhibits its superiority due to the larger supported modal size. In the worst-case scenario

of simultaneous 200nm misalignment in the z-axis and 300nm in the y-axis, the tri-layer SRN achieves a total coupling loss of 1.78dB and back-reflections lower than -54.4dB, demonstrating the resilience of the proposed coupling scheme to large errors in the CVD and lithography processes. All simulations use for the refractive index of the SRN interposer the values extracted from ellipsometry measurements of an ultra Si-rich nitride material (SRN) developed and engineered for this purpose. In addition, the work provides details about this new SRN material that was developed to provide crack free films with thickness up to 500nm. From these samples, cutback structures with waveguides of 500x500nm^2 cross section were formed via e-beam lithography revealing 15dB/cm propagation losses in agreement with other works on amorphous Si-rich nitride (aSi:N) waveguides fabricated by PECVD. However, this value can go down to lower than ~10 dB/cm by further optimizing the deposition and especially the etching process [38]. The Section is organized in the following way: Section 3.3.1 presents the overall coupling scheme aiming to bridge the InP-based source with the Si_3N_4 photonics platform considering a BEOL monolithic integration process. Section 3.3.2 presents the experimental ellipsometry and profilometry results that demonstrate the feasibility of fabricating SRN waveguides with the desired properties for the proposed light interface. Section 3.3.3 describes the detailed design optimization analysis for the interposer SRN waveguiding section by investigating three different cases, i.e. a single-material strip, a single-material rib and the best in terms of performance configuration of the two-material tri-layer rib configuration. Section 3.3.4 incorporates also the simulation results when the InP-SRN interface is angled targeting minimization of the back reflections. In Section 3.3.5 a detailed tolerance analysis for the optimum structures of the three cases is presented, validating the supremacy of the tri-layer design.

3.3.1: Overall III-V to Si_3N_4 interface concept and envisaged integration process

Towards monolithically integrating InP-based gain media with a thick-Si_3N_4 platform in a CMOS compatible and Back-End-Of-Line (BEOL) manner, a suitable SRN material should facilitate light coupling from the semiconductor to the dielectric platform, as a bridging material layer.

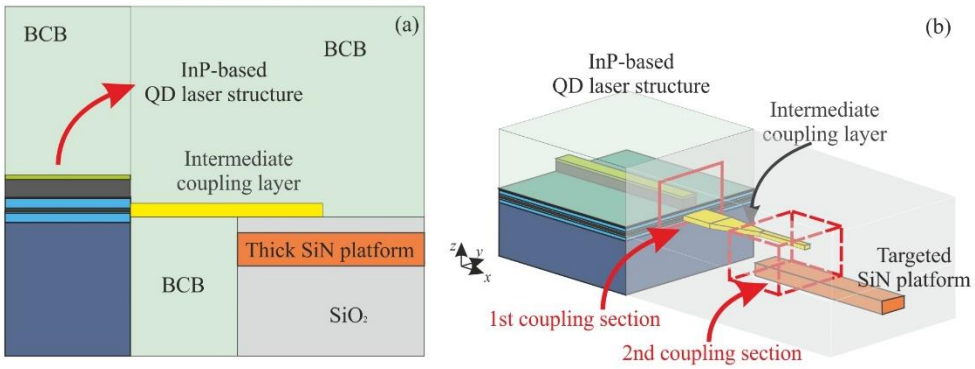


Fig. 3.3.1: (a) Sideview image of the envisaged fully monolithically integrated InP-based QD-laser and Si_3N_4 photonic integrated circuit, (b) 3D geometry model depicting the proposed interfacing scheme. The light is coupled from an InP-based QD emitter to a thick Si_3N_4 waveguide through an intermediate SRN interposer.

Figure 3.3.1(a) illustrates a sideview image of the proposed monolithic integrated QD InP laser that shows also the SRN transition waveguide for light coupling to the Si_3N_4 passive circuitry. The proposed integration platform starts with the 248nm DUV formation of the Si_3N_4 structures on Si-wafers through Low-Pressure Chemical Vapor Deposition (LPCVD). Then local windows are made on wafer level for the deposition of the QD InP-based layer stack of the active components i.e. lasers and SOAs, followed by etching for facet formation. The gaps in the windows are filled by divinyl siloxane benzocyclobutene (BCB) and then SRN films are deposited with Plasma-Enhanced Chemical Vapor Deposition (PECVD) on the whole surface of the wafer covering the facets of the QD InP elements. The SRN transition waveguides are structured then on the film through 248nm DUV lithography and etching. All critical parameters in the proposed coupling scheme are defined by the PECVD process and DUV lithography, in contrast to hybrid integration where the coupling efficiency is dictated by the placement accuracy of flip chip bonders that attach the III-V dies to the Si_3N_4 waveguide.

Figure 3.3.1(b) illustrates in a 3D fashion the proposed Back-End-Of-Line (BEOL) interface that can be described in two cascaded discrete coupling stages: i) an end-fire section from the InP-based QD active region of the laser to the interposer SRN waveguide and ii) an evanescent coupling stage from the interposer SRN waveguide to the targeted Si_3N_4 platform. The design optimization of the overall interface targets two key performance metrics: maximum light transfer to the Si_3N_4 waveguide and minimum back-reflections to the QD-emitter for uninterrupted operation in high speed Directly Modulated Lasers (DMLs) and external cavity Tunable lasers (ECLs). These two parameters are determined by the cross-sectional geometry and the refractive index of the intermediate SRN waveguide at the two interfacing steps (InP-SRN and SRN-Si_3N_4) and are thoroughly investigated in this work by 2D and 3D FDTD simulations. The whole configuration is totally reciprocal and can be employed also for light coupling from Si_3N_4 to InP-based-QD active elements rendering it ideal for two facet devices, such as SOAs.

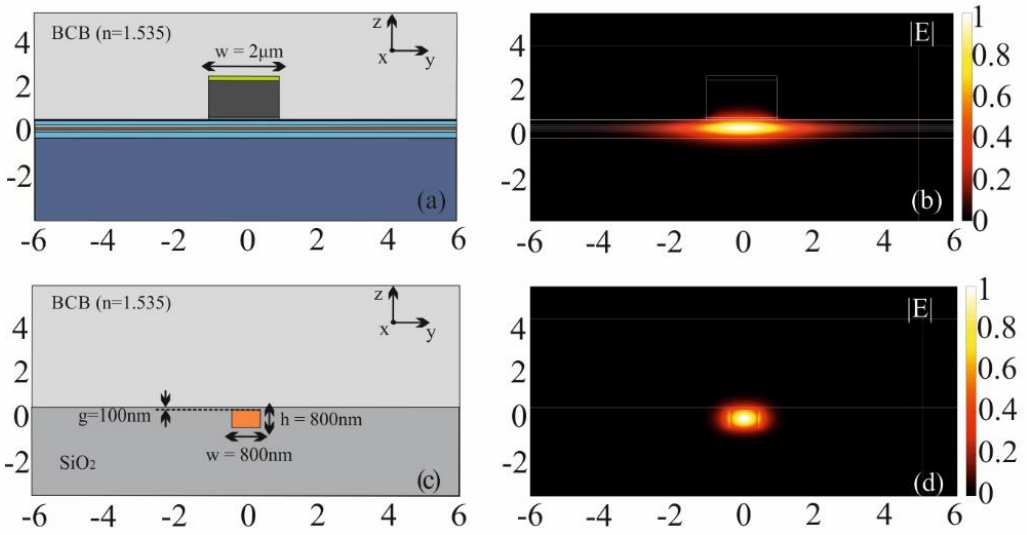


Fig. 3.3.2: (a)Depiction of the InP-based QD-emitter cross-sectional geometry, (b) Electric Field |**E**| distribution of the fundamental TE-mode supported by the InP-based QD-emitter, (c) Depiction of the (800×800) nm^2 Si_3N_4 waveguide cross-section, (d) Electric Field |**E**| distribution of the fundamental TE-mode supported by the Si_3N_4 waveguide.

For optimum light coupling in terms of maximization of the achievable coupling efficiency and the minimization of back-reflections to the semiconductor region, the modal field profile of the targeted mode to be excited in the interposer should approach the profile of the mode supported by the gain medium for spatial matching, at the first coupling stage [39]. Identical modal effective indices between the two modes should also minimize the received back-reflections to the semiconductor region. For the second coupling stage, the propagation constant of the mode propagating at the upper SRN layer should be identical to the propagation constant of the targeted TE mode at the final Si_3N_4 waveguide for phase-matching [39]. Nonetheless, the two conditions that need to be simultaneously met create a trade-off that needs to be dealt with during the design analysis. Optimum spatial matching requires high interposer thicknesses, while optimum phase matching mandates low interposer thicknesses.

The source in the proposed interface scenario is an InP-based QD emitter with a layer stack similar to the one presented in [40] formed by a shallow etched 2μm wide ridge waveguide. BCB acts as the cladding of this QD-emitter. The cross section in the y-z plane of the laser is illustrated in Fig. 3.3.2(a). The interfacing design target is to couple light to the final thick Si_3N_4 waveguide platform. The Si_3N_4 waveguide platform in this work is relying on the 800nm thick standard platform of Ligentec S.A [41]. In this analysis the width has been set at 800nm, supporting both fundamental TE and TM modes with low propagation losses. However, only the TE Si_3N_4 mode is of interest due to the TE polarization of the QD-laser mode. Figure 3.3.2(c) presents the cross-sectional Si_3N_4 waveguide geometry with SiO_2 over- and under-cladding. On top of the 100 nm SiO_2 overclad there is also divinyl siloxane benzocyclobutene (BCB) (n = 1.535 at λ = 1550 nm) for compatibility with the III-V monolithic integration. The refractive index value for SiO_2 at a wavelength λ = 1550 nm has been derived from [42]. For the SRN material, ellipsometry data at 1550 nm have been imported in our numerical simulations. During eigenmode calculations, stretched coordinate PML boundary conditions have been applied at the edges of the computational window to eliminate

artefact modes [43]. Figure 3.3.2(b) depicts the electric field distribution |**E**| of the supported fundamental TE mode of the QD-laser. This TE mode spatially extends about 1μm in the *z* axis and about 4μm in the *y* axis. The calculated effective refractive index of this mode is n_{eff}=3.269. On the other hand, Fig. 3.3.2(d) illustrates the supported TE mode of the 800x800 nm^2 Si_3N_4 waveguide. The electric field magnitude, |**E**| extends to a cross-sectional area around 1μm^2 with an effective index around n_{eff}=1.71.

As mentioned, since the proposed interface scenario involves two discrete coupling stages, the optimization of the SRN intermediate waveguide will come as a result of accordance with the two conflicting dictations. Therefore, the refractive index of the interposer material should be close to 3.2-3.4, aiming to match the effective index of the TE mode by the InP-based emitter towards minimization of the back reflections. On the other hand, the effective index of the propagating SRN mode should approach the effective index in the Si_3N_4 waveguide (*i.e.* n_{eff}=1.71) to satisfy the phase matching condition, approaching the evanescent coupling stage. These two opposing conditions can be met only with a taper structure connecting the two coupling sections. The geometrical characteristics of the transition tapers are presented in the design analysis of the next sections. The gap g, i.e. the thickness of the SiO_2 overclad, of Fig. 3.3.2(c) between the SRN and the Si_3N_4 layers is fixed at 100 nm in all scenarios considering a minimum value where the quality of the surface roughness can be considered almost ideal. In the next section, we present the fabrication and characterization of a SRN material specifically engineered for the monolithic interposer application.

3.3.2: Si-rich Nitride material and waveguide fabrication

The fabricated Si-rich-nitride material that serves the role of the interposer waveguiding layer needs to comply with two fundamental restrictions posed by the utilized coupling schemes. In order to avoid parasitic back-reflections, the refractive index and the thickness of the dielectric material should be large enough, so that the mode in the interposer has a modal profile and an effective index close to those of the mode supported by the QD emitter. The thickness of the material should also be low enough to support a mode with an effective index close to the targeted mode in the Si_3N_4 PIC, at the end of the intermediate layer section. The temperatures in the whole process should also be lower than 300°C in order to prevent group V element adsorption

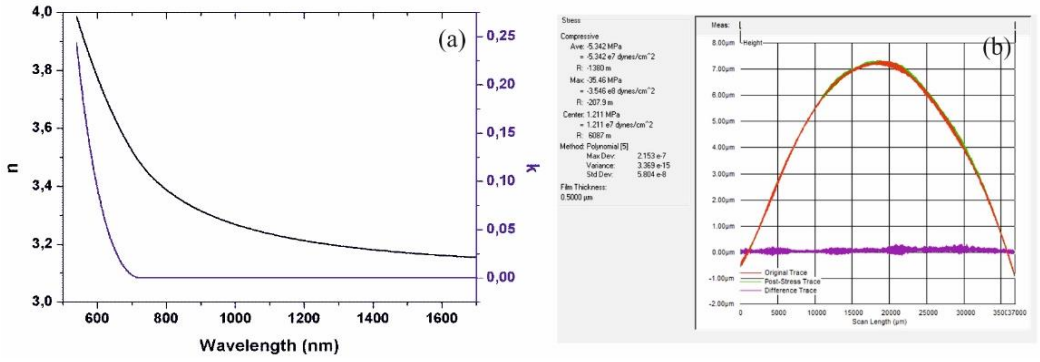


Fig. 3.3.3: (a) Refractive index and absorption coefficient as function of wavelength for the developed SRN material, (b) Stress measurement by profilometer scan before and after the SRN deposition highlighting the low stress of the material.

in the underlying III-V layers. Finally, the material quality should be high enough to prevent intense scattering and absorption phenomena during propagation of the light in this intermediate structure. All these restrictions impose the foundations for the development of the whole fabrication process, that is close to the one presented in [44] for the fabrication of a-Si:H rib waveguides and is described in detail in [36]. One of the key targets in the whole effort was to maintain low-stresses during material deposition, so that high film thicknesses can be obtained without the occurrence of structural defects.

The process is based on deposition with a PECVD tool (Nextral ND200), using monosilane (SiH_3) and nitrogen (N_2) as precursors, and helium (He) as the dilution gas. The gas chamber temperature and RF power were set to 250°C and 100W respectively, while the variables of the process were the precursor ratio and the chamber pressure. These variables were shown to vary the refractive index and the stress of the developed material respectively. The real and imaginary parts of the refractive index of the grown materials were measured by spectroscopic ellipsometry (HORIBA Jobin-Yvon UVISEL) after SRN deposition on 2-inch Si wafers with a 150-nm-thick cladding layer of PECVD SiO_2, while the material stress was derived from profilometry results (KLA/Tencor P17) before and after the material deposition. This has been deduced from the wafer bowing levels before and after the SRN layer deposition. A 500nm-thick Si-rich-nitride film is the targeted thickness, as derived from the design analysis, enabling ranging effective indices from almost 3 to 1.7.

Fig. 3.3.3(a) shows the n-k measurements of the fabricated Si-rich nitride material, revealing a refractive index n as high as 3.165 at the wavelength of 1550nm that matches very well the InP value. According to Fig. 3.3.3(b) also a 500nm thick deposited material on a 2-inch wafer exhibits low stress with less than 1 μm overall wafer bowing. This is achieved for SiH_4, N_2 and He gas flows of 50 sccm, 40 sccm and 200 sccm, respectively and a chamber pressure of 1400 mTorr. The deposition rate is approximately 60 nm/min. The refractive index of the fabricated material can be varied by modifying the SiH_4/N_2 ratio.

The quality of this novel material was evaluated by the formation of 500nm wide waveguides with the 500nm thick low stress SRN. With more details about the fabrication process given in [36], three waveguides of 0.2, 0.5 and 1 cm were fabricated and the propagation losses were estimated from the cutback measurements to be 15dB/cm, in agreement with other works on aSi:N waveguides fabricated by PECVD. This imposes an additional critical target in the design analysis, specifically the minimization of the taper length due to the high propagation losses. In this case, the high propagation losses have very small effect in the overall interface loss if the length is maintained lower than 200 μm, providing an overall loss of 0.3dB at 200 μm. The high propagation losses however can be drastically reduced by optimizing further the whole process reaching values as low as 6dB/cm in similar dimension waveguides deposited by PECVD [38]. In the next section, an extensive design optimization procedure is presented, leading to three types of configurations, with the optimum one providing close to 1dB coupling losses and less than -50dB back-reflections.

3.3.3: Coupling interface design analysis and numerical results

3.3.3.1: Strip Si-rich nitride configuration

The strip SRN waveguide configuration is the simplest solution in terms of fabrication that can be utilized as the medium for bridging the InP-based QD laser with the Si_3N_4 platform. The methodology starts with the identification of the excited targeted fundamental TE mode in the SRN strip waveguide. A 2D-eigenmode analysis reveals the profile of this mode, as well as the rest of the supported modes in the various SRN cross sections. Figure 3.3.4(a) illustrates the waveguide geometry under study composed of a strip SRN waveguide covered by a BCB cladding with a refractive index n=1.535. The parameters under variation were the waveguide width and thickness. Fig. 3.3.4(b) presents the power overlap integral values calculated for the modal profiles of the laser and the targeted mode in the SRN waveguide [39].

The integral was calculated for widths ranging from 1μm to 7μm and for thicknesses of 300nm, 500nm, 700nm, 900nm and 1100nm. Two refractive indices for the SRN material were investigated, one with n=3.18 and one with n=3.41, corresponding to the refractive index values of already fabricated SRN low stress and high stress thin films. It is clear that for both materials and all waveguide thicknesses, the power overlap integral values reach a local maximum near a width value of 4μm. However, the material with n=3.18 seems to provide higher overlap integral values than the material with n=3.41, therefore for the following calculations only the SRN material with n=3.18 was taken into consideration.

In order to identify the optimum waveguide thickness, the evanescent coupling part of the interface was also considered. As mentioned in Section 3.3.1, the proposed waveguide thickness needs to solve the contradiction imposed by the obligation to comply with the needs of both coupling steps. This means that the optimum proposed

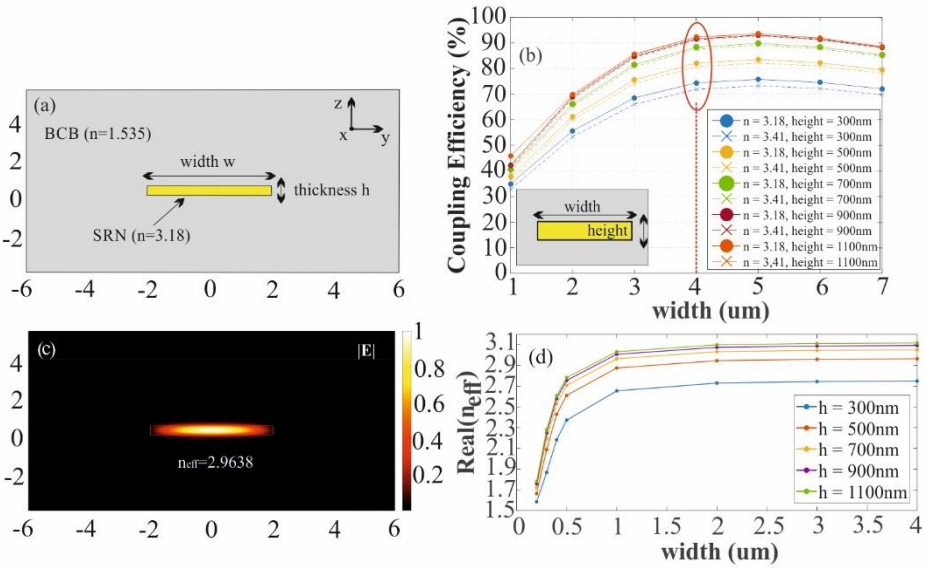


Fig. 3.3.4: (a) Simulated SRN strip waveguide geometry with BCB cladding, (b) Power Overlap Integral values for n=3.18 and n=3.41, and for widths ranging from 1μm to 7μm and for thicknesses of 300nm, 500nm, 700nm, 900nm and 1100nm, (c) |**E**| modal field profile at the yz-plane for the optimum 4μm wide and 500nm thick strip configuration, (d) Effective index values of the fundamental TE mode supported by the SRN strip configuration for varying widths and thicknesses.

thickness should provide effective indices from 3.269, corresponding to the effective index of the mode displayed in Fig. 3.3.2(b), to 1.7, close to the effective index of the targeted underlying Si_3N_4 mode, for a variable range of widths. The effective index of the fundamental TE-mode supported by the SRN waveguide for various widths and thicknesses can be seen in Fig. 3.3.4(d). The optimum waveguide has a thickness of 500 nm because this waveguide thickness supports a fundamental TE-mode with an effective index ranging from 1.69 to almost 3 for widths ranging from 0.2μm to 4μm. The minimum width of 0.2μm is in line with the limitation of 248nm DUV lithography assumed in our calculations towards mass production at low cost.

Thicker SRN waveguides do not support modes with effective indices lower than 1.7 for minimum 200nm taper width and thinner waveguides cannot achieve effective indices close to 3. Therefore, the optimum geometry defined by the 2D-analysis is a 500nm thick structure that supports a fundamental TE-mode portrayed in Fig. 3.3.4(c) with an effective index yielding a real part close to $Re(n_{eff}) = 2.96$ when the width is 4μm for highest spatial matching with the active region of the laser. This is confirmed by the power overlap integral value calculated for this configuration resulting close to 81.4% efficiency or 0.9dB coupling loss.

Having the 2D overlap integral analysis as guide, the power coupling efficiency for the first coupling stage has been extracted through 3D-FDTD numerical calculations. The power of back-scattered light that was obtained yielded back-reflections close to -19.79dB and forward propagation coupling efficiency to the targeted mode of Fig. 3.3.4(c) of 0.9dB in close agreement with the power overlap analysis. From the 2D-analysis it was observed that the optimum configuration for the selected SRN cross section is multi-mode therefore a small percentage of power from the QD-emitter mode transfers to high-order modes. In addition, as it was mentioned in Section 2, the initially excited mode in the SRN waveguide should be converted to a mode with an effective index close to 1.71. This can be derived through a proper taper design with its operation having a twofold effect, first transform the initially excited SRN mode of Fig. 3.3.4(c) to a mode with an effective index of 1.71, and secondly, suppress the high-order modes.

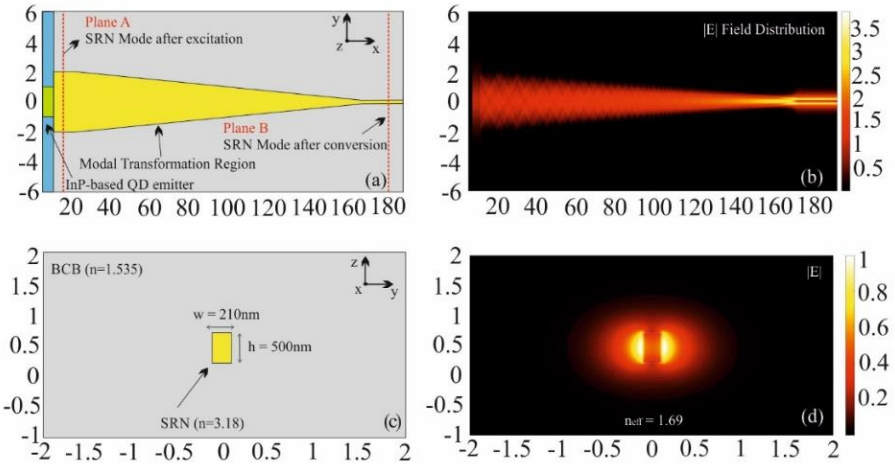


Fig. 3.3.5: (a) xy-plane of the end-fire step, simulated in 3D-FDTD calculations, (b) |**E**| field distribution derived from a 3D-FDTD simulation at 1550nm wavelength (c) yz-plane illustration of the simulated geometry at Plane B of Fig. 3.3.5(a), (d) Modal profile of the targeted TE-mode at Plane B of Fig. 3.3.5(a).

At the end of the taper, the waveguide will guide a suitable mode to enter the evanescent coupling stage.

The xy-plane (top-view) of the simulated tapered structure in 3D-FDTD numerical calculations can be seen in Fig. 3.3.5(a). Figure 3.3.5(b) illustrates the |**E**| field distribution on a xy-plane located at the middle of the SRN waveguide. The source is located at the cross-section of the QD-emitter and is the mode depicted in Fig. 3.3.2(b). Therefore, the |**E**| field includes both phenomena, that of end-fire coupling and that of the adiabatic transition. Intensive simulations revealed that a linear taper with a length of 150μm fulfills the adiabaticity criterion, exhibiting a coupling loss of only 0.037dB during the transformation from the mode of Plane A (Fig. 3.3.4(c)) to the targeted excitation mode of Plane B (Fig. 3.3.5(d)). The excited mode at the taper tip of Plane B is portrayed in Fig. 3.3.5(d) and has an effective index close to 1.69, i.e. close to the targeted mode in Si_3N_4 thereby complying with the phase-matching condition of the evanescent coupling step. The smooth conversion of the initially excited mode from Plane A to Plane B as well as the suppression of the high-order modes can be seen in the electric field distribution |**E**| of Fig. 3.3.5(b), as derived by the FDTD numerical calculation. At the end, before light enters the directional coupling step, the optical mode is deconfined, being closer at the edges of the SRN waveguide cross-section as depicted in Fig. 3.3.5(d).

As in the end-fire case, the directional coupling stage was initially studied with a 2D-eigenmode analysis at the yz- cross sectional plane of Fig. 3.3.6(a) with the 210x500 nm^2 SRN layer lying over the targeted 800x800 nm^2 Si_3N_4 strip waveguide. A small SiO_2 gap with thickness g=100nm separates the two waveguides. The 2D-analysis revealed TE-polarized supermodes of quasi-even and quasi-odd symmetry, supported by the hybrid structure. Figure 3.3.6(b) depicts the $\mathbf{E}_y$-component of the symmetric mode while Fig. 3.3.6(c) depicts the same component for the anti-symmetric supported mode. The beating of the two, a quasi-even and a quasi-odd symmetry supermodes leads to directional coupling of light to the Si_3N_4 waveguide. Theoretically the full amount of the initially inserted energy will pass to the Si_3N_4 at the coupling length L_c, given by the relation [39].

$$L_c = \frac{\pi}{\beta_s - \beta_a} \quad (3.3)$$

By inserting the effective indices of the two supermodes displayed in Fig. 3.3.6(b) and Fig. 3.3.6(c) in equation (3.3), a coupling length of 6.6μm was calculated. This was also verified by a strict 3D-FDTD simulation of the directional coupling step. The side-view of the simulated geometry is depicted in Fig. 3.3.6(d). The SRN taper tip mode of Fig. 3.3.5(d) was set as the excitation in the simulation. Stretched coordinate PML boundary conditions were once again utilized to absorb incoming waves at the edges of the computational window. The calculated |**E**| field distribution depicted in Fig. 3.3.6(e) indicates a seamless transition from the SRN to the Si_3N_4 waveguide with a coupling loss of only 0.13dB for the targeted TE-mode of the final Si_3N_4 waveguide at 1550nm. Finally, taking into account all coupling and conversion steps of the proposed interface, a coupling loss of 1.06dB is achieved for the overall configuration of the strip SRN strip intermediate waveguide.

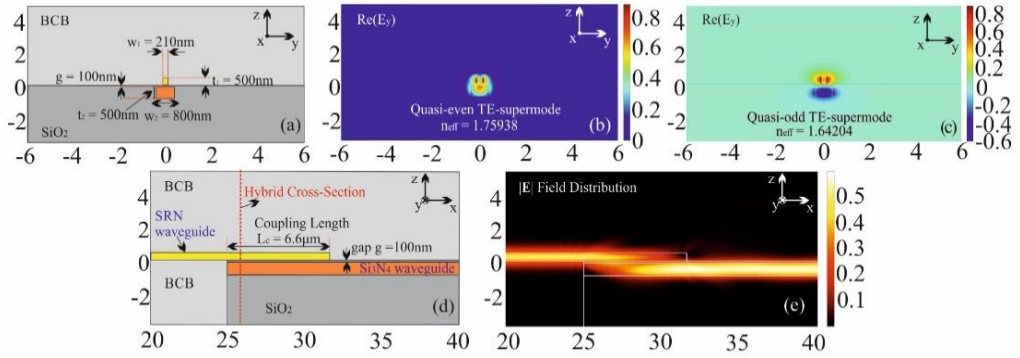


Fig. 3.3.6: (a) Depiction of the hybrid cross-sectional geometry at the yz-plane of the directional coupling step, (b) Real part of the E_y-component of the quasi-even TE-supermode supported by the hybrid geometry, (c) Real part of the E_y-component of the quasi-odd TE-supermode supported by the hybrid geometry, (d) xz-representation of the simulated geometry for the directional coupling step, (e) Electric field distribution |**E**| at the xz-plane derived by a 3D-FDTD calculation, at the middle (y=0) of the computational window.

3.3.3.2: Rib Si-rich nitride configuration

In this section a single material (n=3.18) rib SRN waveguide geometry is investigated as a means to reduce the high back reflection value of -19.79dB exhibited by the simple strip SRN configuration. Initially, two total rib waveguide thicknesses of 900nm and 1100nm were taken into consideration in 2D-eigenmode calculations. A slab thickness h with a value of 300nm, 500nm, and 700nm for a total thickness of H=900nm and a slab thickness h with a value of 300nm, 500nm, 700nm, and 900nm for a total thickness of H=1100nm were initially studied. Fig. 3.3.7(a) depicts cross-sectional yz-plane of the simulated geometry, that was inserted in eigenmode calculations. The power overlap integral value was calculated for all these waveguide

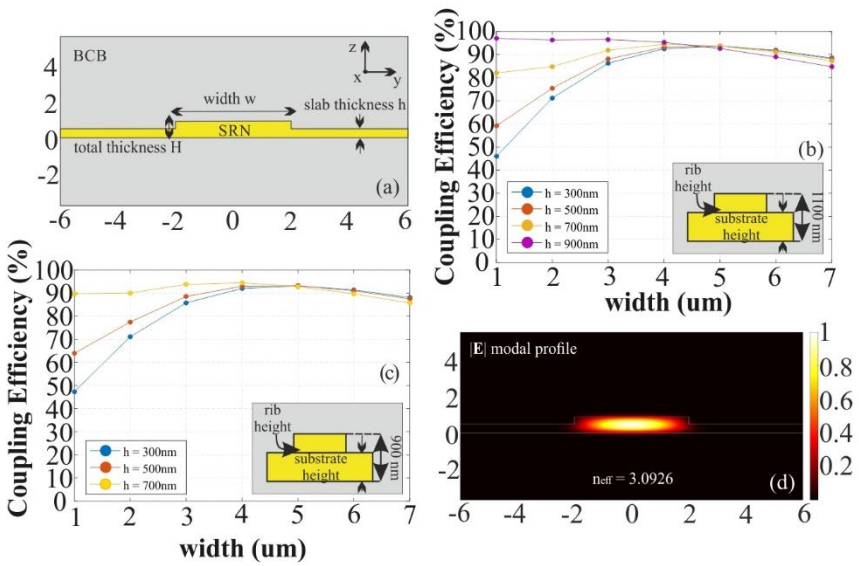


Fig. 3.3.7: (a) yz-plane of the simulated single material SRN rib waveguide, (b) Power overlap integral values for SRN single material rib waveguide and for total thickness H=1100nm, (c) Power overlap integral values for SRN single material rib waveguide, for total thickness H=900nm, (d) Fundamental TE-modal profile |**E**| supported by the single material SRN rib waveguide

thicknesses with waveguide widths ranging from 1μm to 7μm with the results illustrated in Fig. 3.3.7(b) and Fig. 3.3.7(c). For both thicknesses the power overlap integral values reach an optimum point near a 4μm rib width. In terms of slab thickness even though values above 500nm yielded higher power overlap integral values, they are discarded from thereon, since they cannot support directional coupling to the final 800x800 nm^2 Si_3N_4 waveguide with minimum 200nm wide taper tips. More specifically, it was observed that for increasing SRN slab thicknesses the effective index value of the supported TE mode at the double taper tip is increased, thus leading to phase mismatch from the targeted mode at the Si_3N_4 waveguide. Therefore, it was concluded that the optimum rib SRN waveguide features a cross section with a slab thickness h=500nm, total thickness H=900nm, and a rib width of w=4μm. The |**E**| field of the modal profile supported by this waveguide configuration can be seen in Fig. 3.3.7(d).

Due to the 500nm slab thickness, the same directional coupling step exploited in the strip cross section analysis and depicted in Fig. 3.3.6(e) can also be employed through a double taper structure for efficient coupling to the Si_3N_4 waveguide. The initially excited TE mode of Fig. 3.3.7(d) is converted through the taper in the intermediate waveguide to the mode of Fig. 3.3.5(d), that in turn is employed as the excitation in the following evanescent coupling step. The design of the conversion taper was performed utilizing consecutive numerical 3D-FDTD calculations. A minimum tip width condition of 200nm was imposed again to the designed structures for compliance with limitations from 248nm DUV photolithography processes.

A 3D-representation of the SRN based rib waveguide transition utilizing a double taper can be seen in Fig. 3.3.8(a) and the xy-plane of the simulated geometry for the optimization of the double taper structure is displayed in Fig. 3.3.8(b). The double taper is formed by separately etching the slab and the rib of the waveguide. PML boundary conditions were once again employed at the edges of the computational window. In the analysis the upper and lower taper tip widths were fixed at w_1=200nm and w_2=210nm, respectively. The initial dimensions of the rib waveguide before double tapering had

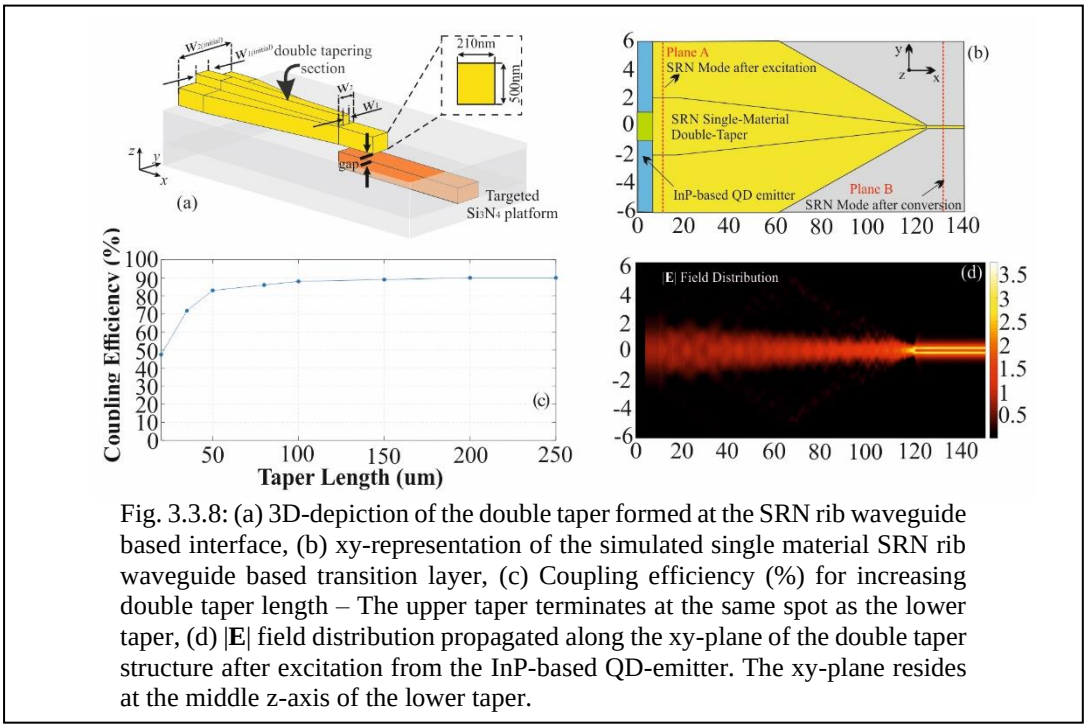


Fig. 3.3.8: (a) 3D-depiction of the double taper formed at the SRN rib waveguide based interface, (b) xy-representation of the simulated single material SRN rib waveguide based transition layer, (c) Coupling efficiency (%) for increasing double taper length – The upper taper terminates at the same spot as the lower taper, (d) |**E**| field distribution propagated along the xy-plane of the double taper structure after excitation from the InP-based QD-emitter. The xy-plane resides at the middle z-axis of the lower taper.

been w_1(initial) = 4μm and w_2(initial) = 20μm. The slab is initially extended out of the PML boundaries. The corresponding results from the 3D-FDTD simulations are illustrated in Fig. 3.3.8(c) where there is a minimization of coupling loss at 0.5dB or close to 89% coupling efficiency for a 100μm taper length. In addition, from the numerical calculation, the coupling loss for the end-fire at the III-V to SRN interface is calculated at 0.36dB in agreement with the correspondent power overlap integral value. The level of back-reflections is near -30dB for a 0° facet angle. Fig. 3.3.8(d) depicts the smooth butt-coupling and transition of the light in the double taper structure, after the initial excitation of the mode that was used as the source. Along with the fundamental TE-mode, higher order TE-modes are initially excited in the taper, but are later suppressed, as in the strip waveguide case. Again, at the end the mode mostly resides at the edges of the SRN waveguide.

3.3.3.3: Tri-layer waveguide configuration

For further performance optimization a tri-layer rib waveguide was investigated, leveraging the versatility of our engineered SRN material. Instead of a single material with a thickness of 900nm and n=3.18, this improved geometry consists of a material with n=3.41 sandwiched by two layers with n=3.18, as shown in Fig. 3.3.9(a). The total waveguide thickness was set to 1100nm targeting supported modes with effective index as close as possible to the ones emitted by the QD laser for minimization of the back reflection. Also, the thicker waveguide enables better tolerances in vertical misalignment during the deposition of the SRN stack at the laser facet. The slab thickness is set to 500nm, corresponding to a SRN waveguide thickness suitable for the directional coupling configuration of Fig. 3.3.6(d). Figure 3.3.9(c) presents the power integral overlap values from scanning the rib width w and middle material thickness t, while Fig. 3.3.9(b) illustrates the corresponding effective index for each cross section. What should be noted is that by increasing the middle material thickness t, a trade off

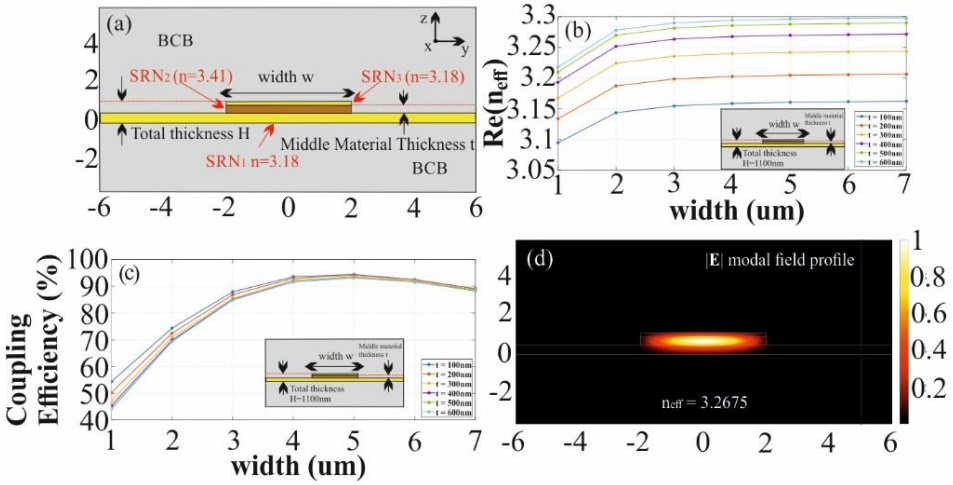


Fig. 3.3.9: (a) Depiction of the simulated tri-layer SRN waveguide at the yz-plane, (b) Effective index of the fundamental TE-mode supported by the tri-layer interposer for varying width w and middle material thickness t, (c) Power overlap integral values for varying width w and middle material thickness t of the tri-layer waveguide, (d) |**E**| field distribution of the fundamental TE-mode supported by the optimum tri-layer SRN waveguide.

appears. While the achievable effective indices increase, the coupling efficiency indicated by the power overlap integral calculations slightly deteriorates. This is attributed to the larger modal confinement in the middle layer from the higher refractive index. Figure 3.3.9(d) reveals that the overlap integrals reach a local maximum near w=4μm rib width. Also, from Fig. 3.3.9(b) it is deduced that for a total thickness of H=1100nm and t=400nm, the effective index of the fundamental-TE mode supported by the tri-layer waveguide is 3.2675, matching very well the laser mode. For this cross section, the power overlap integral calculation yields a transition efficiency value of 91.8% or 0.37dB loss. For the optimum geometry, the modal |**E**| field profile of the fundamental TE-mode presented in Fig. 3.3.9(d) has a spatial distribution very similar to that of the mode of the InP-based QD-emitter shown in Fig. 3.3.2(b).

As already mentioned, the selection of 500 nm total slab thickness allows the employment of the directional coupling step shown in Fig. 3.3.6(d) and (e) coming directly from the strip SRN waveguide. A suitable double taper structure adiabatically transforms the initially excited mode shown in Fig. 3.3.9(d) to the desired mode of Fig. 3.3.5(d), with the corresponding top-view depicted in Fig. 3.3.10(a). The InP-based QD-laser mode was the excitation in cascaded 3D-FDTD calculations that were carried out to determine the double taper shape and geometry. PML boundary conditions were utilized at the edges of the computational window, similarly to the previous configurations. The upper taper has a length of 200um ending to a tip width of w_1=100nm that adiabatically couples light to the targeted cross-section of Fig. 3.3.5(c). The lower taper features a tip width of w_2=210nm and length of 200um. The upper taper geometrical characteristics were selected aiming at optimum coupling performance. The higher thickness of the upper taper requires an upper taper width as low as 100nm in order to be adiabatic. However, the selection of an upper taper with a 200nm width instead, for the same length, would only cause a deterioration of 0.1dB to the overall device performance, based on 3D-FDTD calculations.

The total tri-layer double intermediate taper has a total length of 200μm and achieves a coupling efficiency of 94.2%, or equivalently, induces losses near 0.26dB. A smooth excitation of the targeted TE mode before the directional coupling step is seen at the end of the taper at plane B in Fig. 3.3.10(b). The end fire transition loss at the InP-SRN interface according to the 3D-FDTD calculations is 0.37dB loss (91.8% coupling efficiency), identical to the power overlap integral value. In total by summing the losses from the end-fire coupling, taper transition and directional coupling (0.13dB) of Fig. 3.3.6(d) and 3.3.6(e), the proposed tri-layer based interface yields an overall coupling

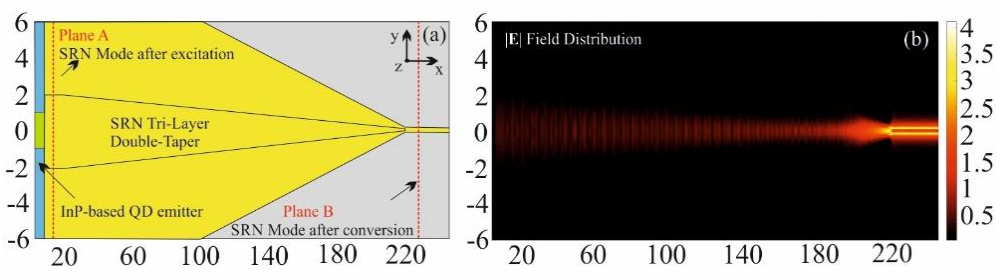


Fig. 3.3.10: (a) xy-plane representation of the tri-layer SRN waveguide-based transition geometry, with the plane being vertical to the z-axis at the middle of the lower taper (b) |**E**| field distribution along the 1st Coupling Stage of the proposed SRN tri-layer rib waveguide based interface.

efficiency of 0.77dB, while back reflections are as low as -34.95dB for a 0° facet angle at a wavelength λ=1550nm. So, overall, the tri-layer SRN waveguide achieves the optimum results compared to the other structures, evidently being a suitable configuration for coupling of light between QD-emitters and thick Si_3N_4 waveguides in monolithic integration scenarios. However, this enhanced performance is coming at the expense of additional fabrication complexity for the deposition of three materials with suitable thicknesses for the formation of the final waveguide geometry.

3.3.4: Investigation of angled facets for the minimization of back-reflections

In the analysis presented so far, the SRN based interface schemes have yielded coupling loss values even at the sub-dB range for the tri-layer interposer. However, the back-reflections were higher than -40dB. In the co-integration of the laser sources in Si photonics PICs, back reflection requirements are highly relaxed by utilizing QDs in the active region. The QDs with recent experimental studies of QD-DMLs on Si have exhibited penalty-free transmission at 10 GHz external modulation even with -7.4dB parasitic optical feedback [45]. However, in order to ensure seamless light transfer under any III-V material an angle was induced to the InP-Si_3N_4 interface for further minimization of the back reflections. Fig. 3.3.11(a) depicts the top-view depiction of the modified simulated interface, repeated for all three configurations. The facet angle was set to 0°, 2°, 8°, 10° and 15°, as dictated by fabrication capabilities, and the corresponding results for back reflections and coupling efficiency are presented in Fig. 3.3.11(b), and (c), respectively. Back-reflections were calculated behind the plane of the source that was used as the excitation. The monitor for the calculation of the coupling efficiency is at plane B, i.e. the end of the modal transformation region. The 3D-FDTD calculations reveal for the strip SRN waveguide reduced back reflection levels close to -49.7dB at 8^0. This comes at the expense of reduced coupling efficiency, where the coupling losses are increased by 0.09 dB at 8^0, as the asymmetry in the end-fire interface leads to the generation of high-order modes in the SRN waveguide structure. This part of light is eventually guided to the cladding and fades away after a few tens of um. In the same graph it is shown that the rib cross section is the most

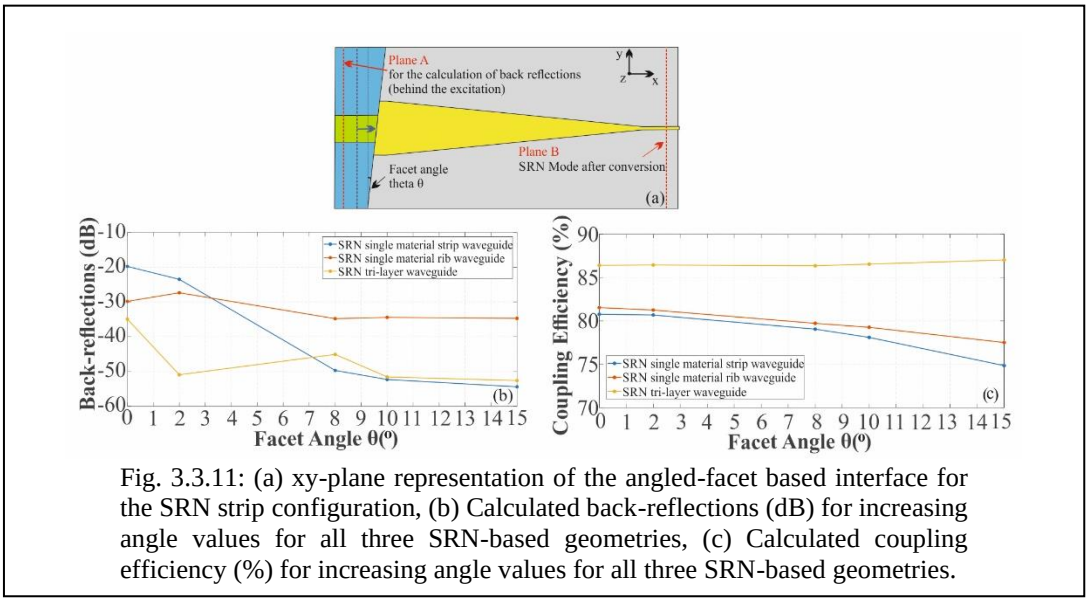


Fig. 3.3.11: (a) xy-plane representation of the angled-facet based interface for the SRN strip configuration, (b) Calculated back-reflections (dB) for increasing angle values for all three SRN-based geometries, (c) Calculated coupling efficiency (%) for increasing angle values for all three SRN-based geometries.

insensitive to the facet angle, based on the studied angles. The back reflections are reduced to -34.81 dB at 8^0 and the coupling losses drop by 0.097 dB at the same angle. On the other hand, for the tri-layer structure the introduction of the angled facet improves the overall performance of light transition, where at an angle of 15º, record low coupling losses of 0.6 dB (equivalent coupling efficiency of 87.02%) and back reflections of -52.6dB are achieved. The small increase of the device performance at 15º is attributed to a more gradual effective index matching between the supported modes in the SRN waveguide and the III-V source after the perturbation of the angle. This leads to slightly higher coupling efficiency at the end-fire stage and lower back-reflections. However even with the small angle of 2^0 the performance is very good with only 0.632 dB losses and -50dB back reflections. Again, the enhanced performance of the tri-layer waveguide is coming with the drawback of the increased fabrication complexity.

3.3.5: Tolerance analysis of monolithic interfaces

In the three coupling interfacing schemes presented in the previous sections, the alignment of the SRN waveguide was considered ideal in terms of matching the center points of the QD laser active region and the SRN waveguide. In this section the resilience of the proposed coupling scheme is demonstrated through a fabrication tolerance analysis performed with 3D-FDTD simulations. In the setup of Fig. 3.3.11(a) for the optimum angle of the three SRN cross sections i.e. 15º for the strip, 8º for the rib and 15º for the tri-layer rib, the SRN waveguide central point was misplaced by up to 300nm in the y-axis (horizontal) and 200nm in the z-axis (vertical). Figure 3.3.12(a) shows the corresponding results in the y-axis, where it is revealed that the performance degradation from optimum value is approximately 0.15 dB (or 3.4%) for all three SRN geometries. This is something that was expected due to the large horizontal dimensions of both the emitted mode from the laser and the initial width of the SRN waveguides, before tapering. Figure 3.3.12(b) presents the results from the misalignment in the z-axis and demonstrates the superiority of the tri-layer SRN rib cross section relative to the single layer strip and rib approaches by supporting larger mode in the z-plane. The coupling efficiency drops for the tri-layer SRN waveguide by only 1dB (20%) for 200nm misalignment, while for the other two, the corresponding values are 1.74dB and

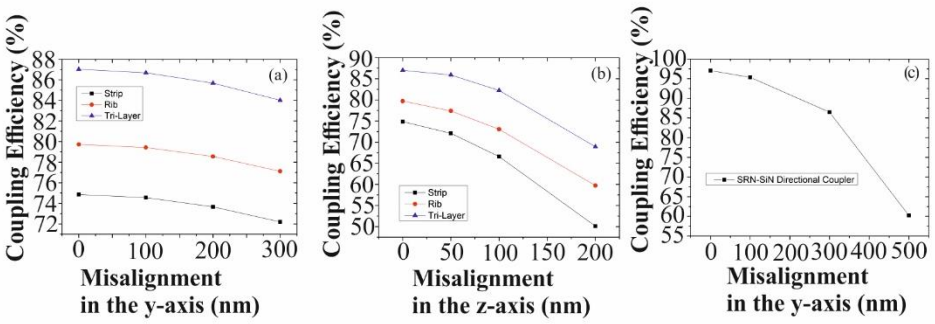


Fig. 3.3.12: Coupling efficiency (%) from Plane A to Plane B of the optimum Strip (black), Rib (red) and Tri-Layer (blue) interfacing configurations for varying misalignment in the (a) y-axis, (b) z-axis, (c) Coupling efficiency (%) of the SRN-to-Si_3N_4 directional coupler for varying misalignment in the y-axis.

1.25dB, respectively and provide total loss of 1.62 dB (68.9%), 3 dB (50.1%) and 2.24 dB (59.7%). At the same time, the back-reflections for 300nm y-axis misalignment are lower than -54dB, -34dB and -52dB for the strip, rib, and tri-layer cross sections, respectively. The corresponding values for 200nm z-axis misalignment are -55dB, -34.9dB and -56.9dB. Finally, for the combination of 300nm in the y-axis and 200nm in the z-axis i.e. the worst-case alignment scenario, the total coupling loss is only 1.78 dB and the back-reflections are lower than -54.4dB for the tri-layer SRN, revealing the high fabrication tolerances of this proposed coupling scheme and fully justifies the required extra fabrication complexity.

Figure 3.3.12(c) illustrates the performance degradation analysis in the SRN-to-Si_3N_4 evanescent coupler and confirms the high tolerances expected from the directional coupling scheme. For 300nm y-axis misalignment the coupled power drops only by 0.5dB with the total coupled power calculated at 0.62dB (86.4%). In terms of z-axis misalignment, the thickness error from the CVD process is typical ±1.5% across a 4" wafer from the targeted value and thus is considered negligible for all three critical parameters i.e. Si_3N_4 thickness, SiO_2 gap, and SRN thickness that determine the coupling efficiency at the SRN-to-Si_3N_4 interface.

Table 1 summarizes the calculated performance of all three proposed coupling schemes with other InP-to-Si_3N_4 experimentally demonstrated coupling interfaces. Our work is the only one compatible with monolithic integration and 248nm DUV lithography for lowest possible PIC fabrication cost in large scale. For the tri-layer SRN the total coupling loss, including the propagation losses in the SRN intermediate waveguide, is 1.03 dB with the potential to go to sub-dB by further improvement in the SRN etching process. Further coupling performance improvement could be obtained from higher resolution lithography for the formation of narrower tip in the SRN.

Table 3.1. Comparison of the presented InP-to-Si_3N_4 Coupling Interfaces with the State of the Art

Reference	Integration Process	Coupling Losses (dB) (InP-to-SRN + SRN-to-Si_3N_4 + SRN Propagation Losses)	Back Reflections (dB)	Minimum Feature Size (nm)
[32]	Hybrid	<1.8	_	_
[33]	Heterogeneous	_	_	150nm
[34]	Heterogeneous	<2	_	120nm
[46]	Heterogeneous	0.66+0.32	-36dB	120nm
[47]	Heterogeneous	0.2	_	100nm
This work - Strip	Monolithic	1.26+0.13+0.3 (Sim.)	-54.4 (Sim.)	210nm
This work - Rib	Monolithic	0.98+0.13+0.3 (Sim.)	-34.81 (Sim.)	200nm
This work – Tri-Layer	Monolithic	0.6+0.13+0.3 (Sim.)	-52.6 (Sim.)	200nm

3.4: Fabrication and Characterization of SRN-to-Si_3N_4 Directional Coupling Interface

Herein, the fabrication and experimental evaluation of the designed SRN-to-Si_3N_4 directional coupling interface is described. A detailed description of the SRN material fabrication and characterization is presented in [36]. The n-k measurements extracted from the fabricated SRN material indicated a high refractive index of n = 3.165 at the wavelength of λ =1550nm. This high value comes from the requirement to match the InP laser effective refractive index. After growth of the 500nm thick SRN films with a low stress recipe, waveguides of 500 x 500nm^2 cross section and length between 1 and 10 mm were formed via e-beam lithography, with the processing of the cutback measurements revealing a value of 15dB/cm propagation losses for these SRN waveguides at λ=1550nm. The next step was the deposition of the SRN material on top of the Si_3N_4 platform and the formation of the directional coupler according to the design of the previous section with e-beam lithography. Figure 3.4.1(a) illustrates a 3D schematic of the designed test structures fabricated for the characterization of the SRN-Si_3N_4 interface. The light goes from the 800 x 800nm^2 Si_3N_4 up to the 210 x 500nm^2 SRN, then follows the tapering to the 4μm x 500nm cross section and after this a straight section of 3mm length. The SRN waveguide is tapered then again to the 210 x 500nm^2 tip and from there the light goes down to the 800 x 800nm^2 Si_3N_4. Both fiber-to-PIC interfaces are grating couplers (GCs). Figure 3.4.1(b) shows a micrograph of the fabricated SRN-to-Si_3N_4 coupler, while Fig. 3.4.1(c) shows a SEM image of the sideview of the fabricated SRN taper tip.

The chip with the test structure was characterized by a tuneable laser source ranging from 1.5μm to 1.63μm. A polarization controller was employed to adjust the polarization of the light injected into the PIC. The corresponding characterization results for both TE and TM polarizations, after subtraction of the losses from a Si_3N_4 straight waveguide with two GCs, used as a reference structure, are presented in Fig. 3.4.1.(d). At 1620nm the ripple is absent for the TE polarization with a total loss value of 7.18dB. Assuming 4.5dB loss for the 3mm straight section also at λ=1620nm, the two SRN-Si_3N_4 interfaces produce in total 2.68dB losses, or 1.34dB/interface. Although this value is quite larger than the one expected from simulations, it is the first one reported for such an engineered material. However, work is ongoing towards the

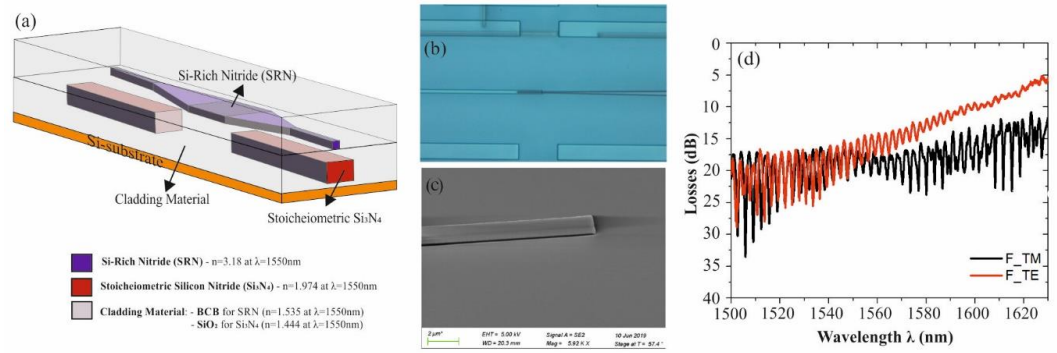


Fig. 3.4.1: (a) 3D-Configuration utilized for the experimental study of the fabricated SRN-to-Si_3N_4 coupling interface, (b)Micrograph of the fabricated SRN-to-Si_3N_4 Coupler, (c) SEM sideview image of the fabricated SRN taper tip, (d) Experimental results for the total transition losses of Fig. 3.4.1(a) after normalization with the input/output structure.

improvement of the SRN material quality and processing for the reduction of propagation and insertion losses in SRN waveguide structures. Additionally, the polarization selectivity is quite low with a maximum value of 6.318dB at 1620nm, while in simulations the obtained value was larger than 20dB. The coupling losses were larger in other chips with the same directional coupling structure. The increased losses with respect to the designed device are attributed to the sensitivity of the coupling efficiency to the coupling length, which is small.

3.5: Design and Evaluation of Alternative Adiabatic SRN-to-Si_3N_4 Coupling Interfaces

To mitigate the increased loss values caused by the sensitivity of the directional coupler to fabrication, a new SRN-to-Si_3N_4 coupling interface based on an adiabatic coupler configuration that targets even better efficiencies and higher tolerances to fabrication variations was realized. A 3D schematic showing the geometry of this structure can be seen in Fig. 3.5.1(a) while a top-view depiction of the coupler including the most crucial design parameters can be seen in Fig. 3.5.1(b). In this way a transition power loss of 0.016dB has been calculated via 3D-FDTD simulations. The optimum geometry was crystallized through consecutive 3D-FDTD calculations with different parameter values. A top-view depiction of the simulated electric field magnitude |**E**| can be seen in Fig. 3.5.1(c), where no significant radiation outside the waveguide core is observed. Finally, the adiabatic coupler consists of a linear adiabatic SRN taper with an initial width of 500nm and a taper tip width of 200nm and a linear adiabatic Si_3N_4 taper with a taper tip of 200nm and an output width of 1μm. For a taper length of only 70μm the coupling losses are lower than 0.147dB, and the equivalent field distribution |**E**| is depicted in Fig. 3.5.1(d). Moreover, the simulated back-reflections of the whole interface are lower than -30dB. In this way, this design achieves the goals of high

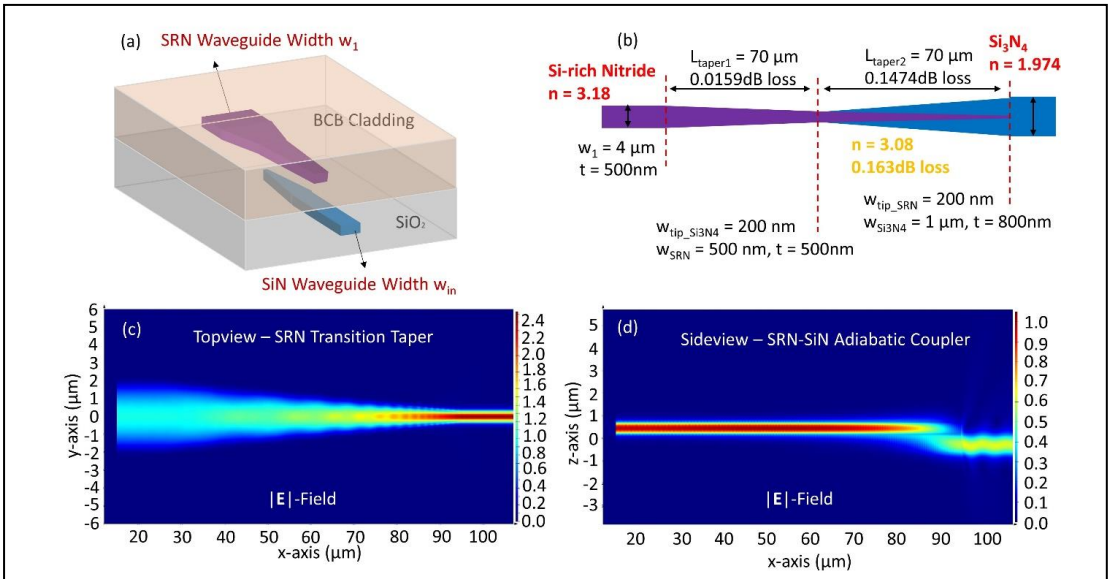


Fig. 3.5.1: (a) 3D-Schematic of the designed SRN-Si_3N_4 Transition Taper, (b) Top-view Image of the designed SRN-Si_3N_4 Adiabatic Coupler with the final design parameters, (c) Top-view |**E**| Field Distribution of the designed SRN transition taper, and (d) Sideview |**E**| Field Distribution for the design SRN-Si_3N_4 Adiabatic Coupler

coupling efficiency and low back-reflections and provides a more tolerant coupling approach at the same time. More specifically, the degradation of the adiabatic coupler's coupling efficiency for even 300nm horizontal misalignment of the SRN taper center with respect to the Si_3N_4 taper center is lower than 0.1dB, while the degradation of coupling efficiency with 30nm deviation from the targeted inter-layer dioxide (ILD) thickness is again below 0.1dB.

The SRN-Si_3N_4 adiabatic coupler design presented above was expected to drastically reduce the measured coupling losses per interface. A 3D-depiction of this structure can be seen in Fig. 3.5.2(a) while a topview of the same structure is depicted in Fig. 3.5.2(b). The parameter w_{in} is the input SRN waveguide width. For the InP-to-Si_3N_4 interfacing target this value is equal to 4μm. The transition taper length here is L_{trans} = 70μm and the adiabatic coupling length is L_{adiab} = 70μm as calculated in numerical 3D-FDTD simulations. The rest of the parameters are also reported: w_{mid} = 500nm, $w_{tip}(Si_3N_4)$ = w_{tip}(SRN) = 200nm, w_{SiN} = 1μm. The normalized measured transfer functions are presented in Fig. 3.5.2(c) with the equivalent cutback measurements reported in Fig. 3.5.2(d). These values correspond to the optimum structure from theory. The SRN propagation losses have been reduced to 5.9dB/cm and the coupling loss per interface for the TE polarization is close to 1dB. The experimentally optimum structure has a transition taper length of L_{trans} = 70μm, and adiabatic coupling length of L_{adiab} = L_c = 60μm. The transfer functions and cutback measurements are shown in Fig. 3.5.2(e) and Fig. 3.5.2(f). The measured propagation losses for the TE polarization are 6.6dB/cm and the coupling losses are equal to 0.8dB per interface. Small ripples can be seen in the transfer functions of Fig. 3.5.2(e) that have been attributed to the existence of degenerate modes that induce another polarization state in the PIC.

To systematically clarify the behaviour of the interface, the same adiabatic coupler designs for input SRN waveguide width of w = 500nm were also included in the mask. A 3D schematic can be seen in Fig. 3.5.3(a). The measured transfer functions can be seen in Fig. 3.5.3(b) and the cutback measurement results can be seen in Fig. 3.5.3(c).

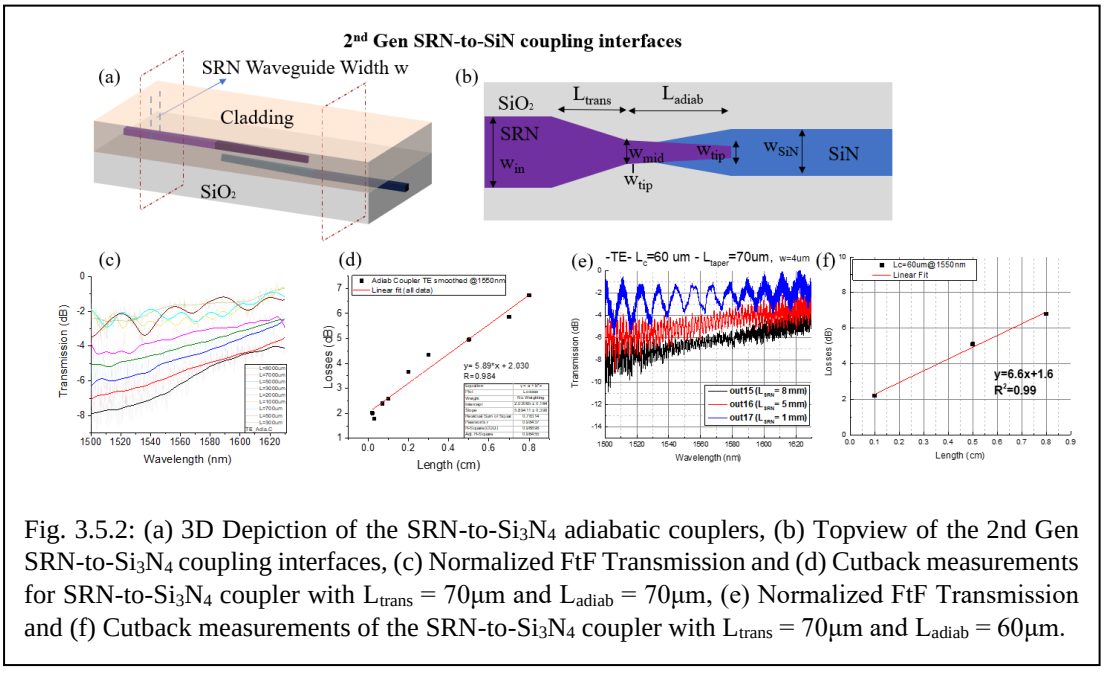


Fig. 3.5.2: (a) 3D Depiction of the SRN-to-Si_3N_4 adiabatic couplers, (b) Topview of the 2nd Gen SRN-to-Si_3N_4 coupling interfaces, (c) Normalized FtF Transmission and (d) Cutback measurements for SRN-to-Si_3N_4 coupler with L_{trans} = 70μm and L_{adiab} = 70μm, (e) Normalized FtF Transmission and (f) Cutback measurements of the SRN-to-Si_3N_4 coupler with L_{trans} = 70μm and L_{adiab} = 60μm.

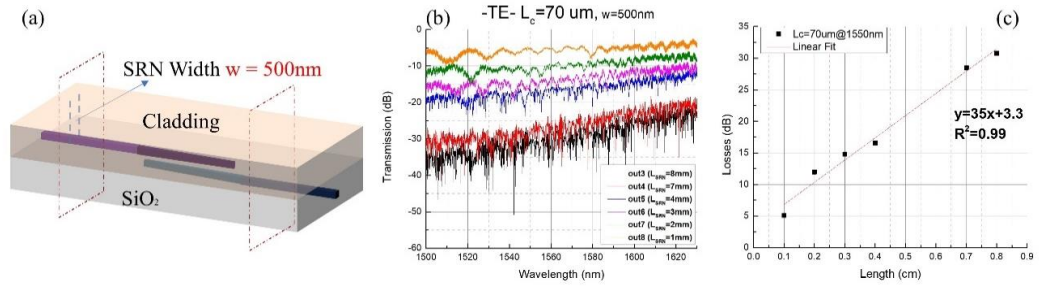


Fig. 3.5.3: (a) 3D Depiction of the SRN-to-Si_3N_4 adiabatic coupler for SRN waveguide width w = 500nm, (b) Normalized FtF Transmission and (c) Cutback measurements for SRN-to-Si_3N_4 directional coupler with w = 500nm, L_{trans} = 70μm and L_{adiab} = 70μm

The measured propagation losses for the TE polarization were 35dB/cm and the estimated coupling losses were close to 1.65dB/interface at 1550nm free-space wavelength. For the TM polarization, the propagation losses were 29.8dB/cm and the coupling losses were close to 1.35dB/interface. In contrast with the 4μm case, only small ripples can be seen in this design. The ripples in the higher width were explained via the utilization 3D-FDTD simulations for sources of both TE and TM polarizations. It was shown that, in the higher waveguide width of 4μm a degenerate mode was excited, for TE+TM mode source, leading to ripples. This behavior can be overcome by modifying the length or the shape of the adiabatic taper further. However, this could possibly cause the losses due to extra propagation length in the SRN taper to increase. Better control of the experimental input polarization could also solve this excitation issue.

3.6: Design of Direct InP-QD-to-Si_3N_4 Coupling Interfaces

To simplify the QD SOA and laser integration process, direct coupling configurations from the QD-lasers to the Si_3N_4 were also designed. In the interlayer light coupling case, the fabrication is more complex but the high yield in the performance of the interface is ensured since this design resolves the high buffer layer thickness issue in QD-devices fabricated with Selective Area Etching (SAE). On the other side, the direct coupling configuration can be fabricated in a much simpler manner, but special treatment needs to be taken to completely negate back reflections and ensure high performance. Additionally, during the application of the envisaged integration for the fabrication of the InP-to-SRN interface by III-V Lab several issues related to the formation of the BCB at the surface and the proper formation of the trenches for the SRN emerged. For this reason, a direct coupling approach was also studied, even though the multi-layer coupling approach exhibited high efficiency values measured for interlayer coupling. A topview of the designed device can be seen in Fig. 3.6.1. For the negation of back-reflections there is a gap between the InP-based emitter and the Si_3N_4 waveguide filled by BCB while the facet angle θ is controlled accordingly to achieve low coupling losses and low back reflections. For the interface design, a 2D-FDE calculation of the overlap integral of the modes supported in the two waveguides for different widths w was followed by 3D-FDTD calculations for various gaps g and angles θ for the extraction of the final results.

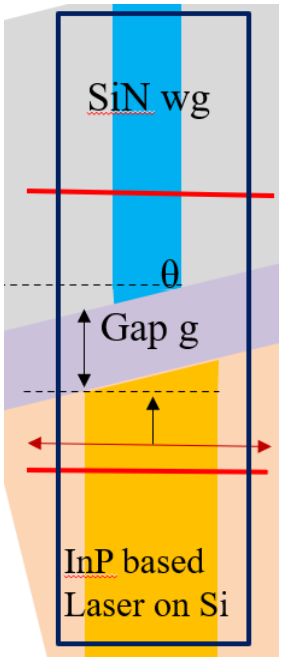


Fig. 3.6.1 Topview of the design of the Direct InP-QD-to-Si_3N_4 coupling interface

The final 3D-FDTD results for the facet angle sweep at the coupling interface configuration can be seen in Fig. 3.6.2. For these calculations, Si_3N_4 waveguides with widths of 4μm and 200nm have been chosen while the gap value chosen was close to 1μm. It is clear that for this gap value the Back-Reflections are lower than -20dB for angles θ>5º. For this gap value the lower width waveguides with w=200nm exhibit poorer values of coupling efficiency. For w = 4μm the coupling efficiency is higher than 3dB at θ = 5º.

For the final design selection, the gap value g was also swept in 3D-FDTD simulations. The results can be seen in Fig. 3.6.3. The facet angle here is set at angle θ = 6º. It can be deduced that for higher gap values the 200nm Si_3N_4 waveguide widths are more tolerant exhibiting higher coupling efficiencies, while for lower gap values, the wider waveguides are of higher interest. At all cases, the back-reflections are lower

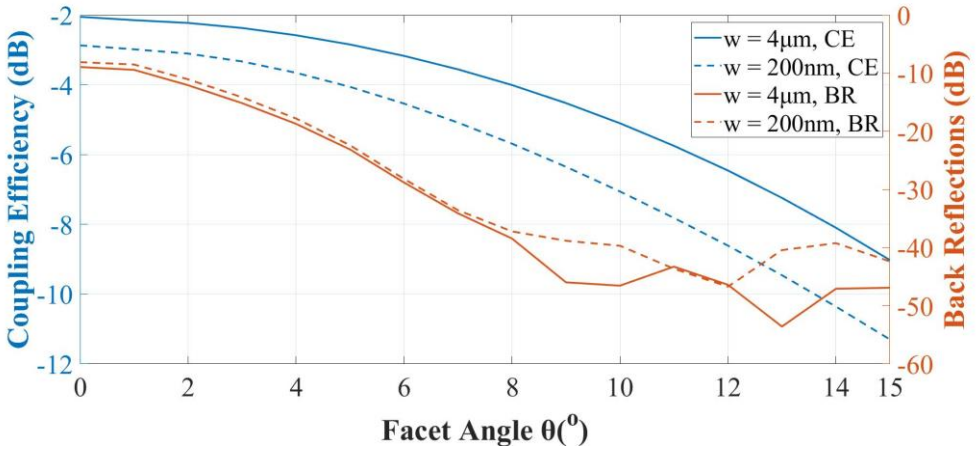


Fig. 3.6.2: 3D-FDTD Results of the Coupling Interface for widths w_{SiN} = 4μm and 200nm and gap g = 1μm. The facet angle θ is swept.

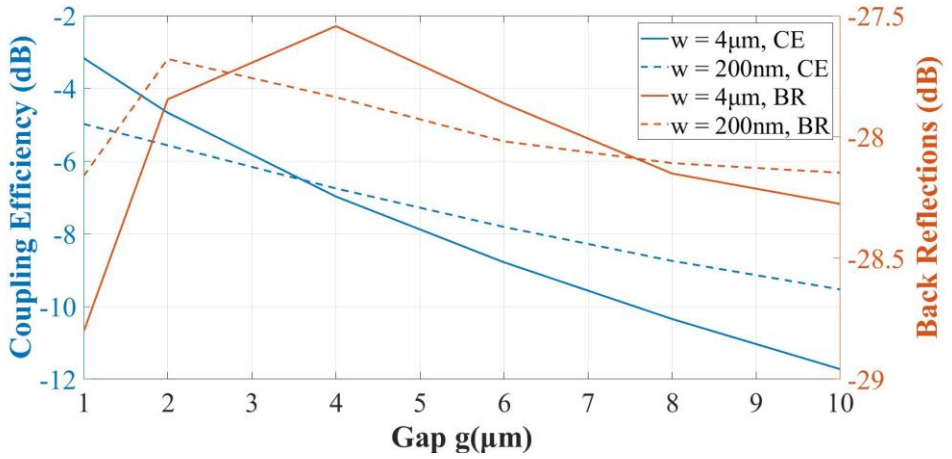


Fig. 3.6.3 3D-FDTD Results of the Coupling Interface for widths w_{SiN} = 4μm and 200nm and facet angle θ = 6°. The gap g is swept.

than -27dB, being excellent for reflection sensitive integrated devices such as low linewidth, spectrally pure, tunable lasers.

The calculated electric field for a coupling interface with Si_3N_4 waveguide width w = 4μm, facet angle θ = 6° and gap g = 1μm is shown below in Fig. 3.6.4. The reflected light in the figure is deflected away from the amplifier waveguide axis, while most of the transmitted light couples to the Si_3N_4 waveguide. The rest is scattered light propagating away to the substrate and top and bottom oxide. The coupling efficiency is higher than -3dB and the back-reflections are lower than -28dB.

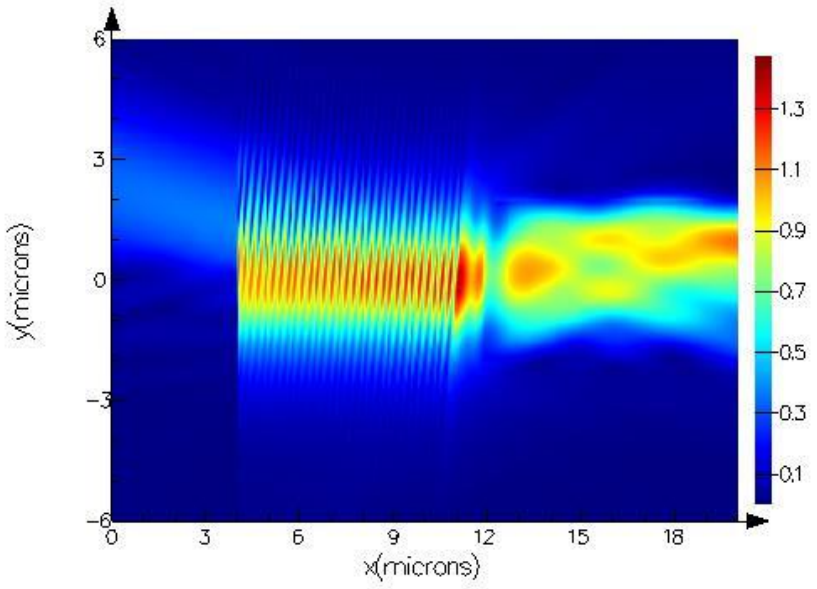


Fig. 3.6.4: Topview of the electric field |**E**| of the coupling interface design with Si_3N_4 waveguide width w = 4μm, facet angle θ = 6° and gap g = 1μm.

3.7: Vernier-Ring Reflectors in Thick Si_3N_4 Platform

3.7.1: Introduction

Tunable lasers over the years have demonstrated their importance for a multitude of applications requiring low linewidths, such as coherent communications [48], microwave photonics [49], and LIDAR [50]. Traditionally, bulky discrete component-based lasers have been utilized for these applications, having increased footprint and low environmental tolerances [51]. However, during the last few years, integrated optical circuits have emerged as a compact, low-cost solution for spectrally pure tunable laser sources [52]. Among the various integration platforms for such lasers, silicon nitride (Si_3N_4) has gained ground [53], offering ultra-low linear and non-linear losses, CMOS compatibility and low-cost production. With such a hosting material, a large variety of exceptional tunable lasers has been developed via the hybrid [54] and heterogeneous integration [55, 56] of Si_3N_4 dielectric waveguides with Semiconductor Optical Amplifiers (SOAs) achieving linewidths as low as 40Hz.

Integrated tunable lasers require specially designed integrated mirrors to form the laser cavity, with the SOA providing the gain medium inside the cavity, and the reflector offering filtered feedback. Vernier-rings, with their high FSR and Q-factors, are excellent candidates for the realization of such mirrors. In many of the devices found in the literature, silicon nitride waveguides have a thickness lower than 300nm with their mode being deconfined from the waveguide core, that imposes major restrictions on their achievable Free Spectral Range (FSR) and the Side-Mode Suppression ratio (SMSR). The thick Si_3N_4 platform, on the other hand, features a strongly confined mode inside the waveguide core and much lower bending radii allowing negligible losses and polarization rotation. Previous demonstrations of Si_3N_4 reflectors consist of devices with triple- and quad-ring reflectors [57] exhibiting a tuning range of 45nm and a side-mode suppression ratio up to 20dB. A dual ring structure was also presented in a hybrid integrated tunable laser with a tuning range of 39nm [58]. A reflector comprising a triple Vernier-ring and a Sagnac mirror was demonstrated [54] providing a 70nm coverage with a SMSR of 12.5dB. Additionally, the use of Coupled Resonator Optical Waveguide (CROW) devices in silicon (Si) photonic Vernier circuitries has recently pushed the boundaries of the achievable FSR values to 124nm and SMSR values up to 64.5dB [59]. Nevertheless, in most of the previous demonstrations based on Si_3N_4 one could not simultaneously achieve adequately high FSR and SMSR values with a simple reflector design.

In this Section we present a Si_3N_4-based dual Vernier-ring resonator reflector providing a FSR value larger than 38nm and a SMSR value as high as 15dB. The SMSR value presented herein is higher than the one by other dual Vernier-Rings of the literature [52]. For the design purpose, we have developed an in-house Transfer-Matrix model in MATLAB inspired by a Transfer Matrix Analysis [60]. The derived FSR and SMSR values have been verified by both a commercial grade solver [61], and through experimental measurements. The developed Transfer Matrix Model is then employed in combination with the analysis [62, 63] in order to design more complex triple ring devices with very sharp central peaks and SMSR values near 20dB, higher than the SMSR of more complex designs that require an extra component [54], and CROW-based Vernier rings demonstrating FSR values higher than 62nm, comparable with such

complex circuits [54]. The rest of the Section is organized as follows: In Section 3.7.2, we present the designed device and the fundamental results of our semi-analytic model. In Section 3.7.3, we report the experimental measurements of our designed dual-ring devices, verifying our model's results. In Section 3.7.4 we demonstrate the design of two more complex Si_3N_4 devices via our Transfer-Matrix based model, one targeting high SMSR and the other targeting ultra-high FSR values.

3.7.2: Modelling of Dual-Vernier Rings with 3D-FDTD Numerical and Transfer-Matrix Techniques

The Si_3N_4 dual-ring reflector investigated herein is presented in Fig. 3.7.1. It consists of a waveguide that acts simultaneously as the input and output waveguide, an 1x2 50:50 MMI coupler and two add-drop ring-based structures possessing radii R_1 and R_2 and ring-bus gaps g_1 and g_2 respectively. When multi-λ light is inserted through the input port it gets split by the MMI and filtered via the two add-drop ring resonators. The drop port of the first ring is connected to the input port of the second ring. This way, the transfer function of the whole reflector is only the result of the dual-ring circuitry. The transfer function of this circuitry comes as a result of the multiplication of the drop transfer functions of each ring. The transmitted power spectral density is simply $T_{dual} = T_1T_2$ where T_1 and T_2 is the power spectral density at the drop port of each add-drop resonator. The filtered output light is obtained by the same input/output waveguide.

A variety of modelling techniques have been reported over the years for the design of ring resonator devices, including numerical Finite-Element (FEM), Finite-Difference Time-Domain (FDTD) Techniques [64] and semi-analytic Coupled Mode Theory (CMT) [65, 66] and Transfer Matrix based Techniques [60] (TMM). In thick Si_3N_4 waveguides the bending radii under study are usually larger than 50μm with gaps being as low as 300nm. For this reason, the utilization of solely numerical techniques for the whole ring-based structure in this case is time consuming and sometimes prohibited by the available computer resources. Additionally, experimentally measured propagation losses of Si_3N_4 waveguides should somehow be taken into consideration in our calculations. For this reason, we opted for a semi-analytic TMM-based technique [60]. The power spectral density of the whole circuitry is derived after obtaining the transfer function of each add-drop configuration, explicitly reported in the literature [60].

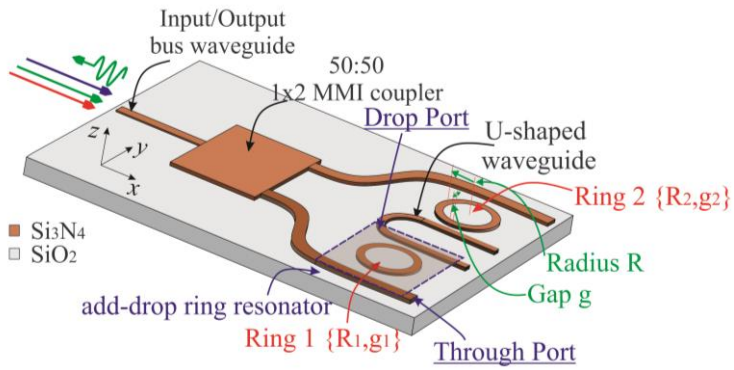


Fig. 3.7.1: Image of the envisaged Dual Vernier-Ring Reflector Circuitry along with Circuit Design Parameters

3.7.3: Numerical Results for Dual Vernier Rings

The power spectral density of the whole circuitry is based on the above relation, where parameters such as the effective index n_{eff} of the TE Si_3N_4 waveguide mode, the total loss factor, *a*, of the resonator, the coupling coefficients, *r* (self-coupling coefficient) and *t* (cross-coupling coefficient) should be inserted. In order to retain an accurate semi-analytic modelling framework, accurate values of these parameters in the whole C-band should be inserted. An initial estimation of the FSR for the whole dual-ring reflector can be given by the relation $FSR = (FSR_1FSR_2)/(FSR_1-FSR_2)$, where FSR_1 and FSR_2 are the FSR values for each add-drop ring. Based on this initial estimation, radii of $R_1 = 50\mu m$ and $R_2 = 55\mu m$ were chosen for each ring. The propagation losses of the Si_3N_4 waveguide for the TE polarization at 1550nm wavelength were obtained via cutback measurements and estimated close to 0.2dB/cm. The rest of the parameters were obtained via a detailed 2D-eigenmode and systematic 3D-FDTD numerical electromagnetic analysis.

From the 2D-analysis, fundamental parameters such as the effective and group index were also obtained. The Si_3N_4 waveguide has a width and a thickness value of 800nm. Its supported TE mode has an effective index of $Re\{n_{eff}\} = 1.71$ for a wavelength near λ=1550nm. Extra bending losses close to 0.2dB/cm were also taken into account for the radii under study. Therefore, the total propagation losses in the rings should be near 0.4dB/cm. The effective and group index in the whole 1500nm-1600nm wavelength window were derived. These values were inserted in the TMM model in order to capture any dispersive effects. The group index with initial ellipsometric measurements of the Si_3N_4 material at λ=1550nm was calculated near $n_g = 2.08$ while recent measurements have slightly shifted the value near $n_g = 2.12$.

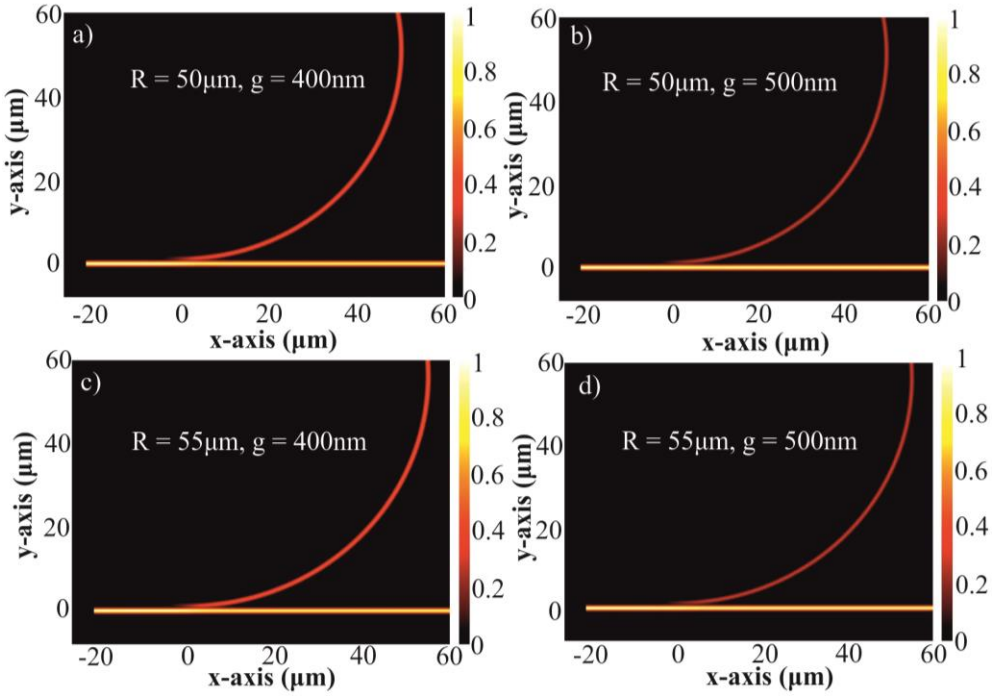


Fig. 3.7.2. |**E**| field from 3D-FDTD Simulations for various ring-bus coupler designs: a) R = 50μm, g = 400nm, b) R = 50μm, g = 500nm, c) R = 55μm, g = 400nm, d) R = 55μm, g = 500nm.

Each of the add-drop structure in the dual Vernier-ring reflector circuitry should operate near the critical coupling regime to obtain efficiently high Q-factors needed for low linewidth and high SMSR values. For this purpose, an accurate value of both propagation and bending losses should be inserted in the TMM model. For this reason, systematic 3D-FDTD numerical electromagnetic calculations were performed for the ring-bus waveguide gaps ranging from 200nm to 700nm in order to accurately evaluate the bus-ring {*r*,*t*} coupling coefficients in the coupler for the whole 1500nm-1600nm wavelength range. The numerical results are shown in Fig. 3.7.2. Ring-bus gaps of 400nm and 500nm were proven to be suitable for the add-drop ring resonator to operate near critical coupling. Adopting a radius of R = 50μm the results are shown in Fig. 3.7.2(a) and Fig. 3.7.2(b). Near 1554nm for g = 400nm the coupling efficiency from the bus to the ring was $|t|^2$ = 18.44% while for g = 500nm the value was $|t|^2$ = 8.59%. The results for R = 55μm are shown in Fig. 3.7.2(c) and Fig. 3.7.2(d). In this case near 1554nm, the efficiency was close to $|t|^2$ = 20.09% for g = 400nm and $|t|^2$ = 9.41% for g=500nm. The ring-bus insertion losses from the coupler were calculated for both radii and g=400nm to be near 0.1dB and near 0.2dB for a 500nm gap. The images in Fig. 3.7.2 showcase the electric field distribution |**E**| from the topview in all four cases, with the plane residing at the middle of the 800nm thick waveguides.

Inserting all the derived parameters in the simple relation of Section 3.7.2 one can obtain an accurate estimation of the expected output transfer function (power spectral density) of the dual Vernier-ring based circuitry for each gap case, i.e. for ring-bus gap g=400nm and ring-bus gap g=500nm. The two transfer functions are shown in Fig. 3.7.3(a) and Fig. 3.7.3(b) respectively. It is clear in both cases that via the multiplication

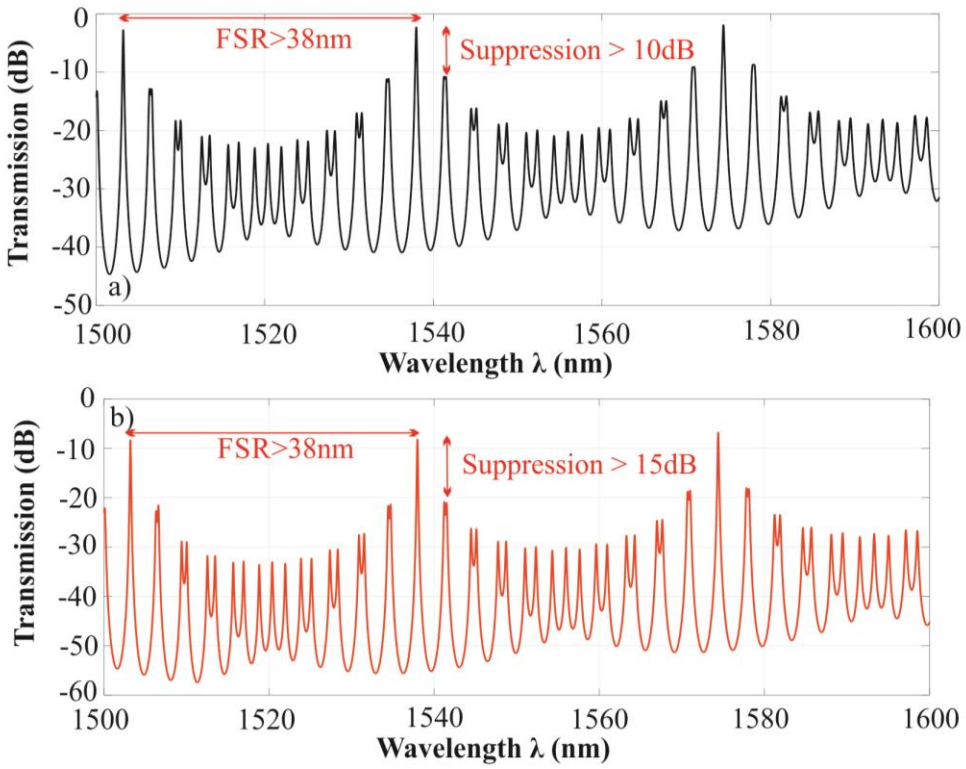


Fig. 3.7.3: a) Calculated Transfer Function for the Dual-Ring Reflector with radii R_1 = 50μm and R_2 = 55μm for ring-bus gap = 400nm, b) Calculated Transfer Function for the Dual-Ring Reflector with radii R_1 = 50μmm and R_2 = 55μm for ring-bus gap = 500nm.

of the transfer functions of the two add-drop ring resonators with different radii the FSR value extends to values higher than 38nm. The main peaks of the transfer function are decreased in terms of power due to the insertion losses of each ring-bus coupler. For this reason, for g=400nm the maxima are near a power level P=-2.4dB and a value near P=-8.2dB for g = 500nm. The side-mode suppression (SMSR) is higher than 10dB in the 400nm gap case and higher than 15dB in the 500nm gap case.

Therefore, so far, we have presented the design of dual Vernier ring inspired reflectors via a semi-analytic TMM model. Parameters regarding the propagation in the Si_3N_4 waveguides were obtained via measurements and 2D-eigenmode calculations, while parameters concerning the coupling coefficients in the coupler were obtained via detailed 3D-FDTD calculations at the ring-bus couplers. In the next sections, we will verify the validity of our calculations through experimental measurements.

3.7.4: Fabrication and experimental demonstration of dual Vernier-ring reflector

In the previous sections, a semi-analytic TMM-based design framework for dual Vernier ring reflectors was presented yielding important estimations. The most important part of the reflector is the dual add-drop ring section imposing the filtering operation. Following the above rationale, in order to verify our calculation results, a mask with variations of the dual ring sections with respect to the gap and radii R_1 = 50μm, and R_2 = 55μm was designed. With the designs in place, an 800nm thick Si_3N_4 layer was deposited on a 4μm thick thermal oxide. 248nm DUV lithography was applied to pattern the overlaying photosensitive layer. The devices were etched, and an upper oxide layer was deposited and planarized. Thermal metallic phase shifters were placed on top of the Si_3N_4 rings to achieve tunability. The phase shifters had a width of 1.5μm and covered half the radii of the rings. The chips were diced and received for measurement.

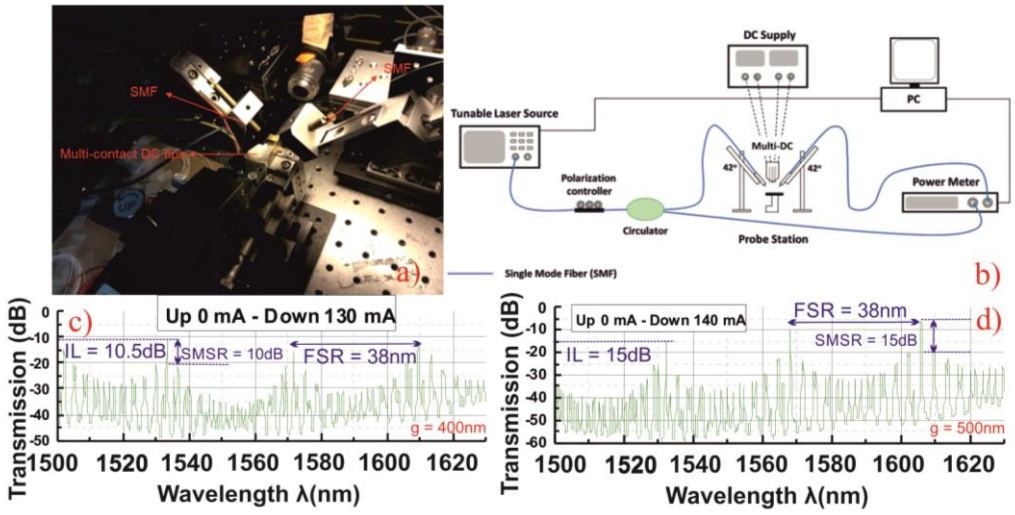


Fig. 3.7.4. a) Image of the experimental setup used during measurement, b) Schematic of the Experimental Setup outlining all the employed equipment, c) Experimental Transfer Function for ring radii R_1 = 50μm, R_2 = 55μm, g = 400nm and, d) Experimental Transfer Function for ring radii R_1 = 50μm, R_2 = 55μm, g = 500nm

For accurate measurements the experimental setup shown in Fig. 3.7.4(a) and Fig. 3.7.4(b) was employed. The light source was a continuous-wave tunable laser emitting light with a wavelength range from 1500nm to 1630nm. For polarization control of the injected light, a polarization controller was placed before the chip, while a circulator was also used to measure reflected light. A computer-controlled DC probe was used to electrically drive the thermal phase shifters on top of the Si_3N_4 rings. A Si_3N_4 grating coupler with losses close to 6.5dB at 1550nm for the TE polarization was designed and included as the input/output device to/from the chip. Reference structures with input/output Grating Couplers and a waveguide section were inserted in the mask for proper normalization of the experimental data.

The normalized transfer functions of the dual Vernier ring circuits for ring-bus gap g=400nm and g= 500nm are shown in Fig. 3.7.4(c) and Fig. 3.7.4(d) respectively. Our measurements are limited by the frequency step during measurement. In each case the transfer function at an applied current value showcasing the best main peak resolution is presented. The results for gap g=400nm can be seen in Fig. 3.7.4(c). It is indicated that the maxima of the peaks are not near the same power value. This is justified by the inclusion of a propagation loss measurement at only a wavelength of 1550nm in our model. The minimum experimental insertion losses near λ=1532nm for g=400nm were close to 10.5dB (the calculated value was 2.4dB). The SMSR value was larger than 10dB in most peaks with the FSR being larger than 38nm, agreeing with the calculated values. The transfer function of the circuitry for ring-bus gap g=500nm is shown in Fig. 3.7.4(d). In this case the minimum measured insertion losses were close to 15dB near λ=1532nm (the calculated value was 8.2dB), with the SMSR being in some cases larger than 15dB and the FSR being larger than 38nm, agreeing again with the theoretical results. The inaccuracies at the insertion losses are justified by the high sensitivity of insertion losses to minimal losses inserted via roughness induced scattering mechanisms at the ring-bus coupler in the high-Q regime. The ring circuitry with a gap value g=500nm provides higher SMSR, as seen in Fig. 3.7.4(c) and Fig. 3.7.4(d) on the grounds that for this gap the individual add-drop rings are closer to critical coupling than for the 400nm gap. Therefore, the 500nm gap value is considered as the most favorable of the two.

Finally, the electrical current value applied on the thermal heaters was swept at the lower and upper ring, elucidating in this way the tunability of our dual Vernier Rings. Our results for gap g=400nm and g=500nm are summarized in Fig. 3.7.5(a) and Fig. 3.7.5(b), respectively. The red line denotes the peak before thermal tuning in each case. The pink and purple lines indicate the position of the peaks after thermal tuning. It is clear that application of current to the lower ring blue-shifts the central peak while tuning the upper ring of the Vernier circuitry red-shifts the central peak. The tunability was revealed to be larger than 30nm for both gap values. A larger metal width at the heater design could upgrade the tunability of our simple dual ring device to cover the whole FSR of 38nm.

In this section we verified the validity of our TMM-based semi-analytic model with respect to the calculated FSR and SMSR values via experimental measurements. In the next section we will expand our model to design reflector devices with higher SMSR and higher FSR values. We will reveal through calculations that by manipulating the Vernier effect in combination with the photonic bandgaps created by the periodical

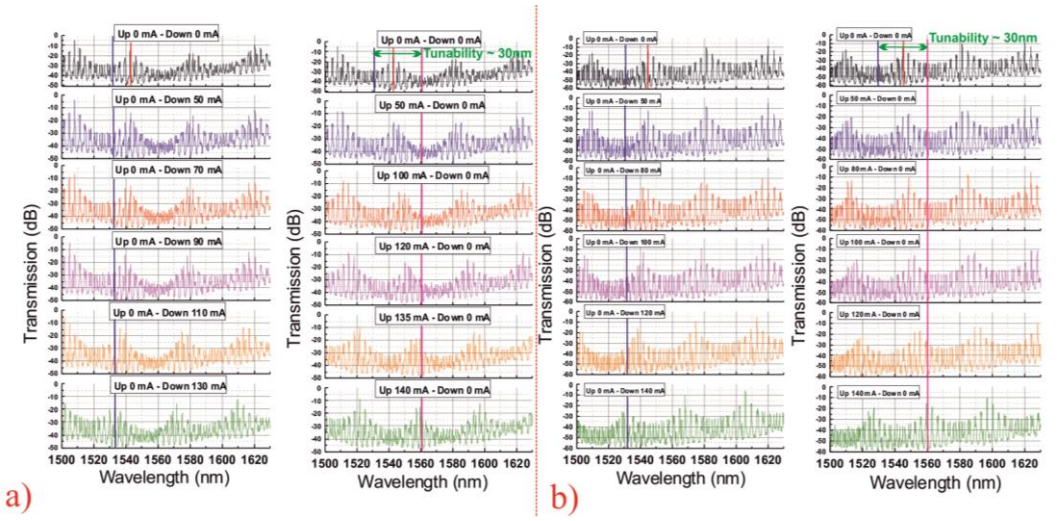


Fig. 3.7.5: Experimental Measurements of Tunability for a) g=400nm, and b) g=500nm

placement of ring resonators in CROWs we can push the boundaries of the FSR to values higher than 62nm in Si_3N_4 devices in the C-band.

3.7.5: Design of triple and high-order Vernier-Rings

In this Section, our design target will be the extension the SMSR and FSR values in Vernier-Ring circuit configurations. Two circuits that can achieve these targets are the Triple Ring (Fig. 3.7.6(a)) and the Coupled Resonator Optical Waveguide (CROW) Vernier Ring Reflector (Fig. 3.7.6(b)). The first can achieve ultra-high SMSR values by adding a third ring at the known dual-ring circuitry. The input port of the third ring

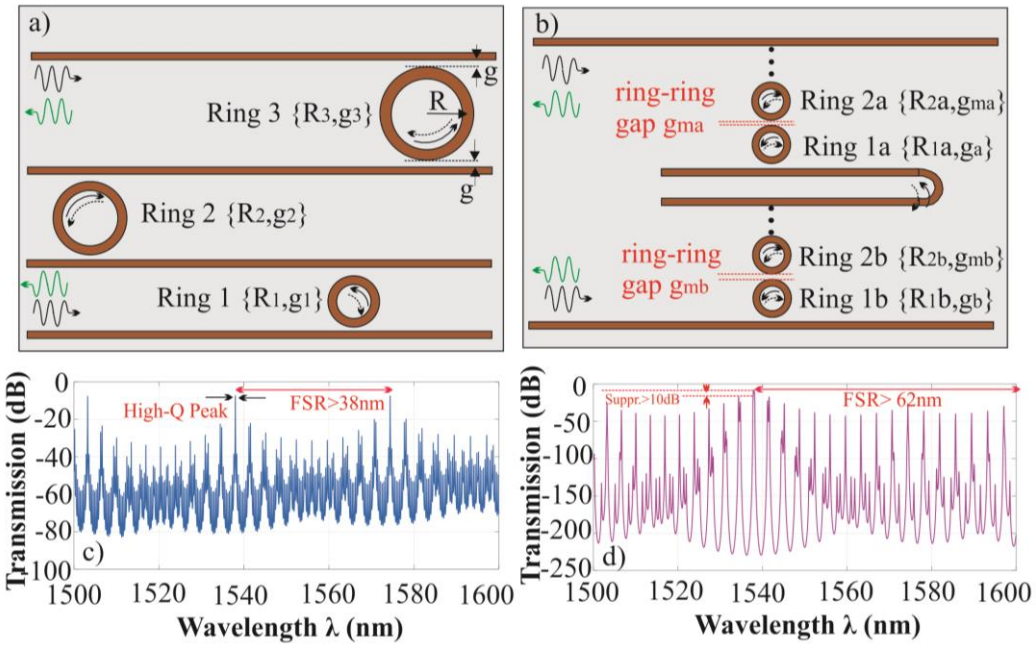


Figure 3.7.6. a) 2D-depiction of a Triple Vernier Ring Reflector, b) 2D-depiction of a CROW-based Vernier Ring Reflector, c) Calculated Transfer Function of the Triple Vernier Ring Reflector through the extended TMM model, d) Calculated Transfer Function of the CROW-based Vernier Ring Reflector through the extended TMM model

is connected with the drop-port of the second ring. Such a configuration is presented in Fig. 3.7.6(a). The first two rings should have radii of R_1 = 50μm, and R_2 = 55μm, based on the analysis presented above. The ring-bus gap should be $g_1 = g_2$ = 500nm to provide a coupling efficiency $|t_1|^2$ = 8.59% for R_1 = 50μm and $|t_2|^2$ = 9.41% for R_2 = 55μm. The third ring should be designed with a much larger radius of R_3 = 500μm, while the gap g_3 should be designed in a manner to achieve a bus-ring coupling efficiency close to $|t_3|^2$=10%. Having taken these parameters into account, the transfer function of the whole circuit can be seen in Fig. 3.7.6(c). The total transfer function is simply the result of the multiplication of the transfer functions of the three rings at the drop ports $T_{triple} = T_1T_2T_3$. The FSR is again larger than 38nm. However, due to the multiplication, the SMSR has been elevated to a value close to 20dB while the central peaks have become sharper.

The CROW-based Vernier Ring is displayed in Fig. 3.7.6(b) and its design target is the extension of the achievable FSR. In this case, we manipulate both the Vernier effect, suppressing some of the resonator peaks due to the multiplication of the transfer functions of the two resonators, as well as the natural phenomenon emerging from the propagation of light through a periodical placement of resonators. More specifically, it can easily be shown that the propagation of light through a series of resonators in a CROW resembles the propagation of an electron through a crystal lattice in solid state physics. As a result, photonic bandgaps can be formed in a CROW suppressing some of the peaks and extending the achievable FSR. The dispersion relation with the possible bandgaps can easily be calculated having Bloch's theorem as a basis [63]. A CROW with m=6 rings and a unit cell with radii R_{1a} = 53μm and R_{2a} = 50μm was used in the upper section, while a CROW with m=6 rings and a unit cell with radii R_{1a} = 55μm and R_{2a} = 50μm was utilized in the lower section. The ring-bus gaps have been designed to provide a coupling efficiency near $|t|^2$ = 18% while the ring-ring couplers are weakly coupled with $|t_{gm}|^2$=5%. The final transfer function of the whole circuitry is shown in Fig. 3.7.6(d). The FSR has been extended to a value higher than 62nm, with

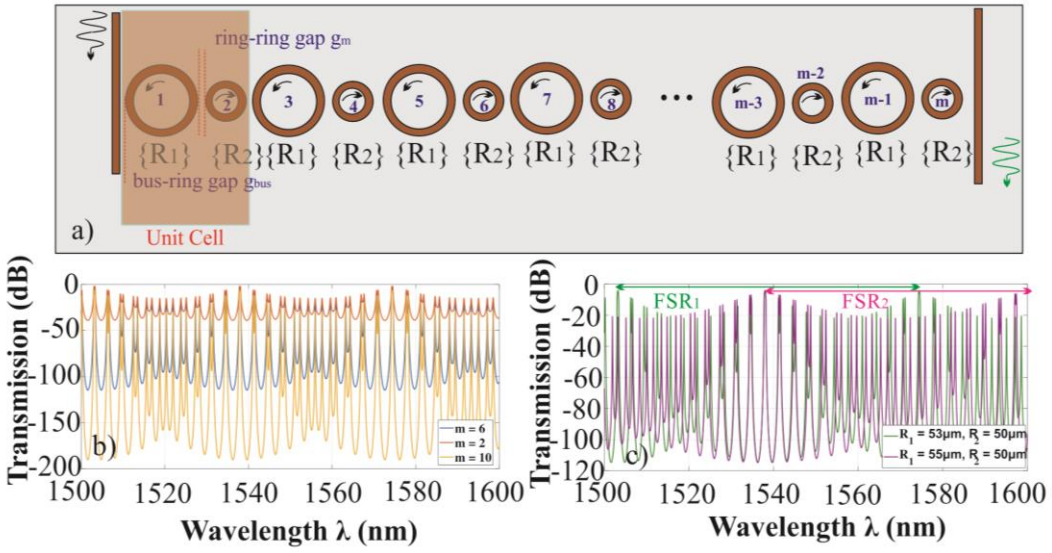


Figure 3.7.7. a) Depiction of a single CROW device. An important parameter is the unit cell of the CROW, b) Calculated Transfer function of a number m = 2, 6, and 10 rings in the crow. The unit cell has two rings with radii 55μm and 50μm, c) Calculated transfer functions for CROWs of two different unit cells.

the SMSR being higher than 10dB. There is only a single main peak in the whole 1500nm-1600nm wavelength band. In reality the 63nm is the distance of the main peak to the 1600nm limit of the calculation. The single peak in the 100nm band signifies a FSR higher than 100nm in the CROW based device.

We demonstrate the design of the individual CROWs below. Figure 3.7.7(a) displays the design of a single CROW. The unit cell consists of two rings with radii R_1 and R_2. The identifier of the ring number is N={1,m} and for our application of interest each CROW should have N=2a rings where a={1,2,..}. Fig. 3.7.7(b) shows that for a large number of rings in a CROW, photonic bandgaps are formed suppressing certain wavelengths. A number m=6 in each CROW is chosen as the optimum for our application, providing higher Q-factors than the m=2 rings, and being a simpler design than the m=10 rings. In Fig. 3.7.7(c) it is shown that by varying the unit cell of the CROW, the FSR is varied. This way, having a combination of two FSR values, one can greatly extend the FSR value via the multiplication of the two transfer functions through the Vernier effect. This was shown in Fig. 3.7.6(d), where a single main peak was observed in the whole 1500nm-1600nm band.

The table below demonstrates the results of this work in comparison with other results in the literature:

Table 3.2. Short State-of-the-art in Vernier Ring Reflector Devices

Publication	Type	FSR	SMSR
[52]	Dual Si_3N_4 Vernier Ring	46.4nm	~5.22dB
[54]	Triple Si_3N_4 Vernier Ring with Sagnac Mirror	70nm	12.5dB
[58]	Dual Si_3N_4 Vernier Ring	39nm	_
[59]	CROW-based Si Vernier Ring	124nm	64.5dB
This work	Dual Si_3N_4 Vernier Ring	38nm	15dB
This work	Triple Si_3N_4 Vernier Ring without Sagnac Mirror	38nm (main peaks)	20dB
This work	CROW-based Si_3N_4 Vernier Ring	>62nm (in reality >100nm)	>10dB

3.8: Chapter Conclusions

In this work we presented a systematic analysis for the design of SRN interposers, suitable for seamless light transfer from QD-InP-based gain sections to Si_3N_4 waveguides in a BEOL process targeting PICs with advanced functionality. Based on

an extended set of simulation results, three types of interposer waveguide configurations have been designed that exchange performance for fabrication complexity. The simplest one relies on 500nm thick SRN strip waveguide with 1.06 dB coupling loss and -19.79 dB back-reflections, while the rib one with the same overall thickness manages to improve the coupling loss to 0.86dB and lowering the back reflections to -30dB. The tri-layer sandwich material provides finally 0.77dB overall coupling loss and back reflections as low as -34.95dB. All these results are for 0^0 angle at the QD InP-SRN interface and by introducing a 15^0 angle, the tri layer stack manages to lower the back reflections and coupling loss to -52.4dB and 0.6dB, respectively. The tri-layer also exhibits the higher fabrication tolerance by imposing only 1.78dB total loss and -54.4dB back reflection for the worst-case scenario of simultaneous 300nm in the y-axis and 200nm in the z-axis misalignment between the central points of the InP QD laser and the SRN waveguide. The proposed concept although assumes an InP active medium can be easily transferred also to monolithically grown GaAs based lasers and to dual facet devices such as SOAs and electroabsorption modulators. In addition, the critical dimensions of all geometries in the three proposed designs are larger than 200nm that render them compatible with 248nm DUV lithography targeting to maximize the low-cost potential of the Si_3N_4 platform. The accuracy of the simulations is enhanced by taking into account for the refractive index of the SRN interposer the values extracted from ellipsometry measurements of an ultra-Si-rich-nitride material (SRN) developed and engineered for this purpose. This new material provides tunability in the real part of the refractive index, while crack free samples were grown up to 500nm in thickness. Test structures with cutback waveguides of $500x500nm^2$ cross section formed via e-beam lithography featured 15dB/cm propagation losses in line with similar SRN materials. The proposed work is expected to pave the way for the next generation of a versatile, low cost, high performance monolithic InP-QD/Si_3N_4 platform on a common Si substrate. Additionally, the design of Si_3N_4 Vernier ring-inspired reflectors based on an 800nm thick Si_3N_4 integrated photonic platform was presented based on a TMM-based semi-analytic model. A dual-ring reflector was designed, and detailed simulation results were presented. The experimental results validated the proposed semi-analytic model indicating FSR values of 38nm and SMSR values of 15dB for the optimum dual-ring reflector. Having validated our theoretical results, we proceeded with the design of more complex triple and CROW-based Vernier rings with simulation results revealing that a triple ring can feature an improved SMSR value close to 20dB, while in the CROW-based devices the FSR is higher than 62nm with a single peak in the 100nm wide (1500-1600)nm band. The incorporation of CROWs in Vernier-Ring circuits was shown to greatly extend the FSR of thick Si_3N_4 reflectors.

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Chapter 4: Interfacing Si_3N_4 waveguides with CMOS plasmonic phase-shifters in the O-band

4.1: Fundamentals of plasmonic waveguides

The optical properties of metals and the important model for the dispersion of metallic materials by Drude have been fundamental for the birth of the field of plasmonics. A surface plasmon polariton is an electromagnetic wave (photon) coupled with the collective oscillations of the sea of electrons at the surface of the metal. Numerous applications have emerged as a consequence of the great development of the field of plasmonics such as sensors, plasmonic transmitters, efficient energy harvesting devices, and artificially designed left-handed materials metamaterials and metalenses. Plasmonic waveguides have been the foundation of many of these applications serving as the fundamental building blocks. As described in detail in Chapter 2, integrated photonic waveguides can support propagating waveguide modes with an effective index $Re\{n_{eff}\} = {}^{\beta}/_{\beta_0}$ and a guided wavelength described as $\lambda_g = \lambda_0/Re\{n_{eff}\}$. The guided wavelength value sets a limit in the minimum dimension Δx that must exist in the waveguide for the fundamental mode of the waveguide to be supported. Therefore, in integrated photonic waveguides the so-called diffraction limit $\Delta x \geq {}^{\lambda_g}/_2$ sets a restriction in the integrated photonic waveguide design methodology. Integrated plasmonic waveguides, on the other side, overcome the diffraction limit being able to support plasmonic modes with an elevated field concentration near the dielectric-metal interfaces. The drawback though is the larger Ohmic loss experienced by the propagating light waves.

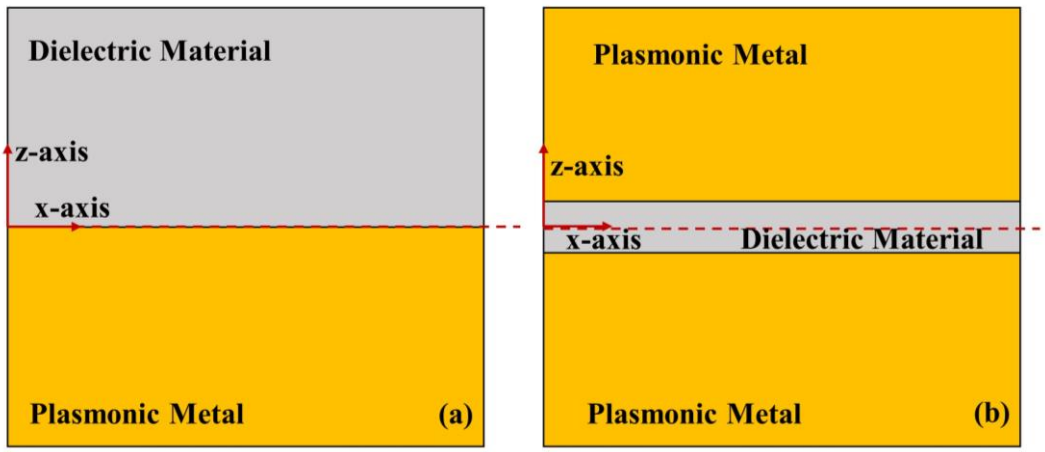


Fig. 4.1.1: Sideview Depiction of Two plasmonic waveguide types: (a) Thin-film plasmonic waveguide, (b) Plasmonic slot waveguide (MIM) with propagation in the x-axis and continuity at the y-axis. The Metal-Insulator-Metal Multilayer resides at the z-axis (thickness direction).

This section will present in detail the analytical electromagnetic solutions of simple common plasmonic waveguides, the thin film plasmonic waveguide and the plasmonic slot (MIM) waveguide. A detailed analysis for the electromagnetic considerations in metallic structures can be found in the excellent reference by Maier [1]. The sideview representations of the simple common thin-film and MIM plasmonic waveguides are presented in Fig. 4.1.1. The first geometry is based on a dielectric layer on top of a metal layer. The device is continuous infinitely in the y direction. The second geometry is based on a multi-layered structure, with a dielectric filling a plasmonic metallic slot in the vertical direction. The propagation direction is the x-axis and the layers are continuous infinitely in the y-direction. It should be mentioned, herein, that in the beginning computation of the electromagnetic fields in plasmonic waveguides was based on analytical work by Rayleigh on diffraction and Sommerfeld's and Zenneck's studies on surface waves. These works were complemented by Drude's improved understanding on the optical properties of metals. In addition, initially radio waves propagating along a conducting surface were the focus of the studies and the excitation of surface waves with visible light utilizing a prism (Kretschmann and Raether) brought the optics to the field of plasmonics, unifying all forms of electromagnetic waves in surface plasmon polaritons. The same with metamaterials, artificial materials with user designed permittivity and permeability values, where fabrication in larger dimensions first allowed the study of their behavior with radio waves and then with waves in shorter wavelengths. Nowadays, numerical methods are the workhorse of modern plasmonic waveguide design. Nevertheless, analytical methods in simple plasmonic structures still provide a greater physical insight on these devices. For this reason, this Chapter will begin from analytical treatments.

The analysis begins from the thin-film plasmonic waveguide of Fig. 4.1.1(a). Beginning from Maxwell's curl equations for the electric and magnetic field one obtains the Helmholtz equation as shown for photonic waveguides in Chapter 2:

$$\nabla \times \nabla \times \boldsymbol{E} = -\mu_0 \frac{\partial^2 \boldsymbol{D}}{\partial t^2} \Rightarrow$$

$$\nabla\left(-\frac{1}{\varepsilon}\boldsymbol{E}\cdot\nabla\varepsilon\right) - \nabla^2\boldsymbol{E} = -\mu_0\varepsilon_0\varepsilon\frac{\partial^2 \boldsymbol{E}}{\partial t^2} \Rightarrow$$

$$\nabla^2\boldsymbol{E} + \beta_0^2\varepsilon\boldsymbol{E} = 0 \quad (4.1)$$

As the direction of propagation is the x-axis, one can make the assumption that the electric field is harmonic with a complex propagation constant β taking the form:

$$\boldsymbol{E}(x,y,z) = \boldsymbol{E}(z)e^{i\beta x} \quad (4.2)$$

Therefore, the Helmholtz equation (4.1) by inserting (4.2), considering also the geometrical characteristics of Fig. 4.1.1(a) becomes:

$$\frac{\partial^2 \boldsymbol{E}(z)}{\partial z^2} + (\beta_0^2\varepsilon - \beta^2)\boldsymbol{E} = 0 \quad (4.3)$$

Two sets of equations therefore ensue for the TM ($E_z = 0$) and TE ($E_y = 0$) polarizations. Considering the geometrical characteristics of the plasmonic waveguide in Fig. 4.1.1(a), one obtains:

For the TM polarizations:

$$\left.\begin{aligned} E_x &= -i\frac{1}{\omega\varepsilon_0\varepsilon}\frac{\partial H_y}{\partial z} \\ E_z &= -\frac{\beta}{\omega\varepsilon_0\varepsilon}H_y \\ \frac{\partial^2 H_y}{\partial z^2} &+ (\beta_0^2\varepsilon - \beta^2)H_y = 0 \end{aligned}\right\} \quad (4.4)$$

For the TE polarizations:

$$\left.\begin{aligned} H_x &= i\frac{1}{\omega\mu_0}\frac{\partial E_y}{\partial z} \\ H_z &= \frac{\beta}{\omega\mu_0}E_y \\ \frac{\partial^2 E_y}{\partial z^2} &+ (\beta_0^2\varepsilon - \beta^2)E_y = 0 \end{aligned}\right\} \quad (4.5)$$

One needs to now use the well-known physical boundary conditions in electromagnetics described in Chapter 2 and especially the continuity for H_y and $\varepsilon_i E_z$ for (4.4) and the continuity of E_y and H_x for (4.5). For the TM polarizations, from the continuity relation one gets:

$$\frac{\beta_2}{\beta_1} = -\frac{\varepsilon_2}{\varepsilon_1} \quad (4.6)$$

The component H_y must also be able to satisfy the wave equation of (4.4). For this reason:

$$\left.\begin{aligned} \beta_1^2 &= \beta^2 - \beta_0^2\varepsilon_1 \\ \beta_2^2 &= \beta^2 - \beta_0^2\varepsilon_2 \end{aligned}\right\} \xRightarrow{(4.6)}$$

$$\beta = \beta_0\sqrt{\frac{\varepsilon_1\varepsilon_2}{\varepsilon_1 + \varepsilon_2}} \quad (4.7)$$

The relation (4.7) represents the dispersion relation for thin film plasmonic waveguides. The relation can be plotted after considering Silica and Air as the dielectric materials [1] and is depicted in Fig. 4.1.2. Straight lines for air and silica are called light lines. For frequencies above the plasma frequency the plasmonic metals are transparent, while below the light lines, surface plasmon polariton modes can be supported. Very

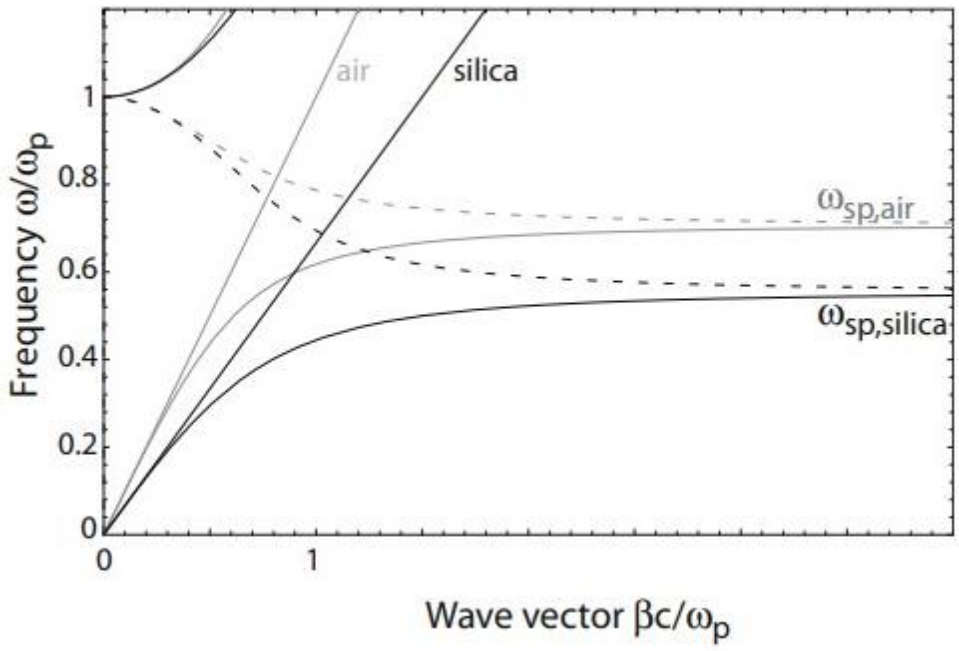


Fig. 4.1.2: Dispersion relation of SPPs at the interface between a Drude metal with negligible collision frequency and air (gray curves) and silica (black curves). Retrieved by [1].

large values for the wavevector β can be achieved by the large confinement of light near the metal surfaces. This leads to a slow-down effect on light, experienced in plasmonic waveguides. The slow-down effect is very useful in devices such as modulators and sensors. Nevertheless, due to the metal Ohmic losses, the slow-down of light is accompanied by excess insertion losses in the designed device.

For the TE modes, the continuity relation leads to the equation:

$$A_1(\beta_1 + \beta_2) = 0 \ (4.8)$$

Since for the confinement on the surface requires $Re\{\beta_1\} > 0 \ and \ Re\{\beta_2\} > 0$ the parameter A_1 needs to be set to zero. Therefore, considering that:

$$A_1 = A_2 = 0 \Rightarrow H_x = H_z = E_y = 0 \ (4.9)$$

It is clear that in this case plasmonic waveguide modes are supported only for the TM polarization. This property is very useful for the design of devices with polarization selectivity such as polarization beam splitters.

Going forward to the MIM design of Fig. 4.1.1(b), the same mathematical controls are used for the derivation of an analytical solution. Considering the 3 domains formed by the three layers of the MIM device. In the first domain z>a:

$$\left.\begin{aligned} H_y &= Ae^{i\beta x}e^{-\beta_3 z} \\ E_x &= iA\frac{1}{\omega\varepsilon_0\varepsilon_3}\beta_3 e^{i\beta x}e^{-\beta_3 z} \\ E_z &= -A\frac{\beta}{\omega\varepsilon_0\varepsilon_3}e^{i\beta x}e^{-\beta_3 z} \end{aligned}\right\} \quad (4.10.1)$$

In the second domain, where z<-a:

$$\left.\begin{aligned} H_y &= Be^{i\beta x}e^{\beta_2 z} \\ E_x &= -iB\frac{1}{\omega\varepsilon_0\varepsilon_2}\beta_3 e^{i\beta x}e^{\beta_2 z} \\ E_z &= -B\frac{\beta}{\omega\varepsilon_0\varepsilon_2}e^{i\beta x}e^{\beta_2 z} \end{aligned}\right\} \quad (4.10.2)$$

In the third domain, where -a<z<a:

$$\left.\begin{aligned} H_y &= Ce^{i\beta x}e^{\beta_1 z} + De^{i\beta x}e^{-\beta_1 z} \\ E_x &= -iC\frac{1}{\omega\varepsilon_0\varepsilon_1}\beta_1 e^{i\beta x}e^{\beta_1 z} + iD\frac{1}{\omega\varepsilon_0\varepsilon_1}\beta_1 e^{i\beta x}e^{-\beta_1 z} \\ E_z &= C\frac{\beta}{\omega\varepsilon_0\varepsilon_1}e^{i\beta x}e^{\beta_1 z} + D\frac{\beta}{\omega\varepsilon_0\varepsilon_1}e^{i\beta x}e^{-\beta_1 z} \end{aligned}\right\} \quad (4.10.3)$$

Applying the continuity relations to all interfaces along with the wave equation to H_y one obtains a linear system of equations, which if solved provides an implicit relation for the dispersion relation in plasmonic MIM waveguides:

$$e^{-4\beta_1 a} = \frac{\beta_1/\varepsilon_1 + \beta_2/\varepsilon_2}{\beta_1/\varepsilon_1 - \beta_2/\varepsilon_2}\,\frac{\beta_1/\varepsilon_1 + \beta_1/\varepsilon_3}{\beta_1/\varepsilon_1 - \beta_1/\varepsilon_3} \quad (4.11)$$

Considering that the metals in areas II and III have the same complex dielectric permittivity values $\varepsilon_2 = \varepsilon_3\ and\ \beta_2 = \beta_3$, the dispersion relation can be split to the following set of equations:

$$\tan(\beta_1\alpha) = -\frac{\beta_2\varepsilon_1}{\beta_1\varepsilon_2} \quad (4.12)$$

$$\tan(\beta_1\alpha) = -\frac{\beta_1\varepsilon_2}{\beta_2\varepsilon_1} \quad (4.13)$$

Relation (4.12) describes supported modes with odd parity while relation (4.13) highlights supported modes with even parity. If these relations are applied to a MIM waveguide where $\varepsilon_2 = \varepsilon_2(\omega)$ refers to the metal dielectric function and ε_1 describes the dielectric function of the dielectric material filling the slot, it can be noticed that this waveguide can confine modes in a very small cross-section without having a cutoff

reducing the slot width. This is very interesting because an electromagnetic wave with guided wavelength λ_g much smaller than the diffraction limit and with a slowed down phase velocity $v_{ph} = c/Re\{n_{eff}\}$ can be propagated. The effective index and wavevector values can be largely increased by decreasing the slot width, providing a more intense slow-down to the propagating wave. However, this is accompanied by increasing propagation losses in the plasmonic waveguide. Finally, it should be mentioned that in this case again only TM polarizations can be supported. Nevertheless, the same plasmonic waveguide (MIM) structure with a horizontal configuration, meaning that the Metal-Insulator-Metal multi-layer resides in the y-axis instead of the z-axis, supports only TE modes, following the calculations presented above. Therefore, the Metal-Insulator-Metal (MIM) waveguide presents a very efficient design for ultra-high light confinement at the expense of increased propagation losses.

To retrieve the benefits of both high-power confinement and light slow-down of plasmonic waveguides and low-losses of photonic waveguides, Hybrid Plasmonic Waveguide (HPW) devices have been proposed as a solution to reap the benefits of both integrated photonic and plasmonic platforms [2]. A schematic of a designed Hybrid Plasmonic Waveguide (HPW) is depicted in Fig. 4.1.3. The platform is based on a CMOS-compatible dielectric or semiconductor waveguide residing on a lower cladding on a Si-based substrate. On top the first layer there is a layer with a lower refractive index than the bottom layer. On top of the layer stack resides a metal layer. Even though analytical solutions have been proposed for the hybrid plasmonic waveguide device [3], due to the multiple material layers the application of the analytical boundary conditions and continuity relations is challenging. For this reason, numerical methods are utilized for the design of such waveguides.

The physical characteristics of hybrid plasmonic waveguides are described in this Section. The hybrid plasmonic waveguide obtains characteristics from both photonics and plasmonics via mode hybridization. This physical process implies coupling of the integrated photonic modes with the plasmonic thin film modes. As a result, two supermodes are usually supported in an HPW device. The first is more plasmonic while the second is more photonic in nature. The plasmonic mode of the HPW exhibits lower propagation losses than the pure plasmonic mode but higher than the pure photonic one.

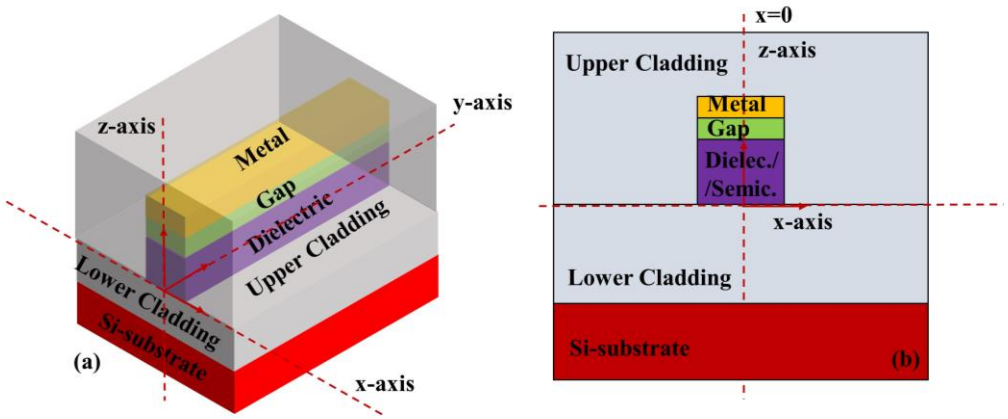


Fig. 4.1.3: (a) 3D-Depiction and (b) Cross-Sectional 2D depiction of a Hybrid Plasmonic Waveguide Device

As a result, the propagation length of the surface plasmon polariton in an HPW, defined as the length where the input power is decreased to 1/e of its value, is higher than the propagation length of a MIM. On the other side, the power confinement in the HPW is lower than the one in the MIM. An amount of power of the hybrid supermode resides in the bottom dielectric layer, while the largest percentage of the optical modal power resides in the gap in the HPW. Finally, the coupling and mode excitation techniques need to fulfill mode selectivity so that only one of the two supported supermodes is excited in the HPW.

In conclusion, integrated plasmonic waveguides can overcome the diffraction limit of typical integrated photonic waveguides, by being able to propagate modes with a very high optical confinement in gaps of ultra-small dimensions. They also benefit devices for a variety of applications by inducing a slow-down effect, determined by the waveguide design. The drawback for all these is the increased propagation loss in the waveguides. For this reason, hybrid plasmonic waveguide designs have been proposed as a great solution set in between photonic and plasmonic configurations.

4.2: Integrated photonics-plasmonics co-integration methods

The previous chapter of this thesis dealt with the monolithic integration and development of integrated light sources, amplifiers, and photodetectors, usually utilizing III-V based materials, on the 800nm thick Si_3N_4 platform on a Si-substrate. A multi-step coupling design technique was proposed to interface the III-V lasing/amplifying element with the passive Si_3N_4 waveguide. This Section will deal with the integration of modulators on the common CMOS platforms, Si and Si_3N_4. In all the Figures that will be displayed in this Section, the strip waveguide, being either Si or Si_3N_4 will have a distinct purple color. Even though a large variety of electro-absorption modulators (EAMs) have been demonstrated in the literature for direct amplitude modulations, the integration of phase modulators/shifters (PMs) on top of passive strip waveguides is of interest in integrated photonics targeting to deliver advanced modulation formats. Additionally, phase modulators in PICs are crucial for the development of systems featuring higher linearity and low power consumption.

High-speed integrated photonic phase-shifters constitute an important part of Si-based and Si_3N_4 photonic transceivers. In monolithic integration, the integration of high-speed photonic phase-shifters has been based on their incorporation in either Si or III-V. Phase-shifters are formed on the same platform as the passive waveguides via doping. On the other side, in hybrid or heterogeneous integration, the active phase-shifting sections are placed on top of the passive strip waveguide section. This involves the bonding of a Pockels electro-optic material on top of the passive waveguide, with a schematic overview of high-speed phase-shifters on passive waveguides provided in Fig. 4.2.1. More, recently, hybrid bonding methods have allowed direct or adhesive bonding of active phase-shifting ferroelectric oxides on top of passive strip waveguide platforms. The active oxide and passive strip waveguide are separated by an interlayer oxide gap (SiO_2). A 3D-Schematic of such a device is presented in Fig. 4.2.1(a). Coupling from the pure to the hybrid photonic waveguide is performed with an end-fire coupling step. The mode resides in both the strip waveguide and the oxide layers and characteristics of the hybrid mode such as the propagation loss and mode confinement in the oxide can be controlled via the strip width. Adiabatic tapers can be

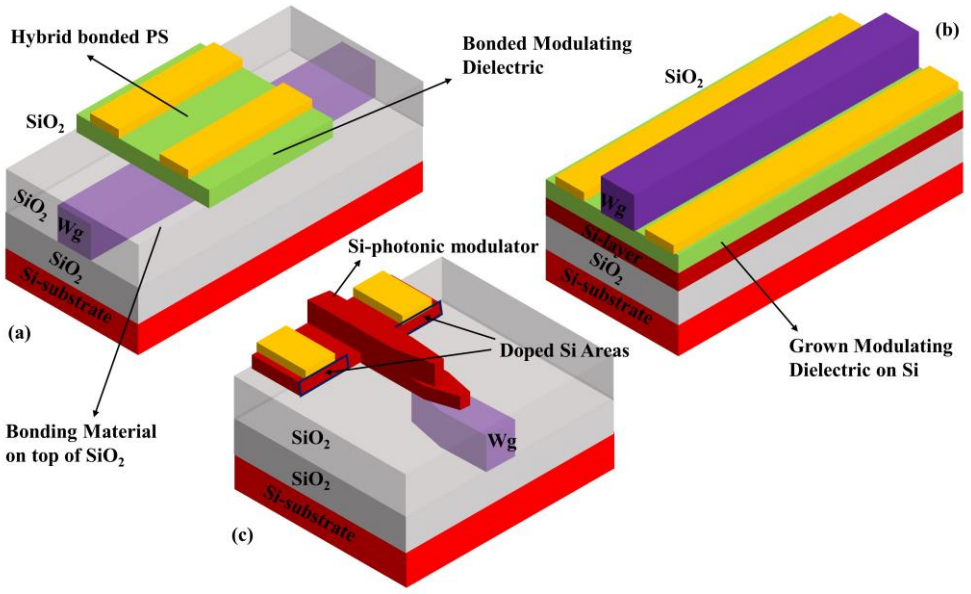


Fig. 4.2.1: 3D-depiction (a) of a phase modulator based on a modulating oxide hybridly bonded on top of a passive strip waveguide such as Si or Si_3N_4, (b) of a phase modulator based on grown modulating oxide on Si with strip waveguide (Si or Si_3N_4) deposited on top of it, and (c) of a Si-photonic modulator interfaced adiabatically with a bottom passive waveguide (Si or Si_3N_4).

employed to transform the initially excited mode from the end-fire interface to a mode more confined either to the active oxide or to the strip. The benefit of such a device is its potential for selective placement on the active section on the chip. The drawback is the increased propagation loss and the partial confinement of the modal field in the oxide. Another possibility is to grow the oxide on top of a CMOS Si layer and then deposit the passive strip waveguide on top of the active oxide, as shown in Fig. 4.2.1(b). Nevertheless, the deposited strip waveguide is usually based on an amorphous Si material (aSi) or deposited Si_3N_4 layer. This would thus lead to the support of a hybrid mode as in the case of Fig. 4.2.1(b), but with increased propagation losses. Finally, Si-photonic sections can form both a passive strip waveguide and an active phase-shifting section based on the carrier dispersion effect, as in Fig. 4.2.1(c). Nevertheless, the ferroelectric material allows a Pockel's effect to induce instantaneous phase-shifting at large bandwidths, unlike carrier-based phase-shifters employed in SOI or graphene. The propagation losses can be decreased in comparison with the hybrid counterpart due to the large percentage of light being confined in the dielectric core. Finally, unlike end-fire interfaces, adiabatic couplers can offer tolerant coupling with ultra-low losses of less than 0.5dB, thus avoiding parasitic reflections to the input waveguide. All the above configurations are promising for future transceiver devices.

In exploring these possibilities further, integrated photonics emerges as a key player, providing modulator devices with remarkable characteristics. They are able to provide modulator devices with half-wave voltage-length products >2Vmm and large bandwidths >50GHz. However, eventually, the mm to cm lengths of photonic phase-shifters can increase the total footprint of a PIC and can introduce bandwidth limitations if the RF wavelength λ_{RF} of the applied electrical signal becomes comparable with the length of the phase-shifter. In these cases, proper design of the RF electrodes is required to achieve large bandwidths and impedance matching. Co-packaged optics, targeting

the very close approximation of the designed electronics with the integrated optical engines of transceivers, requires devices with dimensions at the scale of the electronics ASICs' pin pitch demanding also operation at higher than 100Gbaud symbol rates per channel. For these reasons, plasmonics, offering electrodes that can confine light and the electric field of a RF high-speed signal, has been transformed to a promising technology for co-packaged optics.

Some individual plasmonic waveguides were studied via analytical and approximate techniques. As mentioned above, integrated plasmonic waveguides can confine both the RF and optical field in small gaps and provide an increased modulation efficiency. In addition to this, the slow-down effect to light in plasmonics can be useful for modulators as the propagating light experiences more intense changes in the refractive index of its environment. For these reasons, the length in plasmonic waveguides is sometimes in the field of a few tens of μm. The achievable bandwidth is also elevated by the ultra-small resistance of the plasmonic metals and small capacitances due to the small gaps where the electric field of the applied RF signal is confined. However, pure plasmonic waveguides suffer from increased propagation losses, while hybrid plasmonic waveguides have a tradeoff between propagation losses and light confinement in their gap layer.

Future devices for modulation should leverage both the advantages of photonics and plasmonics. Towards this route, an integrated plasmo-photonic platform could combine the ultra-high modal confinement in a Pockels material, leading to low footprint and power consumption devices, with the ultra-low losses of integrated photonics. Nevertheless, the question that still remains is how to integrate passive Si or Si_3N_4 waveguides with plasmonic sections, in order to demonstrate a complete functional sensor or transmitter. Some options are highlighted in the 3D schematics of Fig. 4.2.2. The simplest plasmonic waveguide, the plasmonic thin film, can be integrated with a

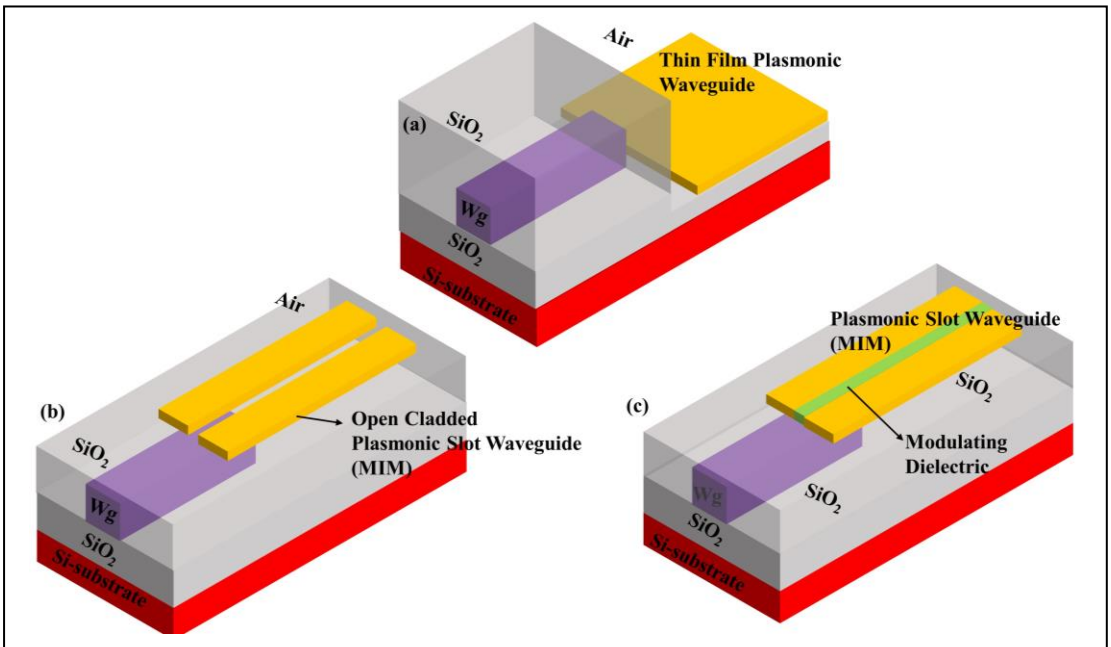


Fig. 4.2.2: (a) 3D-depiction of a Si-to-thin film plasmonic waveguide coupler, (b) a Si-to-open cladding plasmonic slot waveguide and (c) a Si-to-plasmonic slot waveguide filled with a modulating dielectric.

passive strip waveguide, as shown in Fig. 4.2.2(a). This is the case when an end-fire interface is utilized. In end-fire coupling interfaces, a maximum power overlap between the initial photonic mode and the targeted plasmonic mode in the plasmonic waveguide will lead to maximum coupling efficiency.

The drawback of the plasmonic thin films is that the optical mode is loosely confined. Plasmonic slots, seen in Fig. 4.2.2 (b) and (c), on the other side, can strongly confine optical mode in very small cross-sections. To achieve light transfer from the strip part to the plasmonic slot, a directional coupler can be designed. In such a case, the calculated effective indices of the initial and targeted optical modes need to match one another. In such a case, light in the section consisting of the strip waveguide below the plasmonic slot will be periodically transferred from the strip to the plasmonic slot and back to the strip. Maximum power transfer will occur at a length called the coupling length, defined as $L_c = \pi/\Delta\beta = \lambda_0/(2(Re(n_{eff1}) - Re(n_{eff2})))$. The two effective indices correspond to the the two supermodes existing in the cross-section of the hybrid device with both the strip and plasmonic slot. These can be estimated by simple 2D-FDE simulations. Finally, if a phase-modulator is of interest, the plasmonic slot can be filled by an active oxide with a high Pockels coefficient (such as $LiNbO_3$ or $BaTiO_3$), as seen in Fig. 4.2.2(c). The coupling interface between the passive integrated photonic waveguide and the plasmonic slot or metal-insulator-metal (MIM) waveguide filled with the Pockels material could again be a directional coupler. However, the effective index value of the plasmonic mode of the slot increases with the introduction of the active oxide layer. This renders direct phase-matching with a Si_3N_4 waveguide (considering a Si_3N_4 refractive index close to 2) via proper design very difficult.

The hybrid plasmonic waveguide, studied qualitatively in the previous section, can also be interfaced with low-loss photonic strip waveguides. These hybrid waveguides can form phase-modulators via the insertion of the active oxide in the gap of the HPW. Some proper configurations for optical coupling that can be found in the literature are

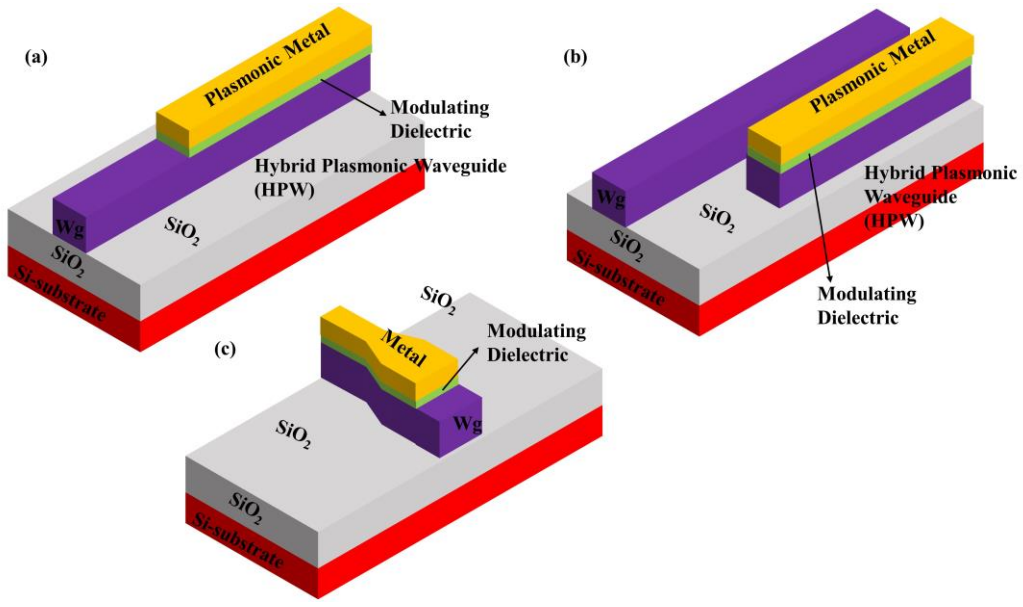


Fig. 4.2.3: (a) 3D-depiction of an end-fire Si-to-HPW coupling interface, (b) of a directional Si-to-HPW coupling interface and (c) of an end-fire+adiabatic Si-to-HPW coupling interface. All the Hybrid Plasmonic Waveguide Devices are filled with a modulating dielectric material.

presented in Fig. 4.2.3. In Fig. 4.2.3 (a) the passive strip waveguide is interfaced with an HPW via a simple end-fire coupling step. This would lead to a broadband coupler with insertion losses determined by the power overlap integral of the photonic and targeted plasmonic mode for a set waveguide width and material thickness values. However, since in most cases hybrid plasmonic waveguides support at least 2 supermodes due to hybridization, these would lead to excitation of multiple modes in the phase-shifter with notable beating effects. Beating of the more plasmonic with the more photonic mode would lead to the increase of the half-wave voltage values in the device. In Fig. 4.2.3 (b) a directional photonic-plasmonic coupler selectively excites one of the two supermodes in the HPW. The coupling efficiency can be controlled via the design of the individual waveguides, taking care of the phase-matching, and the gap between the photonic and plasmonic waveguides. However, in many cases the small gaps, usually lower than 200nm impose stringent requirements from the fabrication side. Finally, in Fig. 4.3.3 (c) an initial end-fire step excites the plasmonic mode in the HPW. A plasmonic adiabatic taper after the end-fire step can be deployed to transform the mode from a more photonic to more plasmonic mode and vice versa. By controlling the length and the final width of the plasmonic section beating phenomena can be greatly suppressed.

In conclusion, plasmonic phase-shifters can offer elevated characteristics to phase-modulator devices in integrated photonic platforms. These include increased bandwidth, ultra-low footprints and increased modulation efficiency, compared to purely photonic phase-shifters. This comes with the sacrifice of increased propagation losses in the system. For this reason, the integration of plasmonic phase-shifters with Si and Si_3N_4 waveguides is desired. The next section will present the design of CMOS plasmonic $BaTiO_3$ phase-shifter interfaces with the 800nm thick Si_3N_4 platform.

4.3: Efficient multi-step coupling between Si_3N_4 waveguides and CMOS plasmonic ferroelectric phase shifters in the O-band

Silicon nitride (Si_3N_4) is gaining ground over the last few years as one of the most common materials found in a CMOS fab promising a wide operation band, ultra-low propagation losses, lower than those of Si-photonic waveguides, and high fabrication tolerances [4]. For this purpose, exploiting hybrid, heterogeneous, or monolithic co-integration strategies, Si_3N_4 has enabled seamless light transition between passive PICs and lasers, amplifiers and/or photodetectors [5-8] both in the O- and C-band. Moreover, a plethora of studies have focused on the realization of modulators on Si_3N_4, via the co-integration of various materials featuring electro-optic properties, such as graphene [9, 10], silicon [11], lithium niobate [12], and ferroelectric lead zirconate titanate [13]. Towards this direction, ferroelectric barium titanate oxide ($BaTiO_3$ or BTO) exhibits a strong electro-optic coefficient, CMOS compatibility and capability to withstand high temperature. For this reason, it is currently the most promising approach for the demonstration of very efficient phase shifters that can be post processed for further packaging and assembly with electronic ICs [14, 15].

In parallel, plasmonic modulators have emerged as a key-enabling technology offering optical devices with ultra-low footprint. A variety of plasmonic waveguides have been presented so far, such as Dielectric-Loaded Surface-Plasmon Waveguides (DLSPW) [16], slot MIM structures [17, 18] and hybrid plasmonic waveguides (HPW)

[19, 20]. The last two offer gaps that can support modes with high power confinement, rendering them an excellent solution for phase modulation [21]. Following this approach, the electro-optic polymer organic material in plasmonic slot phase-shifter configurations is the one with the highest performance reported so far in terms of achievable rates and power consumption [22]. However, temperature stability of the poled organic materials restricts the application range of this technology. Plasmonic ferroelectric modulators employing solid-state lithium niobate [23, 24] and BTO [25-27], with their promising modulation efficiency, tolerance to high temperatures, and compact dimensions, have recently emerged as a more viable solution for compatibility with back end of the line (BEOL) processes [28]. Nevertheless, efforts targeting the integration of Si_3N_4 photonic waveguides with CMOS plasmonic BTO-based phase modulators are still missing.

Towards this direction, the co-integration of Si_3N_4 waveguides with BTO-ferroelectric plasmonic phase shifters requires coupling configurations that will efficiently transfer the light from the Si_3N_4 waveguide to HPW and MIM waveguides. Previous demonstrations of such configurations focused on the design of coupling schemes for light transfer from Si-photonic waveguides to HPW, DLSPW, and MIM plasmonic waveguides based on noble metals [29-32]. Although this approach exhibits the best performance in terms of propagation losses in the plasmonic waveguide, it is incompatible with the CMOS process, raising a big concern about its suitability for production in mass scale. For this reason, devices based on alternative plasmonic metals such as TiN, Al or Cu have been investigated intensively during the last few years [33], with the plasmon propagation losses of Cu shown in [34] to approach those of Au.

Regarding CMOS plasmo-photonic couplers, in [35], a modulator based on the co-integration of silicon with transparent conducting oxides and TiN plasmonic metals was reported with the plasmonic-to-photonic transducer yielding simulated losses of 0.7dB. In [36] a coupling interface based on Si waveguides and Al-based composite hybrid plasmonic waveguide (CHPW) ITO-based amplitude modulators was experimentally demonstrated with measured losses as low as 1.14dB, while simulations indicated 1.61dB. Nevertheless, the above works, are failing to modulate the real part of the refractive index that is necessary for advanced modulation formats and do not solve the problem of incorporating a plasmonic modulator on the Si_3N_4 photonic platform. ITO based plasmonic phase shifters on the other hand [37] are demonstrated so far with only a few tens of GHz operational bandwidth, much lower than that of plasmonic ferroelectric modulators. Going now to plasmo-photonic couplers for non-active devices, the integration of Si photonic and Cu-based air-filled slot waveguides was presented in [38] with the simulated and experimental coupling losses being close to 1.55dB and a metal-silicon gap of only 30nm rendering the whole coupler sensitive to variations. In [39] the induced experimental coupling losses from a Si_3N_4 photonic waveguide to a Cu-based plasmonic one were lower than 2.84dB with 1.9dB simulated coupling losses. Finally, novel efficient coupling configurations for coupling from Si_3N_4 to open-cladded Cu-based slot waveguides have exhibited simulated losses less than 1.15dB [40]. Nonetheless, all the above devices do not incorporate an electro-optic material in the gap, prohibiting any modulation of the transmitted wave.

In this work, we present a thorough analysis for the design of O-band coupling configurations from Si_3N_4 waveguides to Cu/BTO-based HPW and MIM plasmonic

phase shifters. The gap is filled in both waveguides by a ferroelectric BTO material featuring an Electro-Optic coefficient as high as 1000 pm/V [25]. The optimum coupling configuration for the HPW geometry is shown to exhibit total transition losses lower than 1.75dB, while for the MIM case the total transition losses are lower than 1.29dB at the expense of higher propagation losses, being 0.5dB/μm and 0.78dB/μm respectively. Finally, the corresponding light confinement factor in the plasmonic waveguides is 31.5% for the HPW with 20nm thick gaps and 56.2% for the MIM waveguides with 100nm wide slots. As a reminder, the confinement factor at all cases has been numerically calculated via the 2D-FDE Method through the relation:

$$CF(\%) = \frac{\int_0^t \int_{-\frac{w}{2}}^{\frac{w}{2}} |\boldsymbol{E}|^2 dydz}{\int_{-\infty}^{\infty} \int_{-\infty}^{\infty} |\boldsymbol{E}|^2 dydz} \quad (4.14)$$

For both phase shifter configurations, a fabrication tolerance analysis reports the degradation of the coupling efficiency from the optimum structures for varying geometrical parameters, investigating up to a 5% thickness variation from the targeted value. Additionally, increasing lateral misalignment values at the directional and adiabatic coupling steps of each interface configuration were explored. The analysis showcased below is expected to lighten the path towards CMOS compatible, low insertion loss, ultra-compact and low-power CMOS plasmonic phase modulators on Si_3N_4 in the O-band.

This is the first work to the best of our knowledge that is relying on the utilization of a multi-step and multi-layer technology [11], [44]-[47] to seamlessly couple light from a Si_3N_4 waveguide to a CMOS plasmonic MIM slot or HPW structure, for the demonstration of plasmonic active devices. The works presented so far in the literature and summarized in Table 4.1 at Section 4.3.4 of Chapter 4 lack at least one of the following properties: operation in the O-band, employment of CMOS compatible Cu metal for the plasmonic waveguide, incorporation of electro-optic materials for efficient phase-shifting in the plasmonic waveguide or reliance on low loss Si_3N_4 platform. The rationale behind aiming at this wavelength band is related to the high temperature tolerance of the whole waveguide geometry rendering this technology as the ideal optical engine for ultra-short reach (XSR) co-packaged optics, targeting Intra-Data-Center applications. Also, the fact that plasmonic modulators can be driven with sub-volt voltages [47] at rates higher than 100Gbaud, eliminates the need for electrical amplifiers between the ASIC and the modulators, reducing the total power consumption of the package even more. Furthermore, their ultra-low footprint, removes the requirement for fan out of the high-speed signals coming out of the ASIC's pins, saving space, and minimizing propagation losses.

The organization of the rest of the Section is the following: Section 4.3.1 presents the overall multi-step coupling concept from Si_3N_4 to Cu/BTO geometries. Section 4.3.2 focuses on the design of interfaces for efficient light transfer to HPW phase shifters, while Section 4.3.3 investigates coupling schemes for light transition to the MIM type of waveguides. Section 4.3.4 compares the coupling interfaces of this work with other CMOS-compatible interfaces found in the literature. Both sections are completed by a detailed tolerance analysis for the coupling devices to a variety of critical design parameters. Section 4.3.5 concludes the Section with a summary of the obtained outcomes.

4.3.1: Overall concept for coupling between Si_3N_4 photonic and Cu/BTO-based plasmonic modulators.

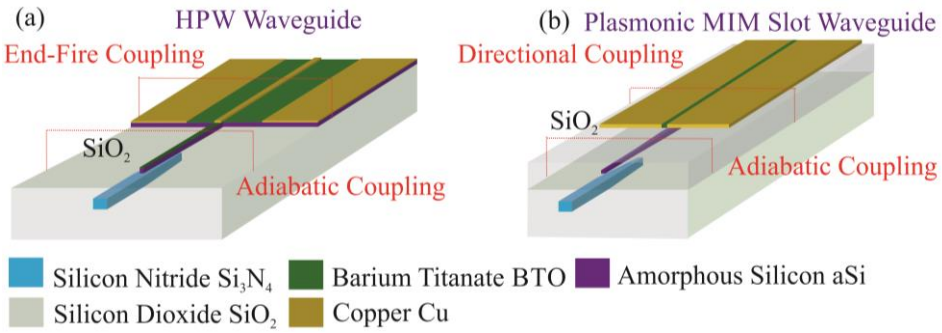


Fig. 4.3.1: (a) 3D-depiction of the multi-step coupler between a Si_3N_4 and an HPW plasmonic waveguide, and (b) between a Si_3N_4 and a plasmonic MIM slot waveguide.

For the co-integration of Si_3N_4 waveguides with BTO-based CMOS plasmonic modulators, two discrete novel coupling structures have been designed and are proposed herein. The first one focuses on light coupling from an 800x800 nm^2 Si_3N_4 waveguide to a hybrid plasmonic waveguide (HPW) structure, whereas the second one on coupling to a plasmonic slot (Metal Insulator Metal-MIM) waveguide. In both cases, a proper 220nm thick aSi interposer waveguide is incorporated to facilitate the phase matching condition between the plasmonic structures and the Si_3N_4 platform. In the HPW case that supports mainly TM plasmonic modes, the aSi layer is an integrated part of the phase shifter, as it provides contact for the RF signal injection and resides at the bottom of the waveguide. For the plasmonic MIM slot structures, the aSi is only utilized as a photonic interposer and the supported plasmonic waveguide modes are TE only, due to the slot symmetry. Fig. 4.3.1(a) illustrates the envisaged Si_3N_4-to-aSi-to-HPW coupler, consisting of an initial adiabatic coupling step from Si_3N_4-to-aSi/BTO, a transition taper to the aSi/BTO waveguide and a final end-fire coupling step from the aSi/BTO to the HPW waveguide. On the other side, Fig. 4.3.1(b) depicts the envisaged Si_3N_4-to-aSi-to-MIM coupler where, the aSi waveguide is solely employed as a vertical interposer efficiently transferring light to the final MIM plasmonic waveguide. This coupler consists of an initial Si_3N_4-to-aSi adiabatic coupling step and an aSi-to-MIM directional coupling section.

To analyze the optical interface of Si_3N_4 waveguides with the proposed CMOS plasmonic MIM and HPW waveguides, a 2D-Finite Difference (FD) eigenvalue

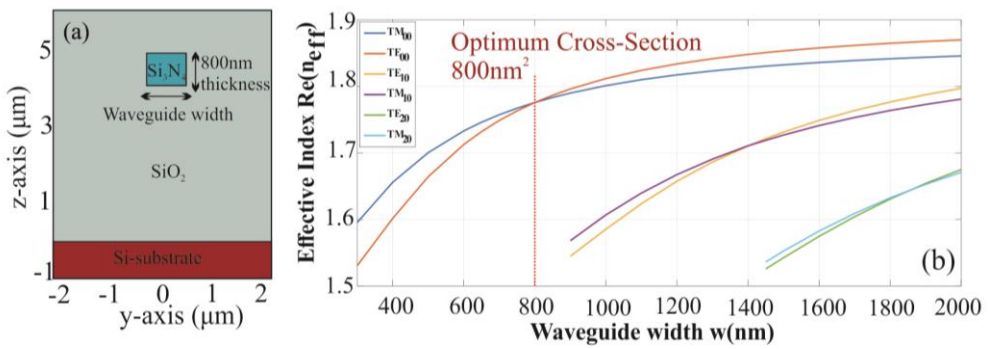


Fig. 4.3.2: (a) Cross-sectional geometry on the yz-plane of the simulated Si_3N_4 waveguide structure, (b) Parametric eigenmode analysis results for a variety of Si_3N_4 waveguide widths.

problem was initially solved for the Si_3N_4 waveguide. Fig. 4.3.2(a) presents the cross-sectional geometry of the Si_3N_4 waveguide with 800nm thickness. At 800nm width, the waveguide supports a fundamental TE-mode with a real part $Re\{n_{eff}\} = 1.775719$ and a fundamental TM-mode exhibiting $Re\{n_{eff}\} = 1.775721$. Figure 4.3.2(b) illustrates the effective index variation versus the Si_3N_4 width for the TE and TM supported modes. It is apparent that higher-order modes are feasible for widths larger than 900nm, while the effective indices of the fundamental TE- and TM-modes become lower for smaller width values. PML boundary conditions were applied at the edges of the computational window to eliminate any non-physical modes.

The coupling analysis was initially focused on the HPW waveguide with Fig. 4.3.3(a) presenting the cross section of the yz-plane. The BTO layer providing the EO phase shift has a thickness of 20 nm and the Cu thin film on top is 100 nm. Higher BTO thicknesses could also be an option, but then there is a trade-off between higher modulation efficiency and lower speed emerging from the higher capacitance of the HPW. The BTO is sandwiched between a plasmonic metal on top with a width of 500 nm and a 220nm-thick aSi layer below that is doped. The final cross-section of the plasmonic waveguide features the same aSi layer with a larger width to facilitate electrode formation for voltage application at the EO plasmonic phase shifter. The refractive index of BTO is derived from [15, 42], while the refractive index of Cu is obtained from ellipsometry results of [34]. The doped aSi has a refractive index with a real part equal to n=3.52 based on ellipsometry measurements, whereas, in this study, its imaginary part was not taken into consideration, as the k value is dependent on the overall process and doping level. The small length of this part being less than 30μm though, will not alter the overall results significantly. This waveguide, as seen in Fig. 4.3.3(b) and Fig. 4.3.3(c), supports a fundamental TM plasmonic mode with an effective index $Re\{n_{eff}\} = 3.284$ and a hybrid high-order mode with a $Re\{n_{eff}\} = 3.072$ at 1.31μm. The rationale behind the 500nm width of the Cu metal will be clarified in

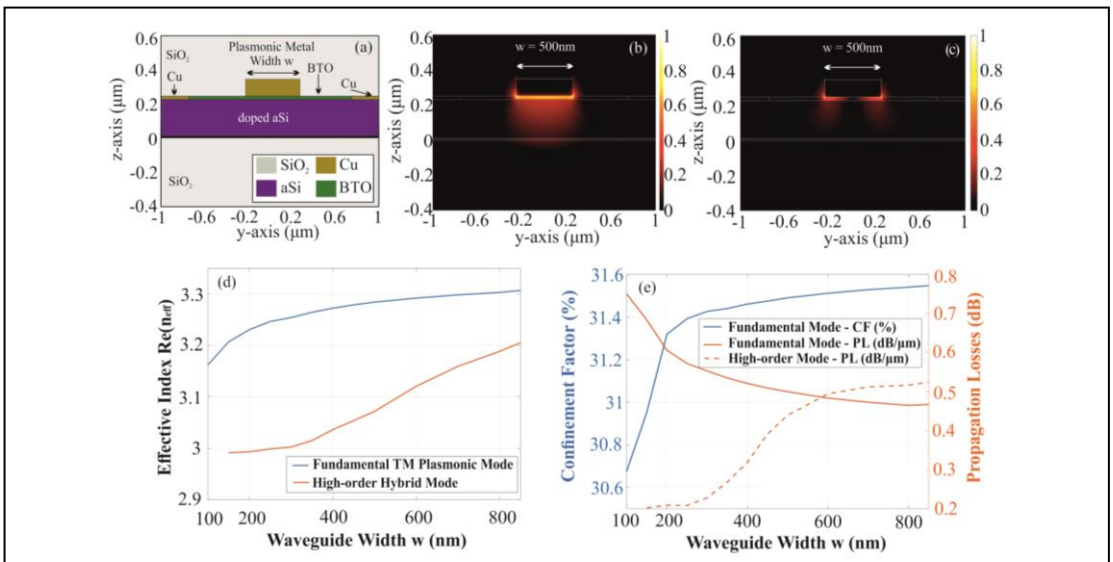


Fig. 4.3.3: (a) Cross section of the designed aSi/BTO/Cu HPW waveguide with variable plasmonic-metal width, (b) yz-image of the |**E**| field distribution of the fundamental TM-mode and (c) yz-image of the |**E**| field distribution of the hybrid high-order mode supported by the 500nm wide plasmonic waveguide, (d) Effective Refractive Index $Re\{n_{eff}\}$ of the fundamental TM- and hybrid high-order mode, and (e) Confinement Factor (%) of the fundamental TM-mode and Propagation Losses (dB/μm) of both modes of the HPW waveguide.

Section 4.3.2, where the results from multiple simulations for various metallic widths will be demonstrated. The chosen width is as a trade-off between optimum coupling efficiency, low propagation losses and minimized beating phenomena between the plasmonic and hybrid modes. As shown in Fig. 4.3.3(d) and Fig. 4.3.3(e), the plasmonic mode is concentrated in the BTO material, with a very small percentage residing at the edges of the plasmonic metal. Thus, according to Fig. 4.3.3(d) its effective index increases, while as seen in Fig. 4.3.3(e) the confinement inside the BTO is also enhanced to a value close to 31.5% for a metal width of 500nm. At the same time, the propagation losses decrease with the rise of the plasmonic width, as for smaller widths a larger field proportion is lying at the edges of the metal. On the other side, the hybrid high-order mode has a cutoff near 100nm, while its polarization is mostly-TE for widths below 300nm and becomes TM for higher values. This polarization variation is highlighted by an increase in the mode's propagation losses as shown in Fig. 4.3.3(e). Finally, it should be noted that this waveguide supports more high-order modes for widths higher than 850nm and due to the even and odd symmetry of the two modes, regardless of the polarization it is expected that the two modes will interfere, as it will be discussed in detail in Section 4.3.2. At 500nm width, for the supported fundamental TM mode shown in Fig. 4.3.3(e), the propagation losses are near 0.5dB/μm.

The second plasmonic configuration investigated in this work is the plasmonic MIM slot waveguide that can support highly confined plasmonic modes in the BTO layer [25], but this comes at the cost of higher propagation losses. Fig. 4.3.4(a) presents the cross-section of the proposed plasmonic-slot waveguide, that is formed by two Cu films with 100 nm thickness spaced apart by a variable distance and filled by BTO of equal thickness. This horizontal MIM waveguide was electromagnetically analyzed through 2D-FD eigenmode calculations by varying the slot width. Fig. 4.3.4(b) illustrates the propagation losses and confinement factor in the slot. According to what was expected

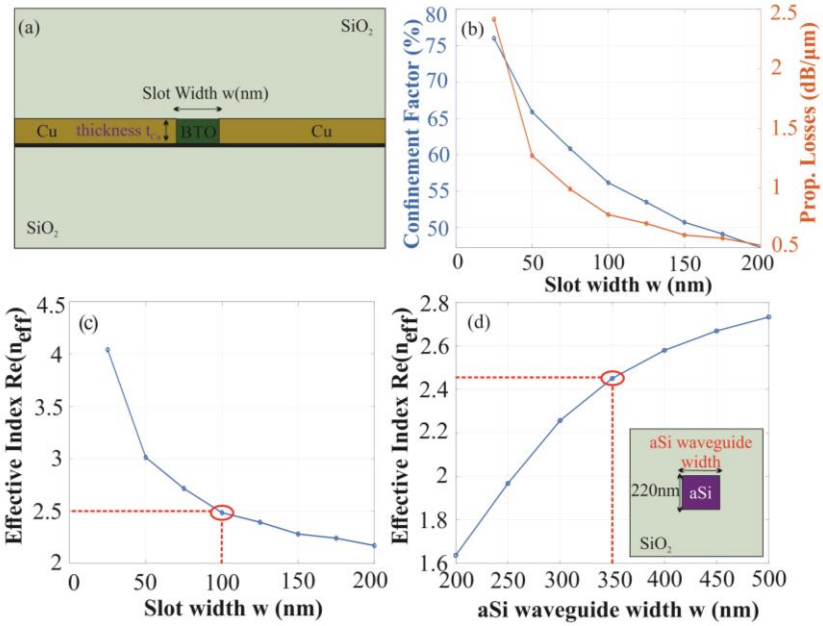


Fig. 4.3.4: (a) Cross section of the designed Cu/BTO/Cu (MIM) plasmonic slot waveguide, (b) Confinement factor (%), and Propagation Losses (dB/μm) for increasing slot width, and (c) Effective refractive index Re{n_{eff}} of the TE plasmonic-slot waveguide mode for increasing slot widths, (d) Effective refractive index Re{n_{eff}} of the TE photonic mode supported by the aSi waveguide for various widths. BTO thickness is 100nm in all graphs.

the confinement factor decreases for widening slots, but, overall, it is higher than 50% for slot widths up to 150nm, whereas the propagation losses are lower than 1.25dB/μm for widths higher than 50nm. For a 100nm slot width value that is compatible with e-beam lithography, the propagation losses of the TE plasmonic mode are 0.78dB/μm, while the power confinement in the slot is close to 56.2% at a free-space wavelength of λ=1310nm. Due to the relatively low loss and high confinement of the mode, this width was selected as the optimum compromise in the following simulations for the evaluation of the photonics to plasmonics interface.

The existing phase-mismatch between the initial TE-mode of the Si_3N_4 waveguide and the plasmonic mode of the MIM slot prohibits direct directional or adiabatic coupling at λ=1310nm. Figure 4.3.4(c) illustrates the plasmonic slot mode's effective index $Re\{n_{eff}\}$ versus slot width, with the value remaining well above 2.28 in the high confinement regime emerging for slot widths lower than 150nm. For this reason, a 220nm thick aSi layer, providing an effective index between those of the Si_3N_4 and the plasmonic modes, can act as an intermediate interposer for seamless light transfer from the Si_3N_4 to the CMOS plasmonic waveguide. This aSi layer is spaced apart 100nm from the top of the Si_3N_4. To couple light, from aSi to the favorable 100nm wide plasmonic slot geometry, eigenmode calculations were performed for the decoupled aSi and plasmonic MIM slot waveguides and the results are shown in Fig. 4.3.4(d). From this graph it is revealed that near optimum phase matching conditions exist for a slot-width of 100nm and an aSi waveguide width of 350nm. This sets the base for directional coupling between the two waveguides [41]. In Section 4.3.2, the design of both multi-step couplers for light transfer from Si_3N_4 to HPW and plasmonic MIM slot waveguides is analyzed in detail. The coupling performance of both configurations is calculated in the following sections only for the 1310nm wavelength, due to the lack of ellipsometry data for the BTO in the whole O-band.

4.3.2: Design analysis for efficient couplers between Si_3N_4 and Cu/BTO/aSi HPW plasmonic waveguides

4.3.2.1: Adiabatic coupling configuration

This section analyzes the design of the adiabatic coupler of Fig. 4.3.1(a), required for efficient light transfer between the 800x800nm^2 Si_3N_4 and the photonic aSi/BTO waveguide for the TM polarization. The interface relies on linear tapers, one for the Si_3N_4 and one for the aSi/BTO layer spaced by 100nm thick SiO_2 interlayer with the complete layout depicted in the 3D schematic of Fig. 4.3.5(a). The width of the Si_3N_4 waveguide decreases linearly, while the width of the aSi/BTO waveguide increases towards the *x* direction, with both tapers featuring the same length for efficient mode

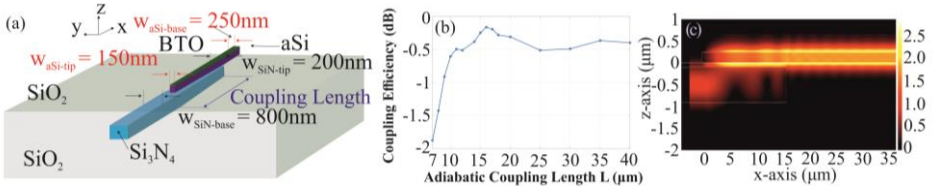


Fig. 4.3.5: (a) 3D-depiction of Si_3N_4-to-aSi/BTO adiabatic coupler, (b) Coupling efficiency (dB) for increasing adiabatic taper length, (c) Sideview of the |**E**| field distribution for the optimum coupling length of 16μm.

conversion towards compliance with the phase matching condition. The initial aSi/BTO waveguide width was 250nm based on 2D-eigenmode calculations for the determination of the tip widths that will enable compliance with the adiabaticity criterion [43]. The Si_3N_4 taper tip has a width of 200nm, while the aSi/BTO taper tip has a width of 150nm. The optimum taper of the linear adiabatic taper was calculated with full-wave 3D-FDTD calculations by sweeping the length from 7μm to 40μm, while PML boundary conditions were set again at the edges of the computational window to artificially absorb outgoing electromagnetic fields. Fig. 4.3.5(b) presents the results of the simulations with maximum efficiency obtained at L_c=16μm, reaching values of only -0.16dB. For this length, the sideview of the electric field distribution obtained at a vertical mid-plane located at the center of both the aSi/BTO and Si_3N_4 waveguides can be seen in Fig. 4.3.5(c), validating the efficient light transfer from the Si_3N_4 to the aSi/BTO platform. Clearly, the optical power couples to the TM-mode of the aSi/BTO waveguide and mostly concentrates at the top and the bottom of the aSi/BTO waveguide. From this point the optical power transfers to the final end-fire coupling step towards the plasmonic structure.

4.3.2.2: End-Fire Coupling Step and Transition Taper

The following section presents the design of the end-fire coupler stage from the aSi/BTO to the Cu/BTO/aSi plasmonic waveguide, with the overall 3D schematic presented in Fig. 4.3.6(a). Its coupling performance is dominated by the optimum spatial matching between the initially excited and the targeted mode and thus should employ a different aSi/BTO waveguide width from the one used in the adiabatic coupling stage. This sub-section is comprised of two parts: coupling from the photonic to the plasmonic waveguide and coupling from the cross-section without the metal electrodes on aSi to the cross-section with the metal electrodes on aSi enabling light modulation. The identification of the optimum geometry for both photonic and plasmonic structures was performed via spatial power overlap integral calculations. Fig. 4.3.6(b) presents the corresponding results for waveguide widths varying between 150nm and 850nm, where clearly the spatial overlap integral is improved for larger widths, as the modal field concentrates better inside the slot. For widths higher than 400nm the coupling losses reach a plateau and are equal to 1.59dB for 800nm width. At this point, the higher order modes are becoming more TM and plasmonic having increased propagation losses according to Fig. 4.3.3(e). This leads to an expected stronger interference with the fundamental plasmonic mode. For this reason, a plasmonic metal width of 500nm was selected as a compromise between high power overlap values and decreased interference phenomena. For this width, the power overlap integral value is close to 1.65dB, whereas the supported modes in the photonic aSi/BTO, narrow HPW and wide HPW geometries are depicted in Fig. 4.3.6(c), (d) and (e), respectively. The transition from the narrow to the wide HPW yielded losses between 0.2dB and 0.3dB for all widths based on the power overlap integral. The validity of the previous results was verified with full-wave 3D-FDTD numerical simulations. The total interface losses are calculated for a yz-monitor plane, at a certain distance L from the beginning of the plasmonic waveguide and include the conversion losses A_{conv} plus the total propagation losses A_{prop_L}, where A_{prop_L}[dB] = A_{prop}[dB/μm] * L[μm], with A_{prop} calculated from the 2D-eigenvalue problem. For the plasmonic

metal width of 500nm, according to the 3D-FDTD simulations, the propagation losses of the targeted plasmonic mode in the HPW waveguide were calculated as 0.5dB/μm and the conversion losses were close to 1.6dB. The slight improvement from the power overlap integral value is attributed to the effective index matching in the two waveguides, taken into account in the 3D-FDTD calculations and the beating phenomena with the hybrid plasmonic-photonic mode. A sideview (xz-plane) of the simulated geometry can be seen in Fig. 4.3.7(a) and the respective electric field distribution |**E**| is presented in Fig. 4.3.7(b), (c) and (d) for widths of 150nm, 500nm and 800nm, respectively. The above analysis reveals that the dominant loss contribution comes from the propagation losses in the plasmonic waveguide, while the modal mismatch of the photonic and the HPW mode is only minor for a width of 500nm. For a width of 150nm, near the cutoff of the hybrid mode, the beating phenomena are less intense as shown in Fig. 4.3.7(b) at the expense of higher total losses getting as high as 6.43dB, with "pure" coupling losses calculated to 4.4dB. All values are calculated at 3μm far from the end of the interface (x = 3μm). This coupling loss value is very close to the power overlap integral value presented in Fig. 4.3.6(b), as beating phenomena with the hybrid mode are becoming almost negligible for this width. Going on the other end now, for an 800nm plasmonic width the beating phenomena are more intense as shown in Fig. 4.3.7(d) and total losses are close to 2.75dB, with the clear mode conversion losses getting down to 1.37dB. However, now there is strong deviation from the power overlap integral value of Fig. 4.3.6(b), because there are more intense beating phenomena with the hybrid mode that lead to a drastic decrease of the total propagation losses. The effect of beating can be avoided though using a directional coupler between aSi and the HPW as in [29], but for such tightly confining cross-sections the coupling loss increases to values higher than 4dB. In parallel, phase-matching conditions are becoming challenging to achieve. Simulations were also performed for an increased 340nm aSi-thickness, where phase-matching with the HPW is easily achieved. This would allow the design of very efficient in-plane directional couplers featuring coupling

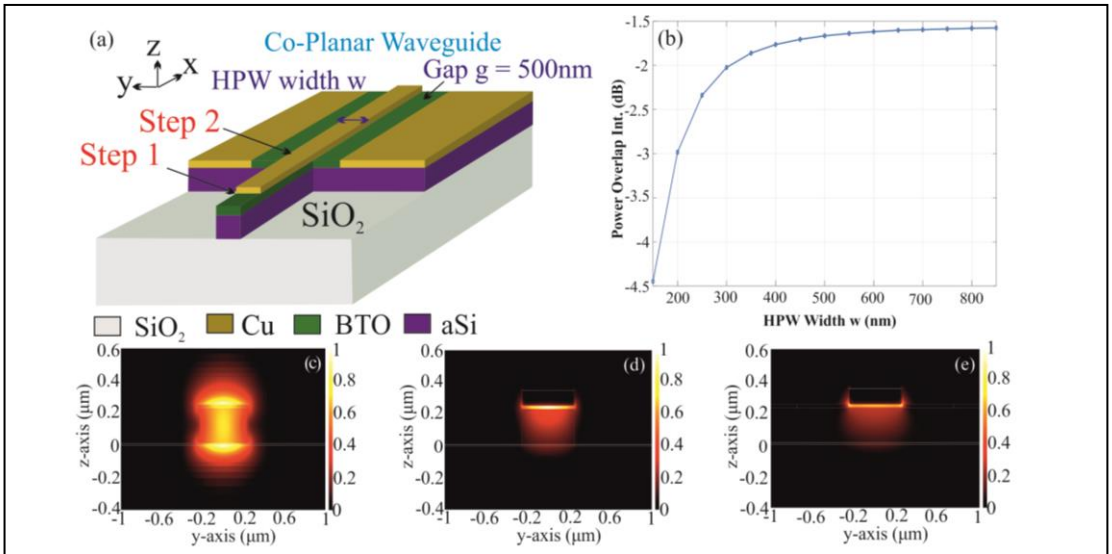


Fig. 4.3.6: (a) 3D-depiction of the end-fire coupler between aSi/BTO and plasmonic aSi/BTO/Cu (HPW) waveguide, (b) power overlap integral values for transition from aSi/BTO to plasmonic HPW geometry for various widths, Fundamental TM-mode of a (c) 500 nm-wide aSi/BTO waveguide, (d) a 500 nm wide finite aSi/BTO/Cu waveguide and (e) aSi/BTO/Cu waveguide geometry with metal electrodes enabling light modulation.

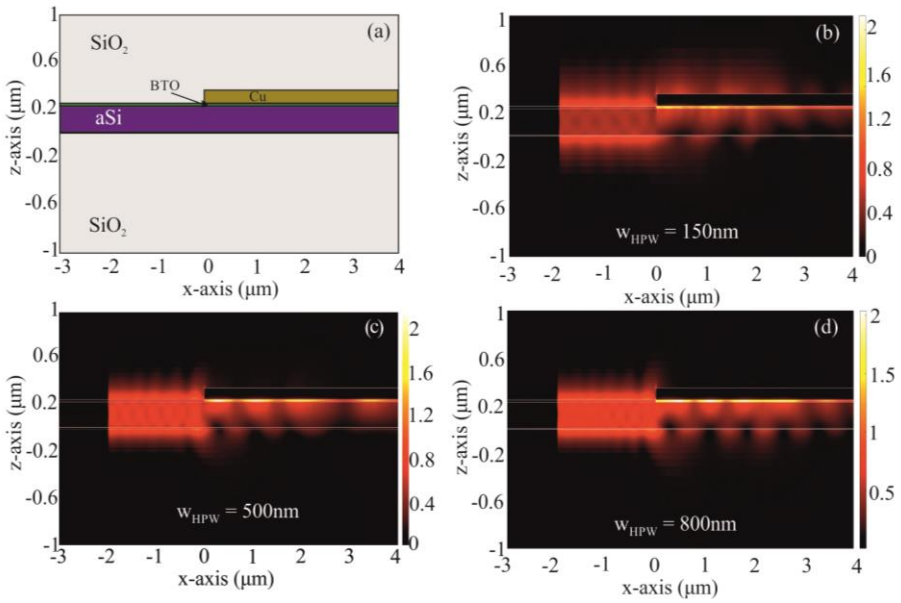


Fig. 4.3.7: (a) xz-depiction of the simulated aSi/BTO to plasmonic aSi/BTO/Cu interface, (b) xz-image of the |**E**| field distribution for HPW width of 150nm, (c) 500nm and (d) 800nm.

losses of only 1.45dB for a gap of 100nm. However, this design transfers a heavy burden to the Si_3N_4-to-aSi adiabatic coupler. The effective index mismatch for a minimum 100nm taper tip results now to 3.43dB insertion loss for this part for adiabatic coupling lengths up to 220μm, bringing the total interface loss to 4.88dB. Finally, it was calculated with 3D-FDTD simulations that, for lengths longer than 20μm, the losses from the linear adiabatic aSi/BTO taper connecting the two sections with the different widths are lower than 0.0055dB. In conclusion, the total coupling losses for this interface are lower than 1.7655dB, with 0.16dB coming from the adiabatic coupling section, 0.0055dB approximately from the transition taper step and 1.6dB from the end-fire photonic-to-plasmonic mode conversion. For futher improvement regarding the elimination of the beating phanomena, an 1μm long plasmonic taper can be inserted in the structure after the end-fire interface. This is achieved at the expense 0.26dB extra losses calculated for the transition from an input 500nm initial HPW width to a final 150nm HPW width value.

4.3.2.3: Tolerance Analysis of Interface Structures

In the analysis presented above, at the adiabatic coupling step the center positions of the aSi/BTO taper and of the Si_3N_4 taper in the y-axis are perfectly aligned. Additionally, at the end-fire step, the plasmonic metal resided exactly on top of the center of the aSi/BTO waveguide in the y-axis with a fixed metal width value. In this section the deterioration of the coupling performance is investigated via thorough 3D-FDTD calculations, for misalignment in the y-axis for 100nm, 200nm, 300nm at the adiabatic section and for metallic width variations of 50, 100 and 150nm in the end fire coupling interface as shown in the inset of Fig. 4.3.8(b). The results for the adiabatic coupling step are presented in Fig. 4.3.8(a), where excess losses close to 0.25dB are induced for a misalignment value of 300nm. Additionally, according to the results of Fig. 4.3.8(b), for the end-fire coupler there are 0.4dB extra losses for a 150 nm difference from the ideal layout. Regarding variations in the z-axis, we investigated the

impact of ±5% material thickness variations of aSi, Si_3N_4, SiO_2 gap, BTO and Cu to the whole interface performance. Herein, between the ±5% values, the worst one is referred. Initially, the thickness of the semiconductor aSi of the HPW was altered by ±5% with the 3D-FDTD simulations pointing to excess coupling losses lower than 1.14dB for + 5%, while a ±5% change in the thickness of the 20nm thick BTO led to deterioration of coupling performance by only 0.1dB for +5%. Similarly, a ±5% thickness difference of the 100nm thick Cu decreased the coupling efficiency by only 0.1dB again for +5%. The coupling efficiency was decreased by only 0.04dB for -5% Si_3N_4 thickness and finally, for a 5% variation at the dielectric SiO_2 gap the coupling losses were increased by only 0.1dB. A yz-simultaneous misalignment between the photonic and the plasmonic waveguide is not expected in this case, since both geometries are lithographically defined in the same layer.

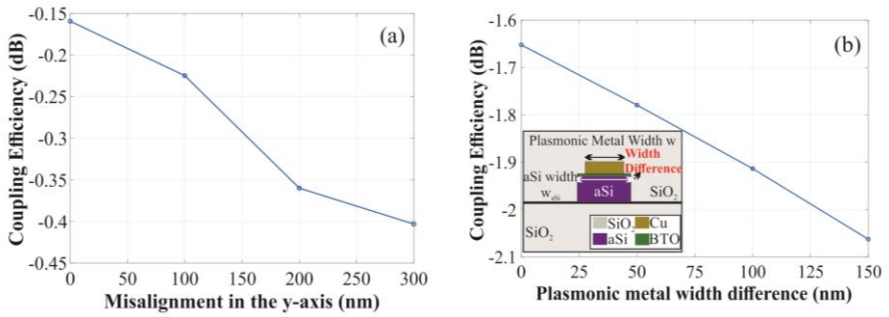


Fig. 4.3.8: (a) Coupling efficiency (dB) for increasing misalignment values (nm) of the aSi/BTO taper from the Si_3N_4 taper in the y-axis (b) Coupling efficiency (dB) for metal width difference values Δw (nm) at the end-fire part.

4.3.3: Design of couplers between Si_3N_4 and plasmonic MIM slot Cu/BTO/Cu waveguides

4.3.3.1: Adiabatic coupling configuration

This section presents the design of a multi-step coupler for efficient TE-dominant mode transition from a Si_3N_4 to a plasmonic MIM waveguide capable of light phase modulation. This multi-step coupler employs an adiabatic Si_3N_4-to-aSi stage and a directional aSi-to-Cu/BTO/Cu step. The adiabatic coupler was designed in a similar manner to the adiabatic coupler of Section 4.3.2.1, having [43] as a basis. Fig. 4.3.9(a) illustrates a 3D-schematic of the optimum Si_3N_4-to-aSi coupler geometry, while Fig. 4.3.9(b) presents the results from the 3D-FDTD simulations for the variation of the

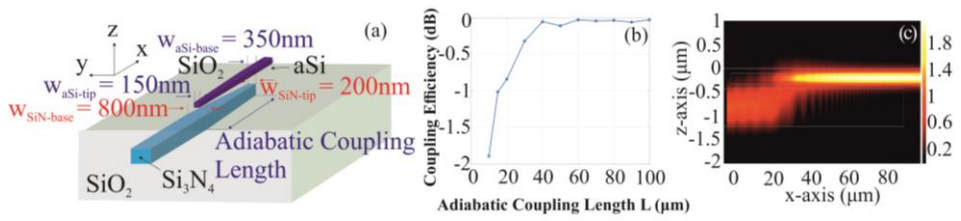


Fig. 4.3.9: (a) 3D-depiction of the designed Si_3N_4-to-aSi adiabatic coupler, (b) Coupling efficiency (dB) for increasing adiabatic taper length, (c) Sideview (xz-plane) depiction of the |**E**| field distribution at the center (y=0) of the adiabatic coupler.

taper length between 10μm to 100μm, where it is evident that the coupling efficiency saturates after 60μm, while a 90μm adiabatic coupling length is almost lossless (losses of 0.06dB). This is verified by the xz-plane of the |**E**| field distribution shown in Fig. 4.3.9(c) demonstrating an efficient light transition from the aSi to the Si_3N_4 waveguide. For this interface configuration, the aSi width of 350nm is also suitable for the next coupling step to the plasmonic MIM waveguide, obeying the phase matching condition for directional coupling to the 100nm wide plasmonic slot.

4.3.3.2: Directional Coupling to the Plasmonic Slot

A directional coupling technique was selected for the light transfer from the aSi to the plasmonic slot targeting the minimization of the coupling length contributing with extra propagation losses. The 3D-schematic of the whole layout is presented in Fig. 4.3.10(a). The 2D-eigenvalue problem was initially solved for the uncoupled aSi and Cu/BTO/Cu slot waveguides and then for the coupled waveguide system to show the existence of supermodes. As shown above in Fig. 4.3.4, for a slot width of 100nm in the MIM waveguide and an aSi waveguide width of 350nm, the two cross-sections in the uncoupled eigenvalue problem support TE-modes with almost the same effective refractive index. For this reason, these geometrical characteristics set favorable conditions for the hybrid plasmonic waveguide system to support supermodes of even and odd symmetry that can efficiently be excited in a directional coupling scheme. These modes are easily detected via eigenmode calculations at the hybrid waveguide yz-plane of Fig. 4.3.10(a). Fig. 4.3.10(b) and 4.3.10(c) show the real part of the y-vectorial components of the electric fields Re{E_y}of the two TE-supermodes exhibiting an even and an odd symmetry, respectively. The even supermode of Fig. 4.3.10(b) mostly resides inside the slot and therefore has higher propagation losses of 0.46dB/μm, while the odd mode of Fig. 4.3.10(c) has a large power percentage residing in the center of the aSi waveguide and thus, the propagation loss value is a little bit lower to 0.33dB/μm. This phenomenon is a direct outcome of the overall photonic-plasmonic mode hybridization in the hybrid waveguide plane. After the excitation of the final plasmonic slot mode by the end of the hybrid waveguide region, the propagation losses of the plasmonic mode are the ones calculated by the 2D-eigenmode solution for the slot.

Utilizing the modes of even and odd symmetries shown in Fig. 4.3.10, the beating of the two supermodes can lead to directional coupling at a coupling length L_c that has been calculated initially to L_c = 2.67μm in a semi-analytic manner. Full-wave 3D calculations were used to reveal the transition losses for such a length with the sideview

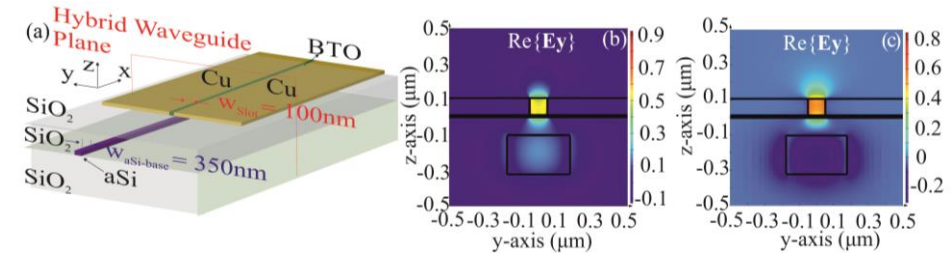


Fig. 4.3.10: (a) 3D-depiction of the aSi-to-MIM directional coupling step, (b) Even and (c) Odd hybrid supermodes supported by the coupled aSi and plasmonic waveguide system.

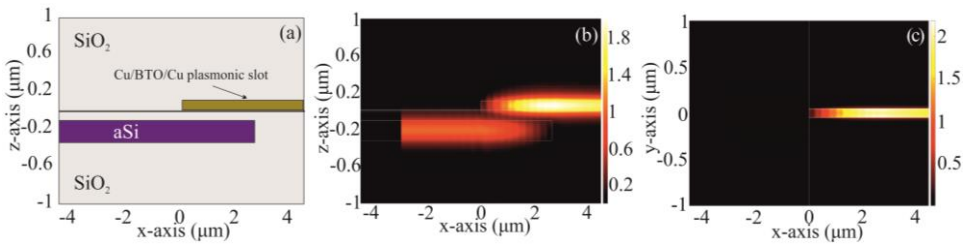


Fig. 4.3.11: (a) Sideview (xz-plane) depiction of the simulated directional coupler geometry, (b) Sideview (xz-plane) depiction of the |**E**| field distribution at the center (y=0), and (c) Topview (xy-plane) depiction of the |**E**| field distribution at the center (z=0.06μm) of the aSi-to-MIM directional coupler

of the directional coupler seen in Fig. 4.3.11(a). The interlayer SiO_2 layer was set again to 100nm, while the coupling length is the overlapping region of the aSi and the slot waveguides. The 3D-FDTD simulations reveal now an efficient transition with total losses close to 1.23dB, coming from a monitor placed at x = 3μm, directly after the coupler, and this value includes the propagation losses of the two supermodes in the hybrid coupler. This is validated by the retrieved electric field distribution at the xz-plane (sideview), of the electric field |**E**| presented in Fig. 4.3.11(b). As shown also in the topview of the |**E**| field in Fig. 4.3.11(c) at the end of the interface, light is tightly confined in the Cu-based plasmonic slot. The light transfers to the slot in a reflectionless and low-loss manner, with the field values increasing due to the ultra-high mode confinement of the final mode inside the CMOS plasmonic slot. In contrast with the hybrid plasmonic waveguide, in the final plasmonic cross-section only the fundamental plasmonic mode has been excited in the phase shifter, highlighting the slot waveguide as the most promising of the two configurations for light modulation.

4.3.3.3: Tolerance Analysis of Interface Structures

As in the HPW case, the waveguides at the adiabatic and the directional coupling step were perfectly centered on top of one another in the y-axis. Figure 4.3.12 illustrates the effect of misalignments of the upper waveguide layer in the y-axis from the center of the propagation axis to the efficiency of the adiabatic and the directional coupling step, respectively. According to Fig. 4.3.12(a) in the worst-case scenario, a 300nm misalignment at the adiabatic step leads to 1dB excess loss from the optimum placement, while from Fig. 4.3.12(b) a y-axis 200nm misalignment at the directional coupler raises the total loss by 2.6dB. The same graph reveals also that for 300nm there

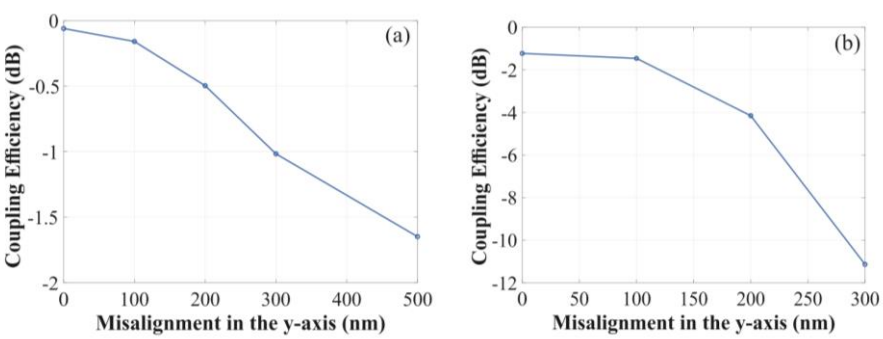


Fig. 4.3.12: (a) Coupling efficiency (dB) for increasing misalignment values (nm) of the aSi taper from the Si_3N_4 taper in the y-axis (b) Coupling efficiency (dB) for increasing misalignment values (nm) of the aSi from the MIM waveguide in the y-axis.

is a sharp drop in the efficiency going down to -11.13dB. The next step in the tolerance analysis is again the investigation via 3D-FDTD simulations of the impact of a ±5% variation on the thickness of the aSi, Si_3N_4, SiO_2 gap, BTO and Cu materials to the whole coupling efficiency performance. Again, the worst case of the two variations is presented. A -5% variation to the SiO_2 gap leads to 0.004dB extra losses to the adiabatic coupler and 0.05dB to the directional coupler, highlighting that the device is quite tolerant to gap thickness alterations. A -5% variation to the thickness of Si_3N_4 increases the losses by only 0.011dB, while the thickness of the aSi was the most critical of the parameters. -5% thickness from the targeted 220nm value, enhances the losses by 0.4dB. Finally, by modifying the thickness of the whole plasmonic slot by 5%, namely the thickness of the BTO and Cu materials, the extra induced losses did not surpass 0.02dB. So, the same interface design can be directly transferred to even thicker BTO plasmonic phase shifters targeting higher efficiency with minimum coupling penalty.

4.3.4: Comparison with other CMOS plasmo-photonic interfaces

Table 1 summarizes the main results of our work and compares them with significant relevant work presented in the literature up until now regarding Si- and Si_3N_4-based photonic to CMOS compatible plasmonic interfaces. The proposed interfacing scheme utilizes a Cu metal for the plasmonic formation and a BTO electro-optic material suitable for pure phase modulation in very simple configurations, unlike [35] and [36] that incorporate an amplitude only modulator. The better overall performance in [35] comes from the large modal overlap and the MOS type of the plasmonic waveguide having large semiconductor layer thicknesses. In active modulation, the capability of such devices for operation at higher than 100Gbaud rates is questionable due to the increased capacitance value. The same is valid for [36], where the simulated Si-to-HPW coupling losses are close to the calculated value for the aSi-to-HPW step in our work. Compared to [40] and [38] also that perform coupling to Cu-based slots filled by air and SiO_2 respectively, the final plasmonic waveguide in this work is a slot filled by an electro-optic material suitable for modulation in the O-band that provides much higher confinement for high modulation efficiency. In [38] the losses are higher than the results of this work. In [40] the slot width is relaxed to 200nm to achieve proper phase-matching and facilitate the photonic to plasmonic transition. The difference in the effective indexes between the initial photonic and targeted plasmonic mode was very small, less than 0.3. Finally, [39] demonstrates a higher confinement factor for efficient high-speed electro-optic modulation, but the interface losses are slightly higher. This comes as no surprise since from the analysis above it is apparent that there is a tradeoff between coupling efficiency and confinement of the mode in the phase shifter. Finally in comparison to [39], the coupling scheme to the examined HPW structure induces lower losses, while it is also capable of electro-optic modulation. Overall, between the HPW and MIM coupling configurations presented herein, the plasmonic slot structure manages to retain high optical confinement and achieves low-loss excitation of a single mode in the final phase-shifter cross-section, thereby being considered the best of the two proposed designs in this Chapter.

Table 4.1. Comparison of the presented Si_3N_4-to-(Cu-Plasmonic) CMOS Plasmo-Photonic Waveguide Coupling Interfaces with the State of the Art (Sim. = Simulated, Exp. = Experimental)

Ref.	Plasmonic Prop. Loss (dB/μm)	Conf. Factor (%)	Coupling Efficiency (dB)	Utilized Materials	Polarization	Band
[40]	0.21	25.24	-1.15 (Sim.)	Si_3N_4-to-Open-cladded-Cu-slot	TE	C
[35]	0.29	_	-0.7 (Sim.)	Si-to-Si/TiN/Si_3N_4/GZO/Si	TM	C
[38]	_	_	-1.55 (Sim. and Exp.)	Si-to-Cu/air/Cu slot	TE	C
[39]	0.78	90	-4.2 (Exp.)/-2.5 (Sim.)	Si_3N_4-to-(Cu/Si_3N_4/Cu HPW)	TE	C
[39]	0.37	87	-2.84 (Exp.)/-1.9 (Sim.)	Si_3N_4-to-(Cu/SiO_2/Si_3N_4/SiO_2/Cu HPW)	TE	C
[36]	0.025	<20%	-1.14 (Exp.)/-1.61 (Sim.)	Si-to-CHPW	TM	C
This work	0.5	31.49	-1.75	Si_3N_4-to-(Cu/BTO/aSi HPW)	TM	O
This work	0.78	56.18	-1.29	Si_3N_4-to-(Cu/BTO/Cu MIM slot)	TE	O

4.4: Alternative aSi/BTO-to-HPW Coupling Configurations

To fully address the question about the optimum design for the aSi-to-HPW interface geometry, the interface was reevaluated through an extensive simulation study covering the four following cases: i) an adiabatic, ii) an end-fire + tapered, iii) a vertical and iv) an in-plane directional coupling configuration, as potential geometries providing better overall performance vs. what is presented in the Section 4.3. It should be mentioned that for the discussion of the results presented herein we are assuming 100nm minimum gaps and waveguide widths, easily compatible with e-beam lithography tools and for parity with the results of Section 4.3.

The four potential coupler designs are illustrated in Fig. 4.4.1. Fig. 4.4.1(a) depicts an adiabatic coupler where the plasmonic metal width is increased, while the aSi/BTO photonic waveguide width is reduced. Fig. 4.4.1(b) on the other side presents a plasmonic taper design right after the end-fire coupler. The rationale behind the

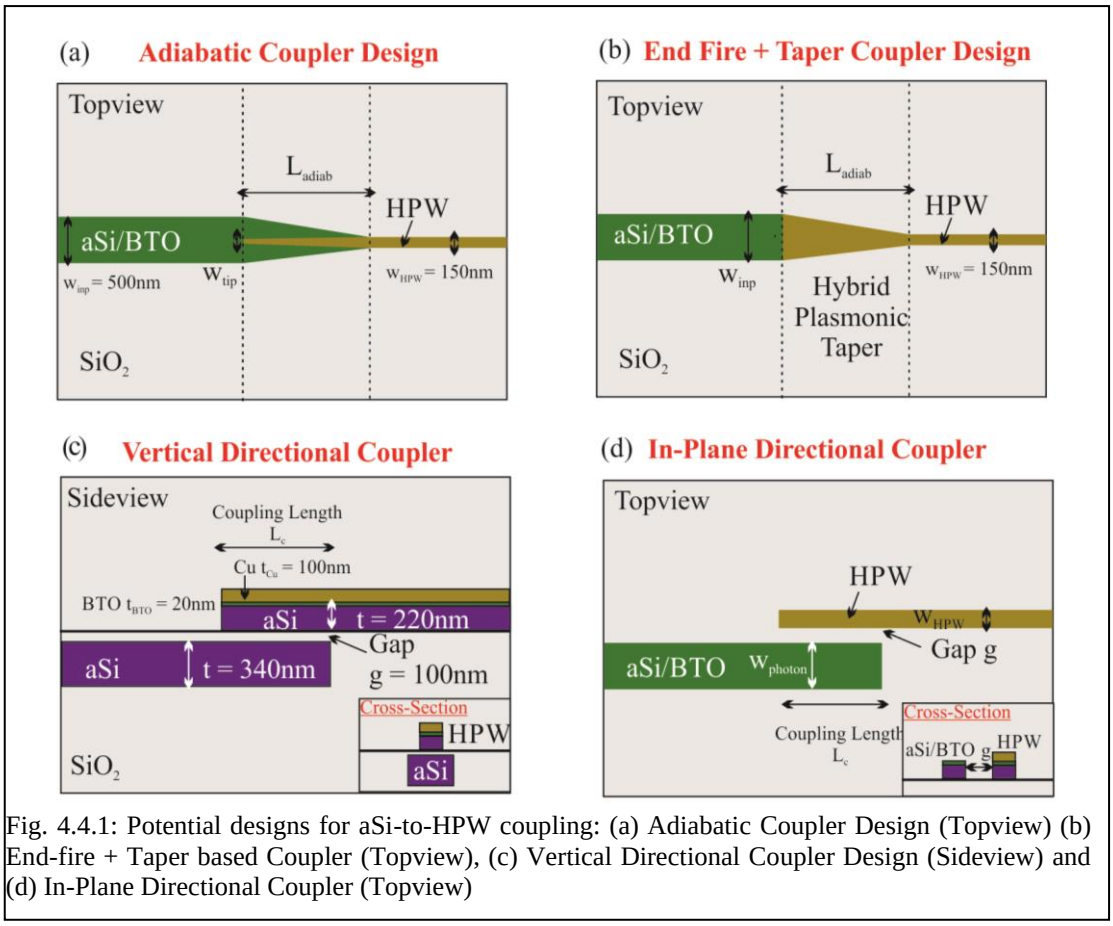


Fig. 4.4.1: Potential designs for aSi-to-HPW coupling: (a) Adiabatic Coupler Design (Topview) (b) End-fire + Taper based Coupler (Topview), (c) Vertical Directional Coupler Design (Sideview) and (d) In-Plane Directional Coupler (Topview)

utilization of a plasmonic taper lies at the potential achievement of both high coupling efficiency and the suppression of beating phenomena. Finally, the directional coupler designs of Fig. 4.4.1(c) and Fig. 4.4.1(d) can achieve the sole excitation of the targeted fundamental TM-mode in the Hybrid Plasmonic Waveguide, thus completely avoiding beating phenomena, as also shown in Section 4.3 for the aSi-to-slot TE configurations. The directional coupler of Fig. 4.4.1(c) is vertical, forming a tri-layer coupler in the overall Si_3N_4-to-HPW design, while the coupler of Fig. 4.4.1(d) is in-plane.

The analysis begins from the adiabatic and tapered coupler designs of Fig. 4.4.1(a) and Fig. 4.4.1(b). The obtained coupling efficiency for an L_{adiab} = 1μm adiabatic coupling length and for an increasing taper tip width w_{tip} is demonstrated in Fig. 4.4.2(a). It is clear that the efficiency is larger when the taper tip width is the same as the HPW width. This is attributed to the fact that the supported plasmonic mode at the narrower metal widths for large underlying material layer widths, is a very confined mode near the surface of the metal attaining an effective index value $Re\{n_{eff}\}>3$. This eventually results to both spatial and phase-mismatch from the initial photonic mode and hindering optimum coupling. It is clear from the simulations that the coupling efficiencies of Fig. 4.4.2(a) are much lower than the calculated -1.75dB reported for the Si_3N_4-to-plasmonic TM coupling interfaces in Section 4.3.

On the other side, the interface of Fig. 4.4.1(b) could potentially be proposed to achieve high coupling efficiency, while suppressing beating phenomena observed and analyzed in Section 4.3.2. A parameter sweep was carried out for input aSi/BTO waveguide widths of 500nm and 800nm, and a final HPW width of w_{HPW} = 150nm, shown to induce weaker beating phenomena in the end-fire configuration of Section

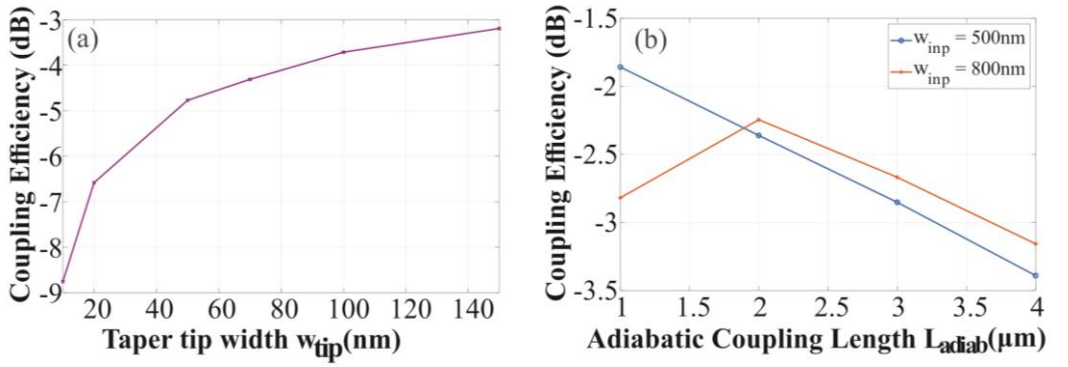


Fig. 4.4.2: (a) Coupling efficiency vs Taper tip width for the adiabatic coupler design of Fig. 1(a), (b) Coupling efficiency vs Adiabatic coupling length for input waveguide widths of 500nm and 800nm for the end-fire + tapered coupler design of Fig. 1(b).

4.3. The coupling efficiency results are obtained for an increasing plasmonic taper length, with a step of 1μm. For all lengths shown in Fig. 4.4.2(b), the obtained coupling efficiency is lower than -1.75dB presented for the whole Si_3N_4-to-HPW interface in Section 4.3. The extra losses come from the plasmonic taper that for an increasing length induces extra light absorption from the metal, leading to a reduced overall calculated coupling efficiency. In reality, this interface includes both an end-fire and a taper-down step and, for this reason, the coupling efficiency is lower than the single end-fire step. However, this is indeed an effective way to eliminate the beating phenomena thoroughly analyzed in Section 4.3.2. For the best-case scenario of an input HPW waveguide width of 500nm, an output HPW width of 150nm and an adiabatic taper length of 1μm, extra 0.26dB losses are induced by the plasmonic taper, after taking out the losses from the 500nm wide end-fire coupling interface. The total coupling losses of the aSi-to-HPW coupler including the photonic-to-plasmonic interface and the optional plasmonic taper are close to 1.86dB.

Fig. 4.4.1(c) and Fig. 4.4.1(d) present two potential configurations for directional coupling. In the first figure, the HPW structure resides on top of an intermediate aSi waveguide, similar to the TE-coupling configuration presented in Section 4.3. The aSi resides on top of the Si_3N_4 layer separated by a 100nm SiO_2 inter-layer distance. The cross-section of the coupled waveguide system can be seen at the inset of Fig. 4.4.1(c). In the second one, an aSi/BTO waveguide resides in-plane next to a HPW structure. Light is side coupled from one waveguide to the other. Directional couplers are mostly dominated by the phase-matching condition, and the effective indices $Re(n_{eff})$ of the initial and targeted modes in the uncoupled waveguides need to approach one another, in order to achieve optimum light transfer from the first waveguide to the second one. For this purpose, the eigenvalue problem was solved at the cross-sectional planes of the independent aSi, aSi/BTO, and HPW waveguides for two aSi thicknesses of 220nm and 340nm and for varying widths. The results can be seen below in Fig. 4.4.3(a) and Fig. 4.4.3(b). As mentioned also in Section 4.3, for the aSi thickness of 220nm both aSi and aSi/BTO waveguides support a TM mode with an effective index $Re(n_{eff})$ very far from the effective index of the targeted TM-mode in the hybrid plasmonic waveguide. For an increased aSi thickness of 340nm, phase-matching conditions can take place between the aSi waveguide and the overlaying HPW structure of Fig. 4.4.1(c) with 220nm aSi thickness and for respective waveguide widths of w_{aSi} = 270nm and w_{HPW} = 100nm. As a next step, as mentioned also in the methodology for the design of the

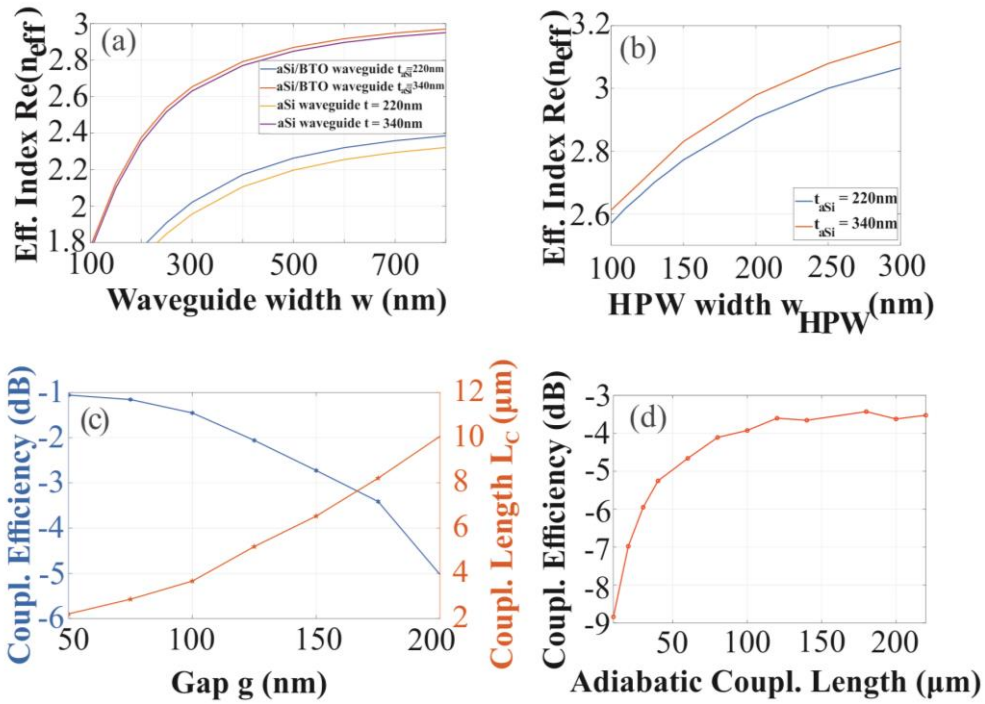


Fig. 4.4.3: (a) Effective index Re(n_{eff}) vs photonic waveguide widths for aSi/BTO waveguides with aSi thicknesses of 220nm and 340nm and aSi waveguides with thicknesses of 220nm and 340nm, (b) Effective index Re(n_{eff}) vs HPW width for aSi thickness of 220nm and 340nm, (c) Coupling efficiency and coupling length vs gap for in-plane directional coupler with w_{aSi} = 300nm and w_{HPW} = 100nm, (d) Coupling efficiency vs Length for adiabatic coupler with aSi thickness of 340nm and minimum aSi taper width of 100nm.

vertical TE directional coupler in Section 4.3, the eigenvalue problem was then solved at the cross-sectional plane of Fig. 4.4.1(c) to identify supermodes of even and odd symmetry. The coupling length is the length needed for the coupled waveguide region to achieve maximum light transfer from the aSi waveguide to the HPW. Nevertheless, the plasmonic nature of the supermodes requires large coupling lengths in the order of 20μm, where those in turn are translated to larger propagation losses and thus a calculated coupling efficiency of -8dB. For this reason, this coupling configuration is abandoned. On the other side the layout of Fig. 4.4.1(d) presenting a side-coupled aSi/BTO photonic and a HPW waveguides with widths of w_{aSi} = 300nm and w_{HPW} = 100nm, is able to achieve phase-matching by also attaining small enough coupling lengths. These results are presented in Fig. 4.4.3(c) for different gap values g and taking again as a reference the 100nm gap, the coupling loss is 1.45dB. However, the 340nm thickness of the aSi, as it shown in Fig. 4.4.3(a) indicates that it is impossible to achieve an effective index Re(n_{eff})<1.8, while maintaining compatibility with 100nm lithography. This of course then induces problem to the adiabatic coupling of the TM mode from Si_3N_4 to the aSi/BTO waveguide that is verified with the results of Fig. 4.4.3(d), demonstrating a coupling efficiency lower than -3.43dB for adiabatic coupling lengths up to 220μm. The saturation of the coupling efficiency near this value for lengths larger than 140μm suggests that higher coupling efficiencies are not achievable even for higher lengths having a minimum taper tip width dimension of 100nm for the aSi. So, overall if we add the two sections, we get a minimum coupling loss of 4.88 dB that is worse compared to the results of Section 4.3. For this reason, the end-fire coupler

is still considered the optimum design in Section 4.3 for the targeted material thicknesses and envisaged overall fabrication process.

4.5: Broadband Response

In Section 4.3 the optical properties (such as the refractive index) of the ferroelectric $BaTiO_3$ material were provided only at 1310 nm wavelength. Therefore, our simulations were focused only on this wavelength, instead of providing broadband transfer functions for the whole O-band. Nevertheless, to address the broadband behavior of the designed interfaces, 3D-FDTD simulations were performed by considering the dispersion of the rest of the materials in the O-band (1260nm-1360nm). The refractive index of aSi was approximated by Si in [48], while the refractive index of Cu was obtained from the refractive index data of [34]. The BTO was considered as a dielectric with a stable refractive index *n* in the whole of O-band. The power transfer functions for each presented optimized coupler design are included herein to demonstrate that the configurations are properly working in the entire O-band.

The transfer function calculation was performed for both steps i.e. adiabatic and directional or end-fire couplers for both TE and TM polarizations, with the corresponding results indicated in Fig. 4.5.1(a)-(f). The graph illustrates that each designed coupling step, as well as the overall TE and TM coupling interfaces discussed in Section 4.3, demonstrate a 1-dB bandwidth exceeding 100nm. This effectively covers the wavelength range of 1260nm to 1360nm.The coupling efficiency at Fig. 4.5.1(b) and Fig. 4.5.1(d) includes small propagation losses in the HPW and MIM slot waveguide respectively, with the monitor in the first case being placed at x = 3μm, and the monitor at the second case being placed at x = 1μm. However, since the BTO is the foundation of the whole platform, the study is neglecting in the whole analysis a very crucial parameter, the material dispersion of the BTO, that can significantly alter the overall performance results, especially for the directional coupler.

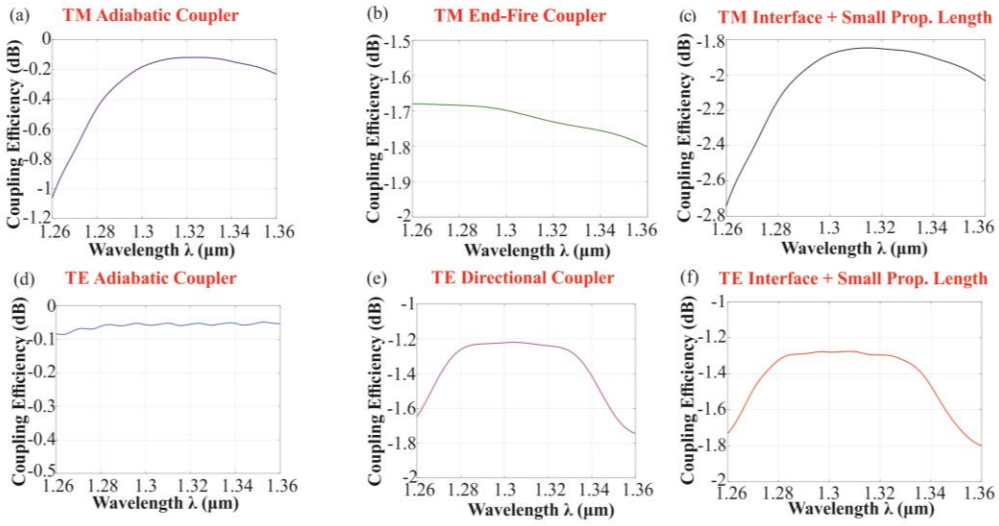


Fig. 4.5.1: Power Transfer Functions for the (a) TM Si_3N_4-to-aSi/BTO adiabatic coupler, (b) TM aSi/BTO-to-HPW end-fire coupler, (c) Overall TM Si_3N_4-to-HPW coupling interface (d) TE Si_3N_4-to-aSi adiabatic coupler, (e) TE aSi-to-MIM Plasmonic Slot Directional Coupler, (f) Overall TE Si_3N_4-to-MIM coupling interface.

4.6: Chapter Conclusions

In this chapter, a multi-step coupling technique was presented for the tri-layer integration of plasmonic slot (Metal-Insulator-Metal or MIM) phase shifters on a thick Si_3N_4 waveguide platform. A tri-layer Si_3N_4-aSi-MIM integration ensures the decoupling of the phase-shifter design from the coupling interface design. First the suitable phase-shifter design is selected and then the coupling interface design is adapted to facilitate optimum coupling for the specific cross-section. In contrast with all publications in the literature the plasmonic waveguides are based on CMOS compatible plasmonic metals. Therefore, the integration scheme presented herein can be easily adapted by the CMOS industry to provide large scale plasmonic modulator devices on the Si_3N_4 platform. More specifically, this work presents a systematic analysis for the design of low-loss multi-step couplers between Si_3N_4 and Cu based CMOS plasmonic waveguides with ferroelectric BTO material for phase shifters operating in the O-band. The HPW waveguides provide lower plasmonic propagation losses of 0.5dB/μm, compared to 0.78dB/μm for the MIM. The MIM though is preferable for light modulation as it features higher field enhancement and modal confinement in the slot calculated to 56.2% versus the 31.5% in the HPW. The calculated loss values coming from 3D-FDTD simulations are 1.75dB for the Si_3N_4-to-HPW interface with 20nm thick BTO-filled gap and 1.29dB for the Si_3N_4-to-horizontal-MIM coupler with 100nm wide slot and 100nm thick BTO. The overall analysis shows that the MIM waveguide is the preferred layout for phase shifters as it achieves the sole excitation of the targeted plasmonic mode that is highly confined in the slot at the expense of higher propagation losses. In the HPW design, an 1μm long plasmonic taper after the end-fire interface can suppress beating at the expense of extra 0.26dB coupling loss. Finally, for manufacturability purposes the work presents a detailed tolerance analysis to lateral misalignment values between all sections for both designs. In addition, performance tolerance results were extended to a ±5% thickness variation of each material, including dielectrics and metals, with 3D-FDTD simulations indicating again almost negligible power penalty for these scenarios. The results are expected to pave the way for the realization of low insertion loss, compact and low-power CMOS plasmonic phase shifters on Si_3N_4 in the O-band.

4.7: Chapter References

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Chapter 5: Beyond 100Gbaud Plasmonic MZIR Modulators on Si_3N_4 in the O-band

5.1: Modulation for Intra-DC and Inter-DC Communications

In the previous two Chapters, we presented a coupling technique for the monolithic co-integration of InP-based QD-laser sources with the passive 800nm thick Si_3N_4 platform. Towards the development of a complete datacom transceiver on the Si_3N_4 platform, we demonstrated a multi-step coupling technique for the integration of passive Si_3N_4 waveguides with plasmonic $BaTiO_3$ ferroelectric phase-shifters. The question that remains is the design and system-level evaluation of novel modulator devices on the co-integrated high-speed Si_3N_4 platform of Chapter 4. Therefore, this Chapter will deal with the introduction and investigation of plasmonic $BaTiO_3$ Mach-Zehnder in a Ring (MZIR) modulators that require lower applied voltage compared to the symmetric MZI counterparts, achieving lower insertion losses, and being able to deliver over 100Gbaud PAM-4 signals.

Before the design of MZIR modulators is discussed, this Section will focus on the fundamentals of modulation for optical communication systems. This Section will only serve as an overview, not going thoroughly to the mathematical details. For a thorough analysis of modulation in optical communication systems references [1-3] are suggested. A simplistic view on optical communication systems can separate and categorize them, even the largest systems, in three fundamental parts: the transmitter, the transmission medium, and the receiver. This is schematically illustrated at the inset of Fig. 5.1.1. The transmitter is based on a laser optical carrier source and a modulator

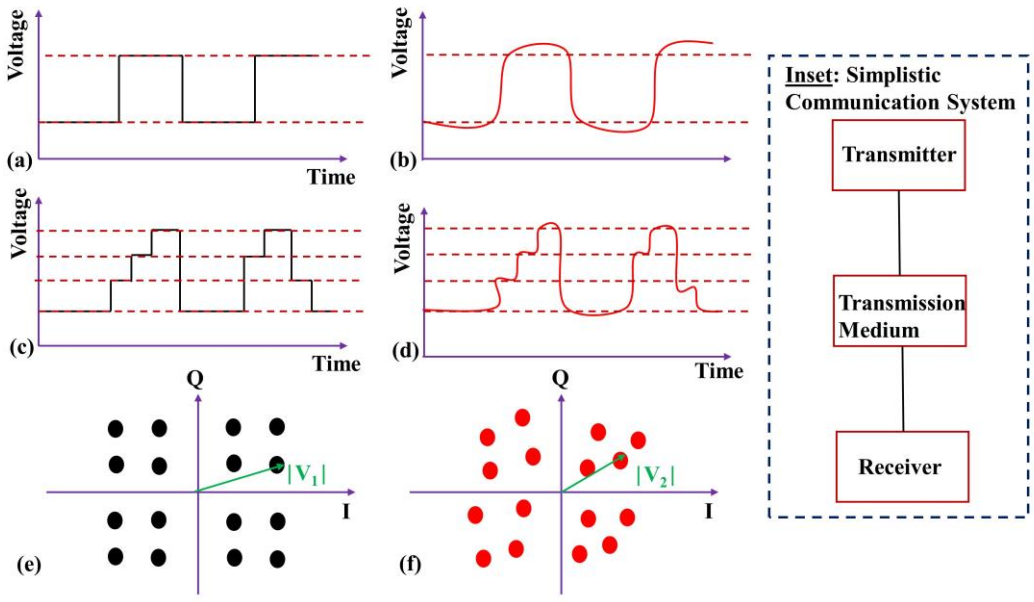


Fig. 5.1.1: Schematic of (a) electrical transmitted OOK modulated signal, (b) received optical OOK signal from modulator, (c) electrical PAM-4 modulated signal and (d) received optical PAM-4 signal from modulator, (e) electrical 16-QAM modulated signal and (f) 16-QAM signal after optical modulator and after coherent reception.

circuit imprinting an applied electrical analog signal to the carrier. The receiver is usually based on a photodiode and the electrical circuitry processing the received signal for subsequent digital processing. The transmission medium can include optical fibers, multiplexers, amplifiers and other elements. All these elements intoduce noise to the propagating signal, while the transmission medium introduces both dispersion and losses. Losses can also be introduced by any interfaces between optical elements and by external elements such as external modulators in transmitters.

A variety of modulation formats can be used in Direct Detection. The simplest modulation format is 2-level Pulsed-Amplitude Modulation (PAM-2) or On-Off Keying (OOK). In this case a bit sequence encoding information to two voltage levels of an electrical signal is applied to an optical modulator as seen in Fig. 5.1.1 (a). In the received optical signal, an electronic circuit eventually decides if the transmitted bit was the one that was transmitted. Due to induced losses, noise, and dispersion in the transmission, the received bit sequence is not directly the same as the transmitted one as shown in Fig. 5.1.1 (b). The response of the optical signal to the sudden increases or decreases in voltage has a curved shape. The response time is connected with the Electro-Optic Bandwidth (EO-BW) of the modulator and the other associated electronic and opto-electronic ICs, e.g. drivers, PDs, TIAs used in the line. This is the simplest modulation format where a single symbol corresponds to a single bit based on the $\log_2(N)$ rule. The NRZ OOK modulator format has been used since the inception of optical communications. However, increases in the required data rates per lane have led to more complex modulation formats.

To take a step further, the applied electrical signal can be imposed to multiple voltages and applied to the optical modulator in a single bit. Information can thus be encoded to multiple bits per symbol. Communication systems such as intra-data-center communications have begun to adapt PAM-4 modulation for short distances to increase the bit rate over 100Gb/s (50Gbaud) per lane. In this case two bits can be transmitted with a single symbol. This significantly decreases the required bandwidth of the devices utilized in the system as the bit rate is double the symbol rate. A PAM-4 modulation is schematically described in Fig. 5.1.1(c). The electrical signal is encoded to 4 distinct voltage levels for a single bit period. In the received signal of Fig. 5.1.1(d) the voltages may be below or above the transmitted voltages. The decision circuit determines if the received bits match with the transmitted ones or not. The rate of error bits compared to the total bit number is called Bit Error Rate (BER) and is used in any type of PAM-N modulation. For PAM-2 and PAM-4 signals a quality factor describing the quality of the transmitted signal by the modulator can be defined and is called the Q-Factor. The Q-Factor value for PAM-2 and PAM-4 signals is related with the expected Bit Error Rate (BER) in a simple transmission system. Eye diagram monitors can record the transmitted and received signal to qualitatively check the quality of the received signal and to measure the Q-Factor and Bit Error Rate. PAM-2 and PAM-4 signals are the workhorse of lower-rate and short-distance intra-data-center networks. Going to higher PAM-N modulation formats there are significant limitations due to the symbols coming much closer to one-another, requiring much higher OSNR and bandwidth across the entire transmission line.

On the other side, inter-DC links require high bandwidth for transmission over large distances. For this reason, coherent transmission systems demand phase and amplitude

modulation used at the same time to deliver even higher data rates. This is the case in N-symbol Quadrature Amplitude Modulation (QAM-N), while Quadrature Phase-Shift Keying (QPSK) modulates only the phase element. This allows the delivery of very large rates to very high distances since both formats efficiently use the available bandwidth by allowing the transmission of multiple bits per symbol. Especially QAM modulation combines amplitude and phase-states leading to a denser information representation in the signal constellation diagram. Finally, unlike PAM-N modulation schemes, phase coherence in QAM and QPSK allows for signal integrity over large distances with robustness to noise and reduced error rates. To ensure the carrier maintains high phase stability, laser sources with an exceptionally low linewidth, typically below 100kHz are necessary. Moreover, the receiver circuit is considerably more intricate than a basic photodiode paired with a transimpedance amplifier (TIA). involving a sophisticated photonic integrated circuit. The transmission setup calls for a coherent transmitter, while the receiving end utilizes a coherent receiver. Signal constellation diagrams are used to represent QAM signals as deployed as shown in Fig. 5.1.1(e) for a QAM-16 signal. In the diagram each point is related with a specific amplitude and phase value. The I axis is related to the amplitude and the Q axis is related to the phase of a signal. In the received signal, due to noise, dispersion, and losses, the symbols are not in their initial position in the constellation diagram, as shown in Fig. 5.1.1(f). The deviation of each symbol from their expected ideal position in the 2D diagram can be measured via the difference of the vector magnitude in the transmitted and received signals. This error to the vector magnitude is called Error Vector Magnitude or (EVM) in coherent communication systems. Complex digital signal processing (DSP) is in most cases applied to improve the quality of the received signal. Coherent communications impose complexity, higher cost, and power consumption to the overall system. For these reasons, they are employed mostly for transmission in very large distances (>50km) in inter-data-center communications, where much higher bandwidth is required between the nodes. Nevertheless, the ever increasing data rate requirements of the last few years have brought forth discussions and proposals on the potential use of coherent optical interconnects inside the data-centers [4, 5].

In all cases, the electro-optic modulator element is fundamental for both intra-data-center and coherent inter-data-center communications. Due to the very high cost and energy requirements for coherent intra-DC, in this thesis, Intensity Modulation – Direct Detection Transmission Systems (IM/DD) will be mainly considered, where the optical modulator is the prevalent bulding block. A schematic of a simple IM/DD system is shown in Fig. 5.1.2. In the case of Fig. 5.1.2(a) the laser emitting light is directly modulated by a PAM-N modulated electrical signal, brought forth by a pseudorandom bit sequence. The signal propagates in a single-mode optical fiber (SMF) and is received by a photodiode. The electrical signal is filtered by a 4th order Bessel Filter to emulate the bandwidth of the PD. The quality of the signal is visualized through an Eye-Diagram Monitor. On the other side, Fig. 5.1.2 (b) employs an external modulator element. The focus will now be on the frequency-domain. The emitted signal by the laser is emitted power at a single frequency (carrier frequency) with a certain bandwidth value, determined by the linewidth of the laser. Modulation of the optical signal results to the emergence of two sidebands (sub-carriers) left and right from the carrier

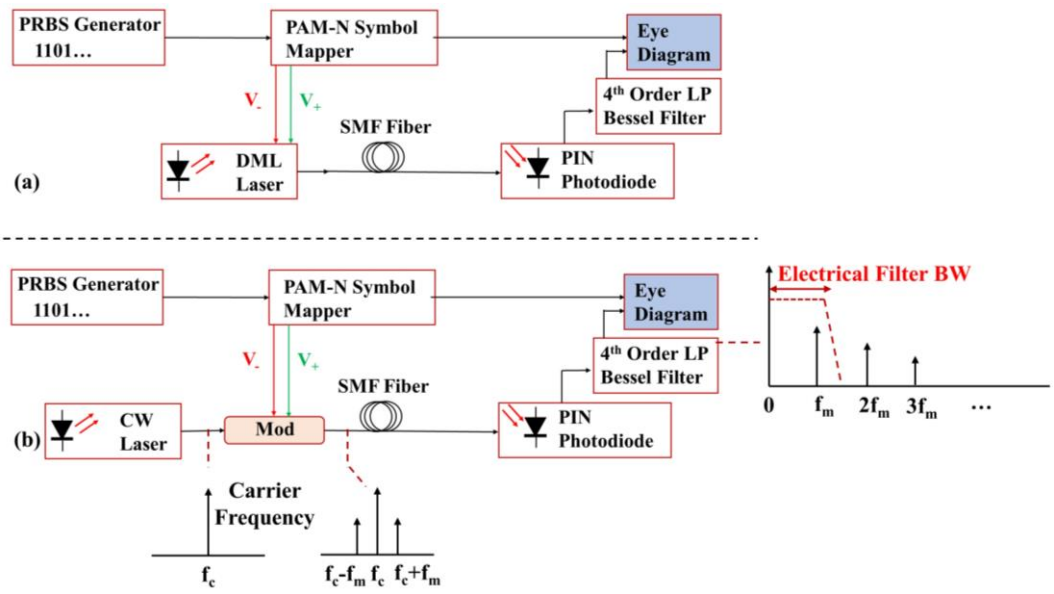


Fig. 5.1.2: Schematic illustration of (a) a DML based transmission setup and (b) a transmission system based on an external modulator.

frequency. The reception of the signal has, as a result, the reception of only the upper sideband. The first harmonic is received while the others are suppressed via the 4th Order Low-Pass Filter.

To conclude, both IM/DD systems and coherent communication systems as shown above require optical light sources, phase-modulating and amplitude modulating elements co-integrated with the Si_3N_4 platform. To easily achieve amplitude modulation, optical interferometric circuits are of interest, since they can deliver more complex modulated signals. Most works so far have focused on the Mach-Zehnder Modulators (MZIs) and Ring or Racetrack Resonator Modulators (RMs) as the fundamental building blocks in such systems. This Chapter aims to introduce plasmonic Mach-Zehnder in Ring (MZIR) modulators on Si_3N_4 as a design to achieve both high performance metrics and low voltage requirements compatible with Co-Packaged Optics (CPO), emerging as an attractive solution for intra-DC communications. The next Section will examine the Ferroelectricity and Pockels Effect in Modulators based on the discussion of [6].

5.2: Ferroelectricity and the Pockels Effect

Pockels modulators have been the workhorse of optical communication systems the last few decades. While other technologies such as magneto-optic modulators and acousto-optic modulators have been utilized and commercialized, Pockels modulators can provide an instantaneous electro-optic (EO) effect, that can reach a bandwidth of over 100GHz, when the transmission lines are properly designed to match the wave velocities of the RF and the optical waves. The most popular EO material has been lithium niobate $LiNbO_3$ that has been used in most commercial modulators. This material exhibits an EO-coefficient close to 30pm/V usually leading to the large device footprints. After many years, the last few years thin-film lithium niobate technology [7] has brought back on-chip Pockels modulators to the forefront of integrated optics. Even

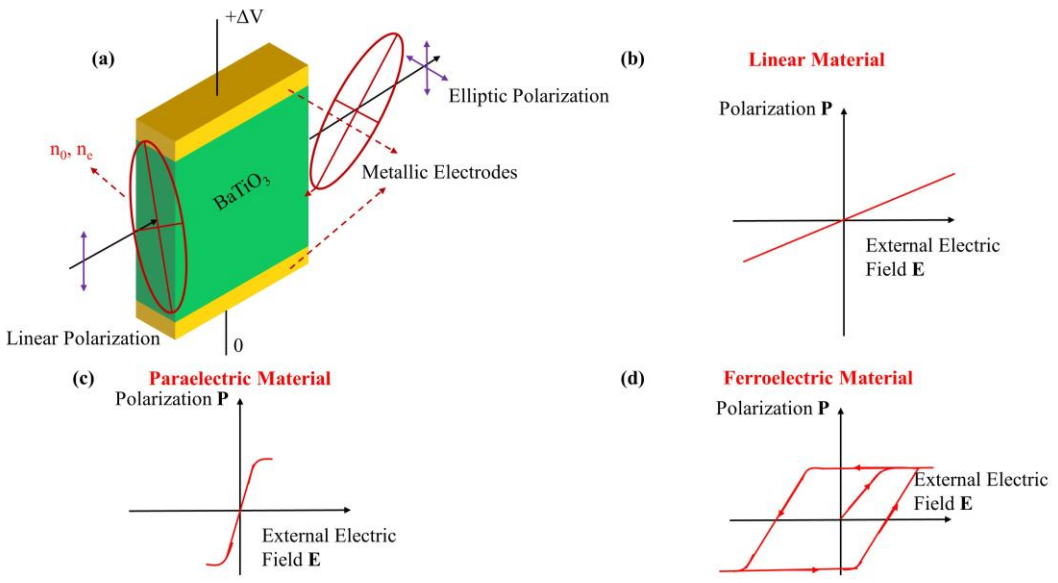


Fig. 5.2.1. (a) 3D-Schematic for the visualization of the operation of a $BaTiO_3$ ferroelectric phase-shifter, (b) Polarization-External electric field relation for Linear material, (c) for Paraelectric material, and (d) for Ferroelectric material.

though the material is not traditionally CMOS compatible, it can be co-integrated via hybrid and heterogeneous integration with CMOS Si-photonic and Si_3N_4-photonic waveguides [8]. Nevertheless, the demonstrated lengths of on-chip $LiNbO_3$ devices have been limited to values over 1cm due to the low EO-coefficient of the material. Barium titanate oxide ($BaTiO_3$ or BTO) has emerged as another very promising ferroelectric material for integrated photonic modulators. BTO exhibits the Pockels effect, having up to 10 times higher EO-coefficient compared to $LiNbO_3$, can be grown on Si [8] in a monolithic manner with a buffer layer, or can be heterogeneously co-integrated with any Si-photonic or Si_3N_4 waveguide [9-11]. This section aims to provide some physical characteristics and the principle of operation of ferroelectric Pockels modulators.

The 3D-Schematic of Fig. 5.2.1(a) elegantly explains the main principle of operation of a bulk Pockels BTO modulator. If one begins from the constitutive relation in electromagnetics, as it was formulated for linear materials:

$$\boldsymbol{D} = \varepsilon \boldsymbol{E} = \varepsilon_0 \varepsilon_r \boldsymbol{E} = \varepsilon_0 (1 + \chi_e \boldsymbol{E}) \quad (5.1)$$

The relation is true for linear materials and the polarization term can be derived as:

$$\boldsymbol{P} = \boldsymbol{D} - \varepsilon_0 \boldsymbol{E} = \varepsilon_0 \chi_e \boldsymbol{E} \quad (5.2)$$

The polarization-external field relation for a linear dielectric material is visualized in Fig. 5.2.1(b). If the material is linear there is a linear dependence of polarization to the external electric field. If the material is also isotropic, then the terms ε_r and χ_e have a single value while if the material is anisotropic these terms are tensors. Such is the case for ferroelectric materials. The materials based on the Pockels effect are formed

only by non-centrosymmetric crystals. In Pockels materials such as $BaTiO_3$ or BTO the refractive index can be physically described by an index ellipsoid:

$$\left(\frac{1}{n^2}\right)_1 x^2 + \left(\frac{1}{n^2}\right)_2 y^2 + \left(\frac{1}{n^2}\right)_3 z^2 + 2\left(\frac{1}{n^2}\right)_4 yz + 2\left(\frac{1}{n^2}\right)_5 xz + 2\left(\frac{1}{n^2}\right)_6 xy = 1 \quad (5.3)$$

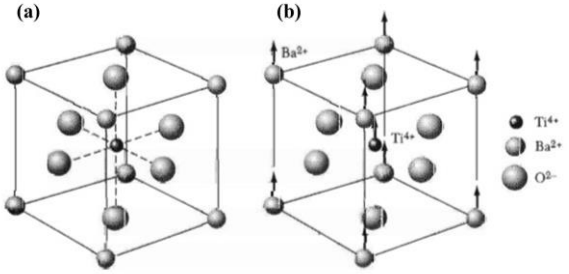
The index i=1,…,6 corresponds to different directions in the Pockels crystal. Relation (5.3) is the general form. This form can be reduced in real situations by exploiting the crystal symmetries (such as the symmetries of the BTO). The application of an electric field in a Pockels material results to the change of its index ellipsoid as seen in Fig. 5.2.1. As a result, this results to the change of the refractive index the propagating light experience, and to the change of the phase. The change is analogous to the applied electric field which can be written in a tensorial form as:

$$\Delta\left(\frac{1}{n^2}\right) = \sum_{j=1}^{3} r_{ij} E_j \quad (5.4)$$

Inversing the matrix, the term Δn as well as the updated $n'(E)$ refractive index with respect to the applied electric field can be calculated. Otherwise, one can make a first-order Taylor approximation to relation (5.4) obtaining the change to the refractive index of the ferroelectric material as:

$$\Delta\left(\frac{1}{n^2}\right) \approx -2n^3\Delta n \Rightarrow \Delta n \approx -\frac{n^3}{2}\Delta\left(\frac{1}{n^2}\right) \Rightarrow \Delta n \approx -\frac{n^3}{2} r_{eff} E \approx -\frac{n^3 r_{eff} V}{2d} \quad (5.5)$$

The terms r_{ij} correspond to all the different electro-optic coefficients to various crystal directions, while the r_{eff} is the approximate effective EO coefficient related with all the r_{ij} elements. As shown in Fig. 5.2.1, the index ellipsoid of the BTO changes with the application of applied voltage ΔV. The input light is polarized to a random axis. The index ellipsoid composed by two main refractive indices n_0 and n_e, for the ordinary and extraordinary axes, that are different from the polarization axis of the input light. Since input light is polarized at a random axis, it is analyzed to the two main axes of the ellipsoid. The direction of the main axes changes with the application of voltage leading to a change in the refractive index of propagating light. The output light has an elliptic polarization since two polarization components exist. By imposing a polarizer set in parallel with the polarization of input light, changes to the light intensity can be directly measured since part of the input power transfers to the other polarization. In the case of the input light being polarized with one of the two axes of the material, the output light experiences phase-changes. In integrated photonic waveguides, unlike the bulk case of 5.2.1(a) with proper coupling configurations, the polarization of input light is maintained to the output and the propagating light only experiences changes to its phase from a Pockels phase-shifter. Also, in waveguides, the confinement factor of the optical mode (%) in the BTO needs to be inserted in eq. (5.5) for accurate calculations, or the matrix calculations of (5.4) need to be performed directly. To insert non-linear



To obtain the spontaneous polarization P_s in the CGS unit of esu cm^{-2}, multiply the value given in μC cm^{-2} by 3×10^3.

		T_c, in K	P_s, in μC cm^{-2}, at T K	
KDP type	KH_2PO_4	123	4.75	[96]
	KD_2PO_4	213	4.83	[180]
	RbH_2PO_4	147	5.6	[90]
	KH_2AsO_4	97	5.0	[78]
	GeTe	670	—	—
TGS type	Tri-glycine sulfate	322	2.8	[29]
	Tri-glycine selenate	295	3.2	[283]
Perovskites	$BaTiO_3$	408	26.0	[296]
	$KNbO_3$	708	30.0	[523]
	$PbTiO_3$	765	>50	[296]
	$LiTaO_3$	938	50	
(c)	$LiNbO_3$	1480	71	[296]

Fig. 5.2.2. (a) Crystal structure of a $BaTiO_3$ material above the Curie Temperature T_c (in the paraelectric phase) and (b) below the Curie Temperature T_c (in the ferroelectric phase), (c) Table of ferroelectric materials with their respective Curie Temperatures in Kelvin. [6]

terms to the constitutive relations, a polyonymic expansion of the polarization is considered:

$$\boldsymbol{P} = \varepsilon_0 \chi_e^{(1)} \boldsymbol{E_i} + \varepsilon_0 \chi_e^{(2)} \boldsymbol{E_i E_j} + \varepsilon_0 \chi_e^{(3)} \boldsymbol{E_i E_j^2} + \cdots \quad (5.6)$$

The field $\boldsymbol{E_i}$ is the electric field of light while the field $\boldsymbol{E_j}$ is the applied electric field through the electrodes. The first term in polarization is the linear term dominant in all materials. The second term expresses the Pockels effect with the dependence of the polarization to the electric field of the external electronic signal. This term exists only in non-centrosymmetric materials such as $LiNbO_3$ and BTO. The third term expresses the Kerr effect, which exists in all materials but is much weaker than the other two terms. In the Kerr effect, the polarization is dependent on the square of the electric field of the applied electronic signal. The strength of the EO-effect depends on the overlap of the light and RF electric field as seen in Eq. (5.6).

Ferroelectric materials are materials that exhibit spontaneous dipole moment and thus spontaneous polarization even at the absence of an applied electric field. Their mode of operation resembles the operation of ferromagnetic materials, whose physical characteristics can be found in references such as [12]. As in the case of ferromagnetic materials, there is a constant "battle" between order and disorder. Below a temperature known as a Curie or transition temperature T_c, materials such as $LiNbO_3$ and $BaTiO_3$ are in their ferroelectric phase. There the materials are composed by ferroelectric domains with spontaneous dipole moments. The domains can be transformed, and the dipole moments can be turned with the application of an electric field. After applying the external electric field if the field is set to zero, the material will keep its previous state by retaining the orientated dipole moments. This is very useful for intelligent memory and computing devices, considering also the non-linear nature of such materials. If an electric field of the opposite direction is applied to the materials, the previously orientated dipole moments developing their own local field resist to the applied external field. For this reason, a hysteresis of the polarization to the external electric field can be observed. Above the Curie temperature, disorder caused by the induced temperature wins and the dipole moments are randomly placed in the material. In this case the material exhibits a paraelectric behavior with the polarization having a non-linear dependence to the applied field, but without any hysteresis observed. This can be exploited also in poling of the ferroelectric materials by applying voltage to

them, when the temperature is above the Curie temperature. This way the initial orientation of the dipole moments can be set, rendering it very useful for ferroelectric modulators. There is a kind of phase-transition happening at the Curie temperature point. This has been studied by scientists such as Landau and details can be found in [6]. The polarization-external field dependence for the paraelectric and ferroelectric phase is depicted in Fig. 5.2.1(c) and (d). In the ferroelectric phase, the polarization forms an hysteresis loop with the applied external electric field. Since the constitutive relations connect the two terms as in the case of (5.6) considering only the Pockels effect, the hysteresis to the polarization-field graph also translates to a hysteresis that can be observed to the phase-voltage graph or, in an interferometer, to the output light power-voltage graph. It should be noted that the Pockels and hysteresis effect co-exist in the ferroelectric material. At low frequencies, (for example DC voltages) the hysteresis effect due to the shift of the ferroelectric domains is dominant, while at higher microwave frequencies (for example over 50GHz) the Pockels effect is dominant.

Figure 5.2.2 showcases the crystal structure of the $BaTiO_3$ in the paraelectric and ferroelectric phase along with a Table on the Curie temperatures of various ferroelectric materials, derived from [6]. In the paraelectric phase, as shown in Fig. 5.2.2(a), the ions of the BTO crystal become symmetric and therefore no spontaneous polarizations and hysteresis effects can be observed. In the ferroelectric phase, the Ba ions but especially the Ti ion at the center of the crystal is displaced, leaning to the upper part of the crystal, as depicted in Fig. 5.2.2(b). This leads the crystal to exhibit spontaneous polarization by the addition of the ionic dipole moments in the BTO. Finally, the Table in Fig. 5.2.2(c) presents that, even though the BTO crystal exhibits a much higher EO-coefficient than the $LiNbO_3$ crystal, the Curie temperature of $BaTiO_3$ is 408K (135°C)

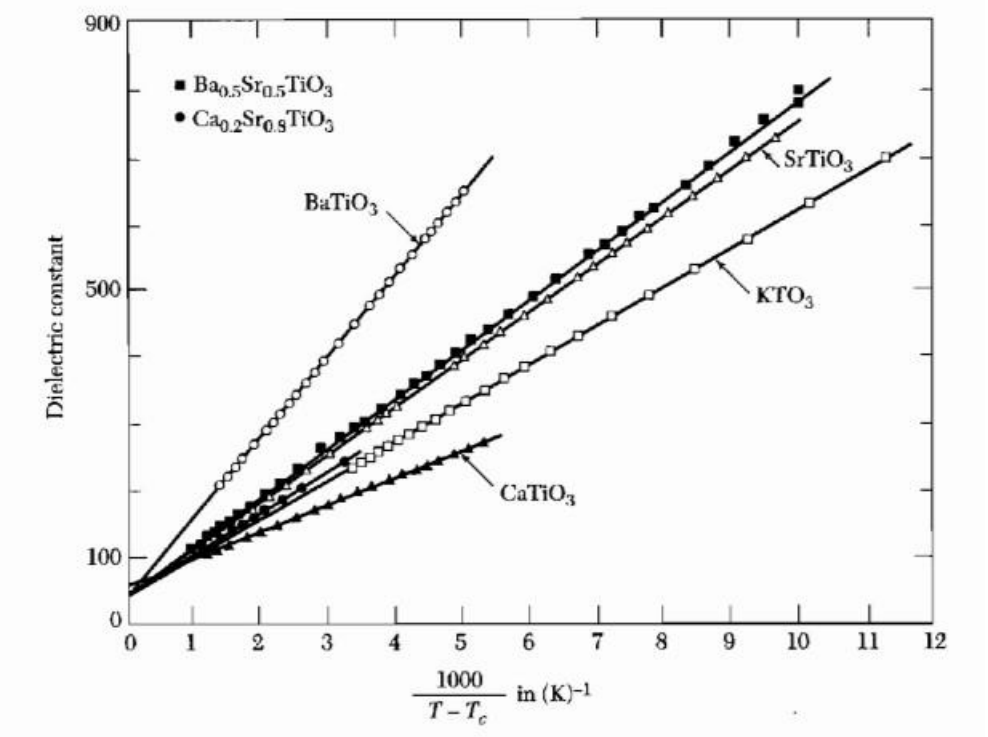


Fig. 5.2.3. Dependence of the dielectric constant ε_r measured at microwave frequencies for increasing $1000/(T-T_c)$ in (K_{-1}) for a variety of ferroelectric materials in their paraelectric phase [6].

while the transition temperature of $LiNbO_3$ is 1480K (1207°C), rendering lithium niobate much more tolerant in very high temperature environments.

Finally, the dependence of various the dielectric constant ε at microwave frequencies to temperature term $1000/(T - T_c)$ in (K^{-1}) for various ferroelectric materials in their paraelectric phase is shown in Fig. 5.2.3, exhibiting a linear behavior. The $BaTiO_3$ material has the largest dependence on temperature compared to the other materials in the figure. The dependence of the dielectric constant to the temperature in the paraelectric phase, considering the Curie Temperature T_c can be written as:

$$\varepsilon \approx \frac{\xi}{T - T_c} \quad (5.7)$$

In conclusion, Pockels ferroelectric modulators have been detrimental over the decades for the delivery of high-speed optical communication systems. The phase-shift experienced by the Pockels Effect can be translated to output power variations utilizing interferometric structures such as symmetric and asymmetric Mach-Zehnder Interferometers (sMZIs and aMZIs), ring or racetrack resonators (RMs), and energy-efficient Mach-Zehnder in a Ring (MZIR) devices. The next sections will deal with the design of plasmonic ferroelectric MZIR elements on the Si_3N_4 photonic platform.

5.3: The Plasmo-photonic MZIR modulator: Semi-Analytic Theoretical Modelling Framework

The purpose of this study is to derive a semi-analytical model for the calculation of the Transfer Function of a Plasmo-Photonic Mach-Zehnder Interferometer in a Ring/Loop Resonator structure. Our model utilizes a Scattering Matrix Method approach to calculate initially the expressions for the circulating and transmitted electric fields and then for the calculation of their respective Transfer Functions. All propagation and interfacing coupling losses have been taken into account in our model while back-reflected fields have been neglected.

In order to initiate the study, the Scattering Matrices of the individual components comprising our plasmo-photonic MZIR circuitry have been calculated. To begin with, in Fig. 5.3.1, the illustrated image of a 50:50 coupler is depicted. No losses in the 50:50 coupler have been taken into account in the initial model.

The individual Scattering Matrix for this coupler is:

$$\begin{pmatrix} b_1 \\ b_2 \end{pmatrix} = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} \begin{pmatrix} a_1 \\ a_2 \end{pmatrix} \quad (5.8)$$

The branches of the MZI comprise of 2 S-bends, 2 coupling interfaces and a plasmonic waveguide in each branch. In the upper arm, a phase shift of $+\Delta\varphi/2$ is applied while in the lower branch a phase shift of $-\Delta\varphi/2$ is applied through voltage and an electro-optic effect in the BTO. The phase of the phase shifter is correlated with the applied voltage through the relation:

$$\Delta\varphi = \frac{\pi}{V_\pi} V \quad (5.9)$$

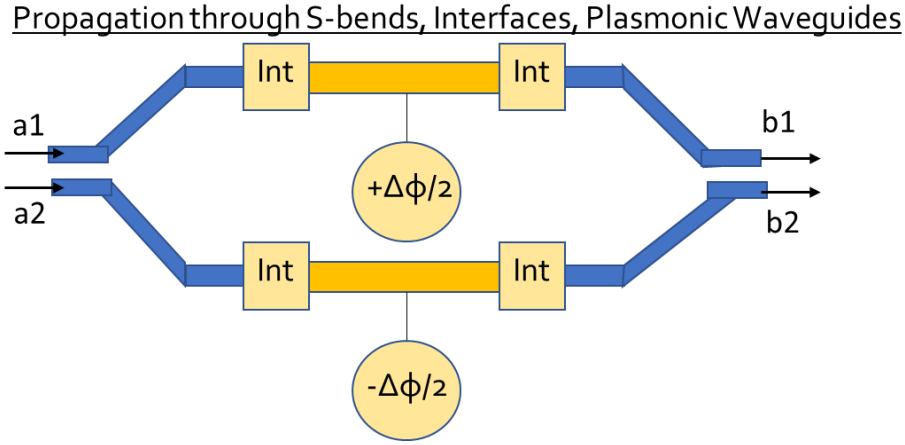


Fig. 5.3.2: Schematic illustration of the MZI branches

The upper and lower branches of the MZI are depicted in Fig. 5.3.2.

The Scattering Matrix expressing the propagation through the 2 MZI branches can be seen below:

$$\begin{pmatrix} b_1 \\ b_2 \end{pmatrix} = A_{Lop''} e^{i\beta_0 L''_{op}} \begin{pmatrix} e^{\frac{i\Delta\varphi}{2}} & 0 \\ 0 & e^{\frac{-i\Delta\varphi}{2}} \end{pmatrix} \begin{pmatrix} \alpha_1 \\ \alpha_2 \end{pmatrix} (5.10)$$

Where:

$$A_{Lop''} = \left(\sqrt{a_{length}}\right)^2 \left(\sqrt{a_{Int}}\right)^2 \sqrt{a_{plasmo}} \ (5.11)$$

$$\beta_0 = \frac{2\pi}{\lambda_0} \ (5.12)$$

$$L''_{op} = 2n_{eff}(SiN)L_{length} + n_{eff}(Plasmo)L_{plasmo} \ (5.13)$$

$$a_{length} = 10^{\frac{Loss_{Length}}{10}} \ (5.14)$$

$$Loss_{Length} = Prop.Loss(SiN) * L_{sbend} \ (5.15)$$

$$a_{plasmo} = 10^{\frac{Loss_{plasmo}}{10}} \ (5.16)$$

$$Loss_{plasmo} = Prop.Loss(plasmo) * L_{plasmo} \ (5.17)$$

$$a_{Int} = 10^{\frac{Loss_{Int}}{10}} \ (5.18)$$

Therefore, the total plasmo-photonic MZI structure can be seen in Fig. 5.3.3 and is modelled with the inclusion of the MMI Scattering Matrices to the left and right side of

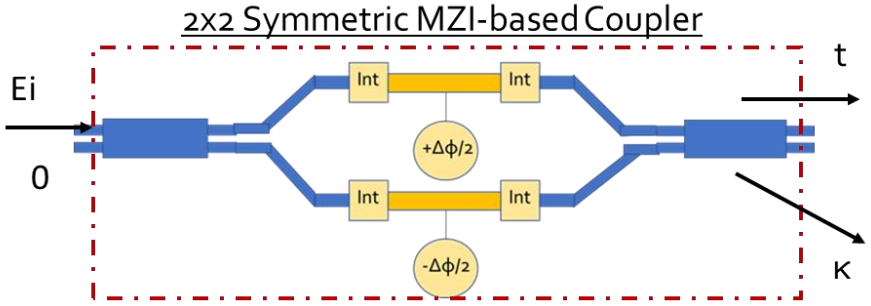


Fig. 5.3.3: Schematic illustration of a plasmo-photonic MZI

the branch scattering matrices. To include the MMI losses, in each side, one can introduce and multiply the expression that will be provided below with the term $\sqrt{a_{MMI}}$ since the equation is written with respect to fields and not power values.

The respective Scattering Matrix of the MZI taking into account Eqs. (5.8) and (5.10) is the following:

$$\begin{pmatrix} t \\ \kappa \end{pmatrix} = \frac{1}{\sqrt{2}}\begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} A_{Lop''} e^{i\beta_0 L''_{op}} \begin{pmatrix} e^{\frac{i\Delta\varphi}{2}} & 0 \\ 0 & e^{\frac{-i\Delta\varphi}{2}} \end{pmatrix} \frac{1}{\sqrt{2}}\begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}\begin{pmatrix} E_i \\ 0 \end{pmatrix} \quad (5.19)$$

The relation reduces to:

$$\begin{pmatrix} t \\ \kappa \end{pmatrix} = \frac{1}{2} A_{Lop''} e^{i\beta_0 L''_{op}} E_i \begin{pmatrix} e^{\frac{i\Delta\varphi}{2}} + e^{\frac{-i\Delta\varphi}{2}} \\ e^{\frac{i\Delta\varphi}{2}} - e^{\frac{-i\Delta\varphi}{2}} \end{pmatrix} \quad (5.20)$$

$$\begin{pmatrix} t \\ \kappa \end{pmatrix} = A_{Lop''} e^{i\beta_0 L''_{op}} E_i \begin{pmatrix} \cos\left(\frac{\Delta\varphi}{2}\right) \\ i\sin\left(\frac{\Delta\varphi}{2}\right) \end{pmatrix} \quad (5.21)$$

Going back to the loop a ring resonator has the general configuration of Figure 5.3.4. The Scattering Matrix of a Ring Resonator has the form given below, based on [68]:

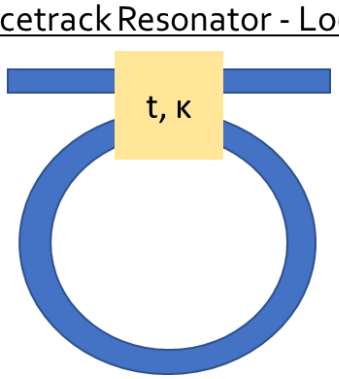


Fig. 5.3.4: Generalized Depiction of a Ring/Racetrack Resonator

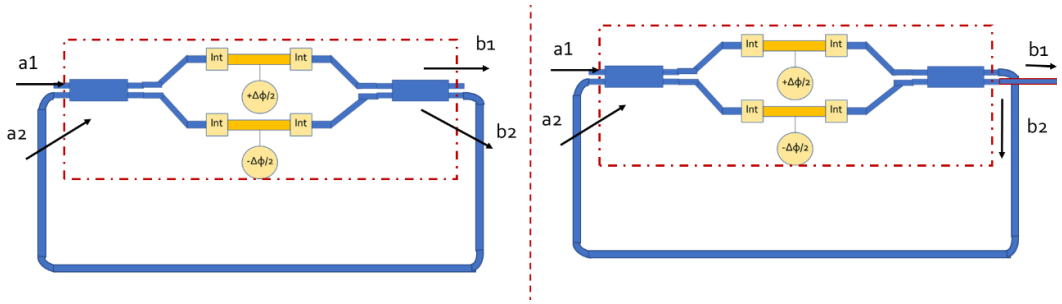


Fig. 5.3.5: Two MZIR cases utilizing (a) the bar (left) and (b) cross port (right) for the output of the MZIR.

$$\begin{pmatrix} b_1 \\ b_2 \end{pmatrix} = S_{ij} \begin{pmatrix} a_1 \\ a_2 \end{pmatrix} = \begin{pmatrix} t' & i\kappa' \\ i\kappa' & t' \end{pmatrix} \begin{pmatrix} a_1 \\ a_2 \end{pmatrix} \quad (5.22)$$

Where S_{ij} is the scattering matrix of the Ring Resonator. Equation (5.22) is the result of the assumption that t, κ are real. The parameter t is the self-coupling coefficient and the parameter κ is the coupling coefficient of the ring-bus coupler.

In the literature two MZIR structures have been found and theoretical models have been proposed but all neglect the losses in the MZI branches and do not provide a clear relation for the calculation of the FSR in the cavity. The two studied cases can be seen in Fig. 5.3.5. In Fig. 5.3.5(a) the output waveguide is at the bar port of the MZIR while in Fig. 5.3.5(b) the output waveguide resides at the cross port of the MZIR. The structure of Fig. 5.3.5(b) also requires a crossing device.

In the structure of Fig. 5.3.5(b) the bar port gives feedback to the MZI-based symmetric coupler. Therefore, by keeping the real parts and normalizing with E_i in Eq. (5.21) we have:

$$t' = \frac{t}{E_i} = A_{Lop''} e^{i\beta_0 L''_{op}} \cos\left(\frac{\Delta\varphi}{2}\right) \quad (5.23)$$

$$\kappa' = \frac{t}{E_i} = A_{Lop''} e^{i\beta_0 L''_{op}} \sin\left(\frac{\Delta\varphi}{2}\right) \quad (5.24)$$

Therefore, from the Scattering Matrix of Eq. (5.22), Eq. (5.23) and Eq. (5.24) one gets the Scattering Matrix of the MZI-based coupler as:

$$S_{ij}^{MZIR_1} = \begin{pmatrix} A_{Lop''} e^{i\beta_0 L''_{op}} \cos\left(\frac{\Delta\varphi}{2}\right) & iA_{Lop''} e^{i\beta_0 L''_{op}} \sin\left(\frac{\Delta\varphi}{2}\right) \\ iA_{Lop''} e^{i\beta_0 L''_{op}} \sin\left(\frac{\Delta\varphi}{2}\right) & A_{Lop''} e^{i\beta_0 L''_{op}} \cos\left(\frac{\Delta\varphi}{2}\right) \end{pmatrix} \quad (5.25)$$

Finally, it is clear that:

$$\begin{pmatrix} b_1 \\ b_2 \end{pmatrix} = S_{ij}^{MZIR_1} \begin{pmatrix} a_1 \\ a_2 \end{pmatrix} \quad (5.26)$$

Therefore, in this case:

$$a_2 = b_2\sqrt{a_{loop}}e^{i\beta_0 L_{loop}^{op}} \text{ (5.27)}$$

$$b_2 = iA_{Lop''}e^{i\beta_0 L''_{op}}\sin\left(\frac{\Delta\varphi}{2}\right)a_1 + A_{Lop''}e^{i\beta_0 L''_{op}}\cos\left(\frac{\Delta\varphi}{2}\right)a_2 \text{(5.28)}$$

From Eq. (5.27) and (5.28) the circulating fields in the cavity are given by the expression:

$$\frac{b_2}{a_1} = \frac{iA_{Lop''}e^{i\beta_0 L''_{op}}\sin\left(\frac{\Delta\varphi}{2}\right)}{1 - A_{TOT}e^{i\beta_0 L_{OPT}}\cos\left(\frac{\Delta\varphi}{2}\right)} \text{ (5.29)}$$

And from (5.29):

$$\left|\frac{b_2}{a_1}\right|^2 = \left(\frac{b_2}{a_1}\right)^*\left(\frac{b_2}{a_1}\right) = \frac{A_{Lop''}^2\sin^2\left(\frac{\Delta\varphi}{2}\right)}{1 + A_{TOT}^2\cos^2\left(\frac{\Delta\varphi}{2}\right) - 2A_{TOT}\cos\left(\frac{\Delta\varphi}{2}\right)\cos(\beta_0 L_{OPT})} \text{ (5.30)}$$

It should be noted that:

$$A_{TOT} = A_{Lop''}\sqrt{a_{loop}} \text{ (5.31)}$$

$$L_{OPT} = L''_{op} + L_{loop}^{op} \text{ (5.32)}$$

Where

$$L_{loop}^{op} = n_{eff}(SiN)L_{loop} \text{ (5.33)}$$

Eventually since from Eq. (5.26)

$$b_1 = A_{Lop''}e^{i\beta_0 L''_{op}}\cos\left(\frac{\Delta\varphi}{2}\right)a_1 + iA_{Lop''}e^{i\beta_0 L''_{op}}\sin\left(\frac{\Delta\varphi}{2}\right)a_2 \text{(5.34)} \overset{(5.27)}{\Longrightarrow}$$

$$\frac{b_1}{a_1} = A_{Lop''}e^{i\beta_0 L''_{op}}\left[\cos\left(\frac{\Delta\varphi}{2}\right) - \frac{A_{TOT}e^{i\beta_0 L_{OPT}}\sin^2\left(\frac{\Delta\varphi}{2}\right)}{1 - A_{TOT}e^{i\beta_0 L_{OPT}}\cos\left(\frac{\Delta\varphi}{2}\right)}\right] \Leftrightarrow$$

$$\frac{b_1}{a_1} = A_{Lop''}e^{i\beta_0 L''_{op}}\left[\frac{\cos\left(\frac{\Delta\varphi}{2}\right) - A_{TOT}e^{i\beta_0 L_{OPT}}}{1 - A_{TOT}e^{i\beta_0 L_{OPT}}\cos\left(\frac{\Delta\varphi}{2}\right)}\right] \text{ (5.35)}$$

Therefore, the Transfer Function is calculated as:

$$\left|\frac{b_1}{a_1}\right|^2 = \left(\frac{b_1}{a_1}\right)^*\left(\frac{b_1}{a_1}\right) = A_{Lop''}^2\frac{A_{TOT}^2 + \cos^2\left(\frac{\Delta\varphi}{2}\right) - 2A_{TOT}\cos\left(\frac{\Delta\varphi}{2}\right)\cos(\beta_0 L_{OPT})}{1 + A_{TOT}^2\cos^2\left(\frac{\Delta\varphi}{2}\right) - 2A_{TOT}\cos\left(\frac{\Delta\varphi}{2}\right)\cos(\beta_0 L_{OPT})}$$

(5.36)

On the contrary, the MZIR configuration of Fig. 5.3.5(a) utilizes the cross port for the loop feedback. For this reason, the t' and κ' in Eq. (5.26) will be different. They will be defined as:

$$t' = \frac{\kappa}{E_i} = A_{Lop''} e^{i\beta_0 L''_{op}} \sin\left(\frac{\Delta\varphi}{2}\right) \quad (5.37)$$

$$\kappa' = \frac{t}{E_i} = A_{Lop''} e^{i\beta_0 L''_{op}} \cos\left(\frac{\Delta\varphi}{2}\right) \quad (5.38)$$

Such that:

$$\begin{pmatrix} b_1 \\ b_2 \end{pmatrix} = S_{ij}^{MZIR_2} \begin{pmatrix} a_1 \\ a_2 \end{pmatrix} \quad (5.39)$$

Where:

$$S_{ij}^{MZIR_1} = \begin{pmatrix} A_{Lop''} e^{i\beta_0 L''_{op}} \sin\left(\frac{\Delta\varphi}{2}\right) & iA_{Lop''} e^{i\beta_0 L''_{op}} \cos\left(\frac{\Delta\varphi}{2}\right) \\ iA_{Lop''} e^{i\beta_0 L''_{op}} \cos\left(\frac{\Delta\varphi}{2}\right) & A_{Lop''} e^{i\beta_0 L''_{op}} \sin\left(\frac{\Delta\varphi}{2}\right) \end{pmatrix} \quad (5.40)$$

Therefore, it can likewise be proved that the expressions derived for the two MZIR configurations of Fig. 5.3.5 are complementary.

For the circulating fields in the MZIR cavity:

$$\frac{b_2}{a_1} = \frac{iA_{Lop''} e^{i\beta_0 L''_{op}} \cos\left(\frac{\Delta\varphi}{2}\right)}{1 - A_{TOT} e^{i\beta_0 L_{OPT}} \sin\left(\frac{\Delta\varphi}{2}\right)} \quad (5.41)$$

$$\left|\frac{b_2}{a_1}\right|^2 = \frac{A_{Lop''}^2 \cos^2\left(\frac{\Delta\varphi}{2}\right)}{1 + A_{TOT}^2 \sin^2\left(\frac{\Delta\varphi}{2}\right) - 2A_{TOT} \sin\left(\frac{\Delta\varphi}{2}\right) \cos(\beta_0 L_{OPT})} \quad (5.42)$$

For the transfer function of our MZIR configuration:

$$\frac{b_1}{a_1} = A_{Lop''} e^{i\beta_0 L''_{op}} \left[\frac{\sin\left(\frac{\Delta\varphi}{2}\right) - A_{TOT} e^{i\beta_0 L_{OPT}}}{1 - A_{TOT} e^{i\beta_0 L_{OPT}} \sin\left(\frac{\Delta\varphi}{2}\right)}\right] \quad (5.43)$$

$$\left|\frac{b_1}{a_1}\right|^2 = A_{Lop''}^2 \frac{A_{TOT}^2 + \sin^2\left(\frac{\Delta\varphi}{2}\right) - 2A_{TOT} \sin\left(\frac{\Delta\varphi}{2}\right) \cos(\beta_0 L_{OPT})}{1 + A_{TOT}^2 \sin^2\left(\frac{\Delta\varphi}{2}\right) - 2A_{TOT} \sin\left(\frac{\Delta\varphi}{2}\right) \cos(\beta_0 L_{OPT})} \quad (5.44)$$

As a comparison, the Transfer Function of a simple All-Pass Ring Resonator is given by the relation [68]:

$$TF_{Ring} = \frac{I_{through}}{I_{input}} = \frac{a^2 - 2atcos\left(\beta_0 L_{ring}^{op}\right) + t^2}{1 - 2atcos\left(\beta_0 L_{ring}^{op}\right) + (at)^2} \quad (5.45)$$

5.4: Si_3N_4-plasmonic ferroelectric MZIR modulator for 112-Gbaud PAM-4 transmission in the O-band

Silicon photonic intra-Data-Center Interconnects (intra-DCI) have flourished during the last decade as cloud computing, high-performance computing, streaming and AI, have spearheaded great alterations to the traffic flow patterns, intensifying bandwidth requirements at ever decreasing distances [13]. In such a demanding environment, co-packaged silicon photonic based transmitters are a promising technology that can scale and evolve in terms of size, bandwidth, and power consumption [14] with demonstrations on legacy SOI solutions providing elevated speed metrics in combination with wavelength multiplexing, filtering or power splitting capabilities [15, 16].

On the other side, the advent of Silicon Nitride (Si_3N_4) has brought forth a large variety of new possible applications leveraging its large transparency window, low propagation losses and excellent performance [17, 18]. The first efforts were focusing on passive components such as filtering and multiplexing/demultiplexing modules for microwave photonics [19-21] and receivers for datacom [22]. Nevertheless, the true strengths of the Si_3N_4 were revealed with the invention of hybrid, heterogeneous [23, 24] and monolithic [25, 26] co-integration providing the realization of complex circuits incorporating also active components via efficient low-loss coupling interfaces [27-29]. In this way recently novel hybrid [30] and heterogeneously integrated lasers [31] and photodetectors [32] have been demonstrated.

However, a complete intra-DC integrated photonic link on Si_3N_4 still requires a modulator element capable of operating at high baud rates beyond 100G per lane in conjunction with low power consumption and low losses. Additionally, CMOS compatibility should be a given targeting replacement of SOI modulators in today's datacom applications where low cost is among the top priorities. As Si_3N_4 is a dielectric and cannot facilitate intrinsic electro-optic activation [17], hybrid and heterogeneous integration are the only ways for the realization of high-speed modulators [23]. These techniques employ a variety of materials in combination with Si_3N_4, so as to achieve the modulation of light in the phase domain, such as silicon [33], graphene [34, 35], ferroelectric lead zirconate titanate [36], electro-optic polymers [37], lithium niobate [38-45] and barium titanate oxide [46-48]. Out of those, Si_3N_4-LNOI is the prevalent solution in most recent publications with results demonstrating simulated bandwidth up to 1200GHz [41], 95GHz experimental 3dB-bandwidth (BW) for a 10mm PS length and V_π of 3.75V [44]. The highest data rates ever recorded on Si_3N_4 is 80Gbps in single-channel [39], 280 Gbps in a quad-channel NRZ array [43] and 128Gbps in a single channel PAM-4 signal transmission [45]. On the other side, ferroelectric barium titanate oxide (BTO) compared to all the other ferroelectric materials, exhibits the largest electro-optic coefficient, is CMOS compatible and can withstand high temperature

variations [46-49] for subsequent post processing during packaging of the devices. The credentials of the high electro optic coefficient of the BTO material were demonstrated in [46] resulting in as low as 106nW/FSR power consumption in a racetrack resonator on Si_3N_4. Nevertheless, these devices, being purely photonic, are bound by large footprint incorporating 150 μm long PSs.

Co-packaged optics on the other side require extremely small dimensions, especially if they are going to be connected directly to the ASIC's pins for driverless operation. Plasmonic modulators fit ideally in this need combining ultra-high bandwidth and a footprint matching ASICs' pin pitch of a few micrometers. Starting from the lithium niobate platform, plasmonic LNOI modulators have demonstrated low half-wave voltage length products of 0.23Vcm for a footprint less than $1mm^2$ and a bandwidth over 10GHz [50]. By integrating Plasmonic Organic Hybrid (POH) PS on SOI photonic waveguides, a C-band modulator was demonstrated for a racetrack design with on-chip BW over 110GHz, 1dB insertion loss (IL), sub-volt operation and enhanced stability [51-53]. Going to BTO, its frequency response was measured up to 270GHz [54]. Relying on the above advantages, in [47, 48], the first Si_3N_4-BTO plasmonic-slot based MZM was demonstrated experimentally delivering 216Gb/s PAM-2 and a 256Gb/s PAM-4 error free signal transmission in the C-band. However, the plasmonic metal was Au, not suitable for CMOS mass fabrication, the Si_3N_4-to-plasmonic coupling losses were over 6.5dB and a V_π x L product of 144 Vμm was measured at 40GHz.

In this work, we numerically analyze the first fully O-band CMOS compatible, in terms of material stack, Copper (Cu) based Si_3N_4-plasmonic MZIR modulator design capable of operating at above 100Gbaud rates and compare its performance versus two identical devices exploiting noble metals i.e. Au and Ag, which are not CMOS compatible but are the most commonly utilized plasmonic metals in plasmonics research. The whole analysis presents in detail the penalty induced in the performance of the modulator for going to CMOS by the higher propagation losses induced by the Cu metal in the plasmonic PSs. In addition, the work highlights the advantages the MZIR layout in terms of power savings versus symmetric MZMs in agreement with a similar conclusion from the LNOI platform [55]. Assuming a 5μm long PS of the MZM or MZIR for all three material platforms, the best performance is obtained as expected for Ag achieving insertion losses of 3.56 dB and V_π x L product 46% lower than the standard MZM. The Au based MZIR is very close to this with insertion losses of 4.74dB, and V_π x L product 39% lower than the conventional MZM. Finally, the CMOS compatible Cu based MZIR exhibits insertion losses of 6.85dB and V_π x L product 30% lower than its symmetric MZM counterpart. The drop in the performance of the Cu based MZIR is coming from the lower Q-factor of the resonance peaks due to the higher propagation losses in the plasmonic phase shifter equal to 0.78 dB/μm versus 0.33 dB/μm for Ag and 0.5 dB/μm for Au. In high-speed operation for the same PS length and $2x1.3V_{pp}$ driving voltage at 112Gbaud PAM-4 modulation format, the signal quality improvement measured by the ΔQ of MZIR versus MZM is calculated as 2.9, 2.4 and 1.3 approximately for Ag, Au and Cu plasmonic metals, respectively. By lowering the driving voltage to $2x0.8V_{pp}$ the ΔQ metric is equal to 0.88 for Cu, 2.2 for Au and 2.78 for Ag. The absolute Q value for the Cu based MZIR is higher than 5.2 even for $2x0.8V_{pp}$ driving voltage confirming error free transmission under FEC. All Q values in these calculations are for 10dBm at the input of the modulator and no biasing.

For a degraded $r_{42}/2$ the proposed MZIR retains a voltage reduction higher than 18% for Cu-, 28% for Au- and 37% for Ag-based designs. These performance metrics render this modulator technology an ideal optical engine for co-packaged optics solutions compatible with next generation 51.2 Tb/s switching routers featuring 224Gbps SerDes interfaces.

The organization of the rest of the Section is the following: Section 5.4.1 presents the overall MZIR modulator layout and demonstrates the device-level analysis and results from all the various elements employed in the circuitry. Section 5.4.2 shows the static and frequency-domain analysis of the plasmonic modulator for the three plasmonic metals, Ag, Au, and Cu and the various design parameters, highlighting all performance trade-offs. Section 5.4.3 describes the results from the time-domain analysis at the system-level, elucidating the capabilities of the MZIR modulator for 112Gbaud PAM-4 Signal Transmission. Section 5.4.4 studies the impact of a degraded r_{42} value of the BTO due to fabrication to the MZIR performance metrics. Section 5.4.5 discusses and compares the results of this work with other modulators in the literature.

5.4.1: MZI-Ring modulator layout and component analysis

5.4.1.1: MZIR layout analysis

The plasmonic ferroelectric MZIR modulator is based on the traditional 2x2 MZI circuit with the adjustment of a dielectric waveguide loop connection from one input of the 2x2 input MMI to one of the outputs of the 2x2 output MMI. Fig. 5.4.1(a) presents the corresponding layout of the MZIR modulator, while Fig. 5.4.1(b) shows the cross section of the plasmonic PS. The design employs three waveguiding levels: the Si_3N_4 dielectric waveguide forming the main device circuitry, the aSi interposer waveguide used as an intermediate layer for optimum light transfer between the Si_3N_4 waveguide, and the plasmonic slot PS filled by a ferroelectric barium titanate oxide (BTO) material. The plasmonic slot waveguide, due to its waveguide symmetry, supports a TE plasmonic mode, therefore the polarization of interest in the MZIR operation is also TE. The Si_3N_4 waveguide has a width of 800 nm and an equal thickness value of 800 nm, supporting both a TE and a TM mode, with the TE one exhibiting an effective index with a real part $Re\{n_{eff}\}$ = 1.773 and propagation losses (PL) of 0.2 dB/cm at a wavelength λ = 1310 nm. The Free Spectral Range of the MZIR is controlled by the total length of the cavity formed by the waveguide loop, the Multimode Interference (MMI) couplers, the Si_3N_4-to-plasmonic-slot coupling interfaces, and the plasmonic slot PS. The total cavity length is kept as low as possible to achieve the highest possible

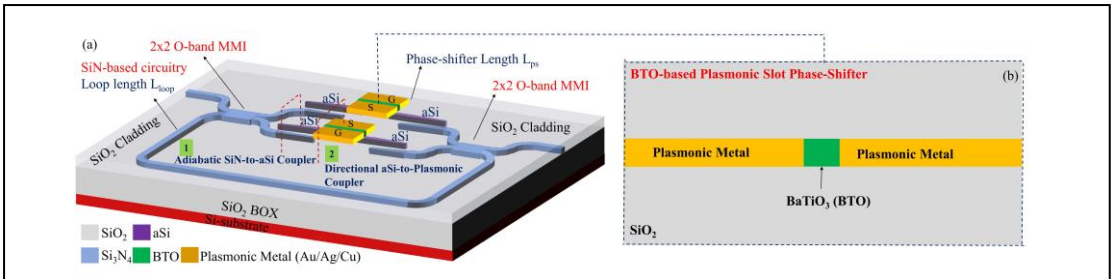


Fig. 5.4.1: (a) 3D Schematic of the tri-layer Si_3N_4-BTO plasmonic slot based ferroelectric modulator and (b) Plasmonic Slot (MIM) Phase-Shifter utilizing a ferroelectric BTO material.

FSR value. The loop length is large enough to enclose two MMIs approximately 300μm long, two coupling interfaces approximately 186μm long and a PS of variable length. Finally, there is a length difference between the two branches of the MZI equal to $\Delta L = L_{FSR}/2 = \lambda_0/2Re\{n_{eff}(Si_3N_4)\}$ for λ_0 = 1310nm inducing an extra π phase shift in the loop for reasons that will be better analyzed and illustrated in Section 5.4.3.

The analysis of the MZIR plasmo-photonic circuit is starting from the optimization through 3D-FDTD simulation of the MMI-Couplers, the plasmonic slot PS and the Si_3N_4-to-plasmonic slot coupling interfaces. Then, the transfer function of these components is inserted in a circuit level simulator towards maximization of the MZIR modulator performance. The whole study also includes a comparison versus normal MZMs, so as to reveal the power savings coming from the MZIR without any penalty in the obtainable bandwidth, contrary to Ring or Racetrack Resonators.

5.4.1.2: MMI coupler design and broadband response

The MZIR device design was initiated with the MMI couplers. Consecutive 3D-FDTD numerical calculations have been utilized in order to obtain maximum coupling efficiency to the two output ports simultaneously with minimum imbalance. The output width, MMI length, and position of the input and output tapers have been varied in a systematic manner to reach the optimum performance in terms of efficiency. These were identified as 6 μm for the MMI width and 108 μm for the MMI length. The taper output width has been increased to 2.5 μm to collect all the field at the two outputs, while the taper length has also been increased to 20 μm. Figure 5.4.2(a) depicts the simulated device in a 3D-manner. The MMI inputs and outputs were placed at a distance dw = 1.5μm from the center of the MMI. The |**E**| field distribution for a wavelength of 1.31 μm for a TE- mode input is shown in Fig. 5.4.2(b) indicating efficient operation in the TE polarization. Fig. 5.4.2(c) shows the corresponding transfer functions extracted from broadband 3D-FDTD simulations. In both graphs the losses per port are below 0.2 dB near 1310 nm, with the imbalance being lower than 2%. The

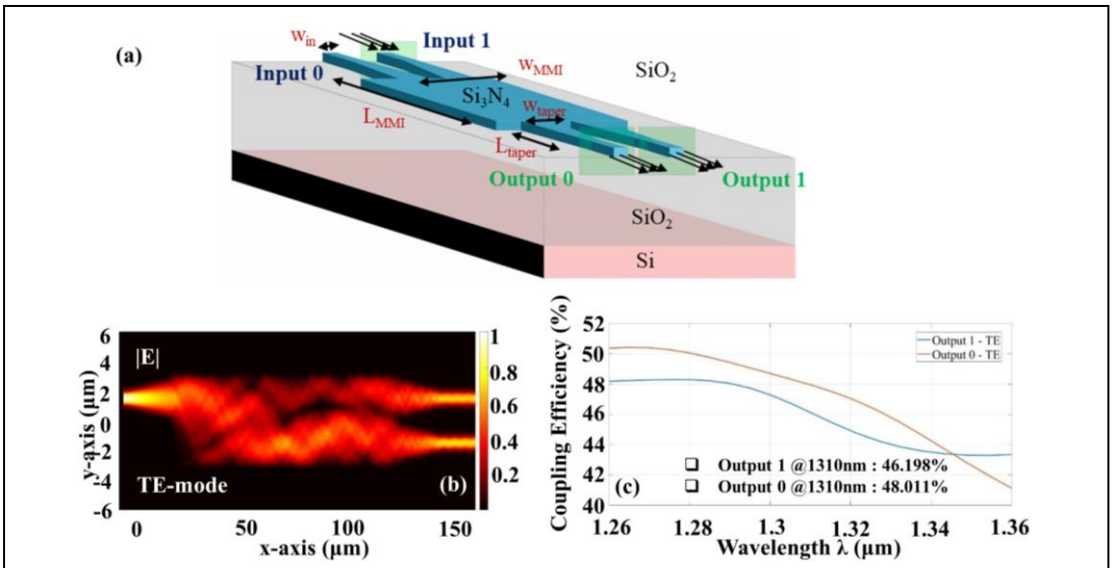


Fig. 5.4.2: (a) 3D Schematic of Designed MMI Coupler, (b) Top view of |**E**| Field Distribution for a TE mode excitation from input 1 to the 2x2 MMI, (d) Numerically Calculated Transfer Function at the two outputs of the MMI for the TE mode excitation.

scattering matrix for this design was obtained and inserted in circuit simulations analyzed in Section 5.4.2.

5.4.1.3: Plasmonic slot phase-shifter analysis

The second element of the MZIR circuit that is thoroughly analyzed in this work is the plasmonic MIM slot PS capable of supporting highly confined plasmonic modes in the BTO. This material features an approximately 3x-10x higher effective electro-optic coefficient versus $LiNbO_3$ [56]. The optical mode in the slot was calculated by solving the eigenvalue problem at λ = 1310nm at the 2D-plane for each voltage V with the Finite Difference Method (FDE).

Towards the analysis of the plasmonic slot PS, by sweeping the applied voltages, Poisson's Equation was initially solved at the cross-sectional plane of the slot employing FEM simulations for the calculation of the RF electric field distribution $|\mathbf{E_{RF}}|$. Then the effective index $n_{eff} = Re\{n_{eff}\} + iIm\{n_{eff}\}$ was calculated at each voltage V, by applying the RF field distribution with the interpolation of the 2D-FDE mesh. The calculation of the n_{eff} considers the following equations with $r_{13} = r_{23} = 8$ pm/V, $r_{33} = 33$ pm/V and $r_{42} = 807.53$ pm/V [56, 57]. The material refractive index elements were derived from [56, 57].

$$n = \begin{bmatrix} n_0 & 0 & 0 \\ 0 & n_0 & 0 \\ 0 & 0 & n_e \end{bmatrix} \quad (5.46)$$

$$\Delta\left(\frac{1}{n^2}\right) = \begin{bmatrix} r_{13}E_z' & 0 & r_{51}E_x' \\ 0 & r_{13}E_z' & r_{42}E_y' \\ r_{51}E_x' & r_{42}E_y' & r_{33}E_z' \end{bmatrix} \quad (5.47)$$

A 2D-Schematic of the plasmonic slot PS can be seen in Fig. 5.4.3(a). The plasmonic metal is either Cu, Au, or Ag and has a thickness of 100nm. The slot width is 100 nm

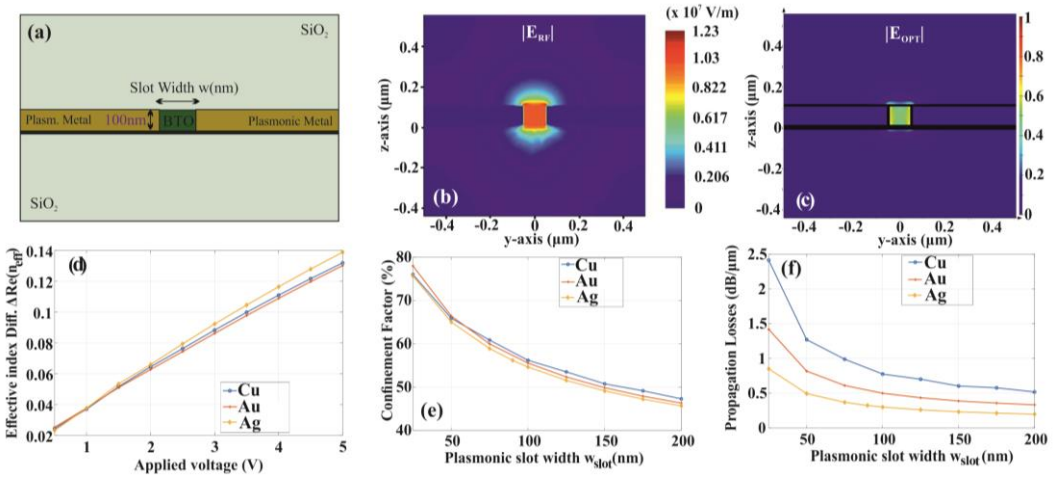


Fig. 5.4.3: (a) Cross section of the designed Cu/BTO/Cu (MIM) plasmonic slot PS, (b) Calculated RF- Electric field $|\mathbf{E_{RF}}|$, (c) and Optical Modal Field $|\mathbf{E_{OPT}}|$, (d) Effective index change $\Delta Re(n_{eff})$ vs Applied Voltage V (e) Confinement Factor (%) of the Optical mode in the plasmonic slot, (f) Propagation Losses (dB/μm) for increasing slot widths.

for Au and Cu, and 90 nm for Ag, considering the practical limitations of standard low cost 193nm DUV lithography [58]. The slight difference in the width of the Ag slot is coming from the optimization of the coupling to the Si_3N_4 layer [28]. The numerically calculated RF electric field for an applied voltage of 1V and optical modal field can be seen in Fig. 5.4.3(b) and (c), respectively, where a great overlap between the RF and optical field in the slot filled by the BTO material is revealed. The effective index change $\Delta Re\{n_{eff}\}$ for increasing voltage values can be seen in Fig. 5.4.3(d). From there it is evident that the three materials exhibit almost the same slope, indicating an almost identical half-wave voltage length product (V_π x L). Indeed, the calculations show for Ag and slot width 90 nm, the V_π x L is 25.51 Vμm, for Au and slot width 100 nm it is 28.06V μm, while for the CMOS Cu and 100 nm slot again, the V_π x L is 27.33 Vμm. These low values are justified by the high RF and optical electric field confinement in the BTO material of the slot waveguide as seen in Fig. 5.4.3(e). The same figure reveals that the confinement factor in the slot ranges from 80% to 45% independent from the type of metal and defined almost exclusively by the slot width. However, Fig. 5.4.3(f) shows a clear tradeoff between CMOS compatibility and propagation performance as the propagation losses vary from 0.33dB/μm for Ag, to 0.5 dB/μm for Au and finally to 0.78 dB/μm for Cu. It should be noted that Cu PS losses is more than double versus the Ag based one with subsequent degradation of the overall performance. Finally, alternative interesting CMOS compatible plasmonic metals, Al and TiN, were examined as possible solutions for plasmonic phase shifters. Nevertheless, the calculated propagation losses of TiN were above 10dB/μm, while the losses for Al were 1.47dB/μm for slot widths of 100nm. For this reason, Copper (Cu) was chosen as the most favorable CMOS compatible metal.

5.4.1.4: Tri-layer low-loss Si_3N_4-to-plasmonic slot coupling interfaces

A tri-layer coupling interface from a Si_3N_4 waveguide to the plasmonic slot waveguide was strategically designed targeting ultra-low coupling losses to the plasmonic slots presented above. The schematic and the numerical results from 3D-FDTD simulations of the configuration for Au, Ag and Cu can be seen in Fig. 5.4.4. The interface follows a two-step approach: (a) an adiabatic coupler from an 800x800nm^2 Si_3N_4 to an aSi interposer waveguide with a width of 350nm and a thickness of 220nm, and (b) a directional coupler from the aSi waveguide to the plasmonic slot PS. Two 100nm thick intermediate dielectric SiO_2 layers are deployed to separate the three layers. Fig. 5.4.4(b) and (c) present the simulated electric-field distribution for the adiabatic coupling and the directional coupling step, respectively. The losses induced from the adiabatic coupler are as low as 0.06 dB per transition, while

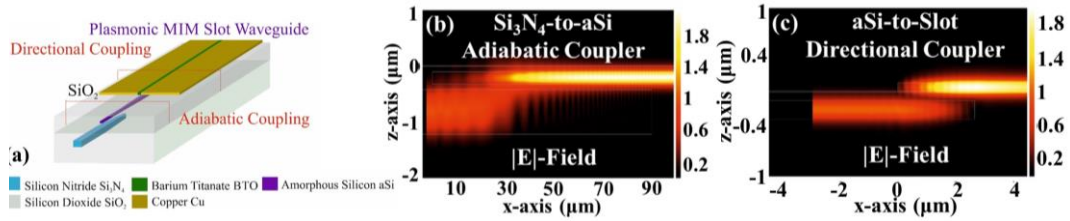


Fig. 5.4.4: (a) 3D-depiction of Si_3N_4-to-plasmonic slot two-step interface geometry, (b) Numerically calculated electric field |**E**| for the adiabatic coupling step from Si_3N_4-to-aSi, and (c) for the directional coupling step from aSi-to-Cu-plasmonic-slot.

for the directional coupler, the lowest coupling losses ensue for Ag having a value of 0.71 dB for a coupling length L_c = 2.78 μm. For Au this value increases to 0.86 dB per directional coupler for L_c = 2.71 μm, while for CMOS Cu the coupling losses increase to 1.23 dB. More details for the coupling interface are provided in our previous work [28], also highlighting the tolerances of the device to fabrication variations.

5.4.2: MZI-Ring frequency-domain analysis

5.4.2.1: Semi-analytic numerical calculations

For the initial study of the physical characteristics of the MZIR modulator, a semi-analytic model was formulated and applied as a starting point towards better understanding of the device physics, as presented in [59] and Section 5.3. The energy efficiency and the voltage reduction of the MZIR with respect to the symmetric MZM for the same material platform and equal PS length is dependent on the total cavity losses. The metric signifying the Voltage Reduction Factor (VRF) of the MZIR with respect to the equivalent MZM modulator is related to the comparison factor CF and is provided by the following relations:

$$CF = \frac{V_{crit}}{V_\pi} = \frac{2\cos^{-1}(A_{tot})}{\pi} \quad (5.48)$$

$$VRF(\%) = (1 - CF) \times 100 \quad (5.49)$$

The voltage reduction factor VRF (%) defined in Eq. (5.48) and (5.49) is the percentage reduction to the V_π value of an MZIR modulator versus an equivalent non-resonant symmetric MZI. The CF value is dependent on the total losses in the MZIR cavity included in the parameter A_{tot} calculated from (5.48) and the V_π x L product for each PS in the MZIR is reduced with respect to the symmetric MZM according to (5.50):

$$V'_\pi L = V_{crit} L = \frac{V_\pi \cos^{-1}(A_{tot})}{\pi} L = \left[\frac{\cos^{-1}(A_{tot})}{\pi}\right] V_\pi L \quad (5.50)$$

Equation (5.50) refers to the voltage that needs to be applied in each of the two PSs for the maximization of the ER in the MZIR. The calculated total device Insertion

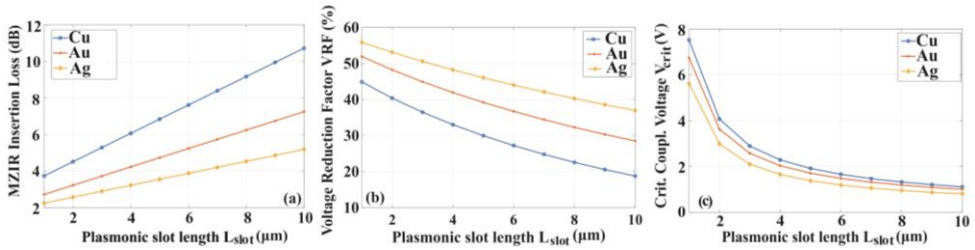


Fig. 5.4.5: (a) MZIR Insertion Loss IL (dB), (b) Voltage Reduction Factor VRF (%), (c) Critical Coupling Voltage V_{crit}(V) for increasing plasmonic slot PS lengths L_{slot} (μm)

Losses (IL), Voltage Reduction Factor (VRF), critical coupling voltage V_{crit} for equations (5.48) and (5.50) are illustrated in Fig. 5.4.5 for different PS lengths. As seen in Fig. 5.4.5(a), the insertion losses are the lowest in Ag-based MZIRs with the value retained below 5 dB for slot lengths up to 10 μm. For Au, the IL are a little higher ranging from 2 up to 7.3 dB and for Cu, the IL increase to 11 dB for a slot length of 10 μm that was expected from the simulations results in the previous Section. Fig. 5.4.5(b) and Fig. 5.4.5(c) present the voltage reduction factor and the critical coupling voltage results, respectively, where it is evident a clear trade-off exists between the two metrics. For lower plasmonic slot PS lengths the total cavity losses are lower, therefore the VRF is higher. However, the critical coupling voltage is also higher as it is inversely dependent on the PS length, originating from the $V_{\pi} \times L$ product of (5.50). For Ag, the reduction factor VRF ranges from 55.9% down to 40% for slot lengths between 1 μm and 10 μm. However, the critical coupling voltage V_{crit} becomes sub-Volt for lengths higher than 7 μm, rendering feasible driving these modulators directly from ASICs. For Au, the VRF varies from 52% to 28.5% with the V_{crit} becoming sub-Volt for lengths higher than 9.5 μm. Finally, for the Cu CMOS metal, the VRF is as high as 44.9% for L_{slot} = 1 μm and decreases to 18.7% for 10 μm long PSs. The critical coupling voltage, on the other side, decreases to 1.1V at L_{slot} = 10μm.

5.4.2.2: Frequency-domain circuit analysis

The broadband scattering matrices for all the designed components presented in the previous Section were inserted in Ansys-Lumerical INTERCONNECT for system level simulation evaluation. The S-matrices of the coupling interfaces allow for the study of any effects that may occur due to potential parasitic reflections. All the photonic and plasmonic waveguides were represented as compact models with a variable length. The model takes into account all the parameters such as effective index n_{eff}, group index n_g, propagation loss PL and dispersive characteristics. Fig. 5.4.6(a)-(c) present the simulated power transfer functions in for all three metals for 5 μm long PS. The calculated Free-Spectral Range (FSR) in all cases is approximately 75.17 GHz. The static responses of Fig. 5.4.6(a)-(c) indicate optimum performance for Ag as it was expected. For a 5 μm plasmonic slot PS, the IL are as low as 3.56 dB with the ER being higher than 30 dB for total absolute differentially applied voltage of 2x1.3 V_{pp}. The VRF is 46% versus the symmetric MZI with the same PS length. For the same length in Au, the IL increase to 4.74 dB, while the ER is 11 dB for the same voltage. Due to the slightly increased cavity losses, the VRF is decreased to 39%. Finally, if CMOS compatibility is mandatory, the Cu-based MZIR with the same length induces IL of

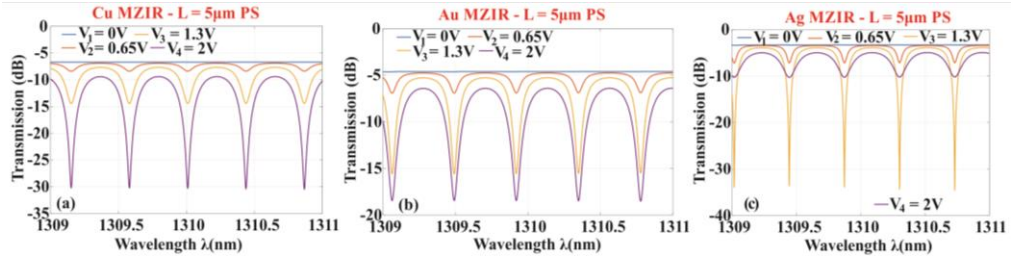


Fig. 5.4.6: (a) Calculated Power Transfer Functions of the MZIR Cu-based slot, (b) Au-based slot, and (c) Ag-based slot for PS length L_{slot} = 5 μm for various applied voltages at each of the two MZI arms.

6.85dB, still below than a 10 dB minimum value common in plasmonic modulators and an ER of higher than 7.7 dB for 2x1.3V_{pp} voltage. The VRF in this case is 0.3 indicating at least a 30% voltage reduction compared to Cu-based MZIs.

5.4.3: System-level Modulator Performance

5.4.3.1: Electro-optic Bandwidth Analysis

This section presents the simulation and circuit analysis in the time-domain. The MZIR modulator bandwidth has been theoretically studied in [60] and applied in [61] for Si-MZIRs and [55] for a low-FSR based $LiNbO_3$ MZIR. As the supplement of [55] mentions in a detailed manner, for an input wavelength aligned at the MZIR wavelength peaks, the MZIR FSR creates resonances at the EO response of the modulator, having a negligible effect on its 3dB cutoff frequency. Even with low FSR, the calculated EO bandwidth of the MZIR device of [55] can extend up to 40 THz. Based on [55] the PS bandwidth is the sole limiting factor contributing to the overall modulator bandwidth. In this MZIR design optimization, the plasmonic slot PSs are lumped elements with a length of 5μm, much lower than the RF wavelength. Their 3dB bandwidth can reach up to 500GHz in plasmonic slots employing the POH technology [62]. The capacitance of the 100nm wide slot is calculated between 44.3 fF for 5 μm to 177fF for 20μm long PSs.

5.4.3.2: System-Level Simulations

A system-level simulation setup illustrated in Fig. 5.4.7 was created for the evaluation of the signal quality at 112 Gbaud PAM-4 targeting compatibility with next generation 51.2Tb/s ASIC routers SerDes interfaces. The S-parameters of the laser input/ fiber output coupling elements were excluded from the model, so as to evaluate the modulator performance exclusively. In the setup, a laser optically feeds the MZIR modulator with its wavelength aligned to a resonance of the MZIR transfer function near 1310nm. The laser output optical power P_{in} is variable with the laser linewidth equal to 1MHz assuming normal low-cost GaAs DFB technology. A PRBS-13 electrical signal from a Pulse Pattern Generator is fed to a PAM-N symbol mapper that correlates each of the PAM-4 symbols to each voltage value differentially applied to

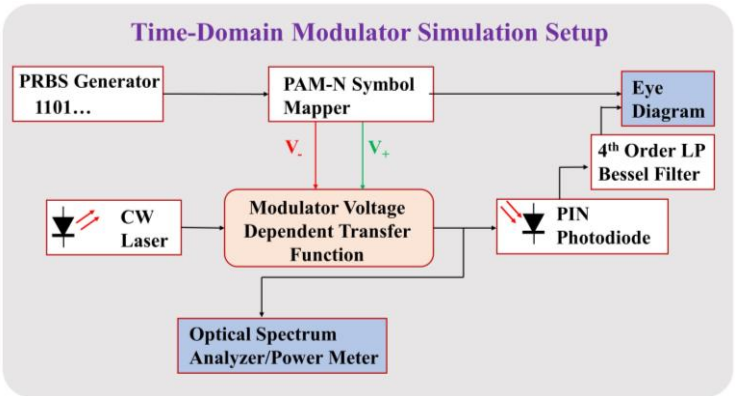


Fig. 5.4.7: Time-Domain System Level Simulation setup for the evaluation of the signal quality received from MZIR modulators.

the MZIR device. The rise and the fall time of the driving electrical pulses is equal to 0.35 of the bit period i.e. 3.3ps. After modulation, the signal is received by a PIN photodiode (PD) with a responsivity of 0.55A/W, a dark current of 10nA, and a thermal noise equal to -173.8dBm/Hz. The signal finally passes from a 4th order Low-Pass Bessel Filter of 84 GHz 3dB bandwidth and then it is recorded on the oscilloscope. An optical spectrum analyzer monitors the signal before the OE conversion at the PD. It should be highlighted that no DSP is applied on the transmitter or receiver side for enhanced signal quality.

Fig. 5.4.8 presents the numerical results of the system level simulations regarding the quality of the signal generated by the MZIR and MZM modulators. Fig. 5.4.8(a) shows the initial calculations of measuring the output power from the modulator by sweeping the applied voltage to the MZMs or MZIRs for the three plasmonic metals. The resonance peaks in the MZIRs assist to the exhibition of significantly higher extinction ratios relative to the MZMs even for low driving voltages. As such the Ag MZIRs present an extinction ratio 22.4dB higher than the Ag MZM, while this value decreases to 19.6dB for Au, and to 16.251dB for Cu. Fig. 5.4.8(b) shows the calculated Q-factor with respect to the input power P_{in} injected by the laser to the modulator for 2x1.3V_{pp} applied to the two PSs assuming a high-power ASIC. The improvement in the signal quality of the MZIR versus the MZM at 10dBm input power is 2.9, 2.4 and 1.3 for the Ag, Au and Cu based modulators, respectively. The bias is equal to 0V in all scenarios targeting equivalent power consumption comparison. In terms of absolute values, the Q-factor is 9.4 for Ag, 7.15 for Au and 5.66 for Cu at 10dBm, easily surpassing the error free transmission threshold under FEC at up to 2km [63]. The Q-Factor has been defined through the BER_Q value as in [63], where N=4:

$$BER_Q = \frac{1}{Nlog_2(N)}\frac{1}{2}\sum_{i=1}^{N}\left[erfc\left(\frac{\widehat{\mu_i} - I_{i-low}^{th}}{\widehat{\sigma_i}\sqrt{2}}\right) + erfc\left(\frac{\widehat{\mu_i} - I_{i-high}^{th}}{\widehat{\sigma_i}\sqrt{2}}\right)\right] \quad (5.51)$$

$$Q = \sqrt{2}erfc^{-1}\left(2BER_Q\right) \quad (5.52)$$

Fig. 5.4.8(c) illustrates the same results under the assumption of 2x0.8V_{pp} driving voltage from the ASIC, considering a more conservative approach for high speed SerDes interfaces. The signal improvement now is 2.77 for Ag, 2.23 for Au and 0.88 for Cu and come in agreement with the results of Fig. 5.4.8(a). The lower voltage swing does not enable now reaching the resonance peak at the MZIRs, however the improvement in the signal quality from the MZIR is still valid. The absolute values for 10dBm input power are now 8.3 for Ag, 6.9 for Au and 5.2 for Cu. It should be highlighted that the Cu based MZIR is error free under FEC even in this highly demanding scenario revealing the high-quality signal generated by the MZIR modulator at such low voltages. The higher drop in the ΔQ of the Cu versus the Ag and Au when driving the modulator with 2x0.8V_{pp} instead of 2x1.3V_{pp} is coming from the lower Q factor of the resonance peaks in the Cu MZIR due to the higher propagation losses in the plasmonic PSs. Again, no biasing is considered assuming direct drive from the ASIC. Finally, Fig. 5.4.8(d)-(f) present the corresponding eye diagrams for the three MZIR modulators for 2x0.8V_{pp} driving voltage and 10 dBm input power. The recorded

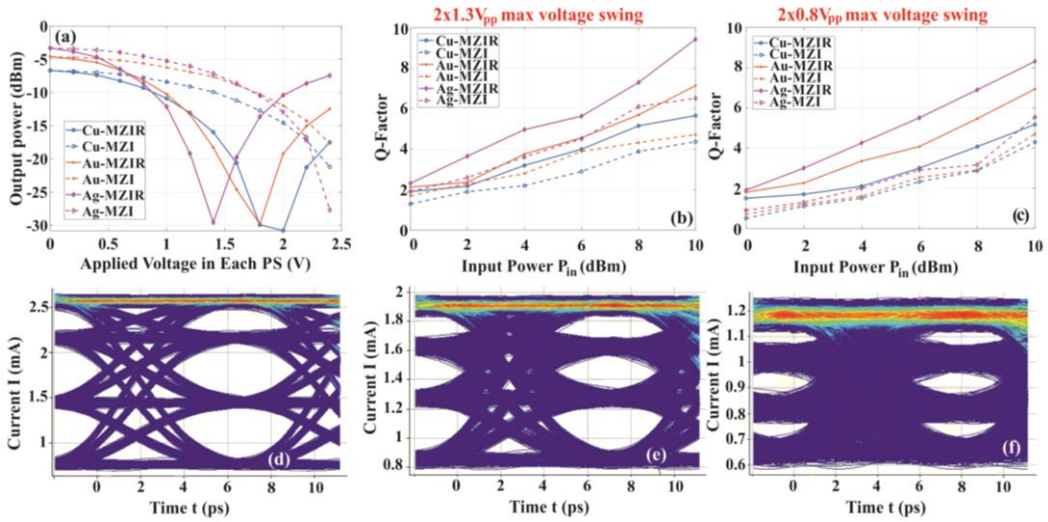


Fig. 5.4.8: (a) Output power (dBm) vs. applied voltage per PS (V) for 0 dBm input power, (b) Q-Factor vs. Input Power P_{in} for Cu, Au, and Ag MZIRs and MZMs with L_{slot} = 5 μm for 2x1.3V_{pp} driving voltage, (c) Q-Factor vs. Input Power P_{in} for Cu, Au, and Ag MZIRs and MZMs with L_{slot} = 5 μm for 2x0.8V_{pp} driving voltage, (d), (e) and (f) eye diagrams of signal generated by the MZIR at 112 Gbaud PAM-4 for 2x0.8V_{pp} applied voltage and 10dBm input power for Ag, Au and Cu metal on the plasmonic slot.

eyes are clearly open in all cases with Ag exhibiting the best performance due to the lowest overall losses. The asymmetry in the eyes for the three metals is due to the manual setting of the four levels from the driver. Higher Q-factor values are expected with proper optimization tools. Also, the effect of the extra π difference induced in the MZI branches of the MZIR is also noted here, setting the initial state of the MZIR to the highest output power.

5.4.4: Degradation of BTO EO-coefficient

In the work presented so far in this article, the main electro-optic coefficient element r_{42} has a value of 807.53pm/V. However, various experimental studies have reported the variability of r_{42}, resulting in different values for the effective electro-optic coefficient, r_{eff}, as mentioned in reference [56]. This is related with the variability of r_{42} due to surface effects and stress induced phenomena [64]. In [56] the reported r_{42} for the stress-free and stressed BTO was 1300±100pm/V and 730±100pm/V, respectively. In [64], 2D-FEM simulations were performed considering a value of 923±215pm/V. This section of Chapter 5 aims to assess the performance degradation of the proposed MZIR modulator design under degraded r_{42} coefficients. Initially, in Fig. 5.4.9(a), the V_π x L product of the phase-shifter is calculated for increasing r_{42} coefficient values as r_{42}/N with N equal to 1,2, and 4. For N=2, corresponding to an effective electro-optic coefficient r_{eff} of 300pm/V, the V_π x L is close to 49Vμm for Ag, 53.65Vμm for Cu and 54.44Vμm for Au, still considerably low compared to other plasmonic modulator demonstrations in the literature. Fig. 5.4.9(b) and (c) illustrate the MZIR insertion loss and the VRF (%) for varying plasmonic slot lengths. Again, the choice of the PS length determines the total losses and the energy advantage of the MZIR versus symmetric MZI-modulators. For Ag, the most favorable plasmonic metal among the three, the losses remain below 10 dB even for lengths up to 20 μm, while the voltage reduction factor exceeds 25% when compared with symmetric MZI designs. Figures 5.4.9(d)-(f)

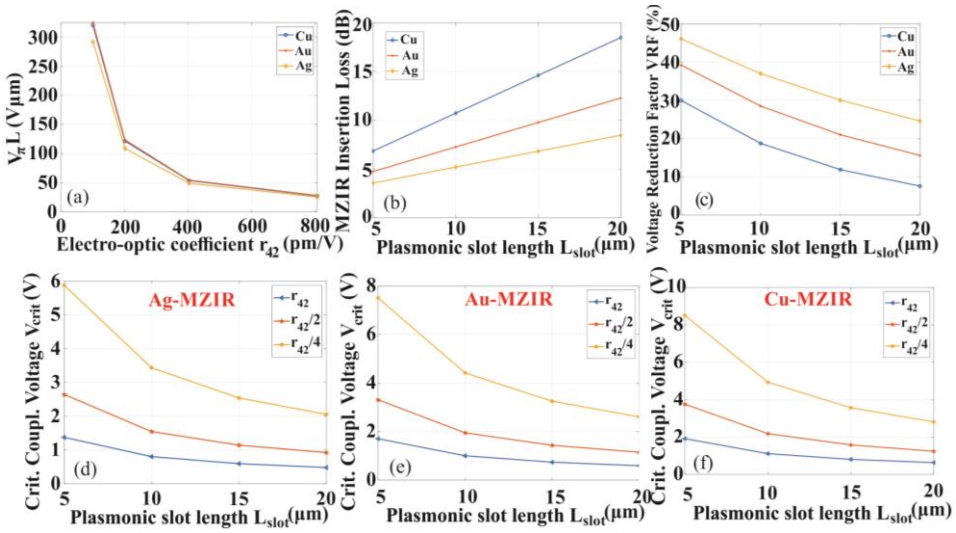


Fig. 5.4.9: (a) Half-wave voltage-length product $V_\pi L$ (Vμm) for varying electro-optic coefficient r_{42} (pm/V) value, (b) MZIR Insertion Loss (dB) and (c) Voltage Reduction Factor (VRF) (%) for increasing plasmonic slot length L_{slot} (μm), and critical coupling voltage V_{crit} (V) for (d) Ag-MZIR, (e) Au-MZIR, and (f) Cu-MZIR for increasing plasmonic-slot length and for r_{42}, $r_{42}/2$ and $r_{42}/4$.

illustrate the voltages required per phase shifter (PS) to achieve maximum extinction ratio (ER) in the resonant modulator for Ag, Au, and Cu, with slot lengths ranging from 5μm to 20μm. The electro optic coefficient of the BTO is varied between r_{42}, $r_{42}/2$ and $r_{42}/4$. For an $r_{42}/2$ and a PS length of 10μm, the corresponding critical coupling voltages for the MZIR are $1.55V_{pp}$ for Ag, $1.95V_{pp}$ for Au, and $2.18V_{pp}$ for Cu. With this slot length the reduction factor would become 18% for Cu, 28% for Au and 37% for Ag. Considering an upper voltage of $1.8V_{pp}$ from typical ASICs, then the length of the PS should be increased to 15 μm or even higher for Cu. The high insertion losses then limit the VRF to 11.88%, drawing the limit of where CMOS compatible MZIRs provide a viable advantage over symmetric MZIs. For $r_{42}/4$ the critical coupling voltages are higher than $2.807V_{pp}$ for Cu even for 20 μm long PSs rendering this technology not compatible with CPO.

5.4.5: Discussion and comparison with other modulator devices

Table 5.1 summarizes the main outcomes of this work and compares them with significant work presented in the literature on high-speed photonic and plasmonic modulators in the Si_3N_4 platform. As noted in the table, resonant modulators in the Si_3N_4-Si [33] and the Si_3N_4-graphene [34] platform, are all based on carrier effects and cannot deliver baud rates over 30Gbaud. Going to ferroelectric modulators on Si_3N_4, the PZT-on-Si_3N_4 platform is limited by a much larger PS length and the baud-rates achieved do not surpass 40Gbaud for NRZ modulation [36]. On the other side, the Si_3N_4-LNOI platform has achieved sub-volt voltages [38], but with the PS length surpassing 2cm and very high single-channel baud rates of 80Gbaud in OOK [39] and 64Gbaud in PAM-4 modulation [45]. Photonic Si_3N_4-LNOI MZMs with an ultra-high bandwidth of 1.26THz numerically [41] and 95GHz experimentally [44] have also been demonstrated. Nevertheless, the designs are bound by either a high required voltage over $7V_{pp}$ for a 3mm PS length, or by 10mm long PS for lower voltage of 3.75V. In [53], a plasmo-photonic Si-POH micro-racetrack design has broken the power-

bandwidth limitations in the C-band achieving 160 Gbaud PAM-4 transmission with only 1.2V_{pp} driving voltage and 1.2dB insertion losses. The POH technology, however, suffers from the aging and maximum temperature envelope of all polymers. The sole work to overcome the footprint, voltage, and bandwidth trade-offs on the Si_3N_4 platform is [47, 48] utilizing plasmonic slot BTO PS, achieving ultra-high baud rates of 128Gbaud with PAM-4 modulation using a voltage of 3.6V_{pp} and a 3V bias. Nevertheless, the design is based on Au and therefore not suitable for the potential mass-production, promised by the CMOS technology. Also, the MZM requires a higher PS length, and the circuit is optimized for the C-band. The simulated insertion loss value of [47, 48] is significantly lower (<6.5dB PS), with the Au propagation loss being only 0.3dB/μm. However, this comes at a cost of a less confining plasmonic slot, leading to larger lengths and a hybrid Si_3N_4-BTO cross-sectional design exhibiting sensitivity of the V_π x L value to material thickness variations. For the first time, herein, the performance of three MZIR modulators featuring plasmonic slot BTO PS based on Cu, Ag, and Au is presented for the O-band. This work demonstrates devices with the lowest footprint, capable of operating at ultra-high baud rates, while requiring low voltages compatible with co-packaging applications in the O-band. The performance of the MZIR is highlighted by the at least 30% lower V_π x L product required versus the symmetric MZM in order to obtain the same extinction ratio confirmed by the time domain simulations revealing a signal improvement of ΔQ between 1.3 and 2.9 for 10dBm input power and 2x1.3V_{pp} driving voltage.

Table 5.1. Comparison of the presented CMOS Plasmo-Photonic Modulators with the State of the Art (Sim. = Simulated, Exp. = Experimental, PS = Measured from Phase-Shifter only (No resonant enhancement))

Ref.	Platform and Design Type	V_π' × L (Vmm)	Phase Shifter Length	Baud Rate and Mod. Format	Insertion Losses (dB)	Extinction Ratio (dB)	Voltage Level (V_{pp})	Band
[33]	Si-on-Si_3N_4 Ring Mod. (Exp.)	2.6 (PS)	~196.35μm	13Gbaud NRZ	2.5	10	1.6	O
[34]	Graphene-on-Si_3N_4 Ring Mod. (Exp.)	–	~251.33μm	22Gbaud NRZ (BW = 30 GHz)	–	15	10	C
[36]	PZT-on-Si_3N_4 Hybrid Ring Mod. (Exp.)	32	524μm	40Gbaud NRZ	~5-6 dB/cm	3.1dB (@ 10Gb/s)	4.2	C and O
[38]	LNOI-on-Si_3N_4 MZI (Exp.)	21.1	2.4cm	_ (BW = several GHz)	5.4	30	0.875	C
[39]	LNOI-on-Si_3N_4 MZI (Exp.)	22.4	7.8mm	80Gbaud OOK	–	20	2.8	C
[43]	LNOI-on-Si_3N_4 Racetrack Mod. (Exp.)	–	1.5mm	70Gbaud OOK	0.5	10.5	4	C
[45]	Si_3N_4-LNOI Heterogeneous MZI (Exp.)	30	7mm	64Gbaud PAM-4 (BW = 37GHz)	1	33	4.3	C
[41]	Si_3N_4-LNOI MZI (Sim.)	23.3	3mm	_ (BW = 1.26THz)	~0.15	–	~7.77	C
[44]	Si_3N_4-LNOI MZI (Exp.)	37.5	10mm	_ (BW = 95GHz)	~2.07	45	3.75	C

[53]	POH on Si Micro-Racetrack (Exp)	0.15 (PS)	7.5μm	160 Gbaud PAM-4 (320Gb/s)	1.2	>12	1.2V_{pp} (0.6 V_p)	C
[47, 48]	Au-based Plasmonic BTO on Si_3N_4 MZI (Exp.)	0.144	15μm	128Gbaud PAM-4 (256Gb/s)	22(Exp.)	–	3.6V_{pp} (V_{bias} = 3V)	C
[47, 48]	Au-based Plasmonic BTO on Si_3N_4 MZI (Sim.)	0.144	15μm	128Gbaud PAM-4 (256Gb/s)	<10dB (Sim.) (<6.5dB from PS)	–	<3.6V_{pp} (V_{bias} = 3V)	C
This work	Ag-based Plasmonic slot BTO MZIR on Si_3N_4 (Sim.)	0.014	5μm	112Gbaud PAM-4 (224Gb/s)	3.56	>30	2x1.3	O
This work	Au-based Plasmonic slot BTO MZIR. on Si_3N_4 (Sim.)	0.017	5μm	112Gbaud PAM-4 (224 Gb/s)	4.74	>11	2x1.3	O
This work	Cu-based Plasmonic slot BTO MZIR on Si_3N_4 (Sim.)	0.019	5μm	112GBaud PAM-4 (224Gb/s)	6.85	>7.7	2x1.3	O

5.5: BTO-based O-band Sub-Volt CMOS Compatible Plasmonic Racetrack Modulator on Si_3N_4

5.5.1: Introduction

Silicon nitride (Si_3N_4) photonic technologies have been recently introduced to telecommunications and microwave photonics applications, harnessing their low cost to outperform conventional SOI based devices. The advancement of the heterogeneous and hybrid integration processes now enables the formation of complete transceivers with the co-integration of lasers, modulators, and photodetectors on the same chip [23, 27]. Although Si_3N_4 lacks inherent modulation capabilities, bonding of Lithium Niobate ($LiNbO_3$) and Barium Titanate Oxide (BTO) thin films, boasting high Pockels coefficient values in combination with high temperature tolerance has boosted the appeal for high-speed Si_3N_4 PICs [65]. In this way, a folded intensity Si_3N_4-$LiNbO_3$ modulator was experimentally demonstrated with a 10mm interaction length, a power extinction ratio (ER) of 23dB and a 3dB-bandwidth (3dB-BW or BW) of 37.5GHz [66]. Also, in [67], a Si_3N_4-loaded $LiNbO_3$ Mach-Zehnder (MZI) Modulator was presented with a half-wave voltage length product V_π x L of 2.24Vcm and a 3dB-BW of 30GHz achieving 80Gb/s OOK signal transmission. Nevertheless, most works so-far have been restricted by high footprint, high driving voltage, or low 3dB-bandwidths.

In this work we present a tri-layer ultra-compact racetrack CMOS plasmonic modulator achieving bandwidths (3dB-BW) over 65GHz, and an Extinction Ratio (ER) higher than 28dB for only 0.5V_{pp} voltage for a 5μm long Cu-based plasmonic slot phase-shifter. A schematic of the proposed racetrack modulator can be seen in Fig. 5.5.1(a) containing the plasmonic MIM slot phase-shifter filled with a ferroelectric BTO material [56], the Si_3N_4 ring passive circuitry with a minimum radius of 50μm, and the tri-layer Si_3N_4-to-plasmonic slot coupling interface of [28]. The BTO is bonded on top of aSi that in turn is deposited on top of the Si_3N_4 passive platform (Fig. 5.5.1(b)). The interlayer SiO_2 gap is 100nm at all cases. The resonator length is the minimum possible to bring forth as high Free Spectral Range (FSR) values as possible.

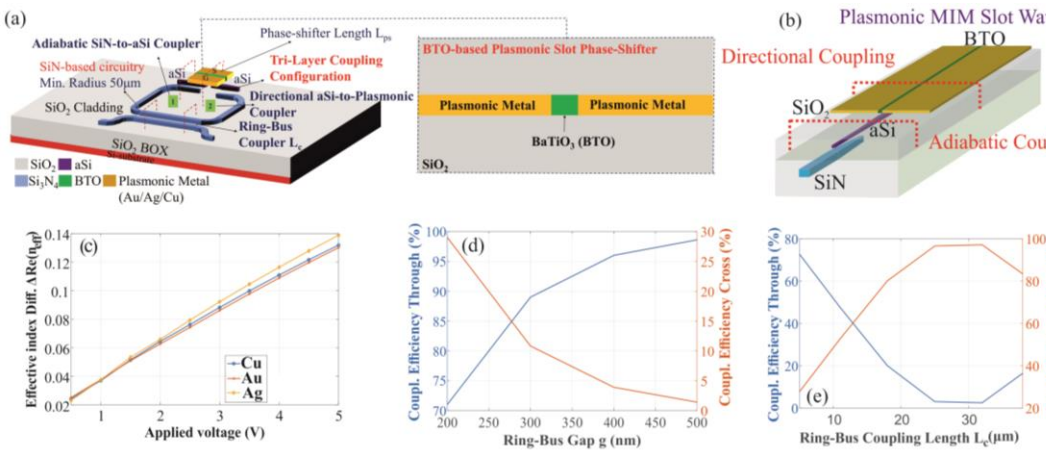


Fig. 5.5.1 (a) 3D Schematic of the proposed Plasmonic Ferroelectric Racetrack Modulator. At the inset at the right side of the image a cross-sectional view of the BTO-based plasmonic slot phase-shifter is provided, (b) 3D Schematic of the utilized Si_3N_4-to-aSi-to-MIM BTO photonic to plasmonic coupling interface, (c) Effective Refractive Index Change $\Delta Re(n_{eff})$ vs Applied Voltage, (d) Coupling Efficiency (%) at the two ports of the ring-bus coupler vs Gap g (nm) for L_c = 0 and (e) vs L_c for g = 300nm.

5.5.2: Computational electromagnetic analysis and results

The DC performance of the modulator device was extracted by the Scattering Matrix Method [68], importing the individual S-matrix parameters obtained from 3D-FDTD, 2D-FEM and 2D-FDE simulations. Fig. 5.5.1(c) shows the electro-optic response of the plasmonic slot phase-shifter vs. increasing voltages for Ag, Au, and Cu for quantitative identification of the performance compromise between noble and CMOS metals. The V_π x L product for an Ag plasmonic slot with w = 90 nm was only 25.51 Vμm, for Au with w = 100 nm was 28.06 Vμm and for Cu for w = 100 nm was 27.73 Vμm. The slot widths were chosen to facilitate optimum photonic-to-plasmonic coupling. For the transition to the Ag plasmonic slot the total simulated coupling loss at λ=1310nm was close to 0.77dB, while for the Au and Cu plasmonic slots the value increased to 0.92dB and 1.29dB per coupler, respectively. Fig. 5.5.1(d) and (e) depict the Ring-Bus Coupler analysis via 3D-FDTD simulations, where it is shown that by adding a straight coupling length in the ring-bus coupler the cross-coupling efficiency ($|k|^2$) can be easily increased to values ranging from 1.4% up to 80%. A 0.5dB insertion loss is considered in each ring-bus coupler to anticipate scattering events.

Figure 5.5.2(a)-(c) demonstrate the passive characteristics of the photonic-plasmonic resonator for an increasing $|k|^2$ value. The phase-shifter has a length of only 5μm for Au, Ag, and Cu. It is clear from Fig. 5.5.2(a) that the insertion losses (IL) for all cases remain lower than 4dB. Fig. 5.5.2(b)-(c) indicate that the simultaneous achievement of high ER and BW can be attained for 5μm long Au, Ag, and Cu phase-shifters near $|k|^2$ of 67%, 55% and 80% respectively. The resonator IL for the three plasmonic metals are 1.75dB, 1.17dB and 2.99dB. For Au a resonance ER of 39.74dB is achieved with a 54.2GHz BW, for Ag an ER of 44.21dB with a 41.96GHz BW and for Cu an ER of 35.76dB is exhibited with a resonance BW of 66.43GHz. Finally, Fig. 5.5.2(d)-(f) showcase that Cu has the highest performance in terms of bandwidth, reaching at the same time 23dB ER for only 0.5V applied voltage. Au and Ag phase

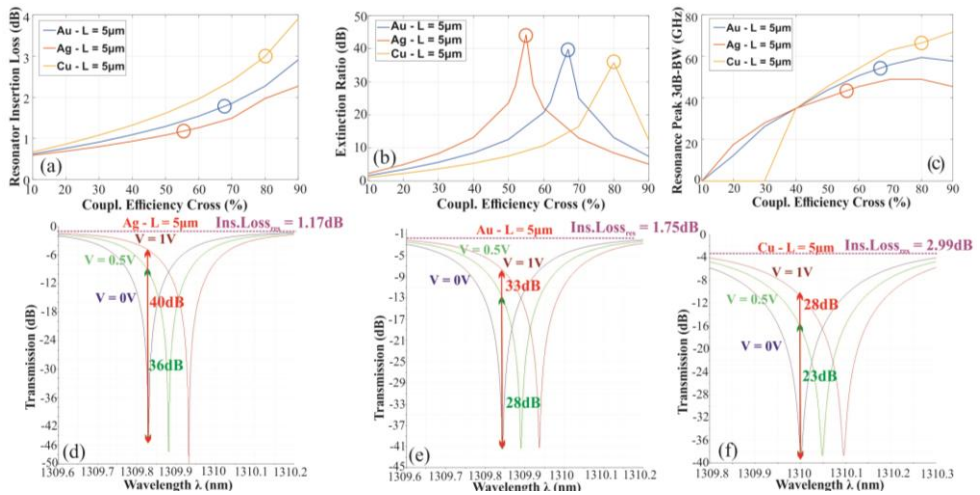


Fig. 5.5.2 (a) Resonator Insertion Loss (dB), (b) Resonance Extinction Ratio and (c) Resonance 3-dB Bandwidth vs Coupling Efficiency (%) at the cross port, Modulator DC response for V=0V, 0.5V and 1V for phase shifter length of 5μm of (d) Ag, (e) Au, and (f) Cu.

shifters achieve higher ER values but at a lower BW, as the Q-factor of the resonant peak is higher from the lower overall losses in the cavity containing the plasmonic phase shifter.

5.6: Chapter Conclusions

This Chapter presented a detailed performance analysis of plasmonic slot ferroelectric Mach-Zehnder in Ring (MZIR) modulator on Si_3N_4 capable of operating at above 100Gbaud in the O-band. The whole study investigates the tradeoff associated with going to CMOS Cu metal for the formation of the plasmonic PS targeting low cost, versus the legacy Ag and Au metals exhibiting the lowest plasmonic propagation losses. In addition, it calculates the energy saving coming from opting for the MZIR layout versus the symmetric MZM in agreement with other recent works in other material platforms. Considering a 5μm long plasmonic PS at each arm of the MZM or MZIR the best results are obtained as expected for Ag, where the insertion losses of the MZIR are 3.56 dB and the V_π x L product is 46% lower than the standard MZM. The Au based MZIR metrics are very close with insertion losses of 4.74dB and V_π x L product 39% lower than the conventional MZM. Finally, the CMOS compatible Cu based MZIR exhibits insertion losses of 6.85dB and V_π x L product 30% lower than its symmetric MZM counterpart. In high-speed operation, the MZIR is generating PAM-4/112 Gbaud signals for 5μm PS length and 2x1.3V_{pp} driving voltage with a ΔQ improvement of 2.9, 2.4 and 1.3 versus symmetric MZMs for the Ag, Au and Cu plasmonic metals, respectively revealing the clear benefits of the MZIR layout. By lowering the driving voltage to 2x0.8V_{pp} the signal improvement is 0.88 for Cu, 2.2 for Au and finally 2.78 for Ag. The absolute Q factor is above 5.2 even for Cu metal and 2x0.8V_{pp} driving voltage indicating an error free link under FEC. All Q factors values mentioned above are for 10dBm at the input of the modulator and no additional biasing. The degradation of the r_{42} value due to thin film effects was also studied through simulations. For $r_{42}/2$ the voltage reduction values become >18% for Cu, >28% for Au and >37% for Ag. Our results pave the way for the adoption of plasmonic modulators as optical engines for

co-packaged optics in next generation 51.2 Tb/s switching ASIC routers. In addition, in this Chapter we numerically proposed a BTO based O-band CMOS plasmonic racetrack modulator on Si_3N_4 with 66GHz BW, an ER higher than 23dB for $0.5V_{pp}$ voltage and IL lower than 3dB. These performance metrics are compared to identical structures fabricated with Au and Ag plasmonic phase shifters, indicating a higher resonance BW for Cu. These modulators achieve insertion losses lower than both the MZI and the MZIR counterparts and lower required voltage for adequately high extinction ratio values. Nevertheless, they are bound by their peak 3dB-bandwidths, with the highest value being 66GHz for Cu. To increase the bandwidth value in a racetrack, one would need to resort to an overcoupled racetrack design thus decreasing the maximum achievable extinction ratio value.

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Chapter 6: Conclusions and Future Work

This PhD Thesis dealt with the simulation and analysis of III-V light emitting and BTO-based plasmonic ferroelectric modulating components in the Si_3N_4 platform. All the building blocks can be a part of an integrated Si_3N_4 transmitter (Tx) module being able to deliver modulated light at >112Gbaud PAM-4 rates per lane for next generation CPO compatible transmitters. This Chapter aims to summarize the main results, to critically examine the impact of the work in each chapter, and to lay out future work that will follow up this PhD Thesis.

6.1: Design of InP-based QD-laser-to-Si_3N_4 coupling interfaces for monolithic co-integration on a Si-substrate

In Chapter 3, Si-rich nitride (SRN) interposer waveguide designs were designed and simulated for efficient coupling from InP-based QD-lasers to Si_3N_4 photonic integrated circuits, exchanging fabrication complexity with optimum performance. From the analysis, three structures, a strip-based, a rib-based and a tri-layer device, were identified as the optimum coupling configurations. Considering the facet angle, the simplest structure was the strip SRN-based configuration exhibiting 1.39dB coupling losses and -54.4dB simulated back-reflections. An intermediate solution between high fabrication complexity and optimum performance was the rib SRN-based device, showcasing 1.11dB coupling losses and -34.81dB back-reflections. Finally, the optimum interface was the tri-layer material rib SRN waveguide structure inducing 0.73dB coupling losses and -52.6dB back reflections, sacrificing the fabrication simplicity of the strip SRN by requiring more lithographic steps. Approximately, 0.3dB propagation losses are also expected in all cases. These values are ideal for the realization of monolithic coherent tunable lasers on Si_3N_4, considering the Vernier-ring based reflector. However, while the SRN waveguides and SRN-to-Si_3N_4 coupling steps have been fabricated and measured, further experimental works are needed for the study of fully co-integrated lasers on Si_3N_4, revealing also any issues due to surface interfacial phenomena that may be observed in fabrication. Recent literature with integrated butt-coupled monolithic lasers on Si indicates the impact of the theoretical work in this thesis. Monolithic Si_3N_4 integrated circuits such as tunable lasers, single sideband modulation transmitters, EMLs, and mode-locked lasers will also need to be explored in both design and experiments when the fabrication processes employed are mature enough.

6.2: Design of Si_3N_4-to-$BaTiO_3$ CMOS-compatible plasmonic waveguide multi-step coupling configurations

In Chapter 4, an efficient multi-step coupling strategy to transfer light from Si_3N_4 waveguides to CMOS-compatible BTO plasmonic phase-shifters was demonstrated. The plasmonic waveguides under study were the plasmonic slot (MIM) and the hybrid plasmonic waveguide (HPW). Two optimum configurations were revealed from the computational electromagnetic analysis. The coupler to the MIM waveguide consisted

of a Si_3N_4-to-aSi adiabatic coupling step and an aSi-to-MIM directional coupling step. The plasmonic slot waveguide boasted a high confinement factor of 56.18% with the propagation losses being 0.78dB/μm. The induced coupling losses were lower than 1.3dB. The coupler to the HPW structure included an initial Si_3N_4-to-aSi/BTO adiabatic coupling step and an end-fire aSi/BTO-to-HPW coupler. The HPW supported a TM mode with 0.5dB/μm propagation losses and 31.49% confinement factor, while the total coupling losses were below 1.8dB. The reported values are important since they render possible the integration of CMOS-compatible plasmonic phase-shifters on the Si_3N_4 platform. However, experimental measurements are expected to highlight the deviations from the expected theoretical values and underline other physical phenomena at play. The experiments should include the observation of interfacial and surface phenomena in the formation of the plasmonic slot in lithography, the measured quality, formation and EO coefficient of the etched BTO material, as well as the yield and deviations due to the differentiation of the SiO_2 gap thickness across the wafer. From a theoretical point of view, more complex coupled hybrid plasmonic waveguides (CHPW) that could lower the propagation loss value of the phase-shifters should also be explored. Additionally, photonic Si_3N_4/BTO modulator structures, incorporating fine-etched BTO and low-loss interfaces with values below 0.2dB per coupler, should be thoroughly further investigated to reduce the total insertion losses.

6.3: Design of Si_3N_4-plasmonic $BaTiO_3$ Mach-Zehnder in a Ring (MZIR) Modulators

Chapter 5 presented the design and analysis of plasmo-photonic MZIR circuits that could deliver >112Gbaud PAM-4 signals per lane with a phase-shifter length of 5μm and a required voltage of $2x1.3V_{pp}$. A comparative analysis between Ag, Au, and Cu was performed with the first being the optimum, in terms of losses and modulator performance metrics, and the third being the solution more compatible with mass-production. A modelling framework from the individual device (phase-shifter, coupling interface, MMI simulation etc.) to the circuit (MZIR frequency domain response simulation) to the system (simplified 112Gbaud PAM-4 intra-DC link performance simulation) was presented via a commercial-grade solver. The numerical models revealed a decrease to the V_π x L product compared to equivalent MZM structures of 46% for Ag, 39% for Au and 30% for Cu versus symmetric MZMs. System-level simulations revealed open PAM-4 eye-diagrams and Q-Factors over 5 in all cases, even with the application of only $2x0.8V_{pp}$ voltages. The simulation-based analysis of this chapter also considered the degradation of the electro-optic (EO) coefficient of the BTO material. For $r_{42}/2$ degraded EO-coefficient, the voltage reduction values became >18% for Cu, >28% for Au and >37% for Ag. These results show the way towards CMOS-compatible plasmonic optical devices for next-generation CPO technologies for >100Gbaud per lane symbol rates. Nevertheless, the work of this chapter studied the standard PAM-4 modulation format encountered in high-speed intra-data center communications nowadays. Future simulations should examine more complex but spectrally efficient techniques, such as PAM-8 modulation, single-sideband modulation, discrete multitone, polybinary modulation and others, using the same optical device. Additionally, the experimental validation of the MZIR and a study of variations to the V_π x L and insertion losses from the theoretical to experimental values

are also fundamental. Improvement of the photonic MZIR counterparts with fine etching of BTO lowering losses should also be investigated.

6.4: System Level Applications

While the traditional application of the individual devices presented in this PhD thesis was the demonstration of optical interconnects and transmitters on Si_3N_4 for low-power intra-data center links, the changing landscape in fields of use for silicon photonics also increases the applicability of the building blocks of this thesis. For example, monolithic tunable lasers on Si_3N_4, composed of coupling interfaces and Vernier ring reflectors (Chapter 3), will be fundamental in future precise clocks, biosensing and gas sensing photonic chip modules. The plasmonic BTO phase-shifters co-integrated with Si_3N_4 (Chapter 4), could be applied for very efficient delay in the different paths of a design in a LIDAR chip, photonic memory cells using the ferroelectricity metrics of the BTO and others. Finally, the hysteresis loop of the plasmonic BTO weight could unlock new application areas such as low-power optical weights and vector multiplication matrices in integrated photonic neural networks. Simulations should be performed in all cases to identify the suitability and system performance metrics after the utilization of the photonic and plasmonic Si_3N_4 devices of this PhD Thesis.

6.5: Chapter Conclusions

In conclusion, this Thesis presented a novel ideal high-speed platform on Si_3N_4 for Tx modules featuring the best in class materials and devices, such as low-linewidth III-V QD-lasers for the light carrier emitters and plasmonic ferroelectric BTO phase shifters for modulation. A low-power Si_3N_4-plasmonic MZIR modulator based on this high-speed Tx platform is shown to deliver over 112Gbaud PAM-4 signals (224Gb/s line rate) with a BER below the FEC limit in simulation. The devices are realized with ultra-low-loss coupling interfaces between the passive Si_3N_4 waveguides and the lasers and phase shifters, brought forth by the best in class reliable and low-cost fabrication processes. All the designs of this thesis are CMOS-compatible and can be mass-produced using low-cost 193nm DUV lithography. The active photonic and plasmonic devices of this Thesis provide a multitude of application scenarios. In all cases, experimental measurements ought to identify the yield of the fabrication processes necessary for the production of all presented devices. These experiments are crucial for gaining a deeper physical understanding of various fabrication-induced phenomena. This includes investigating surface and interfacial phenomena in SRN waveguides, as well as degradation effects in electro-optic plasmonic BTO phase shifters. As the fabrication processes reach a mature stage, the next step involves designing and analyzing innovative circuits. These circuits will be tailored for the specific applications of this thesis, primarily focusing on optical interconnects for CPO. However, the potential applications extend beyond this, encompassing areas such as sensing, AI, and quantum photonics. This comprehensive analysis will cover circuit performance in both the frequency and time domains and will involve the use of computational techniques to examine system performance metrics.

List of Publications

The complete list of publications by the author is provided below. The list includes 3 first authored journal publications and 7 first authored conference publications.

Journal Publications:

1. **Chatzitheocharis, D.**, et al. "Design of Si-rich nitride interposer waveguides for efficient light coupling from InP-based QD-emitters to Si_3N_4 waveguides on a silicon substrate." *Optics Express* 28.23 (2020): 34219-34236.
2. **Chatzitheocharis, D.**, et al. "Efficient multi-step coupling between Si_3N_4 waveguides and CMOS plasmonic ferroelectric phase shifters in the O-band." *Optics Express* 30.15 (2022): 26988-27003.
3. **Chatzitheocharis, D.**, et al. "Si_3N_4-plasmonic ferroelectric MZIR modulator for 112-Gbaud PAM-4 transmission in the O-band." *Optics Express* 31.19 (2023): 30847-30862.
4. Patsamanis, G., et al. "Design and Optimization of a Compact Ultra-Broadband Polarization Beam Splitter for the SCL-Band Based on a Thick Silicon Nitride Platform." *Photonics.* Vol. 9. No. 8. Multidisciplinary Digital Publishing Institute, 2022.
5. Simos, S., et al, "An All-passive Si_3N_4 Optical Row Decoder circuit for Addressable Optical RAM memories." *Journal of Physics: Photonics*, 2023.

Conference Publications:

1. **Chatzitheocharis, D.,** et. al. "Investigation of Silicon Nanophotonic Single-Mode Polarization Insensitive Waveguides", *XXXII Panhellenic Conference on Solid State Physics and Materials Science*, Ioannina, Greece, 2016 (Abstract and Presentation Only).
2. **Chatzitheocharis, D.**, et al. "Low-loss coupling interfaces between InP-based emitters and Si_3N_4 photonic integrated circuits." *Silicon Photonics XIV*. Vol. 10923. SPIE, San Francisco, California (USA), 2019.
3. **Chatzitheocharis, D.,** et al. "Directional Coupling Interface between Si-rich nitride and Si_3N_4 Waveguides Towards the Monolithic Co-integration of QD-InP and Si_3N_4 Photonic Components on Si", *European Conference on Integrated Optics (ECIO),* Online, 2020.
4. **Chatzitheocharis, D.**, et al. "Design of Vernier-ring reflectors in thick Si_3N_4 platform." *Integrated Optics: Devices, Materials, and Technologies XXV*. Vol. 11689. SPIE, Online, 2021.
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6. **Chatzitheocharis D.**, et al., "NEBULA project: Combining BTO based plasmonic modulators with Neuromorphic Augmented Receivers for higher than 100Gbaud Intra- and Inter- Data Center Interconnects", *19th International Conference on Nanosciences & Nanotechnologies (NN22)*, 5-8 July 2022, Thessaloniki, Greece (Abstract and Presentation only)

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17. Papadopoulou, M., et al. "Double-slot CMOS hybrid plasmonic waveguide-based Mach-Zehnder interferometric refractive index sensor", *XXXVII Panhellenic Conference on Solid State Physics and Materials Science*, Thessaloniki, Greece, 2023.
18. Chrysostomidis, T., et al., "Low-Power BTO on SiN MZI Weights for Neuromorphic Photonics", *49th European Conference on Optical Communications (ECOC)*, Glasgow, Scotland (UK), 2023 (Accepted for Oral Presentation).

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List of Abbreviations

AI – Artificial Intelligence

PIC – Photonic Integrated Circuit

CMOS – Complementary Metal Oxide Semiconductor

WDM – Wavelength Division Multiplexing

TDM – Time Division Multiplexing

SMF – Single-Mode Fiber

MZI – Mach-Zehnder Interferometer

RM – Ring/Racetrack Modulator

MZIR – Mach-Zehnder in a Ring

AWGR – Arrayed Waveguide Grating Router

EAM – Electro-absorption Modulator

Tx – Transmitter

Rx – Receiver

TxRx – Transceiver

LNOI – Lithium Niobate on Insulator

TFLN – Thin-Film Lithium Niobate

TMD – Transition Metal Dichalcogenides

EO – Electro-optic

BW – Bandwidth

TCO – Transparent Conductive Oxide

CPO – Co-Packaged Optics

SI - Système International

CGS - Centimetre–Gram–Second

Si – Silicon

SiN or Si_3N_4 – Silicon Nitride

BTO or $BaTiO_3$ – Barium Titanate Oxide

CPO – Co-Packaged Optics

SRN – Si-rich Nitride

CVD – Chemical Vapor Deposition

MBE – Molecular Beam Epitaxy

MIM – Metal-Insulator-Metal

HPW – Hybrid Plasmonic Waveguide

FDE – Finite Difference Eigenvalue

FDTD – Finite Difference Time Domain

FEM – Finite Element Method

OOK – On-Off Keying

NRZ – Non-Return to Zero

PAM – Pulsed Amplitude Modulation

QPSK – Quadrature Phase Shift Keying

QAM – Quadrature Amplitude Modulation

BER – Bit Error Rate

ER – Extinction Ratio

BW – Bandwidth

IL – Insertion Losses

QD – Quantum Dot

SOH – Silicon Organic Hybrid

POH – Plasmonic Organic Hybrid

SD-FEC – Soft Decision Forward Error Correction

HD-FEC – Hard Decision Forward Error Correction

DSP – Digital Signal Processing

TIA – Transimpedance Amplifier

FET – Field Effect Transistor

DOS – Density Of States

MMI – Multimode Interference Coupler

SOA – Semiconductor Optical Amplifier

AR Coating – Anti-Reflective Coating

HR Coating – Highly Reflective Coating

FP – Fabry Perot Laser

DBR – Diffractive Bragg Reflector Laser

DFB – Distributed Feedback Laser

SMSR – Single-Mode Suppression Ratio

LO – Local Oscillator

EVM – Error Vector Magnitude

LP-CVD – Low Pressure Chemical Vapor Deposition

PE-CVD – Plasma Enhanced Chemical Vapor Deposition

BEOL – Back-End Of Line

BCB – Benzocyclobutane

I/O – Input/Output Structure

GC – Grating Coupler

SSC – Spot Size Converter

SAE – Selective Area Etching

DLSPW – Dielectric Loaded Surface Plasmon Waveguide

SOI – Silicon On Insulator